



Rai Technology University

ENGINEERING MINDS

Corporate Finance



SYLLABUS

The Financial Environment

Financial Management: Objectives, Owner Wealth Maximization, Share Price Maximization, Earnings Per Share, Managers and Owner Wealth Maximization, Financial Management, Decisions: Balance Sheet, Asset/Investment Decisions, Net Worth/Financing Decisions, Income Statement, Consolidated Statement of Cash Flows, Financial Ratio Analysis, Profitability Ratios, Debt (Financial Leverage) Ratios, Free Cash Flows, Market Value and Stock Prices Versus Book, Value Organizational forms of Business and Corporate Income Tax: Forms, Advantages and Dividend Income Disadvantages, Finance in the Organization Chart, Corporate Income Tax, Corporate Interest, Dividend Income.

Financial Securities, Markets and Valuation Models

Net Present Value: Time Value of Money, Future Value Calculations, Present Value Calculations, Relationship between Present Values, Future Values and Rates of Return, Interest Factors; Present Value: Sequence of Sums, Annuity; Future Value: Sequence of Cash Flows, Annuities Due, Perpetuities; Net Present Value (NPV): Calculating Interest Rates, Single Payments, Annuities, Unequal Payments, Effective Annual Rate (EAR), Annual Percentage Rate (APR), Internal Rate of Return (IRR); Financial Securities; Common Stock: Classes and Prices, Preferred Stock, Debt Securities: Short Term and Long Term, Bonds, Treasury Bills; Convertible Securities Financial Markets: Markets, Institutions, Private Placements, Capital Markets, International Markets, Eurocurrency Markets, Eurobond Market; Interest Rates Basic Valuation Model: Basic Model, No-Growth Version, Constant Version; Market Efficiency Risk: Calculating Standard Deviation and Variance, Returns and Risk, Diversification, and Beta.

Investment Decision and Capital Budgeting

Investment Analysis: Cash Flows and Discount Rates, Depreciation System, Salvage Value, Book Value and Taxes, Sunk Costs, Mutually Exclusive Investment Opportunities, Break-Even Analysis, Project Proposals; Capital Investment Decisions: NPV, Capital Budgeting Methods, IRR, NPV Profiles, and Payback Methods.

Financing Decisions

Financial Structure: The Product and the Labor Market, Stockholders and Bondholders, Creditors, Agency Costs; Financial Distress: Reorganization and Bankruptcy, Out-of-Court Arrangements, Ethics and Bankruptcy; the Cost of Capital: Cost of Debt, Yield to Maturity; Cost of Equity Capital: Preferred and Common Stock, the SML or Beta Approach, Weighted Average of Cost of Capital (WACC), Financial Leverage, EPS/EBIT Analysis Dividends: Types of Cash Dividends, Share Repurchases, Transactions Cost, Taxes, Dividend Policy, Undervalued Stock, Free Cash Flow Theory, Stock Dividends and Stock Splits, Dividend Reinvestment Plans.

Managing Capitals

Financial Ratios: Ratio Analysis, Liquidity Ratios, Activity Ratios, Accounts Receivable Ratios, Inventory Turnover Ratios, Fixed Asset Turnover Ratio, Debt Management Ratios, Total Debt to Total Asset Ratio, Interest Coverage Ratio, Fixed Charges Coverage Ratio, Profitability Ratios, Gross Profit Margin, Net Profit Margin, Return on Investment, Return on Owner's Equity, Market Ratios, Dividend Payout Ratios, Price to Earnings (P/E) Ratio, Market to Book Value Ratio, Dividend Yield; Working Capital Management: Components, the Cash Conversion Cycle, the Level of Current Assets, Working Capital Financing Strategies, an Integrated Financing Policy; Basic Principles of Bank Financing: 3 Cs (Character, Capital, Capability), Tandon Committee and Chore Committee Guidelines, CAS form of Application, Computation of Maximum Permissible Bank Finance (MPBF).

Suggested Reading:

1. Essentials of Business Finance, Srivastava, Reddy, Appannaiah Himalaya Publishing House.
2. Business Finance, Shashi K.Gupta and RK Sharma, Kalyani Publishers.
3. Financial Management, MY Khan and PK Jain, Tata McGraw Hill.
4. Financial Management, Prasanna Chandra, Tata McGraw Hill.
5. Working Capital Management Strategies and Techniques, Hrishikes Bhattacharya, Prentice Hall.
6. Security Analysis and Portfolio Management, Donald E. Fischer and Ronald J. Jordan, Prentice Hall.
7. Financial Management, PV Kulkarni and BG Satyaprasad, Himalaya Publishing House
8. Marketing of Financial Services and Markets, VA Avadhani, Himalaya Publishing

COURSE OVERVIEW

- a. As all of us are aware Finance plays a vital role in the every day life of individuals, businesses – be it small or a giant corporate firm – and Government. Finance is all the more important in the modern Corporate world. Ultimately, Finance is the very reason of the existence of the Corporate firms which normally involves various stakeholders. It pays to master Finance as a profession, discipline and subject in every walks of life, apart from enabling faster career growth.
- b. To achieve the above, a student is expected to go through the following and the outcomes are also as summarized below:

Summary of Outcomes

To achieve this unit a student must:

- Identify management problems in **the financial environment** and describe framework for tackling them
- **Explain manager's role on owner wealth under** financial securities, markets and valuation models
- **Evaluate projects for** investment decisions and capital budgeting **to provide economic returns**
- **Examine** financing decisions **on sourcing to avoid conflicts of interest between different stakeholders**
- **Construct financial ratios and strategise working capital management for** managing capitals

Description of Unit

The aim of this unit is to introduce the students to corporate finance, which explains how financial theory informs financial management and the development of financial policies. An elementary approach is used in the module, which emphasises the reasons for undertaking financial analysis and their interpretation. An international perspective is maintained throughout the text. The presentation of financial concepts, problems and cases is supported with examples drawn from the financial press and corporate annual reports. The objective of the problems is to enhance the student's understanding of analytical techniques- more emphasis on policy and managerial implications of financial management and de-emphasis on formula and quantitative solutions to managerial problems. The module focuses on financial skills required to become an effective manager. It will develop the student's ability to identify and appreciate effective ways of getting the objective fulfilled. It also develops practical applications ability and knowledge and how these can be used in the decision-making process. The course will cover the financial environment, financial securities, markets and valuation models, investment decision and capital budgeting, financing decisions, and managing capitals

CORPORATE FINANCE			
CONTENT			
	Lesson No.	Topic	Page No.
THE FINANCIAL ENVIRONMENT			
	Lesson 1	Organisational Forms of Business	1
	Lesson 2	Organization Chart & Finance Function	5
	Lesson 3	Finance Function – Organization, Objectives & Decisions	7
	Lesson 4	Financial Statements, Analysis and Information	10
	Lesson 5	Cash Flow Statement	15
	Lesson 6	Corporate Income Tax	19
FINANCIAL SECURITIES, MARKETS AND VALUATION MODELS			
	Lesson 7	Time Value of Money	21
	Lesson 8	The Financial Securities	25
	Lesson 9	The Financial Markets	29
	Lesson 10	International Financial Markets	33
	Lesson 11	Financial Institutions	36
	Lesson 12	Interest Rates	41
	Lesson 13	Basic Valuation Model	44
	Lesson 14	Risk	47
INVESTMENT DECISION AND CAPITAL BUDGETING			
	Lesson 15	Investment Decisions Rules	52
	Lesson 16	Cash Flow	54
	Lesson 17	Incremental Analysis	56
	Lesson 18	Break - Even Analysis	58
	Lesson 19	Project Proposals	61
	Lesson 20	Capital Investment Decisions	63
	Lesson 21	Capital Budgeting Techniques - NPV	66
	Lesson 22	Capital Budgeting Techniques – IRR	69
	Lesson 23	Capital Budgeting Techniques – Other Methods	72
FINANCING DECISIONS			
	Lesson 24	Financial Structure – Capital	75
	Lesson 25	Factors effecting Financial Structure	78
	Lesson 26	Financial Distress – Remedial Options	80

	CORPORATE FINANCE		
CONTENT			
	Lesson 27	Corporate Restructuring Options	84
	Lesson 28	Cost of Debt	89
	Lesson 29	Cost of Capital	91
	Lesson 30	Financial Leverage	95
	Lesson 31	Dividend Policy	98
	Lesson 32	Forms of Dividend	104
	Lesson 33	Dividend Models & its Decisions	107
MANAGING CAPITALS			
	Lesson 34	Financial Ratios	111
	Lesson 35	Working Capital Management	116
	Lesson 36	Bank Credit	121

LESSON 1 : ORGANISATIONAL FORMS OF BUSINESS

Learning Objectives

- To understand various types of business organizations in existence, namely proprietorship, partnership and joint stock companies and public enterprises
- To know basic nature of these organizations and their advantages and disadvantages.
- To compare these different organizations from management point of view.

Introduction

Hello Students,

We shall recollect and look at certain basic things that you might have learnt earlier. This we shall do to be clear about the context in which we will be dealing with the subject 'Corporate Finance'.

Business is the activity with the object of earning an income through profit. As you know, business activities can be production and sale of goods & services or purchase & sale of goods & services. Now you can group these activities into: Commerce, Industry and Service.

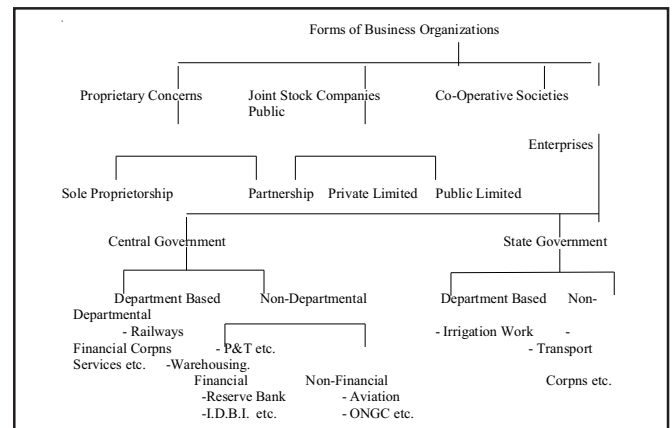
- Commerce involves the activities in the movement of goods and services from the producer to the consumer. It includes procurement, grading, storing, transporting commodities through the wholesalers, retailers etc.
- Industrial activities involve manufacture or some definite treatment of materials from which marketable commodities can be manufactured.
- 'Services' involve intangibles. It can be professional services like in case of doctors, lawyers etc. or it can be other services like transport etc.

Forms of Business Organizations

Having understood the overall view of business activities, you need to relate these activities with how people organize these activities. It will lead to a better understanding if you relate the circumstances in your family business or with the other real life situations where you may be having at least minimum interactions.

A business unit may start as a sole proprietorship (one man ownership) and gradually expand into partnership firm and then into private limited company and into a big public limited company and finally into a giant business house having a number of subsidiary companies. A particular form of business adopted by a business unit at a particular point of time depends upon the various factors like tax advantages, capital requirements, owners' preferences etc.

You may try to visualize each organizational form of business & relate to the business unit known to you, as I explain them one by one. First the overall picture:



Proprietary Concerns

You might have heard people talking about business of a person or a person being a partner in a business. When a business owned by a person or co-owned by few persons we may call it as Proprietary concerns. Full ownership & control of the business under this category remains with one or few individuals. The business or firm's name does not have any legal status or existence.

Sole Proprietorship

This form of business is fully owned by one person and this single person runs the show, may be with the help of family members or relatives or may be employing one or more persons to assist.

Any person who is competent to enter into contract can start a sole proprietary concern. As per law a person, who has attained the age of majority, of sound mind and who is not disqualified (for a period) by insolvency law or any other law, can enter into a contract.

The sole proprietor is entitled for all the profits and shall also bear all the risks and losses.

Advantages

1. Easy formation with formal license only
2. Freedom available to the sole proprietor to take action enables quick decisions and prompt action. There is no need to consult anyone.
3. Personal contacts with customers & employees
4. Business secrecy can be maintained and hereditary business is enabled
5. There is direct relationship between the economical & efficient effort and rewards. Hence costs are also controlled effectively.
6. Since the business is conducted alone or with a tight family structure, there is no necessity for formal organizational structure and hence absolute flexibility is there.

7. Business Tax procedures are relatively simple with all income from business getting reported in the owner's individual tax return. The owner can take maximum benefits of tax concessions given for individuals

Disadvantages

1. The capital structure (capital and borrowings) of the business is limited to personal wealth and credit standing of the proprietor.
2. Limited managerial and technical skills
3. The owner is responsible for all the debts and the liability is unlimited dipping into personal assets like house property, vehicles, furniture, appliances etc.
4. Since the business is dependent upon the health & wealth and life of the owner, there is no continuity.

This form of business is suitable where personal skills & contacts are important.

Partnership

A Partnership is an association of two or more persons (maximum of 10 in case of banking and 20 in case of other businesses) for carrying out a business as co-owners for a profit. In India, Partnerships are governed by the provisions of The Indian Partnership Act, 1932 which defines partnership as "the relation between two or more persons who have agreed to share profits of a business, carried on by all or any of them acting for all." The term business includes all trades, occupation and professions.

The owners of a partnership business are individually called partners and collectively called as firm. The name of the business is called 'firm name' which has no legal status though it might accumulate reputation or build image resulting in goodwill.

Usually a partnership is formed to combine capital, labour and varied specialised skills and abilities through an agreement or deed called partnership agreement or deed which might set out various terms including sharing of profit.

Partners can be sued individually and jointly in respect of the business and the liability is unlimited. So each partner is a principal for the outside world and each partner is an agent to other partners.

Advantages

1. Increased Capital Raising Power and Managerial & Technical expertise compared to sole proprietorship.
2. Easy formation compared to joint stock companies & easy dissolution as agreed by all the partners or 14 days notice to other partners in case of partnership at will.
3. Business Secrecy can be maintained since there is no requirement by law for publication of final accounts

Disadvantages

1. Partnership has a less capital raising power and Managerial or Technical expertise as compared to joint stock companies.
2. Unlimited liability renders the partnership unsuitable to take up large scale operations.

3. Absence of separate legal status for the business and continuity

4. The partnership is not transferable unless all the partners agree for the same

In Europe and America, there is a system of Limited Partnership wherein two types of partners – General and Limited / Special Partners - will be there. The liability of the Limited / Special Partners are limited to the extent of their investment only whereas the liability of General partners is unlimited whose personal assets are also liable for the actions of the business.

Joint Stock Companies

If you are wondering how to have limited liability, let me tell you that the answer is Joint Stock Companies.

A company is an artificial person (an association of minimum 7 natural persons) created by incorporation under law, having common seal and perpetual succession. Once it is created by law, a company gets separate identity and it can enter into legal contracts under its common seal. An incorporated company enjoys perpetual (continuous) life until the law itself winds up the company in accordance with law.

The capital is raised in the form of shares and called share capital. Share means a portion of capital raised by issue of shares. The liability is limited to the amount invested in shares. The shares are generally transferable except under certain circumstances when the transfer may be restricted. In view of limited liability and transferability of shares, large number of persons may come forward to invest, enabling raising of enormous capital. Since a company is an artificial person, operating powers get vested with the elected board of directors and gets delegated mainly to the managing director and the company secretary. In this way most of the shareholders are kept away from the day-to-day operations of the organization. Ultimately, functional experts are employed to manage and carry out the activities. This results in experts doing the respective job and performance gets enhanced.

Advantages

1. Huge capital mobilization
2. Facility of Transfer of ownership through transfer of shares
3. Wide distribution of risk of loss with large membership and limited liability
4. Comparatively lower tax liability

Disadvantages

1. Excessive legal requirements at all stages from the time of formation.
2. In practice fraudulent management and concentration of economic power & wealth are found
3. Slow Decision making
4. Double Taxation – once on the profit of the company and another in the hands of the share holder

If you further want to know how to have limited liability and still have advantages of partnership firm, the answer is in the form of Private Limited Companies.

Private Limited Companies

A private limited company like partnership should have minimum of 2 members and ceiling of 50 members is there unlike no ceiling in public limited company.

In case of private limited companies, there is no public issue of shares and the transfer of shares is restricted. The name of the company will include the words 'private limited'.

This form of business organization enjoys the benefits of lower corporate tax, limited liability and continuity.

Market Value and Stock Prices versus Book Value

Share capital of a joint stock corporation is raised by issue of shares which have a face value. Section 2 (46) of Indian Companies Act, 1956, defines share as "share in the share capital of the companies and includes stock, except when a distinction between stock and share is expressed or implied."

When shares have been fully paid-up they may, if so authorized by the articles, be turned into stock by the company in general meeting (Section 94 (i) C). Stock is merely a name for the aggregate ownership. This is the most convenient method of representing ownership and makes the transfer of ownership easier.

The price at which a share is bought and sold in the stock market is called the Market Price which may be higher or lower than the face value of the share. Whereas Book Value of share is the value arrived at by dividing the Owners' Equity (Share Capital + Reserves & Surplus) by Number of shares.

Co-Operative Organisations

You are aware of the importance of co-operation and team spirit in every sphere of life. One form of business organization which truly practices co-operative efforts is Co-operative organizations.

The Co-operative organizations are formed on the basis of equality for the promotion and furtherance of common economic interest.

The International Labour Organisation had defined a Co-operative organization as "an association of persons who have voluntarily jointed together to achieve a common economic end through the formation of a democratically controlled organization, making equal contributions to the capital required and accepting a fair share of risks and benefits of the undertaking".

Service, Co-operative joint action and Self-help are the essence of a co-operative organization. Co-operative Societies must have at least ten members and must be registered under Co-operative Societies Act'. Examples of Co-operative organizations are Credit Co-operative societies, Consumers' Co-operative organizations, Producers' Co-operative organisations

Advantages

1. This is a democratic and secular form of business organisation. Poor and Persons of limited means can improve their economic conditions
2. Co-operative organizations remove the evils of monopoly and concentration of wealth and power

3. The liability can be limited and separate legal status may be enjoyed.
4. Privileges and exemptions in the form of tax concessions, lower stamp duty etc., are accorded to this form of business organisations.

Disadvantages

1. Co-operative organizations can not raise capital for large scale operations
2. Since the very basis is democratic way of managing, business secrecy can not be maintained.

Public Enterprises

Public enterprises aim at public service in the public utility areas. They are expected to stabilize the supplies of essentials. The strategic and basic industries are taken care by Government only. Public sectors are there in the industrial line to have rapid and balanced economic development.

I am sure that you would be able to imagine the evils, if only non-governmental organizations whose aim is only profit maximization, are there.

Public Enterprises are managed in the following ways:

1. Government Department
 2. Public Corporation
 3. Government owned Joint Stock Company
 4. Mixed-ownership Corporations
1. Government Department – A separate department in the government manages this form of organization which normally caters to the public utility. Examples of this type of organizations are Post and Telegraph, Broadcasting, Railways and Defence industries etc. The minister in-charge of the particular department controls the enterprise. This form is adopted where secrecy and strict control are needed in the areas like defence and strategic industries.
 2. Public Corporation – A public corporation is an artificial legal person created by a special Act which sets out the powers and functions. Public Corporations have financial independence and clear jurisdiction over a specific area, industry or commercial activity. A chosen Governing board manages the affairs of the organization. It enjoys the internal autonomy with no interference from government. Not only its formation is elaborate, any amendment also to be done by amending the Special Act and hence rigid. Public corporations are presently found in insurance, finance, industry, mining, transport, trade etc.
 3. Government owned Joint Stock Company – The entire capital or 51 per cent or more of the share capital is owned by the Central and/or State Government. The concerned ministry performs the functions of shareholders and exercises control over the operations of the company. All or majority of the directors are nominated by the Government. One or few shares are in the name of the officials to take care of legal needs. This form of organization is more autonomous than Public Corporation
 4. Mixed ownership corporations – This is a form of partnership between public sector and private sector

enterprises. These are in the form of Joint Sector / Venture. It is a partnership among domestic partner, government and public. Additionally there may be one more player in the form of foreign collaboration. This enables to bring together capital, technology and skill from abroad, private sector and government. Hindustan Machine Tools is an example of Joint Sector.

Let us now compare various forms of organizations in respect of aspects of our interest:

Comparison among Different forms of Organisations

Proprietary Concerns	Joint Stock Companies	Co-Operative Societies	Public Undertakings
Role of Capital			
Limited to personal capacity	Vast	Secondary Role	No equity interest of shareholders
Ownership & Control			
Proprietor or Partners	Shareholders	Members / Shareholders	Government
Transfer of Ownership			
Consent of all owners required	Shares are freely transferable in case of Public Ltd. Co.; restricted in case of Pvt. Ltd. Companies	Membership can be withdrawn	Transferable in case of joint stock company form
Management			
Limited managerial ability	Expert Management; ownership separated from management	Democratic management with the rule 'one member one vote'	By Government or Governing body

Points to Remember

- Basically the nature, volume of the business and attitude of the business owner/s determine the form of organization.
- Market price of shares of a limited company is highly influenced by the performance of the company, attitudes of the investors and strategies adopted by the competitors.
- Non profit organizations are association of persons, trusts, societies undertaking activities like fine arts, sports etc., not for profit motive.

Review Questions

- 1 Write short notes on
 - i. Sole Proprietorship concern
 - ii. Partnership concern
 - iii. Joint stock companies
 - iv. Co-operative organisations

v. Public enterprises

2. Under what circumstances do you think a sole proprietorship concern gets evolved in to partnership and then to joint stock company.
3. Bring out a comparative study of different forms of organisations

Note

LESSON 2 : ORGANIZATION CHART & FINANCE FUNCTION

Learning Objectives

- To know the various types of organisation structures
- To know about staff and line functions / relationships and authority
- To know the place of finance function in an organisation

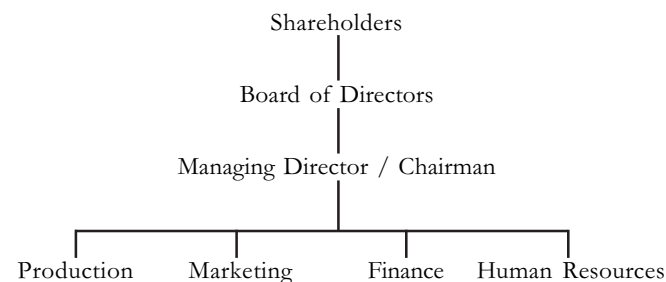
Introduction

Students,

As you are aware all the persons in an organization cannot perform all the activities. The activities need to be organized for performance by the persons in the organization. All the activities in an organization may be divided, grouped and co-ordinated in three major ways:

Functional Organization

Under functional organization, similar or related activities are grouped together as functions. The functions may be Production, Marketing, Finance, Human Resources. It is a traditional organization method, normally adopted by smaller firms or firms with a limited line of products. Functional organization follows occupational specialization.



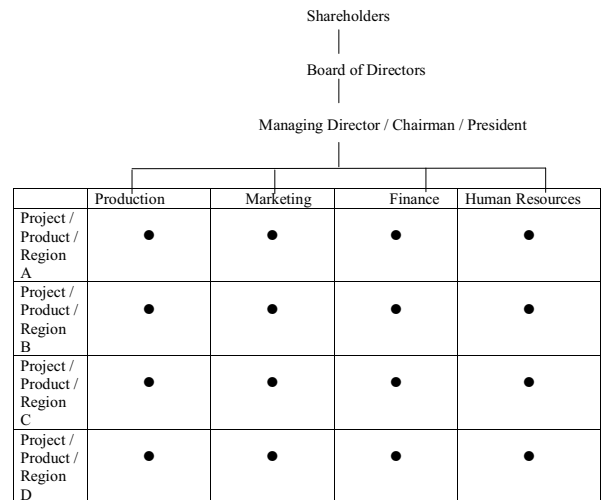
The disadvantages in this type of organizational structures are found when the organization grows geographically or in product lines. It is likely that the functions may develop narrow view of their department only making it difficult for the higher level to co-ordinate.

Product / Market Organization

Under this type of organization structure, activities may be grouped in to product wise or region-wise apart from the core functions as above.

Matrix Organization

Matrix organization is a combination of both functional and product / market organization in one organization structure. Under this organization structure a person will have two bosses - one functional boss and another product / market boss



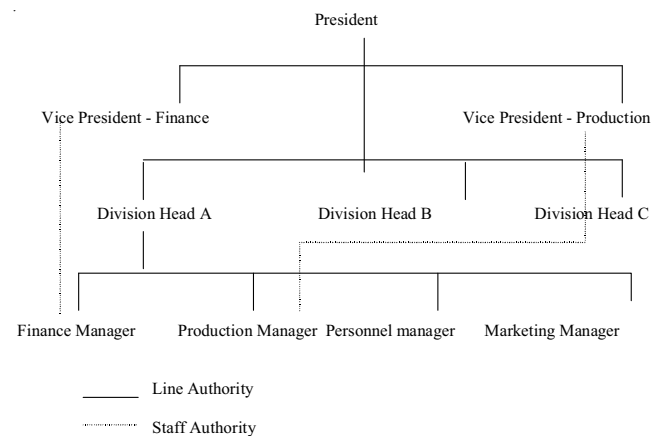
Matrix organization solves the complex problem. Moreover it employs only the number of people required only avoiding duplication.

Line and Staff functions / Relationship / Authority
The line functions are those which are direct responsibility for achieving the organizational goals. Production and Sales (and sometimes finance) are classified as line functions.

The staff functions are those who provide service & help and advice the line to work most effectively in accomplishing the objectives of the enterprise.

Functional and Line Authority

In Line authority, the superior has legitimate power over the subordinate. The nature of the staff relationship is advisory. The authority in this case only out of functional expertise and hence called Functional Authority.



Students,

1. Finance is concerned with flow of cash. The emphasis is on the liquidity aspect. All the policy decisions, objectives and activities / transactions are ultimately expressed in financial terms – present or future.
2. Finance means acquiring / arranging funds for payment of bills
3. Finance is concerned with procurement of funds and their wise or effective use or application

The finance functions can be grouped as below:

These are the finance functions which are carried out in the regular conduct of the business.

- Deciding in advance the quantum of fund requirements and its duration, to achieve the primary objective of the organization
- Deciding the working capital reserve
- Forms in which the total required funds are to be raised gets reflected in capital structure

- Issue of securities using stock brokers, if desired and underwriters, if required.
- Preparation of project plan & negotiating with financial institutions, for borrowing money

- Considering competing uses, immediate requirements, profit prospects and overall management plans
- Capital expenditure in fixed assets, only to meet the current as well as the increased demand for the company's product.
- Efficient administration of financial aspects of cash, receivables and inventories.
- Control of funds, comparing the actual with the plan / estimates and actions for deviations

- To retain the income for financing expansion or to distribute to the owners as return on capital in the form of dividends
- The portion of earnings to be distributed as dividend considering the share price to be maintained, financial position of the company, cash requirements, shareholders preferences etc.

These are the finance functions which are to be performed very rarely i.e. while floating the organization, during liquidity crisis or at the time of merger or reorganization of the organization etc.

- Organisation structure should be flexible to accommodate the changing nature of the business, its expansion, diversification etc.
- Strategic Business Units (SBUs) of an organization function independently as separate profit centers depending upon the degree of decentralization.

1. Explain the nature of finance function in an organisation
2. Highlight the various basic finance functions of a manufacturing company

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

LESSON 3 :

FINANCE FUNCTION – ORGANIZATION, OBJECTIVES & DECISIONS

Learning Objectives

- To know how corporate finance is related to other functions
- To obtain an idea of how finance function gets organized
- To know the main objectives of financial management
- To study financial decisions to be normally taken in a concern.

Corporate Finance

Students,

I have given required basic things to you before we take on the subject in more detail.

Corporate Finance deals with the financial aspects of a corporate enterprise, right from the stage of promotion of the organization. It includes administration during initial stages, stages of growth and expansion. It also deals with financial restructuring during financial crisis.

Corporate Finance includes financial environment, strategies of financial planning, financial markets, capital formation & financial institutions.

Corporate Finance & Other Functions

Finance is omnipresent and is intimately associated with other functions in an organization.

Finance is related to marketing in formulating credit and collection policies. Each aspect of marketing function – sales promotion, advertising, channels selection etc. – has a bearing on finance as well.

In the purchase function, procurement cost, inventory carrying cost etc. are the financial aspects.

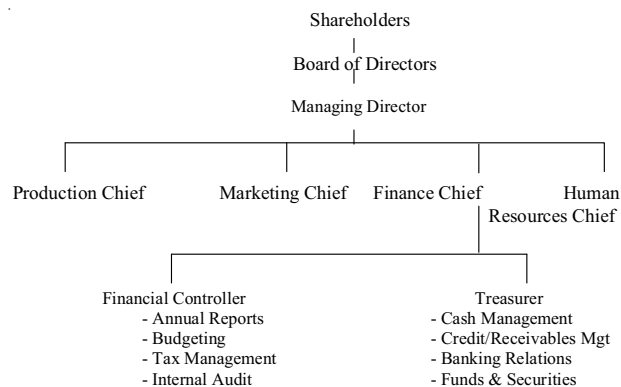
In Production function also, finance plays an important role when decisions like make or buy, maintenance cost, transportation cost, acquisition & replacement of assets etc. Production function takes the greater part of capital investment in fixed assets which should pass through capital budgeting.

As regards personnel function, decisions like over-time payment, incentive schemes, promotions, bonus schemes, retirement schemes, affect finance.

Each activity of an organization has to be financed. The thread of finance is woven into the fabric of the enterprise so well that it is difficult to identify the separate strands of finance.

Organization of the Finance Function

You are aware about organizing the activities of an organization as well as organizing the activities within a function. Since Finance function is very vital for every type of organization, it is necessary to set up a sound and efficient organization. The organization varies depending upon the nature, size of the organization, Though we can not have a standard organization for all the enterprises, we can structure the basic aspects for a corporate as below:



Share holders determine the following while approving the Articles of Association and the bye –laws:

1. The amount and kind of capital
2. Rules governing issue and transfer of stock
3. Powers of directors to declare a dividend, choose a bank, create reserves
4. Sale of firm's assets
5. Plans for reorganization, liquidation, consolidation and mergers etc.

Board of Directors approve the financial policy, declare dividends and translate the aspiration of shareholders into specific goals and objectives and select & appoint the senior officers.

The Controller is concerned primarily with the Planning, Accounting and Control activities. The Treasurer is responsible mainly for financing, management of cash & receivables and investment activities

The organization of financial function depends on the needs of each organization. The financial needs of each organization varies from the other. In some organization working capital requirements may be more, as in the case of manufacturing organizations as compared to service organizations. In some organization the gestation period may be longer before they start generating revenues and earn profits.

I am sure you are able to appreciate the interesting turn the subject is taking now.

Objectives of Financial Management

Financial objectives of a firm must be set within and in congruence with the Corporate Mission & Goals of the business. Most of the organizations, irrespective of the form, seek to maximize immediate Profit for survival and growth. This provides immediate return also to the share holders in the form of better dividend and it increases the value of the share in the stock market. Many firms look for improving the value

of the firm including profit maximization, as the objective. So, essentially we have two major financial objectives:

1. Profit Maximization
2. Wealth Maximization

Profit Maximization

Share holders have invested their money in the organization losing the alternative investment opportunities in anticipation of getting better return here. Other creditors also lent their money in anticipation of getting interest on their money in time. The organization can meet their expectations only when it maximizes its profits.

Profit earning is the main aim of every economic activity. Profits are required to cover the costs & survive, to run the business smoothly taking care of contingencies and to provide funds for growth. Profits serve as the measure of efficiency of an organization and indicate the risk bearing ability during unfavourable times.

The objective of Profit maximization is in congruence with most of the businesses. It helps to maximize the socio-economic welfare.

The following are the drawbacks of Profit Maximization as the objective of finance:

- i. The term 'Profit is vague and is not clear. It means different things. It is not clear as to profits before tax or after tax or total profit or earnings per share. It leaves considerations of timing and duration undefined.
- ii. Profit maximization objective ignores the time value of money.
- iii. It does not consider the element of risk
- iv. It exploits employees, customers and others

Wealth Maximization

Wealth maximization, also called Value Maximization, is the appropriate financial objective of an enterprise since maximizing the value of the firm benefits all the stake holders in the organization - the shareholders, employees, customers, creditors, management and the society.

Wealth Maximization implies the maximization of the market price of shares. The share's market price serves as a performance index. The stock holders wealth is the number of shares held by the stock holders multiplied by the stock price per share. Since higher the price per share means greater as the wealth, the firm should aim at maximizing the current stock price.

Value maximization is simply the extension of profit maximization to a world that is uncertain and multi-period in nature. Value maximization may be achieved through increase in profits, reduction in cost, judicious choice of funds, minimum risks, long-run value approach.

Advantages of Wealth Maximization Objective are:

- It considers the present value of cash in flows & outflows and the net effect of investment and benefits.
- It considers the risk factor by considering discount rate while calculating net present value.

Financial Decisions

You must be aware that basically management is concerned with Decision making. Management of finance function or Financial Management is concerned with financial decisions.

A business may require funds either for funding the long-term assets which yield return over a period of time in future or for day-to-day operations. However the total fund requirement is to be looked at. When the total funds requirement is being considered, it becomes the Investment Decision to be taken. When the sources of finance are tapped for funding the investments, it becomes the Financing Decision. When the profits, as a result of Investment and Financing decisions, are considered for ploughing back into the business or for distributing as dividend, it becomes the Dividend Decision.

Investment Decision

Investment Decision involves deciding about investments to be made in the assets - long-term assets or short-term or current assets which are normally convertible into cash within a year. Deciding about the long-term assets involve Capital Budgeting and the short-term or current assets involve Working Capital Management.

Capital Budgeting

The investment proposals are appraised in terms of relative benefits & returns and magnitude, timing, risk / uncertainty of cash flows associated with it. The standard broadly expressed in investment appraisals is the cost of capital.

Working Capital Management

Working Capital Management or short-term financial management refers to the day-to-day financial activities involving current assets (inventories, trade debtors, short-term holding of marketable securities, and cash) and current liabilities (short-term debts, trade creditors and accruals). It is important because short-term survival is pre-requisite for long-term success.

A proper balance needs to be maintained between liquidity and profitability. Inadequate funds for day-to-day activities renders illiquid and more funds kept for day-day-day activities means not investing the surplus funds in more profitable options. I am sure that you carry money only to the extent required for daily expenses. None of us keep huge amounts in cash form.

Financing Decision

Financing decision is concerned with identification of various sources of finance for funding the investment decisions. Long-term / Capital assets are financed through long-term funds and the current assets are financed through short-term funds. Funds are raised from primary market, financial institutions and commercial banks

Capital Structure

Capital Structure refers to the proportion of Debt and Equity capital. A proper balance between debt and equity (optimal capital structure) is required to ensure a trade-off between risk and return with maximum returns to the shareholders.

Dividend Decision

Declaring more dividend will enhance the image of the enterprise in the stock market, instill confidence in creditors etc. but will reduce the opportunity to plough back the income

which may be required considering the cost of capital of other sources of financing. Optimum dividend pay out ratio needs to be arrived at considering the preference of shareholders and the investment opportunities available.

Points to Ponder

1. Each function in an organisation claims superiority. What are the points favouring finance function to claim superiority.
2. Do you think two posts of Financial Controller and Treasurer are appropriate in the Indian Context?

Review Questions

1. What are the main functions of Financial Treasurer and Controller?
2. Why do you think Wealth Maximization, is the appropriate financial objective of an enterprise?
3. What are the major financial decisions to be taken in a trading concern? Explain.

Note

LESSON 4 :

FINANCIAL STATEMENTS, ANALYSIS AND INFORMATION

Learning Objectives

- To study the general format of balance sheet and profit & loss account
- To know the importance of financial analysis and how to interpret the same.
- To study various tools/methods of financial analysis
- To learn about funds from operations

Introduction

Students,

You have already studied the basics of financial accounting. As you are aware financial accounting ends up with preparation of final accounts consisting of Balance Sheet and Profit & Loss Account. The final accounts along with cash flow statements form the major financial statements.

Financial Statements are often an important source of information for financial decisions. Analysis of Financial Statements of an organization - which includes Balance Sheet and Profit & Loss Account gives more information base for various financial forecasting and decision-making. Preparation of cash flow statement displays the sources and uses of cash during the period.

Major topics of financial management may be related to the financial statements.

Financial statements emerging from recorded facts of business transactions need to be studied and analysed in the light of accounting conventions and assumptions adopted and the justifications for adopting them.

Balance Sheet

The Balance Sheet shows the financial condition of the business as on a particular date. It shows what an enterprise owns and owes. It provides information for assessing Profitability, Liquidity and Solvency. While the profitability is the overall measure of financial condition, Liquidity & Solvency are concerned with assessing the ability of the enterprise to meet its liabilities.

Balance Sheet is a statement of assets and liabilities & capital as on a particular date. It is a static statement or snap shot of resources and allocation of resources to assets. Assets indicate the uses to which the finance of the firm is put. Liabilities show the claims against the assets of the firm. Balance Sheet is a detailed statement of the financial structure of the business. The value of assets are computed on a conventional basis i.e. fixed assets at cost less depreciation and current assets at cost or market value, whichever is less.

Balance Sheet items are related to Finance in the following manner:

Liabilities

Share Capital

- Equity
- Preference

Reserves and Surplus

Secured Loans

- Debentures
- Loan and Advances

Unsecured Loans

Current Liabilities and Provisions

- Trade Creditors
- Provisions

Assets

Fixed Assets (Net)

- Gross Block
- Less Depreciation

Investments

Current Assets, Loans and Advances

- Cash and Bank
- Receivables
- Inventories

Miscellaneous Expenditures & Losses

Corporate Finance Topics

Capital Structure

Cost of Capital

Working Capital Financing

Capital Budgeting

Security Analysis

Cash Management

Receivables Management

Inventory Management

Profit and Loss Account

You must be remembering Profit and Loss Account gives the results of the operations of the firm during a specific period, in the form of profit or loss. Profit and Loss Account gives more detailed report on the retained income shown in the Balance Sheet. By comparing the income statements of successive periods, it is possible to observe the growth or otherwise of the firm

Items of Profit & Loss Account may be related to Corporate Finance Topics in the following manner:

Items of Profit & Loss Account Corporate Finance Topics

Net Sales

Revenue Risk

Cost of Goods Sold

- Stocks
- Wages and Salaries
- Other Manufacturing expenses

Gross Profit	Gross Profit Margin
Operating Expenses	
• Selling and Administration Expenses	
• Depreciation	Depreciation Policy
Operating Profit	
Non-operating Surplus / Deficit	
Profit Before	Business Risk and Reward
Interest and Tax	
Interest	Financial Risk
Profit Before Tax	
Tax	Tax Planning
Profit After Tax	Return on Equity
Dividends	Dividend Policy
Retained Earnings	

Financial Statements - Analysis & Interpretation

You are aware that the main objectives of final accounts are ascertainment of profit and financial soundness of the firm. These financial Statements are helpful in decision-making to a great extent. They help us to understand the financial position and profitability of the business. Analysis of financial statements enables more relevant information for decision-making. Understanding the past is a pre requisite for anticipating the future. Analysis of financial statements means treatment/s of any / all of the following:

1. Breaking financial statements into simpler ones
2. Classifying & comparing the components and establishing the significance of figures
3. Study of relationship among the various financial factors in a single set of statements
4. Study of trends of financial factors as shown in two or more set of statements
5. Regrouping and rearranging of figures.

Analysis is the process of fact-finding and breaking down complex set of facts or figures into simple elements.

Interpretation means grasping the sound and unsound relationships and explaining the real significance of the simplified components.

When the outside users of financial statements analyse, they will have to rely only on the information contained in the published annual reports and financial statements. This External Analysis is of limited purpose since the external people do not have easy access of detailed accounting records and information.

When the financial statements are analysed and interpreted by the internal persons of the organization, they have an easy access to the detailed records as well. They use other managerial reports also in order to formulate policies, establish objectives, evaluate performance and take corrective actions and appropriate actions i.e. Planning and Control. The analysis helps in assessing internal matters of efficiency and external matters of investment etc.

The object of the analysis determines the extent, depth and nature of analysis and varies depending upon the user – Management, Creditors, suppliers, investors etc.

Tools / Methods of Financial Analysis

1. Comparative Financial Statements
2. Common Size Statements
3. Projected Balance Sheet
4. Trend Analysis – Time Series, Index Analysis, Cross-sectional Analysis
5. Statement of Changes in Working Capital – Fund Flow / Cash Flow Analysis
6. Ratio Analysis

I shall explain each one of these tools one by one.

1. Comparative Financial Statements

This is an important device of horizontal financial analysis. Under this method of financial analysis, figures of two or more periods are placed side by side and compared and the magnitude and direction of changes in terms of absolute figures and percentages are ascertained. The exercise can be with figures of Balance Sheet or Income Statement.

When the figures of two firms are compared in this manner, the exercise is called inter-firm comparison.

Recognizing the importance of comparative financial statements, the entire corporate sector show the previous year figures also in the annual report as made compulsory under the provisions of the Companies Act, 1956. Comparisons can be made for lesser periods also like quarter-wise, month-wise etc.

To be comparable, consistent accounting principles, policies, practices and procedures should have been followed.

2. Common Size Statements

Since the absolute figures vary from period to period and from enterprise to enterprise, comparison can be made after converting them to common size.

The Balance Sheet can be converted to common size by converting each item on the assets side as a percentage of total assets. Similarly each item on the liabilities side is converted as a percentage of Liabilities and Capital Fund.

The Profit and Loss Account can also be converted to common size by converting each item of the Profit and Loss Account, including Profit, as a percentage of Sales. With this we will be able to ascertain the extent of influence of each item on the profit and the extent of profit on Sales.

The Common Size Statements can be prepared for different periods or different firms, and compared. It is necessary that the accounting policies & practices are same in the periods / firms under comparison.

The results are interpreted and used in taking decisions.

3. Projected Balance Sheet

Based on the projected sales, we can project the Balance Sheet. The projected assets can be reviewed as to whether there is sufficient liquidity is there.

Projected Total Assets when compared with the total debt and equity, give the extra funds required. While looking at the extra funds required, possibility of smooth repayment of the debt also shall be ascertained.

4. Trend Analysis

a. Time Series Analysis

Time Series Analysis evaluates performance over several years. It is helpful in determining, whether the progress is as planned and in planning the future operations. This will help in detecting the symptoms of major problem if any. Time Series Analysis is helpful in checking the accurate projection of projected financial statements.

b. Index Analysis

The figures of an accounting year is taken as the basis (100%) and figures of other years are compared with that as total percentage or as percentage increase / decrease. The base year may be the earliest year or latest year or any other intervening year. The calculation of trend ratio involves the ascertainment of arithmetical relationship which each item of several years have, to the same item of the base year.

The inefficiency of the base year may reflect as efficiency for the compared years and hence conclusions can not be drawn easily. However, improvements or others over the years can be ascertained.

c. Cross - Sectional Analysis

Often, we need to compare the financial ratios with that of other firms in the industry. In fact a competing firm may be taken as Bench-mark company or industry average may be taken for comparison.

The data for this analysis may be obtained from Industry Association publications or Annual Reports or internet.

5. Statement of Changes in Financial Position (SCFP)
Financial Statements, by themselves, do not provide information regarding changes in the firm's financial position during a particular period of time. They do not indicate the causes of changes or the movement of finances between two periods.

A firm basically generates funds / cash and uses / spends cash. The firm uses funds / cash when it redeems securities, pays interest and dividends, purchases materials, acquires asset etc. The activities that generate funds / cash are called sources of funds / cash and the activities that absorb funds / cash are called uses of funds / cash. A firm generates funds / cash when it increases its liabilities (as well as owners' equity); on the other hand the firm uses funds / cash when it buys assets or reduces its liability (as well as owners' equity)

Sources of Funds / Cash	Uses of Funds / Cash
Increase in liabilities and Owners' Equity)	Decrease in liabilities and decrease Owners' Equity
Decrease in Assets	Increase in Assets

A Statement is prepared to measures the changes (flows) that have taken place in the financial position of a firm between two balance sheet dates. This statement, called Statement of

Changes in Financial Position, enable us to understand the changes. Basically the statement summarises the sources from which funds (could flow) have been obtained and the uses to which they have been applied.

Inflow of funds – All the changes which result in an increase in funds of the enterprise

constitute sources of funds. Generally funds flow in a business concern from the following sources:

- The earnings of the enterprise
- Expansion in liabilities through increased use of borrowed funds or increased trade credit
- Decrease in assets such as liquidation in current assets, the sale of fixed, miscellaneous or intangible assets and earned depreciation on such assets.
- Contribution of additional funds by owners of the company

Outflow of funds – All the changes that cause decrease in total funds of the enterprises are regarded as uses of funds indicating outflow. The following items are included under the head uses of funds:

- Net Losses – When losses are incurred, funds go out of the business
- Decrease in Liabilities – When current liabilities or funds debt decrease, funds go out of the business by the amount of liabilities
- Increase in Assets – With increase in fixed assets, inventory, receivables, investment or other assets, amount of funds decrease indicating application of funds
- Decrease in capital funds – Retirement of stock bringing decrease in owner's capital involves outflow of funds

Fund flows are ultimately related to the changes in the Working Capital through statements called 'Statement of Changes in Working Capital' and 'Fund Flow Statement' looking at the changes in the total financial resources.

Statement of Changes in Working Capital and Fund Flow Statement

a. Statement of Changes in Working Capital

The Net Working Capital (WC) is the difference between Current Assets (CA) and Current Liabilities (CL).
Symbolically, $WC = CA - CL$.

From the above, the following may be deduced:

- An increase in CA causes an increase in WC, which means Funds inflow into Working Capital i.e. sources of working capital. Ex: Sale of buildings for cash
- A decrease in CA causes a decrease in WC, which means Funds outflow from Working Capital i.e. uses of working capital. Ex: Purchase of Machinery for cash
- An increase in CL causes a decrease in WC, which means Funds outflow from Working Capital i.e. uses of working capital. Ex: Purchase of furniture on credit
- A decrease in CL causes an increase in WC, which means Funds inflow into Working Capital i.e. sources of Working capital. Ex: Issue of equity shares to creditors in part settlement of their claims.

A similar increase or decrease in CA and CL does not affect WC and there is no flow of fund. Similarly, in case of increase or decrease in NCA and NCL does not affect WC and there is no flow of fund.

Change in Working Capital and Flow of funds takes place only when there is a transaction involving a Current Account (Current Asset / Current Liability) and Non-Current Account (Fixed Assets or Long-term Liabilities / Owners' Equity).

b. Fund Flow Statement

Working Capital Basis

The major Sources and uses of Working Capital are listed below:

Sources of Working Capital

- Funds from Business Operations (adjusted net income)
- Sale of non-current assets (sale of long-term investments / fixed assets - tangible / intangible)
- Long-term borrowings
- Equity or Preference shares
- Short-term borrowings such bank borrowings

Uses of Working Capital

- Losses from business operations (adjusted net loss)
- Purchase of non-current assets (purchase of long-term investments / fixed assets - tangible / intangible)
- Repayment of long-term debts like debentures and short term debts like bank borrowings
- Redemption of redeemable preference shares
- Dividends to shareholders

Funds from Business Operations

The major source of working capital is the firm's net profit from business operations. Since the Profit and loss account contains certain items which do not affect working capital, the profit / loss as shown in the profit and loss account, per se does not indicate the quantum of working capital provided by business operations. The Profit and Loss account contains a variety of write offs and other adjustments which do not involve any corresponding movement of funds. Therefore, while determining the amount of working capital from operations, appropriate adjustments are to be made to the profit disclosed by the Profit and Loss account.

All the expenses which have been deducted from revenue but do not involve working capital should be added back to net profit. Similarly, the items which have been added to revenue but have not contributed to the working capital or not directly caused by business operations should be deducted.

Adjustment to Profit

Net profit as per Profit and Loss Account	
Add: Items which do not result in outflow of fund	
(i) Depreciation charged during the year	
(ii) Loss on sale of fixed assets / investments (non-current assets)	
(iii) Capital expenditure (like goodwill, preliminary expenses patents, discount on debentures or share issue expenses)	
(iv) Provision for income tax / proposed dividend	
Less: (i) Gains on sale of fixed assets / investment	
(ii) Dividends and interest received on investments	
Funds from business operations	

Adjusted Profit and Loss A/c

	R		R
To Depreciation & Depletion charges		By Balance of P&L A/c B/d (of last year)	
To Amortization of:		By Dividend received	
(a) Intangible assets:		By Retransfer of excess provisions	
- goodwill		By Profit on sale of fixed assets	
- patent rights		By Profit on sale of long-term	
- trade marks		By Funds from operations	
(b) Fictitious assets:		(Balancing figure)	
- preliminary expenses			
- deferred expenditure (if any)			
To Appropriation of retained earnings:			
Transfer to:			
- General Reserve			
- Dividend equalisation fund			
- Sinking fund			
- Compensation fund etc,			
To Dividends paid			
To Provision for taxation			
To Interim dividend			
To Proposed dividend			
To Net profit <i>dd</i> (current year)			
To Funds lost in operations (Bal. Fig.)			
Total		Total	

Note: Proposed dividend and provision for taxation may or may not be a current liability. In case if they are treated as current liabilities, they should be shown in a schedule of changes in working capital only. However, if they are treated as non-current liabilities, then they should be considered as internal appropriation of profits made during the year and should be added back to current year's profit while calculating

funds from operations. But tax paid during the year should be considered as Application of Fund.

Depreciation

Depreciation does not affect the working capital and hence they are added back to profit. Moreover it is considered as the use of funds when the fixed asset was bought. Hence it would be double count to take depreciation again.

Statement of Changes in Total Resources

This statement is comprehensive to disclose all the changes in the financial position of the firm during a period. It shows all the financing and investment activities without omitting anything, including major financial and investment activities like issue of equity shares or debentures for the purchase of building or plant and machinery, conversion of debentures into shares, issue of bonus shares, and the like and gives the result as increase / decrease in Working capital

Specimen of Statement or Report form of Funds of Flow Statements

Sources of Fund:

- i. Sources or funds from operations.
- ii. Issue of shares.
- iii. Issue of debenture.
- iv. Raising of long-term loans.
- v. Sale of fixed or non-current asset.
- vi. Sale of long-term or trade investments.
- vii. Non-trading income - dividends received.
- viii. Decrease in working capital (as per schedule of changes in working capital)

Total

Uses of Fund

- I. Sources or funds lost in operations.
- II. Redemption of preference shares.
- III. Redemption of debentures.
- IV. Repayment of long term loans.
- V. Purchase of fixed or non-current assets.
- VI. Purchase of long-term Investments.
- VII. Payment of dividend in cash.
- VIII. Payment of tax in cash.
- IX. Increase in working capital (as per schedule of changes in working capital)

Total

Cash Flow Statements and Ratio Analysis are discussed as separate Lessons.

Points to Remember

- Ratios are used in the process of bench marking the performance of a company with its industry standards.

Review Questions

1. Name the balance sheet and P& L account items you will consider for

2. Why do you think financial analysis is required by any organization?

3. What ratios you will consider to assess the performance of a FMCG manufacturing company ? Explain with illustration.

4. What are the sources and uses of funds?

5. Write short notes on Trend Analysis

Note

LESSON 5 :

CASH FLOW STATEMENT

Learning Objectives

- To learn about various types of cash flows in an organization
- To discuss briefly about available sources of cash flows
- To learn the changes in the items of current accounts

Introduction

When the changes / flows are related only to change in cash, the statement is called Cash Flow Statement. It reports only the cash receipts and disbursements. It simply explains the change in cash position of a business enterprise during the period by reflecting the flow of cash into and out of the business.

Cash Flow Analysis is useful for short-run planning. A historical analysis of cash flows provides insight to prepare reliable cash flow projections for the immediate future & make suitable arrangements. Cash Flow statement shows inflow - sources of cash (i.e. positive cash flow) and outflow - uses of cash (i.e. negative cash flows) during the period and the difference being 'Net Cash Flow'. This statement analyses changes in non-current accounts as well as current accounts (other than cash) to determine the flow of cash.

Statement of changes in cash position is prepared recording only inflows and outflows of cash, reflecting the net change during the period. Cash received minus cash paid during a period is the cash balance at the end of the period. If the net change in cash position has to be found out from the profit and loss account, comparative balance sheets, adjustments for the non-cash items should be made.

Types of Cash Flow

The flow or movement of cash may be of two types, namely, actual flow of cash and notional flow of cash.

Actual Flow of Cash:

There may be actual or direct flow of cash 'in' and 'out' of the business under the following circumstances:

A. Actual inflow of Cash:

Example: Issue of shares for cash.

Cash Alc Dr

To Share Capital Alc

This transaction results in the actual inflow of cash into the business. Similarly, there is inflow of cash when debentures are issued for cash, loans raised in cash, sale of fixed assets for cash, dividends received in cash etc.

B. Actual outflow of Cash :

Example: Purchase of Machinery for cash. Entry:

Machinery Alc Dr.

To Cash

This transaction results in the actual outflow of cash from the business. Similarly, there is outflow of cash on repayment of

loans, redemption of preference shares or debentures, payment of taxes, dividend, etc. in cash.

Notional Cash Flow:

The indirect movement of cash 'in' and 'out' of the business is referred to as 'notional flow of cash' which may take place under the following circumstances:

A. Notional inflow of cash

Notional inflow of cash takes place whenever a transaction results in increasing current liabilities or decreasing current assets.

Example: Purchase of goods on credit:

Entry: Purchases Alc Dr.

To Creditors Alc

This transaction results in increasing creditors to the extent of credit purchases made. Though there is no actual inflow of cash, goods purchased on credit can be converted into cash. Hence, there is notional inflow of cash i.e., it may be considered as loan taken from the creditors. Similarly, when the transactions result in decreasing current asset such as book-debts, bills receivable, stock etc., it is to be considered as notional inflow of cash [i.e. decrease may be due to collection of book - debts & B/R or the sale of stock].

B. Notional outflow of cash:

Notional outflow of cash takes place whenever a transaction results in decreasing current liabilities or increasing current assets.

Example: Sale of goods on credits.

Entry: Debtors A/c Dr.

To Credit Sales A/c

This transaction results in increasing book-debts / Bills Receivable to the extent of credit sale made. Though there is no actual outflow of cash, goods sold on credit would have been sold for cash and would cost the business in terms of materials, labour and overheads. Hence there is notional outflow of cash i.e. it may be considered as loan advanced to customers. Similarly, when there is decrease in current liabilities, it may be due to part settlement of these dues. Hence, such decrease in current liabilities is treated as notional outflow of cash.

	Rs.	Rs.
Cash or Bank Balance at the beginning		
Add: Cash Inflows:		
Cash from operations		
Proceeds from sale of fixed assets		
Issue of Shares		
Share Premium		
Issue of Debentures		

	Increase in current liabilities		
	Raising of loans		
	Decrease in current assets		
	Non - trading receipts viz:		
	Dividend received and		
	Refund of Tax		
Less:	Cash Outflows:		
	Purchase of fixed assets		
	Repayment of loans		
	Redemption of preference shares		
	Redemption of Debentures		
	Payment of taxes		
	Payment of Dividend		
	Cash lost on Operations		
	Increase in current assets		
	Decrease in current liabilities		
	Cash Balance at the end		

Proforma of an account form of Cash Flow Statement

Cash or Bank Balance at the beginning		Cash Outflows:	
Add: Cash Inflow:		Purchase of fixed assets	
Cash from operations		Repayment of loans	
Proceeds from sale of fixed assets		Redemption of Preference Shares.	
Issue of Shares		Redemption of Debentures	
Share premium		Payment of Dividend	
Issue of Debentures		Payment of taxes	
Raising of loans		Cash lost in operations	
Decrease in Current assets		Increase in current assets	
Increase in current liabilities		Decrease in current liabilities	
Non-trading receipts:		Cash or Bank balance at the end	
Dividend received			
Refund of tax			

Cash flows can be classified into three categories: Cash flows from operating activities, Cash flows from investment activities, Cash from financing activities.

The Accounting Principles Board of The Institute of Chartered Accountants of India recommends a detailed report for cash flow statement, which is insisted in the listing guidelines of stock exchanges of India. The definitions provided by them are as follows:

- Cash comprises of cash on hand and demand deposits
- Cash Equivalents are short-term, highly liquid investments that are readily convertible into known amounts of cash and which are subject to an insignificant risk of changes in value.

- Cash flows are inflows and outflows of cash and cash equivalents.
- Operating Activities are the principal revenue-producing activities of the enterprise and other activities that are not investing or financing activities.
- Investment Activities are the acquisition and disposal of long-term assets and other investments not included in cash equivalents
- Financing Activities are activities that result in changes in the size and composition of the owners' capital (including preference share capital) and borrowings of the enterprise.

Sources of Cash Inflow

Cash from Operations

Cash from operations is a major source of cash flow into the business. Only Cash Sales and Cash Receipts from debtors against credit sales are recognized as a source of cash. Similarly Cash Purchases and Cash Payments to suppliers for credit purchases are regarded as the use of cash. In respect of other expenses and incomes also, no consideration is given for outstanding and prepaid expenses and incomes.

Cash from operations may be calculated in two ways i.e. Cash Sales Method or Net Profit Method.

i Cash Sales Method:

Under this method, cash from operations is ascertained as under:

Cash from operations = Cash Sales - Cash purchases and Cash Operating

Expenses.

Generally, the income statement shows both cash and non-cash items. In order to ascertain cash sales, cash purchases and cash operating expenses based on income statement, the items of current assets and current liabilities given in the balance sheet should also be taken into account along with items of income and expenditure shown in the income statement. This method is also known as 'Income & Expenditure' method. Let us know how to ascertain cash sales, cash purchases and cash operating expenses and cash from operations under this method.

		Rs.
(i)	Cash Sales:	
	Total Sales	
	Less: Debtors / BIR at the end	
	Add: Debtor/B/R at the beginning	
	Cash Sales	
	Alternatively,	
	Total Sales	
	Less: Increase in Debtors or BIR	
	Add: Decrease in Debtors or Bills Receivable	
	Cash Sales	

(ii)	Cash Purchases:	
	Total purchases	
	Add: Creditors & BIP at the end	
	Less: Creditors & BIP at the beginning	
	Cash Purchases	
	Alternatively,	
	Total purchases	
	Less: Increase in creditors and Bills payable	
	Add Decrease in Creditors and BIP	
	Cash Purchases	

Note: Sales made by every firm represent both cash sales and credit sales. On account of credit sales made during the year, the question of collecting debts from debtors arises. As such, that portion of credit sales which remained uncollected at the end of accounting year is termed as 'Debtor or Bills Receivable' and is shown as asset in the Balance Sheet. Hence, the increase in the amount book-debts or debtors and bills receivable is considered as an increase in credit sales and deducted from total sales and decrease in the amount from debtors and bills receivable is added back. However, the collections made from debtors and cash received against bills receivable would be shown as inflow of cash in the cash flow statement.

Similarly, the purchases made by every firm represents both cash and credit purchases. On account of credit purchases made during the accounting year, the question of payment to creditors arises. As such, that portion of credit purchases which remained unpaid at the end of accounting year is termed as creditors and bills payable shown as liability in the Balance Sheet. Hence, increase in creditors and B/P is considered as increase in credit purchases and deducted from total purchases and decrease in amount of credits and bills payable is added back. However, payments made to creditors or cash paid against bills payable should be shown as outflow of cash in the cash flow statements.

	Rs.	
(Hi) Cash Operating Expenses		
Total operating expenses for the accounting year	xxxx	
Less: Non cash items	xxxx	
Increase in O/s expenses	xx	
Decrease in prepaid expenses	xxx	xxx
	xxxx	
Add: Decrease in O/s expenses	xxx	
Increase in pre-paid expenses	xxx	xxx
Cash Operating Expenses	xxxx	

i Net Profit Method

Ascertainment of cash from operations based on the net profit of the business is called as Net Profit Method. Under this method, cash from operations is calculated as under:

			Rs.	Rs.
Net profit	(i.e. of current year)			xxxx
Add:	Non cash or non-operating items which have already been debited to P & L A/c			
	Depreciation on Assets		Xxx	
	Transfers to Reserves		Xxx	
	Intangible assets written - off		xxx	
	Fictitious assets written - off		xxx	
	Outstanding expenses of current year		xxx	
	Prepaid expenses of the previous year		xxx	
	Loss on sale of fixed assets		xxx	
	Provision for taxation		xxx	
	Dividends paid		xxx	xxxx
				xxxxxx
Less:	Non - cash or non - operating items which have already been credited to P & L A/c:			
	Profit on sale of fixed assets		xxx	
	Refund of Tax		xxx	
	Non-trading :- a) Dividend		xxx	
	b) Rent		Xxx	
	c) Interest etc.		Xxx	
	Outstanding income of current year		Xxx	
	Pre- received income of previous year		Xxx	
	Opening Balance of P & L A/c. (i.e. Previous year)			xxxx
	Cash from operations			xxxx

Alternatively,

Adjusted Profit and Loss A/c

Dr.		Rs.		Rs.
To Depreciation	xxx	By Balance Bld	xxx	
" Transfer to reserves	Xxx	" Profit on sale of fixed assets		
" Intangible assets written off	xxx	" Non-trading receipts:		
" Fictitious assets written off		" Dividend	xxx	
" O/s Expenses (current year)	Xxx	" Rent	xxx	
" Prepaid expenses (previous year)	xxx	" Interest etc.	xxx	
" Loss on sale of fixed assets	Xxx	" O/s Income (current yr.)	xxx	
" Provision for taxation	Xxx	" Pre-received income	xxx	
" Dividends paid	Xxx	" (Previous year)		
" Balance Cld	Xxx	" Cash from Operations	Xxx	
" Cash lost on operations	xxx	"		

Important useful points to be noted while calculating cash from operations under the method:

- Outstanding Expenses:** Outstanding Expenses are those expenses which are due to be paid. They are charged, to profit and loss a/c but no cash is paid in this respect during the current year. Hence, they are added back while calculating the cash from operations. However, if any expenses of the previous year is given, they may be assumed to have been paid during the year and are shown as the outflow of cash in the cash flow statement.
- Prepaid Expenses:** Are the expenses of the next or subsequent accounting year paid in the current account year or expenses paid in advance. These expenses are not charges to profit and loss a/c. Hence, they are shown as outflow of cash in the cash flow statement. However, if there are prepaid expenses (previous year) given, they are paid in the last or previous accounting year for the current accounting year, and as such they are added back while ascertaining cash from operation.
- Outstanding or Accrued Income:** It is an income due to be received, it is credited to profit and loss account but no cash is received. Hence, it is deducted from sources from these

operations. However, if there is any outstanding income relating to previous year is given, it may be assumed to have been received during the year and shown as inflow of cash in the cash flow statement.

- iv. **Pre-received Income:** It is the income received in advance. If there is any pre-received income in the current year, it is shown as inflow of cash in cash flow statement. However, if there is any pre-received income (in previous year), it is deducted from sources from operations while ascertaining cash from operations.

Proceeds From Sale of Fixed Assets: Amount received from the sale fixed assets or investments results in the inflow of cash. Hence, it is shown as inflow of cash in cash flow statement.

Issue of Shares or Debentures: When the shares or debentures are issued for cash, there is inflow of cash. As such, it is shown as inflow of cash in the cash flow statement.

Raising of Loans: When the money is borrowed from financial institutions in terms loans or cash credits, it amounts to inflow of cash. As such, it is shown as inflow of cash in the cash flow statement.

Outflow of Cash

Purchase of Fixed Assets : Increase in fixed assets or purchase of fixed assets during the accounting year amounts to the outflow of cash. As such, it is shown as outflow of cash in the cash flow statement. However, purchase affixed assets may have to be ascertained by preparing concerned fixed asset accounts on the same lines as prepared under funds flow analysis.

Repayment of Loans: Loans (if any) repaid during the year amounts to the outflow of cash. As such, it is shown as outflow of cash in the cash flow statement. Loans repaid during the year may be ascertained by facilitating comparison of loans outstanding in the last year with those loans in the current year.

Redemption of Preference Shares of Debenture: Preference Shares or Debentures may be redeemed at par, at premium or at discount. Depending upon the circumstances, repayment of preference share capital or debentures is to be considered as outflow of cash on the same lines as discussed under funds flow analysis and shown as outflow of cash in the statement.

In addition to adjustments of depreciation, gain from sale of non current asset etc. made to ascertain Funds from Operations, changes in current assets and current liabilities are to be analysed in relation to their effect on cash as follows:

Changes in Current Accounts

Changes in Current Assets

- i. All increases in current assets excluding cash (end balance being more than the beginning balance) increase working capital but reduce cash flow from operations and vice versa

Increase in Current Assets (other than Cash), Increase in Working Capital, Decrease in cash	Decrease in Current Assets (other than Cash), Decrease in Working Capital, Increase Cash
Increase in Debtors implies that cash collections from customers are less than the sales figure	Decrease in Debtors implies that cash collections are greater than sales
Increase in Inventory implies that cost of goods purchased is greater than the cost of goods sold figure	Decrease in Inventory means that cost of goods purchased is less than the cost of goods sold
Increase in Prepaid expenses implies that cash outflow is more than the amount of actual expenses	Decrease in Prepaid expenses means that cash outflow is less than the current usage

Change in Current Liabilities

All increases in Current Liabilities cause decrease in Working Capital and leads to increase in cash and vice versa.

Increase in Current Liabilities Decrease in Working Capital, Increase in cash	Decrease in Current Liabilities Increase in Working Capital, Decrease in Cash
Increase in Creditors implies that cash payments to creditors are less than the purchase figure	Decrease in Creditors means more cash payments than the purchase figures
Increase in Income in advance implies greater cash inflow	Decrease in Income in advance means lesser cash inflow

Cash flow Business Operations

A. Working capital from business operation (as per funds flow statement)

B. Adjustment to convert to cash basis:

Add: Decrease in WC (Decrease in CA or Increase in CL)

Decrease in CA other than cash (item-wise)

Increase in CL (item wise)

Less: Increase in WC (Increase in CA or Decrease in CL)

Increase in CA other than cash (item wise)

Decrease in CL (item wise)

C. Cash flow from business operations

Points to Remember

- Knowledge of the types of cash flows helps in profit planning and hence cash budgeting.
- Managing every constituent of working capital leads to optimum capital utilization.
- Sales forecast is the basis for determining working capital requirements.

Review Questions

1. Discuss briefly about the movement of cash in an organisation.
2. What are the methods of calculating cash from operations? Explain.
3. Explain the current account factors that results in changes in working capital.

LESSON 6 :

CORPORATE INCOME TAX

Learning Objectives

- To learn about the impact of taxation on the business of the organization and also personal taxation on the organization.
- To know about corporate income tax and also major tax deductions
- By the end of this lesson, you will understand the impact of taxation on Corporate Financial Management. It can be discussed broadly under two heads as follows.

Business Taxation

The tax payments represent a cash outflow from the business and therefore, these tax cash flows are critical part of the financial decision making in a business firm. In deed in some practical situations, the taxation implications are dominant influences on the final investment decision also. Taxation effects a firm in numerous ways, the most significant effects are given below:

- Corporate taxes on firm's profits.
- Reduction in weighted average cost of capital because interest payments are allowable against tax.
- Dividend will not reduce the tax burden of the firm since it is not a charge on profits.
- Where the firm incurs overall loss, it can be carried forward to a future profit making year.
- Where a unit of the firm incurs loss but the firm gets overall profits from all other units, loss of the loss making unit will reduce the overall tax liability of the firm by set off of losses.
- Impact of depreciation provision on the reduction of taxable income.
- Taxation of capital gains and its impact on profits of the firm.
- Double taxation of firm's earnings like dividends received from other companies and its impact on profits etc.

Financing with debt and making interest payments rather than dividend payments will generate after-tax income for the company's suppliers of equity capital, but simultaneously there is greater risk associated with financing with debt.

Personal Taxation and its Influence on Firm

A business organisation must consider the tax consequences for the investors in company's shares, debentures and bonds in order to attract investors to the firm's securities. The investments in company attract two types of taxes to the investors:

- Earnings in the form of dividends and interest.
- Capital gains from disposal of the investments.

When the investments made in tax free securities like Tax free municipal bonds, will not attract any taxation on the individual investors.

Types of Tax Rate

- A progressive tax rate is a rate that escalates as the amount of income increases.
- A marginal tax rate is to be applied to the next rupee of income.
- The average tax rate equals the amount of tax divided by the taxable income.

Corporate Income Tax

A corporation's taxable income is found by deducting all allowable expenses, including depreciation and interest, from revenues. This taxable income is then subjected to a graduated tax structure.

The taxability of a company's income depends on its domicile.

- Indian companies are taxable in India on their worldwide income.
- Foreign companies are taxable on income that arises out of their Indian operations, or, in certain cases, income that is deemed to arise in India. Royalty, interest, gains from sale of capital assets located in India (including gains from sale of shares in an Indian company), dividends from Indian companies and fees for technical services are all treated as income arising in India.
- The tax, rate the of shares in an Indian company), dividends from Indian companies and fees for technical services are all treated as income arising in India.

A percent of taxable income that must be paid in taxes-that is , applied to each income bracket is referred to as a marginal rate. For example, each additional dollar of taxable income above Rs.50,000 is taxed at the marginal rate of 25 percent until taxable income reaches Rs.75,000. At that point, the new marginal rate becomes 34 percent.

The average tax rate for a firm is measured by dividing taxes actually paid by taxable income. For example, a firm with Rs.100,000 of taxable income pays Rs.22,250 in taxes and, therefore, has an average tax rate of Rs.22,250/Rs.100,000, or 22.25 percent. For small firms (i.e., firms with less than Rs.335,000 of taxable income), the distinction between the average and marginal tax rates may prove important. However, the average and marginal rates converge at 34 percent for firms with tax-able income between Rs.335,000 and Rs.10 million and, finally, converge again, this time to the 35 percent rate, for firms with taxable income above Rs.18,33,333.

Income of companies is taxed at the following rates

Type of company	Old Rate (%)	New Rate (%)
Domestic company		
Widely held company	45	40
Closely held company	50	40
Foreign company	65	55

A surcharge of 15% of the amount of tax payable is levied on domestic companies, if taxable income exceeds Rs.75,000.

Taxable Income

The main source of income of a company is generally from “business”. A company would also earn income from under the following heads:

- Income from house property
- Income from capital gains
- Income from other sources

Taxable income is calculated according to the rules for each class of income and then aggregated to determine total taxable income.

Deductability of Expense

While calculating income from business or profession, expense incurred wholly and exclusively for business purposes are generally deductible. These include depreciation on fixed assets, interest paid on borrowings in the financial year etc.

Interest paid on outstanding corporate debt is treated as an expense and is tax deductible.. Therefore, there are tax advantages associated with using debt financing that are simply not present with either preferred or common stock financing.

Certain expenses are specifically disallowed or the amount of deduction is restricted. These expenses include:

- Entertainment expenses
- Interest or other amounts paid to a non-resident without deducting without tax
- Corporate taxes paid
- Indirect general and administrative costs of a foreign head office.

Set-Off and carry forward of Losses

Business losses incurred in a tax year can be set off against any other income earned during that year, except capital gains.

Unabsorbed business losses can be carried forward and set off against business profits of subsequent years for a period of eight years; the unabsorbed depreciation element in the loss can however, be carried forward indefinitely. However, this carry forward benefit is not available to closely-held (private) companies in which there has been no continuity of business or shareholding pattern. Also, any change in beneficial interest in the shares of the company exceeding 51 per cent disqualifies the private company from the carry forward benefit.

Points to Remember

- Dividends are now taxable in the hands of shareholders.
- The company-declaring dividend, will have to deduct tax at source. However, in case of an individual person, if he receives dividend of up to Rs. 1,000 in a year, from the same company, then tax does not have to be deducted at source

Review Questions

1. Explain the impact of taxation on the business firm.

Note

LESSON 7

TIME VALUE OF MONEY

Learning Objectives

- To learn to compute future and present value at simple and compound interest for single amount and series of payments.
- To find out future and present value of annuities and value of perpetuities
- To know what is Net Present Value (NPV) and Internal Rate of Return (IRR)

Introduction

Students,

Given a chance, any rationally thinking individual would prefer to have an amount of money now, as compared to having the same at a later date. Why do you think we do this? This is because we give due weightage for 'Time Preference for Money' since we may otherwise lose the opportunity to earn additional income / interest.

Hence 'Time Preference for money' is considered very valuable due to the following reasons:

- Investment opportunities / opportunities cost and the element of cost
- Preference for current consumption to future consumption
- Inflationary trend
- Uncertainty

Due to the above reasons money loses value with time and hence a rupee on hand presently has a more value compared to a rupee receivable at a later date. Considering the interest and futuristic risk, the preference will be to have the even future value now itself, discounting the same at the rate of return. For instance, one may prefer Rs.100/- instead of Rs.105/- after one year. This means that the value of the higher amount of future is equivalent to a lesser present amount. In the example future Rs.105/- after one year is treated as equal to present Rs.100/-.

Financial decisions are to be made comparing the cash outlays / outflows and the benefits / cash inflows. A financial decision – financing or investment – taken today has implications for number of years in terms of cash flows. For a meaningful comparison, the variables – cash outflows and cash inflows at various points of time periods – shall be converted to a common period of time.

Simple Interest

Simple interest is a function of three variables: Original amount borrowed or Principal, Interest Rate, Number of time period.

$$SI = PV (r) (n)$$

Where, SI = Simple Interest

PV = Principal or Original Value

r = rate of interest per time period

n = number of time periods

Future Value at Simple Interest

Future Value = Principal + Interest

$$FV_n = PV + SI$$

$$FV_n = PV + PV (r) (n)$$

$$FV_n = PV [1 + (r) (n)]$$

Translating a value to the future is referred to as compounding.

Present Value

Present Value i.e. Principal or Original amount (PV) can be ascertained if the future value, interest rate and the number of time periods are known.

$$PV = \frac{FV_n}{[1 + (r) (n)]}$$

Translating a value to the present is referred to as discounting.

Compound Interest

For Single Amount

Future Value

The term Compound Interest merely implies that interest paid (earned) on a loan (an investment) is periodically added to the principal. Consequently, interest is earned on the interest as well as the Principal or Original Amount. This interest-on-interest is compounding.

$$FV_1 = PV (1 + r)$$

$$FV_2 = FV_1 (1 + r) = PV (1 + r) (1 + r) = PV (1 + r)^2$$

$$FV_n = PV (1 + r)^n$$

For the compounding factor $(1 + r)^n$ for Re.1, compound interest tables or Future Value Interest Factor (FVIF_n or terminal value interest factor) tables are designed.

$$FV_n = PV (FVIF_m)$$

Present (or Discounted) Value

Present Value of future cash flows allows us to place all cash flows on a current footing and enables to compare them in terms of present rupees

$$PV = \frac{FV_n}{(1 + r)^n} \quad \text{or} \quad FV_n [1 / (1 + r)^n]$$

The component $[1 / (1 + r)^n]$ is the Present Value Interest Factor (PVIF_m) and the reciprocal of the Future Value Interest Factor (FVIF_m)

$$PV = FV_n (PVIF_m)$$

For Series of Payments

We may be interested in the future value of a series of payments made at different periods. Translating a series of payments into future or present value is similar to translating a single amount.

Future or Compound Value

For finding future value of each cash flow in case of series of cash flows, we can use the formula $FV_n = PV(1+r)^n$ or $FV_n = PV(FVIF_m)$ and add up to ascertain the Future Value.

Annuities

An annuity is a series of equal cash flows occurring over a specified number of periods. When cash flows occur at the end of each period, it is called an Ordinary Annuity.

Future or Compound Value of Annuity

$$FVA_n = A(1+r)^{n-1} + A(1+r)^{n-2} + \dots + A(1+r)^1 + A(1+r)^0$$

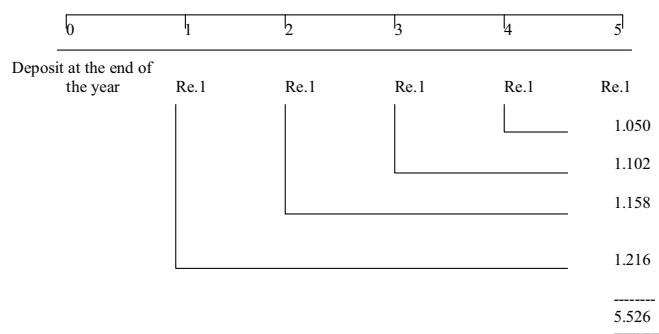
$$= A \left\{ \frac{(1+r)^n - 1}{r} \right\} \quad \text{*mathematical derivation is given at the end of the chapter}$$

The term $\left[\frac{(1+r)^n - 1}{r} \right]$ is referred to as the Future Value Interest Factor for an Annuity,

$$FVA_n = A(FVIFA_m)$$

Where FVIFA stands for the Future Value Interest Factor of an Annuity at $i\%$ for n periods.

Time Line: The series of cash flows can be represented on a time line as below, PV of Re.1 at 5% interest:



In the above Time Line, the future value for each cash flow is ascertained either by manual calculation of individual item with formula $FV = PV(1+r)$ or by using the Future Value Interest Factor table. Similarly when the cash flows are of equal value at the end of each period as in the

above example i.e. Ordinary Annuity, we can ascertain Future / Compound Value Interest Factor of the Annuity (FVIFA) using the table with the interest rate and the number of years.

Present (or Discounted) Value

For finding out the present value of a series of cash flows in future, we may ascertain the present value of each cash flow in future with the formula $PV = FV_n / (1+r)^n$ or $FV_n [1 / (1+r)^n]$ and add up.

Alternatively the formula $PV = FV_n(PVIF_m)$ may be used for each cash flow and added to ascertain the future value.

As we know, the present value of a single cash flow in future is

$$PV = FV_n [1 / (1+r)^n]$$

For a series of cash flows the formula to be expanded as

$$PV = FV_1 [1 / (1+r)^1] + FV_2 [1 / (1+r)^2] + \dots + FV_n [1 / (1+r)^n]$$

In case of Annuity,

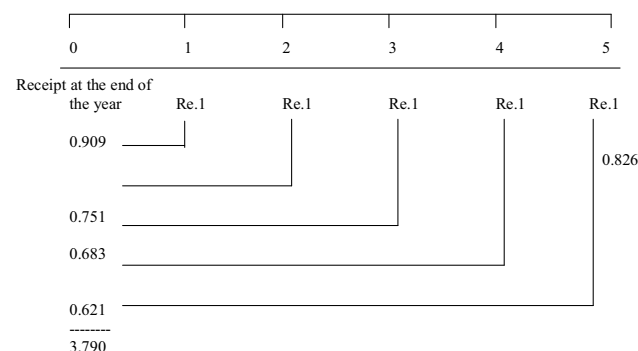
$$PV = A [1 / (1+r)^1] + A [1 / (1+r)^2] + \dots + A [1 / (1+r)^n]$$

Since the future cash flow of annuity is same for every period, the formula is,

$$PVA_n = A \left\{ \frac{1}{(1+r)^1} + \frac{1}{(1+r)^2} + \dots + \frac{1}{(1+r)^n} \right\}$$

= $A \left\{ \frac{(1+r)^n - 1}{[(1+r)^n \times r]} \right\}$ * mathematical derivation is given at the end of the chapter

It will be more clear for you looking at Graphically the Present Value of an annuity of FV, say, of Re. 1 with discounting rate of 10%



In the above time line discounting is done individually using the formula $PV = FV_n / (1+r)^n$. Alternatively, table for Present Value Discounting Factor of an Annuity (PVDFA) may be referred to replace the term $\left[\frac{(1+r)^n - 1}{[(1+r)^n \times r]} \right]$ and calculate.

Perpetuities

A Perpetuity is an Ordinary annuity whose cash flows continue indefinitely.

This special type of annuity will have to be determined in case of irredeemable preference shares with out maturity where preference dividend is expected to be paid perpetually. Looking at the formula for PVA_n i.e. $A \left\{ \frac{(1+r)^n - 1}{[(1+r)^n \times r]} \right\}$, this can be rewritten as

$$PVA = A \left[\frac{1 - \frac{1}{(1+r)^n}}{r} \right]$$

In case of perpetuity since the time period is taken as infinite,

$$PVA = A \left[\frac{1 - \frac{1}{(1+r)^n}}{r} \right]$$

Since any denominator of infinite nature, including any term raised to the power infinity, is zero,

$$PVA = A \frac{1 - 0}{r} = \frac{A}{r}$$

Annuity Due

A series of equal cash flows starting at the beginning of each period is called an Annuity Due.

While calculating future value, Annuity due is simply equal to the future value of a comparable three year ordinary annuity compounded for one more period. This is because last cash flow also accrues interest i.e. for one year., since the cash flow is in the beginning of the year.

Using the formula for future value and providing interest for last cash flow the formula for Annuity Due for future value is

$$A \left\{ \frac{(1+r)^n - 1}{r} \right\} (1+r)$$

Alternatively, the Future / Compound Value of Annuity Due may be ascertained by using the Future Value Interest Factor table as below

$$FVAD_n = A (FVIFA_m) (1+r)$$

Similarly Present Value of Annuity Due can be calculated. Since the first cash flow is immediate at the beginning, the Present Value is equal to absolute value and hence the cash flow series will be as below:

$$PVAD_n = A \left\{ 1 + \frac{1}{(1+r)^1} + \frac{1}{(1+r)^2} + \dots + \frac{1}{(1+r)^{n-1}} \right\}$$

A $\left\{ \frac{(1+r)^n - 1}{r} \right\} (1+r)$ * mathematical derivation is given at the end of the chapter

Multi-period or Semi-annual and other compounding Normally interest rate is expressed for a year. In case, compounding is done semi-annually or quarterly or monthly or weekly, interest rate & compounding to be adjusted accordingly to alter the number of time periods. The formula $FV_n = (1+r)^n$ will get altered as below:

$$FV_n = PV \left[1 + \frac{r}{m} \right]^{mn}$$

where m is the number of compounding per year.

The interest rate specified on annual basis is known as the Nominal Interest Rate. If compounding done in a year is more than once, the actual rate of interest is called the

Effective Interest Rate (EIR) / Annual Percentage Rate (APR), which can be ascertained as below:

$$FV_n = PV \left[1 + \frac{r}{m} \right]^{mn} - 1$$

Similar to Future Value, Present Value of cash flows more than once a year can be discounted with the following formula:

$$PV = \frac{FV_n}{\left[1 + \frac{r}{m} \right]^{mn}}$$

Net Present Value

While Financial Objective is concluded as Wealth Maximization / Shareholders' Wealth, Wealth is defined as Net Present Value.

NPV of a Financial decision is the difference between the Present Value of Cash Inflows and the Present Value of Cash Outflows:

$$NPV = \left[\frac{FV_1}{(1+r)} + \frac{FV_2}{(1+r)^2} + \dots + \frac{FV_n}{(1+r)^n} \right] - C_0$$

Where C_0 is Cash Outflow and r may represent the Opportunity Cost of Capital.

Internal Rate of Return

The Internal Rate of Return is the rate which equates the Present Value of Cash Inflows with the Present Value of Cash Outflows of an investment. It is the rate at which the Present Value of the proceeds and the outlays are equal. In other words, it is the rate at which the Net Present Value is (PV of Cash Outflows – PV of Cash Inflows) is zero.

As you know we use the following formula to find the Present Value of Cash inflows:

$$FV_1 \left[\frac{1}{(1+r)^1} \right] + FV_2 \left[\frac{1}{(1+r)^2} \right] + \dots + FV_n \left[\frac{1}{(1+r)^n} \right]$$

Since PV of Cash Outflows (C_0) is also known, we may use the following equation to find out r:

$$\left\{ FV_1 \left[\frac{1}{(1+r)^1} \right] + FV_2 \left[\frac{1}{(1+r)^2} \right] + \dots + FV_n \left[\frac{1}{(1+r)^n} \right] \right\} - C_0 = 0$$

Mathematical Derivation of Formula for Future/Compound Value of Annuity

$$FVA_n = A (1+r)^{n-1} + A (1+r)^{n-2} + \dots + A (1+r)^1 + A (1+r)^0$$

When we take out A in every item, the series is as below:

$$(1+r)^{n-1} + (1+r)^{n-2} + \dots + (1+r)^1 + (1+r)^0$$

We may rewrite the series as below also:

$$(1+r)^0 + (1+r)^1 + \dots + (1+r)^{n-2} + (1+r)^{n-1}$$

This series of n terms is in Geometric Progression with a common ratio. The Common Ratio is arrived at by dividing the following term by the preceding term in the series.

Let us take the first term $(1+r)^0$ which is 1 (anything raised to the power of 0 is equal to 1). Dividing the second term by the first term 1, we get $(1+r)$ i.e. $(1+r)^1 / 1$. Next we shall divide the third item $(1+r)^2$ by $(1+r)^1$ which is again $(1+r)$. Hence the Common Ratio in our series is $(1+r)$.

We get S_n by adding the above n number of terms in the series
 $S_n = (1+r)^0 + (1+r)^1 + \dots + (1+r)^{n-2} + (1+r)^{n-1} \dots \dots \text{Eqn.1}$

Multiplying both sides by '(1+r)' we have,

$$\begin{aligned} (1+r)S_n &= (1+r)(1+r)^0 + (1+r)(1+r)^1 + \dots \dots \dots \\ &\dots \dots \dots + (1+r)(1+r)^{n-2} + (1+r)(1+r)^{n-1} \\ (1+r)S_n &= (1+r) + (1+r)^2 + \dots \dots \dots \\ &+ (1+r)^{n-1} + (1+r)^n \dots \dots \dots \text{Eqn. 2} \end{aligned}$$

Now Subtracting Equation 2 from Equation 1, we have

$$\begin{aligned} S_n - (1+r)S_n &= 1 - (1+r)^n \\ S_n [1 - (1+r)] &= 1 - (1+r)^n \\ S_n &= \frac{1 - (1+r)^n}{[1 - (1+r)]} = \frac{-[(1+r)^n - 1]}{1 - 1 - r} = \frac{(1+r)^n - 1}{r} \end{aligned}$$

Multiplying with A which was removed in the beginning, the formula is

$$A \{ [(1+r)^n - 1] / r \}$$

Mathematical Derivation of formula for Present Value of an Annuity

$$\begin{aligned} PVA_n &= A [1 / (1+r)^1] + A [1 / (1+r)^2] + \dots \dots \dots \\ &\dots \dots \dots + A [1 / (1+r)^n] \end{aligned}$$

We may rewrite the equation as follows:

$$\begin{aligned} PVA_n &= A [(1+r)^{-1}] + A [(1+r)^{-2}] + \dots \dots \dots \\ &\dots \dots \dots + A [(1+r)^{-n}] \end{aligned}$$

When we take out A in every item, the series is as below:

$$(1+r)^{-1} + (1+r)^{-2} + \dots \dots \dots + (1+r)^{-n}$$

This series of n terms is in Geometric Progression with a common ratio. The Common Ratio is arrived at by dividing the following term by the preceding term in the series.

Dividing the second term by the first term, we get $(1+r)^{-1}$.

Next we shall divide the third item $(1+r)^{-3}$ by $(1+r)^{-2}$ which is again $(1+r)^{-1}$. Hence the Common Ratio in our series is $(1+r)^{-1}$.

We get S_n by adding the above n number of terms in the series
 $S_n = (1+r)^{-1} + (1+r)^{-2} + \dots \dots \dots + (1+r)^{-n} \dots \dots \text{Eqn.1}$

Multiplying both sides by '(1+r)^{-1}' we have,

$$\begin{aligned} (1+r)^{-1}S_n &= (1+r)^{-1}(1+r)^{-1} + (1+r)^{-2}(1+r)^{-1} + \dots \dots \dots \\ &\dots \dots \dots + (1+r)^{-n}(1+r)^{-1} \end{aligned}$$

Using the concept of $a^m \times a^n = a^{m+n}$, we get

$$\begin{aligned} (1+r)^{-1}S_n &= (1+r)^{-2} + (1+r)^{-3} + \dots \dots \dots + (1+r)^{-n-1} \\ &\dots \dots \dots \text{Eqn. 2} \end{aligned}$$

Now Subtracting Equation 2 from Equation 1, we have

$$\begin{aligned} S_n - (1+r)^{-1}S_n &= (1+r)^{-1} - (1+r)^{-n-1} \\ S_n [1 - (1+r)^{-1}] &= (1+r)^{-1} - (1+r)^{-n-1} \\ S_n &= \frac{(1+r)^{-1} - (1+r)^{-n-1}}{[1 - (1+r)^{-1}]} \end{aligned}$$

Simplifying the denominator,

$$1 - (1+r)^{-1} = 1 - \frac{1}{1+r} = \frac{1+r-1}{1+r} = \frac{r}{1+r}$$

Simplifying the numerator,

$$(1+r)^{-1} - (1+r)^{-n-1} = \frac{1}{(1+r)} - \frac{1}{(1+r)^{n+1}} = \frac{(1+r)^n - 1}{(1+r)^{n+1}}$$

Using the simplified numerator and denominator,

$$\begin{aligned} (1+r)^n S_n &= \frac{1 - (1+r)^{-n-1}}{\frac{r}{1+r}} = \frac{[(1+r)^n - 1]}{(1+r)^{n+1}} \times \frac{(1+r)}{r} \\ &= \frac{[(1+r)^n - 1]}{(1+r)^n (1+r)^1} \times \frac{(1+r)}{r} \\ &= \frac{[(1+r)^n - 1]}{(1+r)^n \times r} \end{aligned}$$

Multiplying with A, which was removed in the beginning, the formula is

$$A \{ [(1+r)^n - 1] / [(1+r)^n \times r] \}$$

Mathematical Derivation of formula for Present Value of an Annuity Due

$$\begin{aligned} PVAD_n &= A \{ 1 + [1 / (1+r)^1] + [1 / (1+r)^2] + \dots \dots \dots \\ &\dots \dots \dots + [1 / (1+r)^{n-1}] \} \end{aligned}$$

Multiplying and dividing the whole expression by (1+r), we get

$$\begin{aligned} A(1+r) \{ 1 + [1 / (1+r)^1] + [1 / (1+r)^2] + \dots \dots \dots + [1 / (1+r)^{n-1}] \} \\ \frac{(1+r)}{(1+r)} \\ A(1+r) \{ 1 + [1 / (1+r)^1] + [1 / (1+r)^2] + \dots \dots \dots + [1 / (1+r)^{n-1}] \} \times [1 / (1+r)] \\ A(1+r) \{ [1 / (1+r)] + [1 / (1+r)^2] + \dots \dots \dots + [1 / (1+r)^n] \} \end{aligned}$$

Keeping the term $A(1+r)$, the other term is in geometric progression with common ratio as $1/(1+r)$. Hence applying the formula for sum of geometric progression i.e. $S_n = a(r^n - 1) / r - 1$ where a is the first term and r is the common ratio, we get

$$(1+r)^n - 1 / [(1+r)^n \times r]$$

Adding the term already removed we have $A(1+r) [(1+r)^n - 1] / [(1+r)^n \times r]$

Points to Remember

- Capital budgeting methods includes NPV and IRR which are said to be the discounted methods as annual cash flows are discounted using time value of money.
- Project Appraisal and Selection is made considering present value of future cash flows.

Review Questions

- If you invest Rs.3000 today at a compound interest of 9%, what will be its future value after 50 years?
- What is the present value of Rs.80,000 receivable 15 years from now, if the discount rate is 10%?

LESSON 8 :

THE FINANCIAL SECURITIES

Learning Objectives

- To know important sources of funds
- To study the features, advantages, disadvantages and types of important securities for raising funds

Introduction

Students,

Can you think of leading our life without money, in these modern times ? We need to invest money while taking major decisions, we need to spend money every day to carry on with our daily life. Just as we require to invest money, spend money, organizations need to invest and spend money. Organisations need to invest money in long term projects as well as short term requirements and spend money for its every day activities.

While Finance is providing money at the time when it is required, tapping the right source of finance is very important. We need to arrange finance for the long term finance from those who are willing to invest as owners and look for sharing profits or lend as creditors and wait for a long period to get repayment of the principal amount while interest is also paid periodically. Similarly Short term finance may be arranged from those who are willing to provide funds for a short period. Hence the finance provider may be an investor or a creditor.

Following are the important sources of funds:

1. Issue of Equity shares
2. Issue of Debentures
3. Issue of Preference Shares
4. Loans from Banks & Financial Institutions
5. Deposits from public
6. Ploughing back of profits
7. Trade Credit
8. Trade Advances
9. Factoring
10. Instalment Credit

We shall discuss below the important securities involved while raising funds:

1. Equity Shares

Issue of Equity Shares (also known as Ordinary or Common shares) is undertaken by a joint stock corporation for raising long term capital from the investors who are interested in the participation / sharing of the final profit or loss. Hence the funds raised through issue of equity shares are called Owners' capital since the holders of these shares are the real owners.

Equity Shares is a source of permanent capital and are not redeemed during the life time of the company. The equity shareholders are entitled for the dividend, rate of which is decided by the board of directors of the company. Hence it is known as Variable Income Security.

Features of Equity Shares

i. Maturity

Equity Shareholders can demand their capital only at the time of liquidation of the company and after the claims of others, including preference share holders are settled.

ii. Claims / Right to Income

Equity shareholders have a residual claim on the income of the company. The distribution and the rate of dividend to equity shareholders, is left to the discretion of the Board of Directors of the company under the Companies Act, 1956. There may be occasions of no residue after paying others including preference shareholders. Even if there are sufficient funds, equity shareholders can't force for dividend. The residual income not distributed will also benefit the equity shareholders by way of enhanced earning capacity and capital appreciation.

iii. Claim on Assets

In the event of liquidation of the company, priority is given to creditors and preferred stock holders in settling the claims. The equity shareholders are entitled for all whatever left only after meeting all the other claims. Hence, equity share capital provides a cushion for creditors to absorb losses on liquidation. Ultimately, equity shareholders may or may not get paid fully.

iv. Voting Rights and Control

Every Equity shareholder has voting right equivalent to the number of shares held on resolutions in the meetings of the company. Since a company is managed by Directors who are appointed in the Annual General Meeting by majority votes, shareholders exercise indirect control in the working and management of the company. The voting rights may be exercised by a shareholder through a proxy also.

v. Pre-emptive Right

This is a right of the existing shareholder to purchase new shares in proportionate to the number of shares held. As per Section 81 of the Companies Act, 1956, the new shares shall be offered in proportion to existing equity shareholders. This is done to protect the interests of the existing shareholders from diluting the share holdings investment or accumulated profits.

Advantages of Equity Shares

1. Equity shares do not create any obligation to pay fixed rate of dividend.
2. Equity Shares provide permanent source of capital since it need not be repaid till the company is in existence.
3. Equity shares do not create any charge over the assets of the company

Disadvantages of Equity Shares

1. More Equity Share Capital issued means less utilization of trading on equity
2. As equity capital can not be redeemed, there is danger of over capitalization
3. Investors desirous of investing in safe securities with fixed income do not go for equity shares

2 Preference Shares

Preference shareholders have preferential rights and privileges with respect to income and assets over equity shareholders. In respect of dividend, preference shareholders are given preference before dividends are given to equity shareholders. Also in the event of liquidation, preference is given to preference shareholders in repayment of capital before equity shareholders are paid. Since the rate of dividend paid on preference shares is fixed, those who want to have a constant return on their investment, will purchase preference shares.

Features of Preference Shares

- i. Maturity - As per the Companies (Amendment) Act 1988, preference shares can not be issued as irredeemable or redeemable after the expiry of a period of ten years from the date of its issue.
- ii. Claims on Income – Preference shareholders have a priority in getting their dividend paid before the equity shareholders.
- iii. Claims on Assets – In the event of winding up of the company, Preference shareholders have a superior claim on the general assets of the company, as compared to the equity shareholders.
- iv. Control – Ordinarily, preference shareholders do not have any voting rights and do not have any say in the management or control of the company. However, under Section 87 of the Companies Act, 1956, preference shareholders can vote on resolutions which directly affect the rights attached to their preference shares. In these situations, preference shareholders voting right is in the same proportion of paid up preference share capital bearing to the total paid up equity capital of the company.
- v. Hybrid form of security – The preference shares represent a hybrid form of corporate security which combines the features of equity as well as debt financing. It has characteristics of equity shares by providing for dividend out of profits only, not deductible for company tax purposes and company need not mortgage its assets to secure preference share capital. Also, preference shares have the characteristics of constant rate of return like interest, have preference in repayment over equity shareholders and normally do not have voting right.

Types of Preference Shares

The important types of Preference Shares are as follows:

1. Cumulative – Irrespective of the fact of profit or no profit in a year, the Dividend has to be paid and in case of no profit in a particular, the same gets accumulated
2. Non-cumulative – The dividend is not paid in case of no profits in a particular year and arrear can not be claimed in the subsequent years.

3. Participating Preference Shares – The holders of these shares are first paid the fixed rate of dividend and then the equity share holders are paid a reasonable rate of dividend. The mode for dividing surplus profits between preference and equity shareholders is given in the Articles of Association
4. Non-participating Preference Shares – The preference shares which are paid a fixed rate with no additional right in the surplus profits are called non-participating preference shares.
5. Convertible Preference Shares – The holders of this kind of preference shares are given a right to convert their holdings into equity shares after a specific period.
6. Non-Convertible Preference shares – These are the preference shares which can not be converted into equity shares.

Advantages or Merits of Preference Shares

1. Preference Share Capital provides long-term capital to the company without any fixed charge on the assets.
2. There is no legal compulsion to pay dividend though there are profits. This is helpful for new or growing concerns requiring large funds for short run.
3. Redeemable preference shares can be repaid whenever there are surplus funds with the company.
4. Trading on equity or leverage benefits are available when the expected earnings of the preference shares are more than the stipulated rate of the dividend.
5. Since there is no normal voting rights to preference shareholders, controlling power of the current stockholders are kept in tact and not diluted.

Disadvantages of Preference Shares

1. Preference shares are expensive source of finance as compared to debt. Since the risk is more in case of preference shares as compared to debentures, generally higher rate of dividend may have to be given compared to the rate of interest on debentures.
2. Preference share have a tax disadvantage since dividend on preference shares is not a deductible expense whereas interest on debentures is deductible expense.
3. In case of non / delayed payment of dividend, there is a chance of dilution of control when these shareholders get voting rights.

3 Debt Securities – Debentures or Bonds

A company may borrow funds from the public by issue of Debentures or Bonds to meet its requirements for long-term funds requirements. A Debenture is a document issued under the company's seal promising to pay periodic interest and return the principal sum at a fixed date in future. Debentures may be secured against specific assets. Even if they are not specifically secured, debenture holders have a general lien on the assets of the company.

Features of Debentures or Bonds

- i. Maturity: Debentures are to be repaid at a definite time stipulated. The company may have to set aside funds out of the earnings of the company at periodic intervals for retiring the debentures.

- ii. **Claims on Income:** Debenture or Bond holders have priority of claim on income of the company over equity and preference shareholders. Irrespective of the level of earnings, the company is liable to pay interest to the debenture holders. The default in payment of interest may lead to debenture holders resorting to law and causing winding up of the company.
- iii. **Claims on Assets:** In the event of liquidation of the company, the debenture holders have priority to get their principal amount plus interest before the shareholders are paid. While even the unsecured debentures have general lien over assets, the debenture holders have no right in the surplus assets, if any.
- iv. **Control:** Debenture holders have no voting rights either to elect directors or in any other matter. They have no say in the management of the company, so long as their interests are paid regularly and covenants if any in indenture are met. If there is any default in the payment of interest or principal amount, debenture holders may take control over the company.

Types of Debentures

The important types of Debentures are as follows:

1. **Cumulative –** Irrespective of the fact of profit or no profit in a year, the interest has to be paid and in case of no profit in a particular, the same gets accumulated
2. **Non-cumulative –** The interest or the dividend is not paid in case of no profits in a particular year.
3. **Zero Coupon Bonds –** These bonds do not carry interest, instead they are sold at a deep discount from its maturity value. The difference between the maturity value and the issue price is the gain or interest for the investor.
4. **Convertible Debentures –** Convertible Debentures give the debenture holders an option to exchange the debentures for equity shares after a specific period.
 - a. **Fully Convertible Debentures –** These debentures are converted into equity shares after the lapse of the specified period.
 - b. **Partly Convertible Debentures –** In this kind of debentures, a portion of the debentures is converted into equity shares and the balance is not converted into equity. The unconvertible part of the debentures is redeemed after the lapse of the specified period.

Advantages of Debentures

1. Debentures are cheaper source of funds due to the following reasons:
 - Interest cost is usually less than the dividend paid on shares
 - Issuance cost is less than the cost for any other form security
 - Tax benefits are available
2. Debentures do not dilute the control of existing shareholders since debenture holders do not have any voting rights
3. Debentures are preferred for the fixed rate of interest and since they can be redeemed when there is surplus
4. Debentures allow trading on equity

Disadvantages of Debentures

1. The fixed interest charges become fixed burden on the company, which have to be paid even if there are no profits. Non-payment may lead to legal complications and may even end up in winding up of the company.
2. Use of debt financing increases the risk perception of the firm
3. A company whose earnings are not stable and involve high initial investment can not use this source of funds.

Treasury Bills (T-Bills)

T-Bills are one of the most important instruments in virtually all the money markets in the world. T-Bills are issued by the Government for periods ranging from 14 days to 364 days through regular auctions. They are highly liquid instruments and demand is largely from banks, financial institutions and corporations. Fundamentally, T- Bills are short term instruments issued by RBI on behalf of the Government of India to tide over short term liquidity shortfalls. T-Bills market is much more liquid than that for dated Government of India securities because of shorter tenure of T-Bills.

There are three categories of T-Bills.

- **On-To Bills:** Can be bought from the RBI at any time at an interest yield of 4.663 per cent. But, with the deregulation of the interest rates, they have lost much of their relevance.
- **Ad hoc T-Bills:** are created to replenish the Government's cash balances with the RBI. Thus, they essentially are just an accounting measure in RBI's books. They have a maturity period of 91 days, but can be redeemed prior to the final date of maturity, because for them the dealing is only between the Government and the RBI.
- **Auctioned T-Bills:** First introduced in April 1992 are the most active of the three categories. In effect, they are the only one among the three categories which can actually be called an active money market instrument.

Treasury bill short term money market instrument as well as short term security by which the Government raises finance to meet its short term requirements. The investment in the Treasury Bill is reckoned for the purpose of Statutory Liquidity Reserve (SLR) requirements. The periodicity is 14 days, 28 days, 91 days, 182 days and 364 days. Periodically, RBI comes out with the T-Bills auctions, whereas only in the case of 91 days T-Bills, the amount is notified.

T-Bills transactions are routed through the Special General Ledger (SGL) accounts. Being a short term instrument, it has a good secondary market. Apart from others, all the participants in the money market can sell the T-Bills, including the parties who participate in the T-Bills auctions to the non-competitive bids. In absence of a developed money market, the T-Bill becomes a reference rate for the relative maturities.

Points to Remember

- Shareholders regard a bonus issue as a firm indication that the prospects of the company have brightened and they reasonably look for an increase in total dividends.
- The stock invest provides protection for investors against loss of interest and delay in allotment..

- Debenture interest is tax deductible.
- Derivatives namely options and futures trading volume and value is in the upward movement.

Review Questions

1. As an investor, which long term security you will prefer over others and why?
2. Write short notes on treasury bills.

Note

LESSON 9 :

THE FINANCIAL MARKETS

Learning Objectives

- To understand the role of financial markets in developing the economy of the nation.
- To know existing methods of floating securities including private placement
- To know about major classification of capital market, market participants and Indian capital market scenario and transaction costs in capital market

A financial market is a market for creation and exchange of financial assets.

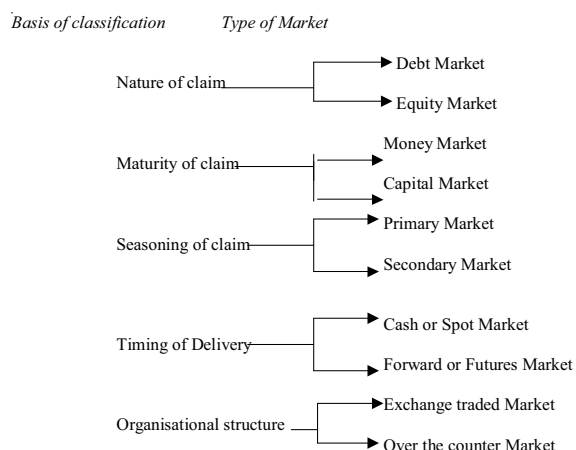
Functions of Financial Markets

Financial markets play a pivotal role in allocating resources in an economy by performing three important functions:

1. Financial markets facilitate price discovery: the continual interaction among numerous buyers and sellers who through financial markets helps in establishing the prices of financial assets. The value of a financial asset can be obtained just by looking at its price in the financial market.
2. Financial markets provide liquidity to financial assets: Investors can readily sell their financial assets through the mechanism of financial markets. Companies can raise long term funds from investors by negotiation and transferability of securities through financial markets
3. Financial markets considerably reduce the cost of transacting. The major costs associated with transacting are search costs and information costs. Search costs include explicit costs such as the expenses incurred on advertising when one wants to buy or sell an asset and implicit costs such as the effort and time one has to put in to locate a customer. Information costs refer to costs incurred in evaluating the investment merits of financial assets.

Classification of Financial Markets

There are different ways of classifying financial markets. They are as follows:



The debt market is the financial market for fixed claims (debt instrument) and equity market is the financial market for residual claims (equity instruments)

Money market is the market for short term financial claims.

Capital market is the market for long term financial claims. Since short term financial claims are almost invariably debt claims, the money market is the market for short term debt instruments and the capital market is the market for long term debt instruments and equity instruments.

Third way to classify financial markets is based on whether the claim represents new issues or outstanding issues. The market where issuers sell new claims is referred to as the primary market and the market where investors trade outstanding securities is called the secondary market.

A cash or spot market is one where the delivery occurs immediately and a forward or futures market is one where the delivery occurs at a pre-determined time in future.

An exchange-traded market is characterized by a centralized organization with standardized procedures. An over the counter market is a decentralized market with customized procedures.

Methods of Floatation

Initial issues are those floated by new companies for the first time, while further issues are subsequent issues floated by the existing companies.

Issues can be classified as those given for cash, for exchange of technical know-how, exchange of shares of another company or exchange for any other services rendered by the agencies or promoters.

The placement of the issues may be through (i) prospectus (ii) offer of sale (iii) private placement (iv) rights issue

- i. Offer through prospectus: Offer through prospectus involves inviting subscription from the public through issue of prospectus. The price at which the securities are offered for sale is at the face value of the share in the case of new companies and may be at a premium or discount in the case of old companies.

The prospectus is a document, which is a notice, or circular or advertisement inviting offers for subscription or purchase of any shares or debentures from the public. The prospectus should contain authentic data such as name of the company, address, activities, board of directors, location of industry, authorized, subscribed and paid up capital, dates of opening and closing of the subscription list, names of brokers and underwriters etc. and should be delivered to the Registrar of Companies for registration. The draft of the prospectus should be approved by board, solicitors, lending financial institutions and the stock exchanges in which they are to be listed.

ii. Offer of sale: This method consists of outright sale by the company instead of offering shares to the public through intermediaries such as issuing houses or share brokers. The company sells shares en bloc at an agreed price to brokers or merchant bankers who in turn resell them to the investing public. Offer of sale of shares takes place in the case of existing shareholders purchasing en bloc and then reselling them to the public. Similarly, foreign collaborators or promoters may sell their shares to the Indian public through offer of sale which may be either through brokers or through prospectus.

iii. Private Placement: Private placement is defined as sale by the issuing house or broker to his own clients of securities previously purchased by him. The issuing houses or financial intermediaries buy them outright with the intention of placing them with their clients afterwards. The brokers would make their profit in the process of reselling to the public. This method of private placement is used to a limited extent in India but is very popular abroad.

Before the issues are placed in the market by the promoters, they sell to their friends and well wishers as the Govt. guidelines require a minimum contribution from the promoters and their friends. Private placement is also made by bargaining with financial institutions for their share of contribution.

Private placement in India is mostly of equity or equity related instruments of unlisted companies and debt instruments of listed companies.

Appeal of private placement: The phenomenal growth of private placement of debt may be attributed to the following reasons.

1. Accessibility: Private placement market can accommodate issues of smaller size whereas the public issues market does not permit an issue below a certain minimum size.
2. Flexibility: There is greater flexibility in working out the terms of issue. e.g. When a non-convertible debenture issue is privately placed, a discount may be given to institutional investors to make the issue attractive. There may be more latitude to renegotiate the terms of the issue and even roll over the debt.
3. Speed: A private placement requires lesser time, perhaps a month or two, as the elaborate procedure followed in a public issue is largely bypassed.
4. Lower issue costs: The issue cost for a private placement is substantially less as a public issue entails several statutory and non-statutory expenses associated with underwriting, brokerage, printing, mailing, announcements, promotion and so on. whose costs are quite high.

Institutional investors, before participating in a private placement, need to look into a number of factors like the net worth of the company, the interest cover, the level of profit before tax, the asset cover, the debt equity ratio, the dividend record, the percentage of unsecured deposits/borrowings, the price history of the company's stock, and the existence of carried forward loss.

iv. Rights Issue: is an offer with a preemption right to the shareholder of existing companies to contribute to the share capital or its debt capital in the form of debentures. An existing company can issue rights to augment its equity base, if necessary. These are offered to the existing shareholders in a particular proportion to their existing share ownership. The rights are themselves transferable and saleable in the market.

Under the Companies Act, where a company increases its subscribed capital by the issue of new shares either after two years of its formation or after one year of the first issue of shares whichever is earlier, these have to be first offered to the existing shareholders with the right to reserve them in favour of a nominee.

A company issuing rights is required to send a circular to all existing shareholders. The circular letter should provide information on how the additional funds would be used and their effect on the earning capacity of the capital. The company may normally give a time limit of at least 1 month or two to shareholders to exercise their exercise their right before it is offered to the public. Rights may also be offered through underwriters. If the company is doing well, rights will be received well by the shareholders and the need for underwriting may not be felt.

Main condition is that the shareholders be given an opportunity to apply for additional shares. If the rights are not fully taken up, the balance is to be distributed among the applicants for additional shares. Any balance still left over after making such allotment is required to be disposed of by the company in the market at the ruling price or the issue price.

Capital Market

The capital market is a market in which the securities with maturities of more than one year are bought and sold. The capital market refers to the market for long term securities, such as corporate stocks and bonds, for financing long term assets. The major players being Corporates, Banks, Financial Institutions, Mutual Funds etc. The capital market is further classified as follows:

a. Primary Capital Market

The 'primary capital market' is one in which new issues of common stock, bonds and preferred stock are sold by companies. The proceeds raised in this market is the new capital funds.

b. Secondary Capital Market

The secondary capital markets are also called 'stock markets'. In the secondary markets, the outstanding issues are permitted to trade. In this market, a stock or bond issue which has already been sold to the public, and it is traded between current and potential owners. The proceeds from a sale in the secondary market do not go to the issuing organisation but to the current owner of the security. The stock markets are classified as follows:

i. Stock Exchanges

The Stock Exchanges will have a physical location where stocks buying and selling transactions take place in the stock exchange floor.

ii. Over the Counter Exchange

Where shares, bonds, and money market instruments are traded using a system of computer screens and telephones is called 'Over the Counter Exchange'

Capital market is the barometer of the economy and represents the macro economic affairs of the country. It is an index of economic and industrial development of the country. Capital market discounts the future and it is reflection of future of the economy. In the long run it is a true measure of the health of any economy.

In the capitalistic economy, the capital markets play a pivotal role by bring the common investor to invest in corporate securities. The global trend is that even socialist countries like China and Russia are moving towards capitalistic economy by inviting the private investments into the industry.

Financial disintermediation along with diversification of financial sector is one of the important developments in the era of liberalisation world over. The term financial disintermediation' refers to the transformation of a financial system from institution based to market based.

In an institution based financial system, the role of financial intermediaries like Bank is prominent in mobilisation and allocation of financial resources. In the current capital market system, the role of banks as intermediaries is diluted in the market based system where the investor and user of funds are expected to come into direct contact with the borrowers i.e. the corporates.

A tendency was developed among the corporate houses to access savings directly through public deposits, commercial paper, Equity and Debenture issues etc. This process of transformation has been further facilitated by the introduction of a set of new financial instruments with varied degrees of liquidity, risk and returns.

Among the instruments, mutual funds, bonds and derivative instruments are more active which have grasped a substantial share in resource mobilisation giving a challenge to traditional monetary assets such as bank deposits. Similarly, instruments like deep discount bonds, zero coupon bonds and other bonds with very long maturity period compete with traditional term saving instruments. The nature, extent, and efficiency of disintermediation varies from country to country. The USA and UK have a market based system while economies like Germany and Japan are dominated by bank based financial system.

In the era of globalisation and liberalisation, the capital markets assume a greater importance. The smooth functioning of the capital market depends on the regulators, participants and investors. The past decade has been a golden age for securities market in India. It is now a far more important source of finance than traditional financial intermediaries for corporate sector. It is poised to dominate the future of corporate finance in India.

The process of reforms has led to a pace of growth of markets almost unparalleled in the history of any country. Securities market in India has grown exponentially as measured in terms of amount raised from the market, number of stock exchanges and other intermediaries, the number of listed stocks, market

capitalisation, trading volumes and turnover on stock exchanges, and investor population.

Along with this, the profiles of the investors, issuers and intermediaries have changed significantly. The I market has witnessed fundamental institutional changes resulting in drastic reduction in transaction costs and significant improvements in efficiency, transparency and safety. Indian market is now comparable to many developed markets.

The state of primary market of late has not been very encouraging as far as the industry is concerned. The number of issues and the amounts mobilised through the primary market is predominantly by financial institutions and banks as opposed to industry. This is a clear reflection of recessionary trends in the economy over the 5 years.

Indian Capital Market Participants

The current state of the Indian capital markets can be summarised as under:

Market Participants	Number as on 31st March, 2001
Securities Appellate Tribunal	1
Regulators (DCA, DEA, RBI & SEBI)	4
Depositories	2
Stock Exchange	
• with Equities Trading	23
• with Debt Market Segment	1
• with Derivative Trading	2
Listed Securities	9922
Brokers	9782
Foreign Brokers	38
Corporate Brokers	3808
Sub-Brokers	9957
FILs	527
Portfolio Managers	40
Custodians	14
Share Transfer Agents	186
Primary Dealers	15
Merchant bankers	233
Bankers to an Issue	69
Debenture Trustees	37
Underwriters	56
Venture Capital Funds	35
Foreign Venture Capital Investors	1
Mutual Funds'	39
Collective Investment Schemes	4

Source: Indian Securities Market - A review (NSE)

SEBI-Securities Board of India

Scenario of Indian Capital Market

Reforms in the securities market, particularly the establishment and empowerment of SEBI, determined allocation of

resources, screen-based nation-wide trading, dematerialisation and electronic transfer of securities, rolling settlement and ban on deferral products, sophisticated risk management and derivatives trading, have greatly improved the regulatory framework and efficiency of trading, settlement. Indian market is now comparable to many developed markets in terms of a numbers, qualitative parameters. Over the past several years the securities market has witnessed a sea change. The market has been more in terms of infrastructure, adoption of best international practices and introduction of competition.

With the maturity of the regulatory framework and increased market surveillance, the market has also become safer and investor is better protected. The extensive reforms introduced by SEBI over the last few years have enhanced the integrity, transparency and efficiency of the operations of the securities market. The introduction of electronic trading and "order matching" system in all the 23 stock exchanges, have led to reduction in transaction costs, speedier execution of trades and gains in liquidity.

Transaction Costs in Capital Market

Transaction cost for a market is important for several reasons. It is a key parameter to measure the impact of modernisation of market infrastructure, institutionalisation of the market and market regulation. Transaction costs can be classified into two categories viz.,

- Explicit costs are observable and measurable. They can be easily measured and are directly borne by the investors. These costs include:

- Brokerage commission
- Stamp duty
- Service tax
- Custody charges
- Regulatory charges etc.

Some of these charges, such as stamp duty and service tax are also levied in few other markets also .

- The Implicit costs include:
- Bid-ask spreads
 - Realised spreads

Opportunity cost of delayed execution (timing costs) or non-execution.

Like implicit costs, certain additional explicit costs also arise from the nature of the trading and settlement systems, existence of physical paper and the registration process. The transaction costs are influenced by the following important factors:

- Trading, Clearance and Settlement
- Systems Physical and
- Dematerialised Shares Investor
- Profile
- After- Trade Practices
- Margins
- Rolling Settlement

In the Indian capital market, transaction costs in respect of most of the components especially brokerage, are charges as a "percentage of the value of trade" (ad valorem) as is the practice in several other markets, viz. Hong Kong, Australia, Singapore, UK and not on "per transaction "basis as in the US markets. Both principles of charging are valid and have their own logic and merit.

The components of explicit transaction costs for institutional investors include brokerage, stamp duty, safe keeping which includes custody charges for physical stock and charges for dematerialisation. Each of these components vary depending on whether:

- The stock which is traded is in compulsory dematerialisation list, which implies that trading and safe keeping have necessarily to be in the dematerialised form or,
- The stock which is traded is in the physical segment, in which case both trading/delivery will be in physical form while safe keeping may be in the physical or dematerialised form or,
- The existing stock has not been fully dematerialised and hence is still being held in the physical form..

The transaction cost is different for each of the above situations. Variations on account of squaring off or carry forward of transactions do not occur for institutional investors as they have to operate on delivery basis.

Points to Remember

- OTCEI has a new tier in the stock trading operations, through the electronic media.
- Badla trading was banned in March 1994 and reintroduced again following the G.S.Patel Committee recommendations. SEBI came out in July 1995 with a modified Badla proposal.
- The function of the central depository is as a trust to hold the physical custody of shares and effect transfers by book entries without the need to deal and transfer the physical certificates between parties.

Review Questions

1. State the reasons for the companies preferring private placement as the most appealing method of floatation of securities.
2. Explain the classification and basis of classification of financial markets.
3. Discuss briefly about the transaction costs in capital market.
4. Discuss the role of capital market in the development of nation's economy

Note

LESSON 10 :

INTERNATIONAL FINANCIAL MARKETS

Learning Objectives

- To make ourselves aware of the International Financial System and Foreign Exchange Market.
- To know what is Eurocurrency market and what its functions are.
- To know what constitutes Euro bond market and features of Euro bond issue and brief procedure in Euro bond issues

Foreign Exchange as a subject deals with the means and methods by which rights to income and wealth in one country's currency are converted into similar rights in terms of another country's currency. It involves the investigation of the methods by which the currency of one country is exchanged for that of another, the causes which render such exchange necessary, the forms in which exchange may take place and the ratios or equivalent values at which such exchanges are effected.

Such exchange may be in the form of one currency to another or of conversion of credit instruments denominated in different currencies such as cheques, drafts, airmail transfers, fax and telegraphic transfers, cable transfers, bills of exchange, trade bills, bankers' bills or any promissory notes. It is through these instruments and foreign currency accounts of banks that the banks are able to effect such exchange of currencies or claims to currencies.

RBI has, however, responded to emerging situation in the foreign exchange market quickly and promptly. Since rupee convertibility was launched in March 1992, RBI is publishing a reference rate around which it operates in the market to stabilise the rate of exchange of rupee in terms of US dollar.

International Financial System and Foreign Exchange Market :

One of the important components of the international financial system is the foreign exchange market. The various commercial and financial transactions as between countries result in receipts and payments as between them. The items of balance of payments leading to receipts and payments are trade, travel, transportation, royalties, fees, investment income, unilateral transfers, short-term and long-term capital receipts and payments etc. Such receipts and payment involve exchange of one currency for another.

Each economy has a foreign sector representing that economy's external transactions. All such external transactions are either economic and commercial or financial transactions. These result in receipts into and payments out of the domestic economy. Such receipts and payments involve exchange of domestic currency as against all foreign currencies of countries with which domestic economy has dealings. Such exchange transactions are put through in the exchange market for that currency. For each nation, therefore, there is an exchange market for that country's currency vis-a-vis other currencies. These national foreign

exchange markets are components of the international financial system

There are three major components of this market, depending upon the level at which transactions are put through:

1. Transactions between the public and the market at the base level involve receipts from any payments to the public or purchases from and sales to public.
2. Transactions as between the banks dealing in foreign exchange involving conversion of currencies taking place in the same centre or as between centres in the same country (inter-bank market) or as between countries involving the correspondent or agent banks abroad or branches of the domestic banks abroad.
3. Transactions between banks and central bank involving purchase and sale of foreign currencies for cover or final disposal of excess foreign balances

The rate of exchange is determined at any moment by the forces of demand for and supply of a currency, which in turn depends on the demand for and supply of commodities and services as between India and the USA.

Market Features of International Markets

1. Both investors and borrowers are well-known names in the international market.
2. These loans are unsecured and no government guarantee either. Only creditworthy borrowers are generally approaching this market.
3. Country's credit rating and borrowing party's credit rating rate being looked into; rating by standard and poor or moody's are popular for assessing the creditworthiness of borrower.

Regulations

The market is functioning outside the countries and is offshore in nature. No regulation of any national and international nature exists. The market practices in the secondary market trading are based on the rules laid down by Association of International Bond dealers.

Same parent companies and some foreign governments do guarantee the interest and payment of principal. The interest payment is provided in the agreement itself and stricter controls are exercised in the collection of interest, although these rates are lower than those on national bonds of any country.

Euro Currency Market

The Euro currency market refers to funds channeled via financial intermediaries from international lenders to international borrowers. The Eurocurrency markets provide the short to medium term debt required by banks, corporate and government borrowers.

The source of these funds is domestic bank deposits whose ownership is transferred to bank outside the controlled domestic monetary systems. The deposits are in large denominations, frequently \$ 100,000 or more, and the banks use them to make Eurocurrency loans to quality borrowers. The Eurocurrency deposit rate is usually slightly higher than the rate paid by the domestic banks.

The Euro Currency loans in a particular currency are priced according to a 'LIBOR (London Inter Bank Offering Rate) plus basis' with the margin over LIBOR depending on market conditions and the credit quality standing or riskiness of the borrower. The banks generally arrange syndicated loans in this market. Thus the risk of a particular borrower are distributed across several banks.

The Euro currency market is not regulated by any government and therefore, is an international currency. The 'Euro' prefix refers to the high volume of these funds circulating in Europe, mainly through London. However, this market is world wide, and sectors of the market exist in the Middle East and Far East (Asia dollar market)

Functions of Eurocurrency Markets:

The Euro currency markets serve three basic functions.

- They are extensively used for foreign exchange hedging purposes as the banks seek to balance out their foreign assets and liabilities. The banks, therefore, take positions in the Eurocurrency markets to cover the forward commitments they have made with their customers.
- Euro currency markets can at times bypass domestic channels of financial 'intermediation' especially when governments impose tight credit policies. For example, US Corporation can acquire Eurodollars in London.

These deposits may be US domestic dollar deposits that have been transferred abroad during a US domestic credit squeeze.

- Function of the market is the full international intermediation role of channelling surplus liquid resources from, say, OPEC countries or corporations who need to borrow.

The most important Euro currency markets are Eurodollar, Euromark and Eurosterling. Eurocurrency market operates only in currencies that are freely convertible into other currencies, such as the US dollar and the German mark, and only outside those currencies-Country of Origin. Eurobanks are, therefore, located in countries which do not regulate their foreign currency banking activities.

The Eurocurrency markets are money markets rather than foreign exchange markets. Their importance in the context of international financial management is that they provide an alternative source of funds to domestic banks. A wide range of participants make use of the Eurocurrency markets.

Types of Eurocurrency Loans - The following types of Eurocurrency loans are available in the Eurocurrency markets:

- Fixed interest loans are usually over a medium term of up to five years, and the borrower knows in advance what the interest payments will be.

- Stand-by credits are overdraft facilities offered by banks to customers in Eurocurrency.
- Most Eurocurrency loans are roll-over loans, where the bank agrees to make funds available over a given period of time, say five years, but the interest rate is open to renegotiation every three or six months.
- A number of banks may form themselves into a syndicate in order to provide a large loan to a single customer over a period that is longer than usual. Such loans are called 'syndicated loans':

The advantages to the borrower of using Euromarket are that lenders on the domestic market may discriminate, or larger sums may be required than are available in the home markets, or it may be deemed necessary to avoid the possibility of exchange control barriers.

Eurobond Market

The second major sector of the unregulated capital market is the Eurobond market. The Eurobond market is characterised by direct transaction between borrowers and lenders, although the large banks play a major role in underwriting and placing the bonds. The market permits lenders to lend directly to borrowers across national borders, without the intermediation of a bank. Bank can, however, play a major role in arranging the transaction. Eurobonds differ from foreign bonds in that the latter are issued in one country in its currency by a foreign borrower. By contrast, the Eurobond is issued in one currency such as US \$, the yen, the pound, the mark or the Swiss franc for sale in many capital markets with the sale managed by an international syndicate of bank Underwriters.

The borrowers or issuers of the bonds include public sector organisations and commercial banks, established bluechip companies. The lenders or purchasers of the bond securities are individual and institutional investors, with the latter group dominating. A secondary market for Eurobonds has grown rapidly as these financial institutions continuously seek to adjust their portfolios, and as a result the original lenders are not committed to a borrower until the final maturity of the bond issue.

Generally, purchasers buy Eurobonds in currencies other than their own, and a major attraction of these bonds for individuals is that in most currencies they are exempt from withholding taxes. Further more, the unregulated nature of the market means that its costs are lower and its issuing process is much faster domestic or foreign bond counter parts. These bonds market provide a wide variety of medium to funds of 3 to 15 years maturity. Fixed coupons or interest rates are common but floating rate notes are also important.

Features of Eurobonds Issue:

The distinctive features of Eurobonds issues are as follows:

- The issue of Eurobonds is in the form 'placing' rather than formal issuing, to avoid national regulations on new issues.
- Eurobonds are placed simultaneously in many countries through multinational syndicates of underwriting banks who sell them to an investment clientele throughout the world.

- Euro bonds are sold principally in countries other than the country of issue.
- The interest on Eurobonds is not subject to withholding tax, which is an advantage over domestic bonds owned by non-residents, on which tax deduction is made before interest is paid.

Brief Procedure in Eurobonds Issues

A Eurobond issue is arranged as follows:

The borrower will appoint a bank to manage the bond issue. This managing Bank will probably be a merchant bank or an investment bank.

This managing bank invites one or more other banks or financial institutions to co-manage the issue.

- The managing banks try to ensure the success of the bond issue by inviting a group of banks and other financial institutions to underwrite the issue.
- Once the issue is underwritten, the managing banks will try to place the issue with a number of selling banks and other financial institutions.
- The selling banks will in turn try to place all or some of their allotment with clients. The 'Placing Power' of the selling banks will be a vital factor in the successful launch of a new issue, i.e., it is crucial that the selling banks should be able to place the bonds with buyers. To help the selling process, a 'Prospectus' will be issued giving information about the terms and conditions of the bond, the name and details of the issuer, the purpose of the loan and selling information.
- Eurobond issues in the UK are usually constituted under a trust deed. The bank which is responsible for making payments to the bondholders will act as their trustee

Points to Remember

- Petro-dollars like other Euro currencies affect the volume and geographical pattern of capital flows among countries.
- Asian currency market facilitates the use of dollar balances in the Asian continent for balance of payments purposes as well as for investment in development projects.
- Banks resort to currency swaps, options and futures in foreign currencies to cover their exchange risk.

Review Questions

1. Discuss the role of foreign exchange market in the international financial system.
2. What is Euro currency market? What are its functions.
3. What are the types of Eurocurrency loans? Explain.
4. What are the distinctive features of the Euro bond issue?
5. Brief the procedure of Euro bond issue

Note

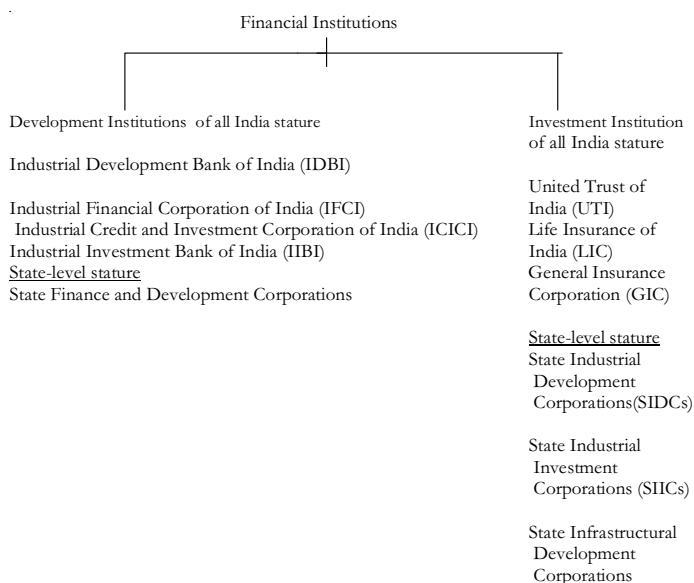
LESSON 11: FINANCIAL INSTITUTIONS

Learning Objectives:

- To know objectives, main functions, major schemes of important financial institutions in India and credit rating agencies.

Functionally, there are two types of financial institutions in Indian capital market namely, developmental institutions and investment institutions.

They are:



Of the major players, the investment institutions like LIC, GIC and UTI, the bulk of the activity is generated by the UTI by its mobilisation of savings from the public through its various schemes.

Now we shall see various institutions and their functions enriching the capital market.

Unit Trust of India (UTI)

Main objectives are:

- To mobilize savings particularly from the low and middle income groups
- To channelise these savings into productive investment
- To provide an assured income to savers

It provides the savers with expert investment service, portfolio management and assured income. The UTI mobilizes the savings through the sale of its units and operates a number of schemes with varied characteristics to suit the requirements of savers and investors. These characteristics include income, capital appreciation, liquidity through repurchase, an element of insurance etc.

Activities of UTI includes

- Underwriting new issues
- Make direct subscription to public issues or private placements
- Keep deposits with banks and companies
- * Operate in the money market by subscribing to Treasury bills, commercial bills etc

It can partake in the inter bank market as a lender or borrower of short term funds. It can borrow from RBI or the Government in case of need.

UTI has more than 64 schemes.

1. Unit Scheme 1964: is a perpetual scheme, each unit with a face value of Rs.10, transferable and repurchased by the UTI. Dividend income from the UTI qualifies for deduction of income tax under Section 80L of the Income Tax Act and eligible for Wealth Tax exemption up to Rs.5 lakhs. Reinvestment facility as an alternative to dividend distribution is available.
2. Unit-linked Insurance Plan (ULIP) 1971: Schemes of 10 year and 15 year periods are available both with life insurance and accident insurance. The target amount is a maximum of Rs.60,000 and the age limit is 55 years. The life coverage is secured from the LIC by the contributions of UTI. The contributions of the insured are payable yearly or half yearly.
3. Capital Gains Unit Scheme (CGUS): Investment in part or in full of the capital gains due to sale of any long term capital assets in the Capital Gains Units Scheme would qualify for exemption from capital gains tax under Section 54A of the Income tax Act. This investment is to be made within 6 months of the date of the sale of capital assets and there is no maximum limit. This is to be held for a minimum period of 3 years. Dividend income along with others is exempt under Section 80L of Income tax Act up to Rs.115,000 and from the wealth tax up to Rs.5 lakhs
4. Children's Gift Growth Plan: Units with a face value of Rs.10 are gifted to a minor child under this scheme, maturing at the age of 21 by the child. Gifts up to Rs.20,000 were exempt from gift tax, at that time and Rs.30,000 now.
5. Various Close ended Schemes: Such schemes as monthly income unit scheme, growing income unit scheme etc, were started in various years since 1982 for meeting the varying needs of investors.
6. Master Shares floated for Indian investors and India Fund and India Growth Fund etc. for foreign investors: The Mutual Fund Subsidiary Unit Scheme (Master Shares) 1986 are quoted on the stock exchanges and are treated as shares. A third offshore fund was initiated in 1989-90 with the help

of a Japanese securities firm and AB, in addition to the above two.

The first 3 schemes account for 70 % of the total investible funds of the UTI. The second venture capital fund managed by the Technology Development and Information Company of India Ltd. Was set up in 1990 with a contribution of Rs.35 crores from the UTI and contributions from other financial institutions. This helps the financing of Greenfield ventures from new entrepreneurs to promote industrial growth.

Industrial Development Bank of India (IDBI)

IDBI is the leader in the Indian capital market. It has both regulatory and developmental functions.

The role of IDBI in the market is to promote and develop industries as per the planned projections of the Government, to provide technical and consultancy assistance for the expansion and modernization of industry and for rediscount and refinance facilities to financial institutions.

IDBI as the leader in the market coordinates guides and monitors the credit facilities by all India and State level financial institutions and development corporations. It undertook the responsibility for promoting and financing the small and cottage industries and also the import-export trade of the country until separate bodies namely Exim Bank and Small Industries Development Bank of India (SIDBI) were set up.

Major schemes of IDBI are

- i. Textile Modernisation Fund Scheme for the purpose of promoting modernization of jute and cotton textile industries
- ii. Venture capital Fund Scheme for the purpose of developing new ventures and promoting risky ventures by new technicians and new entrepreneurs
- iii. Technology upgradation Scheme for the purpose of upgrading technology in selected capital goods industries
- iv. Energy conservation Scheme for the purpose of financing of energy saving proposals of industries and
- v. Seed Capital Scheme for the purpose of providing initial seed capital in the form of promoters' contribution to new ventures. The IDBI operates direct assistance schemes whereby project appraisal and financing for medium and large scale projects is undertaken. It provides refinancing under the Industrial Loans Scheme for the loans extended by commercial and cooperative banks and SFCs. This refinancing is for assisting SSIs, transport operators and units in the backward areas etc.

IDBI also borrows abroad and secures funds under arrangements with foreign financial institutions, foreign banks and foreign governments and operates foreign lines of credit, secured by the Government.

Industrial Credit and Investment Corporation of India (ICICI)

The objectives are:

- To encourage industrial investment in the private sector and to promote participation of private capital, both external

and internal with a view to widening the base of industrial development and of capital market in India.

It has been providing

- Full merchant banking services, venture capital financing and allied financial services.
- Assistance to the private and joint sector enterprises with rupee and foreign currency loans and advances, underwriting, guaranteeing of loans from other agencies, equity participation and other forms of assistance particularly to the new enterprises.
- Services to the development of backward areas through industrial growth, promotion of investment priorities of the Government and for the growth of private corporate sector.

Venture capital funds for the promotion of green field companies and risk capital investment and joined other financial institutions in setting up SHCI, CRISIL and OTC Exchange of India Ltd.

Sources of Funds

It paid up capital, borrowing from Government, IDBI, RBI etc. and raises funds from the capital market by issuing bonds or equity and borrow from abroad with the permission of the Government.

It has set up its own bank and a mutual fund like UTI.

Industrial Finance Corporation of India (IFCI)

IFCI was as set up for the purpose of assisting the medium and long term projects for capital expenditure on projects, expansion, modernization and technological upgradation etc.

It plays a promotional and developmental role, particularly in relation to the small and medium scale entrepreneurs.

Main functions are providing:

- Underwriting and guaranteeing facilities to the industrial concerns.
- Term loans or contribute to the debentures or equity
- Rupee loans and foreign currency loans are through foreign credit lines from foreign banks and foreign governments etc.
- Financial assistance to limited companies either in the private, public or joint sectors and cooperative societies engaged in
 - a. manufacturing, preservation and processing of goods
 - b. shipping
 - c. mining
 - d. hotel industry
 - e. electricity generation and distribution etc

It concentrates mostly on medium and large scale projects. By an agreement with other Financial Institutions and IDBI, the IFC specializes in financing of certain industries like jute, cement, sugar etc.

Industrial Reconstruction Bank of India (IRBI)

Main purpose is to provide assistance to sick but potentially viable units and extend finance for their modernization, expansion or diversification.

Sources of Funds

Government, borrowing from the IDBI, RBI etc. issue of debentures or bonds to the public or raise deposits from the public or raise deposits from the public.

Industrial Investment Bank of India (IIBI)

It has shed off its outfit of financing sick industries into one financing industrial investment for growth, expansion, diversification etc. It has now become a development financial institution with a five year tax holiday in addition to completely restructured capital structure.

In addition to the above development institutions, we have National Housing Bank that is the apex bank for Housing Finance to provide finance and refinance to Housing Finance Corporations like HDFC. Shipping Credit and Investment Corporation of India (SCICI) help financing and refinancing shipping companies.

Insurance Companies

Life Insurance Corporation (LIC)

LIC provides cover of risk of life, retirement, disability etc.

Sources of Funds

Paid up capital, premium collections, borrowing from government and the RBI and now the mobilization of savings by insurance companies accounts for 79% of the financial savings of the household sector.

The proportion of funds to be deployed at their discretion was hardly 25% of the total investible funds, while the rest are earmarked for specified government and semi government securities (50%) and for lending or socially desirable projects like housing finance loan, mortgage loans, loans to approved bodies for electricity, housing, water etc (25%)

LIC invests in shares/debentures of companies, provide rupee loans to companies, makes initial subscription and underwriting of new issues of shares etc.

It has set up a Mutual Fund for operating various schemes for mobilization of savings from the public, particularly from the rural and urban areas and channel these funds to the stock market and capital market.

General Insurance Corporation (GIC)

Sources of Funds

Paid up capital, reserves, profits and premium income.

It operates in the money and capital markets, earn on underwriting and making of long term loans and investments.

Main Objectives are

- To carry on the general insurance business other than life such as accident, fire etc. to aid and advise the subsidiaries to conduct insurance business and to help the conduct of investment strategy of the subsidiaries in an efficient and productive manner.

SFCs/SIDCs

Sources of Funds

Issue of bonds and debentures in the market, fixed deposits from the public, borrowings from RBI, IDBI, state Government etc.

These bodies assist small scale industries, units in backward areas, new entrepreneurs etc. Industrial Development Corporations, Industrial Investment Corporations were set up by some states for promotion, infrastructural development, financial, managerial and entrepreneurial guidance to the industrialists, provision of industrial sheds/plots for setting up of small units etc.

Stock Holding Corporation of India Ltd. (SHCIL)

It was intended to facilitate quicker share transfer among financial institutions, and their transfers with other investors, members of stock exchanges etc.

It is offering its services to its own seven promoter institutions. It acts on the basis of Letter of Authority/Power of Authority issued by the financial institutions and render the following services

- a. Clearing services – It receives and/or delivers the securities on behalf of the institutions after each settlement of the Stock Exchange. Monitors the execution of the purchase/sale orders of the institutions by the brokers.
- b. Depository services-including services related to the collection of dividend warrants, and interest cheques; it attends to the necessary consolidation or splitting of shares, redemption and collection of bonus and rights and other privileges attached to shares, debentures and other instruments.
- c. Support services-arranges to monitor interest/dividend payments due to the institutions and ensure that the warrant
- d. Management/Information services
- e. Development services

Credit Rating and Information Services of India Ltd (CRISIL)

Main objective is to rate the debt obligations of Indian companies on a voluntary basis with a view to providing the investors a guide as to the risk of timely payment of interest and principal by the company. The rating is confined to fixed deposits, debentures, preference shares and short term instruments like commercial paper. CRISIL does not rate equity shares or the performance of the company as a whole.

CRISIL Rating and Investor Protection

Investors should be familiar with the ratings given by the Credit Rating and Information Services of India Ltd. (CRISIL) for protecting their interests. The CRISIL ratings are given only for debt instruments of companies, namely, Commercial Papers (CP), debentures, bonds and fixed deposits. Since early 1992, the ratings of Investment Information and Credit Rating Agency (ICRA) are also used by companies.

The ratings in use by the companies are published by the CRISIL in their "CRISIL Rating Scan." The companies use them only when the ratings are favourable to them and when they are making any public offer for deposits or debentures, etc. The investors should examine these symbols with regard to their benefits and their implication. Since a rating given can be revised upwards or downwards, the investors should keep a watch on any such revisions.

Rating of company deposits is made Compulsory under the new companies Bill 1993.

The implications of the rating used are as follows:

For Debentures, simple 'A' and 'B' are used; for preference shares, 'pr' is prefixed.

For fixed deposits and short-term instruments, 'F' and 'P' are prefixed.

Debenture Ratings	Implication
Triple A - (AAA)	- Highest Safety
Double A - (AA)	- High Safety
Single A - (A)	- Adequate Safety
Triple B - (BBB)	- Moderate Safety
Double B - (BB)	- Inadequate Safety
B, C & D	- High risk and default-prone
Fixed Deposits Ratings,	
F Triple A - (FAAA)	- Highest Safety
F Double A - (FAA)	- High Safety
F Single A - (FA)	- Adequate Safety

FB, FC and FD are inadequate, high-risk and default-prone instruments.

Information and Credit Rating Agency (ICRA)

Another Credit Rating Agency was set up later, namely, The Investment Information and Credit Rating Agency of India Limited, registered in 1991, sponsored by the IFCI and a number of other financial institutions. It has its headquarters at Delhi, while the CRISIL has its headquarters in Mumbai.

Credit Rating is now mandatory for all debt instruments - issued by the companies both short-term and long-term. The debentures of both convertible and non-convertible nature require credit rating, if they are not convertible within 18 months.

Debentures, bonds, preference shares and fixed deposits are eligible to be rated. The rating is instrument wise but not for the whole company. The ratings given by the credit rating agency like ICRA are indicators of the correct opinion of the relative capability of timely servicing of debts and obligations by the company.

The ratings given are AAA, AA or A for safety, BBB, BB or B for less safe investments C and D indicate no safety and default respectively. But as equities are essentially risky and investor is willingly entering the risky investment of equity, credit rating for equity is neither possible nor necessary. Only in respect of debt instruments, investor wants to avoid risk and so he should know the extent of the risk, he is taking. Equity holder is the owner and takes risk voluntarily while the debenture holder is creditor and he does not want to take risk in respect of interest or in payment of the principal.

Factors Considered

In evaluating the instruments in question, CRISIL takes into account the various risks involved in servicing of instruments. They undertake business analysis, financial analysis, manage-

ment evaluation and fundamental analysis. They also take into account the regulatory and competitive environment faced by the company and other relevant factors.

The CRISIL also evaluates the management risk as judged by the track record of the management, planning and control system, managerial expertise, tax planning and expansion plans of the company. In general, the goals and objectives of the company, its sales and asset growth, capacity to overcome adverse economic and market conditions, etc. will be assessed by the CRISIL.

Under fundamental analysis, evaluation is carried out for the equity earnings per share, book value, capital and sales turnover, management, debt-equity ratio, current assets to liabilities ratio, contingent liabilities, etc. Besides, the asset quality will be judged by the risk element of various assets and interest and tax sensitivity of the company

Ratings

All the above types of analysis will be done by experts after discussions with the company executives, its suppliers, customers and market competitors. The information is gathered on a confidential basis and assessed with a view to examine the capacity of the company to service the instruments issued to the public.

Broadly, there are three categories of ratings:

- a. Long-term Instruments
(Debentures and Preference Shares) Triple A to triple B
- b. Medium-term Instruments
(Fixed Deposits) FAAA to FA
- c. Short-term Instruments
(C.P., etc.) PI + to P₃

The rating indicates the degree of risk and the highest safety possible is under the rating of AAA or F AAA or P I' The credit worthiness is indicated in the ratings and reflects the capacity of the company to service them adequately with a view to ensure the timely payment of interest and principal.

Advantages

It will help raise funds in the capital market. Although many companies have got their instruments evaluated by CRISIL to get an independent view of their credit standing, a few only have publicised them. For the investors these rating help them in making a decision of their investment in various instruments.

Such ratings have a demonstration effect and improve competitiveness of the companies to outperform each other.

Duff and Phelps

This is another credit rating agency in the private sector, set up in 1995-96. It has started rating of companies with regard to all the debt instruments and is offering fair competition to all the above rating agencies.

Points to Remember

- National Small Industries Corporation (NSIC) helps SSIs with promotional services and National Industrial Development Corporation (NIDC) provides consultancy services and designing project management kit.

Review Questions

- 1. Highlight the major schemes by UTL.
- 2. Brief the main functions of development and investment institutions.
- 3. Write short notes on credit rating agencies.

Note

LESSON 12 :

INTEREST RATES

Learning Objectives

- To learn about the role of interest rates, computation of nominal and real rate of interest.
- To understand how changes in interest rates influence financing decisions and also its effect on share prices
- To know what is term structure basically and factors influencing interest rates
- To see how interest rates are fixed for different securities in India and also in international scenario.

Interest is a return to the capital as a factor of production. Savings is the basis of investment and capital formation. Investment is a function of interest rate and the quantum and directional flow of investment determine income. Interest Rate and interest income have a vital role in the economy and plays a significant part in the growth dynamics of the economy. Capital formation and capital accumulation are very much dependent upon the level and rate of savings and investment in the economy.

The savings and investment in the economy which are influenced by the interest rates are the real economic variables. It is these savings and investment activities of the real sector which are influenced and facilitated by the flow of funds from the financial system development.

Role of Interest Rates

Interest rate has diverse roles in the economy such as a reward to capital which is a factor of production, a return to saving and a cost to investment and an instrument of monetary policy for credit. These and other aspects of interest rates are relevant for policy makers.

At the macro level, interest is a reward to the capital, which is a factor of production. In the developing economy, capital formation and capital accumulation are of vital importance to economic development. Interest rates influence both savings and investment in the economy and serve a vital role for the mobilisation of savings. Besides, interest rate has a role in the allocation of investment funds as between different uses. Thus, in addition to influencing their cost, and availability of funds from the supply side, interest rates also influence the quantum of investments from the demand side and thus determine the income and employment in the economy.

While the overall household sector savings are mildly influenced by interest rates, household sector savings in the form of bank deposits are highly sensitive to interest rates. It would, therefore, appear that investment rates have a positive role in the savings mobilisation, particularly with regard to savings in the form of bank deposits

Interest Rates is an important consideration for a Modern Finance Manager in taking investment decisions.

Interest rates are the measure of cost of borrowing. The interest rates of a country also influence the foreign exchange value of its own currency. Interest rates are taken as a guide in investments into shares, debentures, deposits, real estates, loan lending etc.

The interest rates in different market segments due to the following reasons:

- Risk: Borrowers carrying high risk will pay higher rates of interest than the borrowers with less risk .
- Size of Loan: The higher amounts of deposits carry higher interest than small deposits.
- Profit on re-lending : Financial intermediaries make their profits from re-lending at a higher rate of interest than the cost of their borrowing.

Nominal and Real Rates of Interest

- Nominal rates of interest are the actual rates of interest paid.
- Real rates of interest are the rates of interest adjusted for the inflation. The Real rate is, therefore, a measure of the increase in the real wealth, expressed in terms of buying power, of the investor or lender.

The Real rate of interest is calculated as follows:

$$\text{Real rate of interest} = \frac{1 + \text{Nominal rate of interest}}{1 + \text{Rate of inflation}} - 1$$

Also, Real rate of interest = Nominal rate of return – Rate of inflation

The Real rate of interest will usually be positive, although when the rate of inflation is very high, because the lenders will want to earn a real return and will therefore want Nominal rates of interest to exceed the inflation rate. A positive real rate of interest adds to an investor's real wealth from the income he earns from his investments.

Interest Rates, Capital Gains and Losses - The increase or decrease in the value of stock is calculated as follows:

$$\text{Real value of stock} = \text{Face value of stock} \times \frac{\text{Nominal rate of stock}}{\text{Market Nominal rate}}$$

Changes in Interest Rates and Financing Decisions:

The changes in interest rates will have; strong impact on financing decisions taken by a Finance Manager .

Financial strategy to be followed when interest rates are low:

- Borrow more moneys at fixed rate of interest to increase the company's gearing and to maximise return on Equity.
- Borrow long term funds rather than short term funds.
- Replace the high cost debt with low cost debt.

Financial strategy to be followed when interest rates are higher:

- Raise funds by issue of equity shares and to stay away from raising debt finances.
- Debt finance can be taken for short-term rather than long-term.
- Surplus liquid assets can profitably be invested by switching of investments from equity shares to interest bearing investments.
- Reduce the need to borrow funds by selling unwanted and inefficient assets, keep the stocks and debtors balances at lower levels etc.

New projects need to be given careful consideration, which must be able to earn the increased cost of financing the projects.

Interest Rates and Share Prices

The shares and debt instruments are alternative ways of investment. If the interest rates on debt instruments fall, shares become more attractive to buy. Demand for shares increases, their prices rise too, and so the dividend return gained from them falls in percentage terms. If interest rates went up, the shareholder would probably want a higher return from his shares and share prices would fall.

Term Structure and Interest Rates

The most commonly quoted interest rates in the financial markets are:

- a. The bank's base rate
- b. The inter-bank lending rate
- c. The treasury bill rate
- d. The yield on long-dated gilt-edged securities.

The term structure of interest rates describes the relationship between interest rates and loan maturities. The maturity pattern of debt is an important factor for interest rate determination. The maturity pattern is particularly relevant in the gilt-edged market, small savings schemes and deposit and lending rates of banks. The longer the maturity, the larger is the risk and hence the risk premium.

The interest rates on debentures, PSU bonds and term lending rates by banks are higher due to high risk involved. These rates depend on the characteristics of these instruments like ownership risk, variability of return, maturity period, safety, marketability etc. Therefore the upward and downward shifts in the structure of rates depend upon the level of risk, time period of maturity and other features of the financial instruments involved.

Factors Influencing Interest Rates

These financial assets have a structure of interest rates which should be rationally based on the degree of risk of ownership, uncertainty of return, period of maturity and a host of other factors.

The factors which influence interest rates on various financial investments are the nature of the instrument, maturity period, risk, tax status, marketability and liquidity.

Interest Rates in India

Most of the interest rates in India are controlled and administratively fixed. It would, therefore, be difficult to isolate the free

market forces which influence the interest rates in India. So far, interest rate policy has been used as a support to other instruments of monetary policy in India. Even under these controlled conditions, the basic postulates that interest rates are influenced by the money, liquidity and the rates of inflation etc. hold good particularly in uncontrolled unorganised market.

Bank Rate and Refinance Rates

In the organised sector of the financial system for which data are available, interest rate policy is set by the actions of the Central Bank as its discount rate (viz., Bank rate is the leader of interest rates). In India, the bank rate is the discount rate at which the central bank of the country makes advances to the commercial and co-operative banks against eligible assets.

Section 49 of the RBI Act defines Bank Rate as the standard rate at which the Bank is prepared to buy and discount bill of exchange or other commercial paper eligible for purchase under the Act. In view of the limited role of the money market and bill market in India, the Bank rate is not the real leader among the interest rates. Under Section 17 of the RBI Act, the Reserve Bank provides refinance through various windows at different rates. It is these refinance rates which are the discount rates of the central bank in India.

The weighted average rate of interest on these various types of refinance would be the effective rate of discount by the RBI. This is different from the Bank rate. The multiplicity of rates is necessitated by the social obligations of the banking system and the nature of our financial system.

Money Market Rates

Next to the Bank rate, the other short-term rates of significance are the Treasury Bills rate and Call Money rate. The 91-day Treasury Bill rate has remained frozen for years at 4.6% but from January 1993 this rate was freed. This is because there is no market for Treasury Bills in India and the instrument is used for the purpose of financing government expenditure. As such this rate has no significance for the general structure of interest rates. Treasury bills of 182 days sold on auction basis since 1987 bear higher rates which are more realistic and market-oriented.

As regards call money rate, it has no doubt some relevance to the money market conditions, as the stringency or otherwise of the banks' funds is reflected in this market. The ceiling rate was fixed at 10% in April 1980 but was removed in May 1989. Since then the rates fluctuated daily depending on the demand and supply conditions.

Rates in Gilt-edged Market

In the government securities market, interest rates are set by the government in the form of coupon rates. The yields on these securities are also influenced by the RBI through their purchase and sale operations in the market. In India, there is a captive market in the government securities as the banks are required to invest a proportion of their funds in government securities and other approved trustee securities as fixed by the statutory liquidity ratio (SLR).

Similarly, the non-bank financial institutions, provident funds, trusts etc. are also required to invest a proportion of their funds by law in the government securities. From the point of view of risk, these securities are least risky and if the return is to be

positive in real terms, the yield has to be higher than the actual inflation rate in the economy. The yields are also related to the maturity period and a differential is maintained as between the Center and States and government-guaranteed bonds. As part of reforms in gilt edged markets their rates of interest are decided through auction system.

The auctioning system of 182 days of Treasury bills was extended to Treasury bills of 364 days since April 1992. This has become an additional money market instrument between 91 day Treasury bill and dated Government security. The Discount and Finance House of India Ltd. was allowed to operate in Government securities since April 1992, in addition to their operations in Money Market instruments like Treasury bills and Commercial paper and in the inter bank call market.

Private and Government Bond Rates

So there is need for two types of adjustments in gilt-edged market; firstly, the long end of the spectrum is to be made more attractive, consistent with the expected rate of inflation. Secondly, the yields on government securities should be attractive vis-a-vis yields on private securities, which is reflected in gaps and reverse gaps. A gap is positive if there is risk aversion in the economy which has to be corrected.

These adjustments were made over the last decade of the eighties with result that the gap between them is narrower at around 2 percentage points. The coupon rates offered in the gilt-edged market during the last few years have been jacked up to bring them to realistic levels of the expected rate of inflation. Besides, this has improved the earnings of banks and made them more attractive to banks and non-bank investors.

Rates of Borrowings and Lending by Commercial Banks

In view of the limited role of the interest rate policy in India and the weak linkages as between the various segments in the market, it has become necessary for the RBI to resort to direct controls on banks' lending and borrowing rates. Minimum lending rates of scheduled commercial banks were first fixed in 1960 and the maximum lending rates in 1964. Similarly, interest rates on deposits have been fixed by the Reserve Bank of India since September 1969. The control encompasses almost all categories of borrowers and different maturities of deposits.

The rationale of direct controls on interest rates may be briefly set out here.

1. In view of the weak linkages between the Bank rate and/or refinance rate of the RBI with other money market rates, the effectiveness of monetary and credit policy can only be ensured by the direct fixation of various categories of interest rates.
2. The banking structure of India is such that there is uneven competition amongst the banks, namely, as between the weaker and smaller banks and dominant bigger banks. It is, therefore, necessary that competition in matters of interest rates has to be regulated by the Reserve Bank.
3. With the acceptance of social obligations by the Banking system following the nationalisation of banks in 1969, the multiplicity of lending rates has become necessary to encourage lending to priority sectors, neglected segments of

the economy and backward classes of people. The geographical and sectorial credit gaps have to be filled in through price and non-price incentives.

4. The developmental needs of the country have to be met at the lowest possible costs by the banking system through bank credit to government borrowings through bonds, treasury bills, etc. This has also necessitated the differential interest rate system in India.

Interest Rates on Small Savings

Interest rates on saving deposits, cumulative time deposits and other categories of deposits in post offices are fixed from time to time by the government in consultation with the Reserve Bank. These rates are marginally higher than those offered by commercial and co-operative banks in order to attract the small savers and to encourage the flow of savings into the government sector. Similarly, interest rates on various categories of national savings certificates, social security certificates etc. are fixed higher, as they are for lump sum investments and meant for small savers.

Export Credit

Interest rates on export credits have been subject to a ceiling with a view to encouraging exports from India. These rates have been uniformly lower than those applicable to domestic creditors. Term credits and those of deferred payments arrangements are kept still further lower.

In India, many of these interest rates are fixed by the government. The banks' deposits and lending rates and those of financial institutions are fixed by the RBI, while the rates on P.O. instruments, PSU bonds and those on government securities are all fixed in consultation with the RBI.

The rationale of regulating interest rates in India is as follows:

- i. It keeps the interest cost low for the government borrowings and their investment.
- ii. It promotes the socio-economic objectives of providing cheaper finance to weaker section and priority sectors of the economy.

International interest Rates

The rate of interest may vary from country to country due to differing rates of inflation, government policies and regulations, foreign exchange rates etc.

Points to Ponder

- The level and structure of interest rates will be determined by the policy of the government and the RBI and not by the market forces of supply and demand.
- The nominal interest rates according to the neo-classical theories depended also on the expectations of the public and the rate of inflation in the economy.

Review Questions

1. What are the factors influencing interest rates in India and also in International market?
2. Explain how interest rates influences financial decisions.
3. Discuss briefly about the interest rates in India

LESSON 13 : BASIC VALUATION MODEL

Learning Objectives

- To learn to compute valuation of important securities based on discounted cash flows at a discount rate for certain period of years.

As you know bonds and debentures carry fixed rate of interest and preference shares carrying fixed rate of dividend, the value of these securities can be computed on discounting the estimated cash flows at a discount rate.

Valuation of securities are done as follows:

Bond Valuation

The value of a bond- or any asset, real or financial is equal to the present value of the cash flows expected from it. Hence determining the value of a bond requires

- An estimate of expected cash flows
- An estimate of the required return

The following assumptions are made to simplify the analysis of bond valuation

- The coupon interest rate is fixed for the term of the bond
- The coupon payments are made every year and the next coupon payment is
- Receivable exactly a year from now.
- The bond will be redeemed at par on maturity

The cash flow for a non callable bond comprises an annuity of a fixed coupon interest payable annually and the principal amount payable at maturity. Hence the value of a bond is:

$$P = \frac{CF_1}{(1+r)^1} + \frac{CF_2}{(1+r)^2} + \frac{CF_3}{(1+r)^3} + \dots + \frac{CF_n}{(1+r)^n} + \frac{M}{(1+r)^n}$$

$$= \sum_{t=1}^n \frac{CF_t}{(1+r)^t} + \frac{M}{(1+r)^n}$$

where P = value (in rupees)

n = number of years

CF_t = annual coupon payment or annual cash flows
(in rupees)

r = periodic required return

M = Maturity value

T = time period when the payment is received

Since the stream of coupon payments is an ordinary annuity, we can apply the formula for the present value of an ordinary annuity.

Hence the bond value is given by the formula

$$P = C \times PVIFA_{r,n} + M \times PVIF_{r,n}$$

Illustration:

Consider a 10 year, 12 % coupon bond with a face value of Rs.1000. The required yield on this bond is 13%. The cash flows for this bond are:

10 annual coupon payments of Rs.120

Rs.1000 principal repayment 10 years from now

Therefore the value of the bond is

$$P = 120 \times PVIFA_{13\%,10yr} + 1000 \times PVIF_{13\%,10yr.3}$$

$$= 120 \times 5.426 + 1000 \times 0.295$$

$$= 651.1 + 295 = \text{Rs.}946.1$$

Debenture Valuation

$$P = \frac{I}{(1+r)} + \frac{I}{(1+r)^2} + \frac{I}{(1+r)^3} + \dots + \frac{I}{(1+r)^n} + \frac{M}{(1+r)^n}$$

$$= \sum_{t=1}^n \frac{I}{(1+r)^t} + \frac{M}{(1+r)^n}$$

where P = value (in rupees)

n = number of years

I = annual coupon payment i.e Debenture Interest
(in rupees)

r = periodic required return

M = Maturity value

T = time period when the payment is received

Valuation of Preference Shares

$$P = \frac{D_p}{(1+r)} + \frac{D_p}{(1+r)^2} + \frac{D_p}{(1+r)^3} + \dots + \frac{D_p}{(1+r)^n} + \frac{M}{(1+r)^n}$$

$$= \sum_{t=1}^n \frac{D_p}{(1+r)^t} + \frac{M}{(1+r)^n}$$

where P = value (in rupees)

n = number of years

D_p = Preference Dividend

r = periodic required return

M = Maturity value

T = time period when the payment is received

Stock Valuation

According to the dividend discount model, the value of an equity share is equal to the present value of dividends expected from its ownership plus the present value of the sale price expected when the equity share is sold.

Following assumptions are made for applying the dividend discount model.

- Dividends are paid annually – it is a common practice for business firms in India.
- The first dividend is received one year after the equity share is bought.

There are two models of stock valuation: They are

Single Period Valuation Model

When an investor expects to hold the equity share for one year, the price of the equity share will be

$$P_o = \frac{D_1}{(1+r)} + \frac{P_1}{(1+r)}$$

where P_o = current price of the equity share
 D_1 = dividend expected a year hence
 P_1 = price of the share expected a year hence
 r = rate of return required on the equity share

Illustration:

Uranus Ltd.'s equity share is expected to provide a dividend of Rs. 2.00 and fetch a price of Rs.18.00 a year hence. What price would it sell for now if the investors' required rate of return is 12%?

The current price will be

$$P_o = \frac{2.0}{(1.12)} + \frac{18.00}{(1.12)}$$

If the price of the equity share is expected to grow at a rate of g per cent annually, the current price being P_o becomes $P_o(1+g)$ a year hence and hence we get

$$P_o = \frac{D_1}{(1+r)} + \frac{P_o(1+g)}{(1+r)}$$

On simplifying, we get,

$$P_o = \frac{D_1}{r-g}$$

Given the current market price and forecast values of dividend and share price, the expected rate of return is equal to

$$R = D_1/P_o + g$$

Multi-period Valuation Model

Since equity shares have no maturity period, they may be expected to bring a dividend stream of infinite duration.

Hence the value of an equity share will be

$$P_o = \frac{D_1}{(1+r)} + \frac{D_2}{(1+r)^2} + \dots + \frac{D_\infty}{(1+r)^\infty}$$

$$= \sum_{t=1}^{\infty} \frac{D_t}{(1+r)^t} \longrightarrow \text{Eqn. (1)}$$

where P_o = price of the equity share today
 D_1 = dividend expected a year hence
 D_2 = dividend expected 2 years hence
 D_∞ = dividend expected at the end of infinity
 r = expected return

An investor who plans to hold it for n years and sell it therefore for a price of P_n will get his equity share having a value

$$P_o = \frac{D_1}{(1+r)} + \frac{D_2}{(1+r)^2} + \dots + \frac{D_n}{(1+r)^n} + \frac{P_n}{(1+r)^n}$$

$$= \sum_{t=1}^n \frac{D_t}{(1+r)^t} + \frac{P_n}{(1+r)^n}$$

For practical applications it is helpful to make simplifying assumptions about the pattern of dividend growth as follows:

- The dividend per share remains constant forever, implying that the growth rate is nil (the zero growth model)
- The dividend per share grows at a constant rate per year forever (the constant growth model)
- The dividend per share grows at a constant extraordinary rate for a finite period, followed by a constant normal rate of growth forever thereafter.
- The dividend per share, currently growing at an above normal rate, experiences a gradually declining rate of growth for a while. Therefore, it grows at a constant normal rate

Zero Growth Model:

If we assume that the dividend per share remains constant year after year at a value of D , then the eqn.(1) becomes

$$P_o = \frac{D}{(1+r)} + \frac{D}{(1+r)^2} + \dots + \frac{D}{(1+r)^n} + \dots \infty$$

On simplification, we get

$$P_o = D/r$$

Constant Growth Model

One of the most popular dividend discount models assumes that the dividend per share grows at a constant rate (g). The value of a share, under this assumption is

$$P_0 = \frac{D_1}{(1+r)} + \frac{D_1(1+g)}{(1+r)^2} + \dots + \frac{D_1(1+g)^n}{(1+r)^{n+1}} + \dots$$

Applying the formula for the sum of a geometric progression, the above expression simplifies to

$$P_0 = \frac{D_1}{r-g}$$

Illustration:

Arvind Engg. Ltd. is expected to grow at the rate of 6 % per annum. The dividend expected on Arvind's equity share a year hence is Rs.2.00. What price will you put on it if your required rate of return for this share is 14 % ?

The price of Arvind's equity share would be

$$P_0 = \frac{2.00}{(0.14-0.06)} = \text{Rs.}25.00$$

Stock Valuation: The P/E Ratio Approach

An approach to valuation, practiced widely by investment analysts, is the P/E ratio or earnings multiplier approach. The value of a stock is estimated as follows:

$$P_0 = E_1 \times P_0/E_1$$

where P_0 = estimated price

E_1 = estimated earnings per share

$$P_0/E_1 = \text{justified price-earnings ratio} \quad (1-b)$$

$$\text{And } P_0/E_1 = \frac{1}{r - \text{ROE} \times b}$$

where b stands for the plough back ratio and $g = \text{ROE} \times b$, ROE stands for Return On Equity. Therefore the factors that determine the P/E ratio are:

- The dividend payout ratio (1-b)
- The required rate of return, r
- The expected growth rate, ROE x b

Market Efficiency

An efficient market is one in which the market price of a security is an unbiased estimate of its intrinsic value. Market efficiency is defined in relation to information that is reflected in security prices.

Eugene Fama suggested that it is useful to distinguish three levels of market efficiency:

Weak form efficiency: Prices reflect all information found in the record of past prices and volumes.

Semi strong form efficiency: Prices reflect not only all information found in the record of past prices and volumes but also all other publicly available information.

Strong form efficiency: Prices reflect all available information, public as well as private.

Points to Remember

- As the expected growth rate in dividend increases, other things being equal, the price earnings ratio increases.
- The yield to maturity of a bond is the interest rate that makes the present value of the cash flows receivable from owning the bond equal to the price of the bond.

Review Questions

1. A bond of face value Rs.100 bearing a coupon rate of 15% will mature after 5 years. What is the value of the bond, if the discount rate is 10%?
2. A bond of face value Rs.100 bears a coupon rate of 10% and matures after 5 years. Interest is payable semi annually. Compute the value of the bond if the required rate of return is 12%?
3. The equity stock of M/s. Venus Ltd. is currently selling for Rs. 50 per share. The dividend expected next year is Rs.4.00. The investors' required rate of return on this stock is 15%. If the constant growth model applies to M/s. Venus Ltd., what is the expected growth rate?
4. The current dividend on an equity share of Xenics Ltd. is Rs. 12. The present growth rate is 30%. However, this will decline linearly over a period of 5 years and then stabilize at 10%. What is the intrinsic value per share of Xenics Ltd., if investors require a return of 15% from its stock?
5. Explain the P/E ratio approach of stock valuation.
6. Define market efficiency.

Note

LESSON 14 :

RISK

Learning Objectives

- To learn about risk and types of risk based on the factors affecting the securities.
- To know the measures of risk and advantages of standard deviation over variance
- To learn Risk-Return relationship, diversification and risk, benefits of diversification
- To understand what is beta, how to calculate beta and using beta to estimate return

The risk of an investment refers to the variability of its rate of return.

Forces contributing to variations in return – price or dividend (interest) constitute elements of risk. Some influences are external to the firm and affect large number of securities and are uncontrollable in nature. Such forces are called sources of systematic risk, whereas controllable factors internal to the firm and peculiar to industries are called sources of unsystematic risk.

Systematic or NonDiversifiable Risk

It refers to that portion of total variability in return caused by factor affecting the prices of all securities. The sources of such a risk are economic, political and sociological changes. This results in prices of all common stocks to move together in the same manner.

Systematic Risk Includes

1. Market Risk: There are many causes, but is mainly due to a change in investors' attitude toward equities in general, or toward certain types or groups of securities in particular.
2. Interest rate risk: refers to the uncertainty of future market values and of the size of future income, caused by fluctuations in the general level of interest rates.
3. Purchasing power risk: is the uncertainty of the purchasing power of the amounts to be received. In short, purchasing power refers to the impact of inflation or deflation of an investment.

Rising prices on goods and services are normally associated with what is referred to as inflation. Falling prices on goods and services are termed deflation.

Unsystematic or Diversifiable Risk

It is the portion of total risk that is unique to a firm or industry. Factors causing such risk are management capability, consumer preferences, and labour strikes. These factors affect one firm and they must be examined for each firm.

Unsystematic Risk

The uncertainty surrounding the ability of the issuer to make payments on securities has two sources (i) the operating environment of the business and (ii) the financing of the firm. They are

1. Business Risk : is a function of the operating conditions faced by a firm and the variability these conditions inject into operating income and expected dividends. The degree of variation from the expected trend would measure business risk.

Business risk can be divided into two categories. They are:

- i. External business risk is the result of operating conditions imposed upon the firm by circumstances beyond its control.
 - ii. Internal business risk is largely associated with the efficiency with which a firm conducts its operations within the broader operating environment imposed upon it.
2. Financial risk: is associated with the way a firm finances its activities. By engaging in debt financing, the firm changes the characteristics of the earnings stream available to the common stock holders, as interest on debt is paid before dividend payment.

The relationship between total risk, diversifiable risk and non-diversifiable risk is given by the equation

$$\text{Total risk} = \text{Diversifiable risk} + \text{Non-diversifiable risk.}$$

Diversifiable risk can be eliminated by holding a large enough portfolio of securities, the only relevant risk to be considered is non-diversifiable risk.

Non-diversifiable risk is unavoidable, and each security possesses its own level of non-diversifiable risk, measured using the beta coefficient.

Measures of Risk

A simple measure of dispersion is the range of values, which is simply the difference between the highest and lowest values.

Other measures commonly used are as:

- i. Variance: It is the mean of the squares of deviations of individual returns around their average value
- ii. Standard deviation: It is the square root of variance
- iii. Beta: This reflects how volatile is the return from an investment in response to market swings.

Variance and Standard Deviation are defined as

$$\sigma^2 = \left(\frac{\sum_{i=1}^n (R_i - \bar{R})^2}{n-1} \right)$$

$$\sigma = \sqrt{\sigma^2}$$

where σ^2 = variance of return
 σ = standard deviation of return
 R_i = return from the stock in period I (I=1,...,n)
 –
 R = arithmetic return
 N = number of periods

Illustration:

Consider the following example of the return from a stock over a period of 6 years

$$R_1=15\%, R_2=12\%, R_3=20\%, R_4=-10\%, R_5=14\%, R_6=9\%$$

The variance and standard deviation of returns are calculated as follows:

Period	Return	Deviation	Square of deviation
	R_i	$(R_i - \bar{R})$	$(R_i - \bar{R})^2$
1	15	5	25
2	12	2	4
3	20	10	100
4	-10	-20	400
5	14	4	16
6	9	-1	1
$\sum R_i = 60$			$\sum (R_i - \bar{R})^2 = 536$
$\bar{R} = 10$			

$$\sigma^2 = \frac{\sum (R_i - \bar{R})^2}{n-1} = 107.2$$

$$\sigma = (107.2)^{1/2} = 10.4$$

Therefore you can see that

- The difference between the various values and the mean value are squared. This means that the values which are far away from the mean value have a much more impact on standard deviation than values which are close to the mean value.
- The standard deviation is obtained as the square root of the average of squared deviations. This means that the standard deviation and the mean are measured in the same units and hence the two can be directly compared.

Criticism of Variance as a Measure of Risk

Two main criticisms of the use of variance as a measure of risk are:

1. Variance considers both positive and negative deviations. Some researchers argue that only negative deviations should be considered while measuring risk. Markowitz suggests a measure of downside risk called semi variance. Semi variance is calculated in the same way as variance but considers only negative deviations. As long as the returns are disturbed symmetrically, variance is simply two times semi-variance and it does not make any difference if variance is used or semi-variance is used.

2. When the probability distribution is not symmetrical around its expected value, variance alone does not suffice. In addition to variance, the skewness of the distribution should also be used. Markowitz model justifies the use of variance on the grounds that the historical returns of stocks have been approximately symmetrical.

Rationale for Standard Deviation

Notwithstanding the above criticisms standard deviation (the square root of variance) is commonly used in finance as a measure of risk

The reasons are:

1. If a variable is normally distributed, its mean and standard deviation contain all the information about its probability distribution. The probability that a particular outcome will be above or below a specified value can be determined.

68.3% of the values fall within a band of one standard deviation on either side of the arithmetic mean. This % increases to 95% for 2 standard deviations and 99% for 3 S.D.s.

2. If the utility of money is represented by a quadratic function (a function commonly suggested to represent diminishing marginal utility of wealth), then the expected utility is a function of mean and standard deviation.
3. S.D is analytically more easily tractable.

Rate of Return

Rate of Return on an asset for a given period (usually a period of one year) is defined as

$$\begin{aligned} \text{Rate of return} &= \frac{\text{Annual income} + \text{Ending price} - \text{Beginning price}}{\text{Beginning price}} \\ &= \frac{\text{Annual income}}{\text{Beginning price}} + \frac{\text{Ending price} - \text{Beginning price}}{\text{Beginning price}} \\ &= \text{Current yield} + \text{Capital gains/loss yield} \end{aligned}$$

Illustration:

Price at the beginning of the year : Rs.75

Dividend paid at the end of the year : Rs.3

Price at the end of the year : Rs.90

Therefore the rate of return on this stock is

$$\begin{aligned} &= \frac{3.00 + (90-75)}{75} \\ &= 0.24 \text{ or } 24\% \end{aligned}$$

This rate of return may be broken down as follows

$$\begin{aligned} &= \frac{3}{75} + \frac{(90-75)}{75} \\ &= 4\% + 2\% \end{aligned}$$

thus we have 4 % current yield and 2% capital gains yield

Probability Distribution

An investment in a stock may result in various possible values. Consider two equity stocks, Mercury Company Ltd and Zenith Company Limited. Mercury Company Ltd may provide a return of 15%, 20% or 25% with certain probabilities associated with them, based on the state of the economy. The stock Zenith Company Limited, being more volatile, earn a return of - 18%,15% or 38% with the same probabilities, based on the state of the economy.

The probability of an event represents the likelihood of its occurrence. Suppose you say that there is a 4 to 1 chance that the market price of a stock A will rise during the next fortnight. This implies that there is an 80% chance that the price of stock A will increase and a 20 % chance that it will not increase during the next fortnight. Your judgement can be represented in the form of probability distribution as follows:

Outcome	Probability
Stock price will rise	0.80
Stock price will not rise	0.20

When you define the probability distribution of rate of return (or for that matter any other variable) remember that:

- The possible outcomes must be mutually exclusive and collectively exhaustive.
- The probability assigned to an outcome may vary between 0 and 1. (An impossible event is assigned a probability of 0, a certain event a probability of 1, and an uncertain event a probability somewhere between 0 and 1.)
- The sum of the probabilities assigned to various possible outcomes is 1.

Probability Distributions of the Rate of Return on Mercury Company Limited Stocks and Zenith Company Limited Stock

State of the Economy	Probability of Occurrence	Rate of Return (%)	
		Mercury Co.Ltd.	Zenith Co. Ltd.
Boom	0.30	25	38
Normal	0.50	20	15
Recession	0.20	15	- 18

Based on the probability distribution of the rate of return, you can compute two key pa-rameters, the expected rate of return and the standard deviation of rate of return.

Expected Rate of Return

The expected rate of return is the weighted average of all possible returns multiplied by their respective probabilities.

In symbols,

$$\bar{R} = \sum p_i R_i$$

where $\bar{R}$ = expected return

R_i = return for the i th possible outcome

P_i = probability associated with R_i

n = number of possible outcomes

From the above eqn. it is clear that $\bar{R}$ is the weighted average of possible outcomes-each out-come is weighted by the probability associated with it. The expected rate of return on Mercury stock is:

$$R_m = (0.30) (25\%) + (0.50) (20\%) + (0.20) (15\%) = 20.5\%$$

Similarly, the expected rate of return on Zenith stock is:

$$R_z = (0.30) (40\%) + (0.50) (10\%) + (0.20) (- 20\%) = 13.0\%$$

Standard Deviation of Return

Risk refers to the dispersion of a variable. It is commonly measured by the variance or standard deviation. The variance of a probability distribution is the sum of the squares for the deviations of actual returns from the expected return, weighted by the associated probabilities. In symbols,

$$\sigma^2 = \sum p_i (R_i - \bar{R})^2$$

where σ^2 = variance

R_i = return for the ith possible outcome

p_i = probability associated with the ith possible outcome

$\bar{R}$ = expected return

Square root of variance namely $\sigma = \sqrt{\sigma^2}$

Continuous Probability Distributions

The probability distribution of returns on Mercury Co. Ltd. and Zenith Co. Ltd. Is a discrete distribution because probabilities have been assigned to a finite number of specific values. In finance, probability distributions are commonly regarded as continuous, even though they may actually be discrete. In this case, probabilities are assigned to intervals between two points on a continuous curve. e.g. What is the probability that rate of return will fall between 25% and 50%?

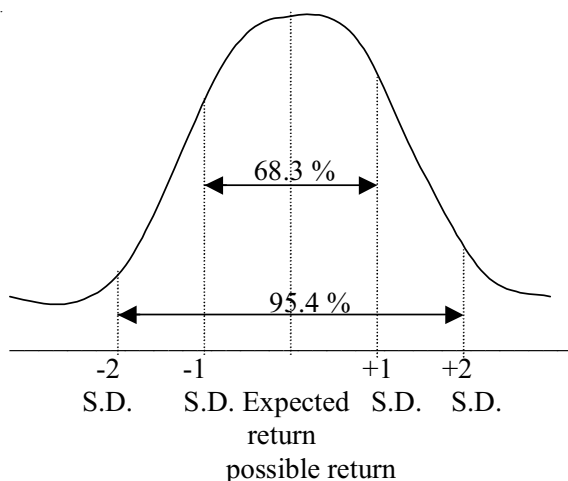
The Normal Distribution, a continuous probability distribution, is the most commonly used probability distribution in finance. The normal distribution resembles a bell shaped curve. It appears that stock returns, atleast over short time intervals are approximately normally distributed.

The following are the features of a normal distribution:

1. It is completely characterized by just two parameters viz. Expected return and standard deviation of return.
2. A bell shaped distribution, it is perfectly symmetric around the expected return.
3. The probabilities for values lying within certain bands are as follows:

Band	Probability
+ 1 S.D	68.3 %
+ 2 S.D	95.4 %
+ 3 S.D	99.7 %

The same can be represented graphically as follows



Risk-Return Relationship

The entire scenario of security analysis is built on two concepts of security: return and risk. The risk and return constitute the framework for taking investment decision. Return from equity comprises dividend and capital appreciation. To earn return on investment, that is, to earn dividend and to get capital appreciation, investment has to be made for some period which in turn implies passage of time. Dealing with the return to be achieved requires estimate of the return on investment over the time period.

Risk denotes deviation of actual return from the estimated return. This deviation of actual return from expected return may be on either side -both above and below the expected return. However, investors are more concerned with the downside risk.

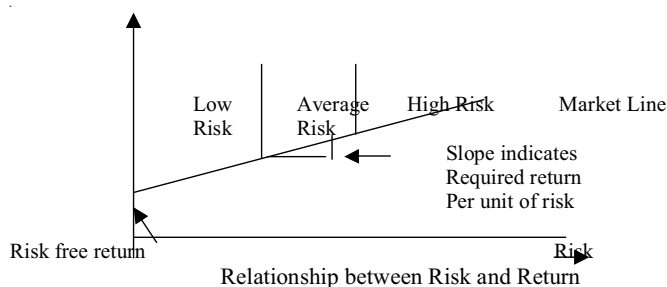
The risk in holding security deviation of return deviation of dividend and capital appreciation from the expected return may arise due to internal and external forces. That part of the risk which is internal that is unique and related to the firm and industry is called 'unsystematic risk'. That part of the risk which is external and which affects all securities and is broad in its effect is called 'systematic risk'.

The fact that investors do not hold a single security which they consider most profitable is enough to say that they are not only interested in the maximisation of return, but also minimisation of risk. The unsystematic risk is eliminated through holding more diversified securities. Systematic risk is also known as non-diversifiable risk as this cannot be eliminated through holding more securities and is also called 'market risk'. Therefore, diversification leads to risk reduction but only to the minimum level of market risk.

The investors increase their required return as perceived uncertainty increases. The rate of return differs substantially among alternative investments, and because the required return on specific investments change over time, the factors that influence the required rate of return must be considered.

The following figure represents the relationship between risk and return. The slope of the Market line indicates the return per

unit of risk required by all investors. Highly risk-averse investors would have a steeper line, and vice versa.



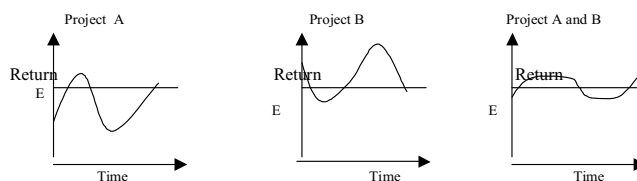
Given the composite market line prevailing at a point of time, investors would select investments that act consistent with their risk preferences. Some will consider low risk investments, while others prefer high risk investments.

Diversification and Risk

The important principle to consider that in an efficient capital market, investors should not hold all their eggs in one basket; they should hold a well-diversified portfolio.

In order to diversify risk for the creation of an efficient portfolio (one that allows the firm to achieve the maximum return for a given level of risk or to minimize risk for a given level of return), the concept of correlation must be understood. Correlation is a statistical measure that indicates the relationship, if any, between series of numbers representing anything from cash flows to test data. If the two-series move together, they are positively correlated; if the series move in opposite directions, they are negatively correlated. The existence of perfectly correlated - especially negatively correlated - projects is quite rare.

In order to diversify project risk and thereby reduce the firm's overall risk, the projects that are best combined or added to the existing portfolio of projects are those that have a negative (or low positive) correlation with existing projects. By combining negatively correlated projects, the overall variability of returns or risk can be reduced. The figure below illustrates the result of diversifying to reduce risk.



It shows that a portfolio containing the negatively correlated projects A and B, both having the same expected return, E, also has the return E, but less risk (i.e., less variability of return) than either of the projects taken separately. This type of risk is sometimes described as diversifiable or alpha risk.

The creation of a portfolio by combining two perfectly correlated projects cannot reduce the portfolio's overall risk below the risk of the least risky project, while the creation of a portfolio combining two projects that are perfectly negatively correlated can reduce the portfolio's total risk to a level below that of either of the component projects, which in certain situations

may be zero. Combining projects with correlations falling between perfect positive correlation (i.e., a correlation coefficient of + 1) and perfect negative correlation (i.e., a correlation coefficient of - 1), can therefore reduce the overall risk of a portfolio.

Measurement of Portfolio Risk

In order to measure risk in investment analysis, we may express estimates of the results of a project as probability distributions by giving a list of the possible cash flows with an estimate of the relative likelihood (probability) of the occurrence of each one. This leads to a characterization of the results of a project in two measure:

- The expected value: the weighted average of the present values of the various possible outcomes, using probabilities as weight; and,
- A measure of the dispersion of possible outcomes (the risk of the project), conveniently summarized in a statistical quantity, the variance or standard deviation.

Benefits of Diversification

The gains in risk reduction from portfolio diversification depend inversely upon the extent to which the returns on securities in a portfolio are positively correlated. Ideally the securities should display negative correlation. This implies that if a pair of securities has a negative correlation of returns, then in circumstances where one of the securities is performing badly the other is likely to be doing well, and vice versa in reverse circumstances. Therefore the 'average' return on holding the two securities is likely to be much safer than investing in one of them alone.

Beta

It measures non-diversifiable risk. Beta shows how the price of a security responds to market forces. The more responsive the price of a security is to changes in the market, the higher will be its beta. Beta is calculated by relating the returns on a security with the returns for the market. The beta for the overall market is equal to 1.00 and the market return is measured by the average return of a large sample of stocks.

- If the beta is more than 1, it is more sensitive to the market or systematic risk than the average investment.
- If the beta is 1, then it has the same risk profile as the market as a whole, the average risk profile.
- If beta is less than 1, it is not as sensitive to systematic or market as the average investment.

The beta factor is the correlation co-efficient between the returns on a market portfolio of investments and the returns on a particular stock or investment.

Calculation of Beta

The expected return of an individual security is equal to the risk free rate plus an additional amount for bearing systematic as measured by beta.

$$R_s = a + B_s R_m$$

Where R_s = estimated return on the stock

a = estimated return when the market return is zero

B_s = measure of stock's sensitivity to the market index

R_m = return on the market index

Illustration:

Star Finwest Ltd., a mutual fund manager having highly sensitive securities in his portfolio. Its return on securities will vary twice the average market returns.

Suppose, if the market went up or down by 5%, then the returns on the securities of the above investor will increase or decrease by 10%. The beta factor of securities of the mutual fund investments is 2.0

Using Beta to Estimate Return

Capital Asset Pricing Model (CAPM) uses beta to link formally the notions of risk and return. CAPM was developed to provide a system whereby investors are able to assess the impact of an investment in a proposed security on the risk and return of their portfolio

CAPM is used to understand the basic risk –return tradeoffs involved in various types of investment decisions.

The model is used to define the required return on security according to the following equation:

$$R_s = R_f + B_s(R_m - R_f)$$

where

R_s = the return required on the investment

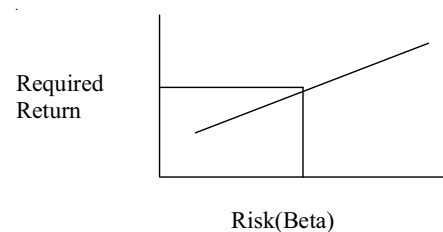
R_f = the return that can be earned on a risk free investment

R_m = the average return on all securities

B_s = the security's beta (systematic) risk

Security Market Line

When the capital asset pricing model (CAPM) is depicted graphically, it is called Security Market Line (SML) which is a straight line. It tells us the required return an investor should earn in the market place for any level of unsystematic(beta) risk.



Points to Remember

- Just as the risk of an individual security is measured by the variance, the risk of a portfolio is measured by the variance of its return.
- Expected return on a portfolio is simply the weighted average of the expected returns on the securities comprising the portfolio.

Review Questions

1. Define risk. Explain the types of risks.
2. How do you design a portfolio of securities that optimizes risks involved.
3. Explain the concept of rate of return
4. What is beta? Explain the beta models followed to estimate return.

LESSON 15

INVESTMENT DECISIONS RULES

Learning Objectives

- To understand various investment decision rules.
- To know what are the good investments decisions rules.
- To know the category of investment decision rules.

Introduction

In the earlier units, you have learnt about the financial environment as well as financial securities, markets and valuation models while knowing about the corporate finance. Now, you need to have the clarity about how to take investment decisions and also the proper capital budgeting in achievement of our goals.

You are aware that any investor has to take decision about the investment, as own funds are involved in the investment. So, you can take investment decision only after analyzing entire process of investment that starts with funds contribution and ends with getting expectations fulfilled. In the sense, you will take decision only after ensuring that the required expectations in terms of returns are ensured at any cost. You are yet to decide whether to invest or not, whenever, you have faced with new investments and projects, though earlier units has lead to take decision.

In this lesson, the investment decision rules allow you to formalize the process and specify what condition or conditions need to be met to accept the project. For example, an investment decision rule may specify that only projects that the amount invested in them, in less than five years will be accepted or only projects that earn a return on capital greater than their cost of capital are good projects.

The characteristics of good investment decision rules are: -

1. It should maintain a fair balance between allowing manager to analyze the project and bring-in subjective assessment into the decision and ensuring that different projects are judged consistently.
2. It must maximize the value of the firm
3. It should work across a variety of investments.

So, an investment decision rule should lead to different conclusions on whether the project should be accepted or rejected. Main rule can be made as primary rule.

Category of Investment Decision Rules

The investment decision rule can be categorized into two broad ways based on the basis of calculation of return: -

- Accounting Income-based decision rule
- Cash flow-based decision rule.

Accounting Income-based Decision Rule

It is the oldest and most established investment decision rules. Here, accounting statements in particular accounting measures of income have been used. I.e., return on capital and return on equity.

Return on Capital

The expected return on capital on a project is a function of both total investments required on the project and its capacity to generate operating income. Return on capital is calculated both pre-tax and after-tax.

So,

Return on capital (Pre-tax) =

$$\frac{\text{Earnings before Interest \& Tax (EBIT)}}{\text{Average Book value of total investment in project}}$$

Return on capital (After-tax) =

$$\frac{\text{Earnings before Interest \& Tax (EBIT) (1 - t)}}{\text{Average Book value of total investment in project}}$$

Decision Rule:

If return on capital exceeds the cost of capital, the project is viewed as good.

If return on capital > cost of capital → Accept the project and

If return on capital < cost of capital → Reject the project

Return on Equity

The return on Equity means return to the equity investors using the accounting net income as a means of this return.

So,

Return on Equity =

$$\frac{\text{Net Income}}{\text{Average Book value of total investment in project}}$$

Decision rule :

If return on Equity > cost of Equity → Accept the project and

If return on Equity < cost of Equity → Reject the project

Cash Flow-based Decision Rule

This is nothing but the cash flow return on capital & equity. One simple modification to the accounting return on capital and equity is to use the cash earnings to compute these returns rather than the accounting earnings. Adding non-cash accounting expenses to operating earnings yields cash operating earnings.

Therefore,

Cash Operating Income =

$$\text{EBIT (1 - t) + Depreciation \& other non-cash charges}$$

Cash Return on capital =

$$\frac{\text{Cash Operating Income}}{\text{Average Book value of total investment in project}}$$

Decision rule:

If Cash Return on capital > cost of capital → Accept the project and

If Cash Return on capital < cost of capital → Reject the project

This can be generalized to look at the return to equity investments. Adding depreciation & other non-cash charges to net income, you can derive the cash equity income.

Therefore,

Cash Equity Income =

Net Income + Depreciation & other non-cash charges

Cash Return on Equity =

Cash Equity Income

Average Book value of total investment in project

Decision rule :

If Cash Return on Equity > cost of Equity → Accept the project and

If Cash Return on Equity < cost of Equity → Reject the project

Points to Ponder

- Characteristics of good investment decision rules
- Category of Investment decision Rules
- Accounting Income-based decision rule
- Cash flow-based decision rule

Review Questions

1. What is investment decision?
2. List the characteristics of good investment decision rules?
3. What are the bases for investment decision rules?
4. Describe the various ways of calculating the returns used in investment decision?

Note

LESSON 16 :

CASH FLOW

Learning Objectives

- To understand various investor expectations.
- To know the need for cash flow statement.
- To estimate cash flow for an investment.
- To aware about the various components of cash flow.

Introduction

Till now, you have learnt that any investment decisions must be taken within the investment decision rule. The investment decision rule ensures the required expectations in terms of returns from any cost.

The investor's expectations will be

- To get Maximum Return.
- To have Minimum Risk.
- To have Safety for the funds.
- To have Liquidity through marketability, so that funds can be withdrawn at any moment of time.

You know that the cash movements in the project will give immense assurance for reaching the expectations. It is in this context, you need to know how cash flow takes place in the investment and also how to estimate the same.

Cash Flow

You can understand cash flow better by means of

1. Definition
2. Need for cash flow statement
3. Estimation of cash flow
4. Components of cash flow

Definition

Cash flow is the changes in financial position on cash basis, commonly known as the cash flow statement. It summarizes the cause of changes in cash position between dates of two balance sheets. It indicates the source of cash & uses of cash.

It is similar to fund flow statement, but it focuses attention on cash instead of working capital (funds). Thus this statement analyses the changes in the non-current accounts as well as current accountants.

Need for Cash Flow Statement

A project is an activity that involves investing sum of money now in anticipation of benefits spread over a period of time in the future. So, you encounter how you will determine whether project is financially viable or not? Immediately, you will sum up the benefits accruing over the future period and compare the total value of the benefits with the initial investment. If the aggregate value of the benefits exceeds the initial investment, the project is considered to be financially viable.

So, sound investment decisions should be based on the cash flow. But as you have studied earlier time value of the money

will affect the cash flow generated. Hence, you have to calculate the net present value (NPV) of future returns. So, you will encounter with a first problem of what should be discounted? It is very clear that the returns of the future, needs to be discounted. The returns will be received by the investor in terms of flow of cash received from the investment made by means of proper investment decision.

Discount Rates

The second major problem will be in applying NPV rule, i.e., what rate should be used to discount cash flows? As a principle we use the opportunity cost of capital as the discount rate. You will understand more about the concept of opportunity cost of capital and its measurement in the Unit 4, while discussing about the aspect of financial decisions in the corporate finance. The Discount rate is the rate at which the cash flows will be discounted to know the net present value of the future cash flows.

Estimation of Cash Flow

The future cash flow of any investment has to be estimated. In reality estimating cash flow is the most difficult task. It is because of uncertainty & accounting ambiguities. Event affecting the investment opportunities change rapidly and unexpectedly. Anticipation of change in the events is not an easy task. Mostly accounting data forms the basis for estimation of cash flows. Accounting data are the result of arbitrary assumptions, choices and allocations. Hence it needs to be adjusted as errors may occur while estimating cash flow.

Now, you are aware that cash flow estimation is the most crucial step in the investment analysis. Therefore, management of a company devotes considerable time, effort and money in obtaining correct estimates of cash flows. As financial manager, you have to prepare the cash flow estimates on the basis of the information supplied by the experts in accounting, production, marketing, and economics and so on after checking the information for relevance and accuracy.

Depreciation & Taxes

In the cash flow estimation, the effect of depreciation & taxes makes lot of difference. The computation of after-tax cash flows requires careful treatment of non-cash expenses items such as depreciation. Depreciation is an allocation of cost of an asset. It involves accounting entry and does not require any cash outflow, the cash outflow occurs when the assets were acquired.

Depreciation (as per income tax rules) is however deductible for computation of taxes. Indirectly it influences the cash flow, since reduces the firm's tax liability. Cash outflows for tax saved is in fact an inflow of cash. The savings resulting from depreciation is called depreciation tax shield.

Depreciation is a non-cash item and should be added to profit to compute actual cash flows.

Components of Cash Flow

During estimation of cash flow, you have to know the various components of cash flow so that you can develop a complete cash flow. A typical investment will have three components of cash flows:

1. Initial investment
2. Annual net cash flows
3. Terminal cash flows

Initial Investment

Initial investment is the net cash outlay in the period in which an asset is purchased. A major element of the initial investment is gross outlay or original value (OV) of the asset, which comprises its cost including cost of accessories & spare parts and freight and installation charges.

Book Value

Original value of an asset is included in the exiting block of assets for computing annual depreciation. Original value minus depreciation is the asset's book value (BV).

Initial investment = Cost of asset + Cost of accessories & spare parts + installation charges.

However, when assets are purchased for the expanding revenues, it may require lump sum investment in net working capital. Thus, initial investment will be equal to : gross investment plus increase in net working capital.

$$\text{Initial Investment} = \text{Gross investment} + \text{increase in net working capital}$$

In case of replacement decisions, the existing assets will have sold if the new asset is acquired. The sale of the existing assets provides cash inflow. So the cash proceeds from the sale of existing assets should be subtracted to arrive at the initial investment.

$$\text{Initial Investment} = \text{Gross investment} + \text{increase in net working capital} - \text{Cash inflow}$$

So, depending upon the situation, we can calculate the initial investment using the above equations.

Annual Net Cash Flows

An investment is expected to generate annual cash flows from the operations after the initial cash outlay has been made. Cash flow should always be estimated on an after-tax basis. However, some are advocating for cash flow before-taxes and discount them before-tax discount rate to find NPV. But, in practice there is no easy and meaningful way for adjusting the discount rate on a before-tax basis.

$$\text{Net Cash Flow (NCF)} = \text{Revenue (REV)} - \text{Expenses (EXP)} - \text{Taxes (TAX)}$$

Terminal Cash Flows

Annual net cash flows will continue to be received till the last or terminal year. In this year, the value received as salvage value along with incremental net cash flow for the last or terminal year will amounts to the terminal cash flow. Therefore, terminal cash flow is the cash flow received during the last or terminal year of investment.

Salvage Value

Salvage value may be defined as the market price of an investment at the time of its sale. The cash proceeds from the sale of the assets will be treated as cash inflow in the terminal (last) year. As per the existing tax laws in India, no tax liability will arise on the sale of an asset. The value of the asset sold is adjusted in the depreciable base of assets.

In case of a replacement decision, in addition to the salvage value of the new investment at the end of its life, two other salvage values have to be considered. They are

1. The salvage value of the existing assets now (at the time of replacement decision)
2. The salvage value of the existing assets at the end of its life, if it were not replaced.

Points to Ponder

- Expectations of Investors
- Various values in relation to cash flow.
- Components of cash flow.
- Initial investment
- Annual Net Cash flow.
- Terminal cash flow.

Review Questions

1. What the expectations of an Investor?
2. Define Cash Flow?
3. Explain the components of cash flows?
4. What is the need for the estimation of cash flow and how it is done?
5. Explain the effect of depreciation & taxes on the cash flow?
6. Explain the meaning of Book Value & Salvage value with illustration?

Note

[illegible]

LESSON 17: INCREMENTAL ANALYSIS

Learning Objectives

- To understand Incremental analysis.
- To know about sunk cost & other costs involved in cash flow.

Introduction

In earlier lessons of this unit, you have learnt about the investment decision, cash flow and other related aspects. Now, you need to know about additional aspects of incremental analysis in investment opportunities.

You have to analyze investment opportunities to the different levels of achievement in the investment. The incremental principle should be carefully used in determining an investment's cash flows. All cash flows occurring because of the investment under consideration should be included. Cash flows, which would occur otherwise, whether or not the project is under-taken, should not be taken into account. Similarly, cash flows which have occurred before the consideration of an investment are irrelevant in taking the decision now.

The following are the examples of some more aspects of incremental cash flow analysis:

Allocated Overheads

Firms generally have a practice of allocating budgeted general overheads to projects, including the new projects under consideration. Since the general overheads will be incurred whether or not the new projects are undertaken, those allocated overheads should be ignored in computing the net cash flows of an investment. However, some of the overheads may increase because of the new project; these should be charged to that project. The incremental cash flow rule indicates that only incremental overheads are relevant.

The allocation of overheads is a difficult question in practice. One or two investment projects may not cause any change in overhead items such as supervision, rent, employees' welfare, or accounting. But the cumulative effect of many investments may ultimately result in increase in overheads. This creates a problem of cash flow estimation. It is difficult to know when the overheads will change. Efforts should be made to identify such changes so that they may be included in the calculation of net cash flows.

Opportunity Costs

Sometimes a proposed investment project may use the existing resources of the firm for which explicit, or adequate, cash outlays may not exist. The opportunity costs of such projects should be considered. Opportunity costs are the expected benefits, which the company would have derived from those resources if they were not committed to the proposed project. Assume, for example, that a company is considering a project, which requires 7,000 cubic feet of area. Also suppose that the firm has 10,000 cubic feet of area available. What is the cost of the area available within the firm if it is used by the project? One

answer could be that since no cash outlay is involved, therefore, no charges should be made to the project. But from the point of the alternative investment opportunity foregone by transferring this available area to the project, it seems desirable to charge the opportunity cost of the area to the project. Suppose that the company could rent the area at Rs 18 per cubic feet, then Rs 1,26,000 should be considered as the opportunity cost of using the area. The opportunity cost of other resources can also be computed in the same manner. It may be sometimes difficult to estimate opportunity cost. If the resources can be sold, its opportunity cost is equal to the market price. It is important to note that the alternative use rule is a corollary of the incremental cash flow rule.

Incidental Effects

An investment project under consideration may influence the cash flows of other investment opportunities, or the existing projects or products. The incremental cash flow rule applies here; it tells us to, identify all cash flows, direct or incidental, occurring as a result of the acceptance of an investment opportunity. It is, therefore, important to note that all incidental effects, in terms of cash flows, should be considered in the evaluation of an investment opportunity. Let us take some examples to illustrate this point.

A company is contemplating setting up of a chemical plant in a remote, backward area of the State of Gujarat. The company can attract the working force for the plant only if it provides basic facilities such as residential houses, approach roads, schools, hospital and so on to the employees. The estimates of cash flows of the chemical plant would include cash outlay to be incurred in creating the basic facilities for the employees.

A soft-drink manufacturer is considering the introduction of a new soft drink, a low-calorie drink. The market research has indicated that the total market for soft drinks would be growing at 10 per cent. The Company's product, "Sip-It", is anticipated to capture 20 per cent of the total soft drink market. It is also revealed that one-sixth of the total volume of the new product would come from erosion of the company's one of the existing products, "Fresh Cola", a high-calorie drink. The cash flow estimates of "Sip-It" should include the lost contribution of the existing product, "Fresh Cola." Let us add some more information in the present example. The managing director of the soft drink company is not convinced that sale of "Fresh Cola" will be eroded because of the introduction of the new product, "Sip-It". His opinion is that if they do not introduce a low-calorie drink like "Sip-It", competitors would do so; therefore, sale of "Fresh Cola" would be eroded, whether "Sip-It" is introduced or not. If this is a correct argument, the lost contribution of "Fresh Cola" cannot be considered in the evaluation of "Sip-It".

Consider yet another example. A state government is considering the construction of a railroad bridge. In itself the construction of the bridge may not be beneficial. However, if the incidental effects, such as the operation of railroad, are considered, the proposal may become enormously profitable. The cash flow estimates of constructing the bridge should include the net benefits of operating the railroad.

Sunk Costs

Sunk costs are cash outlays incurred in the past. They are the results of past decisions, and cannot be changed by future decisions. Since they do not influence future decisions, they are irrelevant costs. They are unavoidable and irrecoverable historical costs; they should simply be ignored in the investment analysis.

To illustrate, let us assume in our preceding example of the soft drink that before deciding to introduce a new product, the company has conducted a market test. The results of the market test were found to be favourable. Should the company include the market test costs in the evaluation of the new product? The answer is no. The costs of the market test have already been incurred and they are sunk costs; the decision to introduce a new product cannot affect them. They are, therefore, irrelevant to the decision of introducing a new product

Consider another example. A company set up a plant for a cost of Rs 20 crore to manufacture ball bearings. The project proved to be bad for the company, and it started accumulating losses. The total outflows to-date are Rs 30 crore. The company is thinking of abandoning the plant. Some executives consider it suicidal to abandon a plant on which Rs 30 crore have already been spent Others feel it equally unwise to continue with a plant which has been incurring losses and offers no possibility of any satisfactory return on that money spent. The arguments of both the groups do not make sense. The Rs. 30 crore spent by the company is a sunk cost; therefore, it is irrelevant. It is also not correct to discard the plant since it is not earning a satisfactory return on a sunk investment. The company should take the decision to sell or not to sell the plant today in light of the future cash flows mid return.

Points to Ponder

- Allocated Overheads
- Incidental Effects
- Sunk Costs

Review Questions

1. What do you mean by Incremental analysis?
2. Explain how opportunity cost will affect the cash flow?
3. Define Sunk Cost?
4. Illustrate how the sunk cost will affect the cash flow?
5. What do you mean by Incidental effect?

Note

LESSON 18 :

BREAK - EVEN ANALYSIS

Learning Objectives

- To understand Break-even analysis.
- To know about various costs involved in cash flow.

Introduction

You have learnt about the investment decisions, cash flow & its estimation along with the incremental analysis of the investments. Having known about the proper investment, you will be keen to know how early investments will start make profits.

In this lesson, you will understand the same by means of Break-even analysis and its relevant information, which can be adapted to any of the investments. So, you need to have a close look into break-even analysis regularly.

The determination of break-even point of a firm is an important factor in assessing its profitability. It is a valuable control technique and planning device in any business enterprise. It depicts the relation between total cost and total revenue at the level of a particular output. If an entrepreneur is aware of the product cost and its selling price, he can plan the volume of his sale in order to achieve a certain level of profit. The break-even point is determined as that point of sales volume at which the total cost and total revenue are identical. This will facilitate you address the following questions: -

- How do expenses (cost) behave in relation to volume of production / sales?
- What minimum sales are to be achieved to reach the point of no profit – no loss?
- How profits are sensitive to variations in sales?
- What is the quantum of business risk of unit?
- Whether the expansion or diversification reducing the business or not?

Break-Even Analysis

To perform the break-even analysis behavioural classification of cost is required. On the basis of behavioural classification cost can be divided into variable, fixed and semi-variable cost.

Variable Costs

The cost that varies according to changes in volume of sales is called variable cost. Cost of raw material, spares, power and fuel, factory wages and other manufacturing expenses are examples of variable cost.

Fixed Cost

Fixed cost remains fixed for a given level of output and period of time viz, salaries, administrative expenses, occupancy overheads like lighting. In deciding the particular cost as fixed cost, you should look behaviour of the cost but not the quantum of money.

Examples for the fixed cost will be rent of premises, depreciation on fixed assets and interest on loan. All these costs though

vary year to year will be treated as fixed cost because these will not vary with the sales. However, there are opinions that in some of the cases the interest will be treated as variable cost and in some cases the same will be treated as fixed cost. This is due to the effect of interest on working capital. Since, now-a-days working capital is being financed as demand loan, commercial paper and bills. In such cases interest is a fixed cost.

Semi-variable Cost

These are costs, which are partly variable and fixed. Most expenses fall under this category. Telephone expenses, repairs and maintenance, administrative expenses, etc., are treated as semi-variable costs.

Contribution Margin

The difference between sales and variable cost is called contribution. Contribution margin is the relationship between contribution and sales. It indicates the portion of amount for fixed expenses. The decline in the ratio indicates that product is losing the market; it may be due to general obsolescence that occurs during the life cycle of the product.

$$\text{Contribution Margin} = \frac{\text{Contribution}}{\text{Sales}} \times 100$$

Definition

The simple definition of a break-even point is the volume of sales needed for a business to generate zero profit.

$$\text{Break-Even Point (Sales in Rs.)} = \frac{\text{Fixed Cost}}{\text{Contribution}} \times \text{Sales}$$

Wherein Contribution = Sales – Variable Cost

$$\text{Alternatively, Break-even Point is } \frac{\text{Fixed Cost}}{\text{Contribution Margin}} \times 100$$

Therefore a decline in break-even point annually is a good sign of improvement.

Further, you may also need the cost data to be converted into per unit basis, because you would like to know at what stage of units of production break-even can be had. For this, Break-Even Point in units (in terms of capacity utilization) can be computed as under:

$$\text{BEP (Sales in units)} = \frac{\text{Fixed Cost}}{\text{Selling price per unit} - \text{Variable cost per unit}}$$

In another way, A break-even analysis shows, you when you've started to make a profit. One useful tool in tracking your business's cash flow will be break-even analysis. It is a fairly simple calculation and can prove very helpful in deciding whether to make an equipment purchase or just to know how close you are to your break-even level.

Here are the variables needed to compute a break-even sales analysis:

- Gross profit margin
- Operating expenses (less depreciation)
- Total of monthly debt payments for the year (annual debt service)

Since we are dealing with cash flow and depreciation is a non-cash expense, it is subtracted from the operating expenses.

The break-even calculation for sales is:

$$\text{Break-even Sales} = \frac{(\text{Operating Expenses} + \text{Annual Debt Service})}{\text{Gross Profit Margin}}$$

Let's use ABC Clothing as an example and compute this company's break-even sales for Years 1 and 2:

$$\text{Break-even Sales for Year 1} = \frac{(\text{Rs.1,70,000} + \text{Rs.30,000})}{0.25} = \text{Rs.8,00,000}$$

$$\text{Break-even Sales for Year 2} = \frac{(\text{Rs.2,45,000} + \text{Rs.30,000})}{0.30} = \text{Rs.9,16,667}$$

It is apparent from these calculations that ABC Clothing was well ahead of break-even sales both in Year 1 (Rs.1 million sales) and Year 2 (Rs.1.5 million).

Break-even analysis also can be used to calculate break-even sales needed for the other variables in the equation. Let's say the owner of ABC Clothing was confident he could generate sales of Rs.7,50,000 and the company's operating expenses are Rs.1,70,000 with Rs.30,000 in annual current maturities of long-term debt. The break-even gross margin needed would be calculated as follows:

$$\text{Break-even gross margin} = \frac{(\text{Rs.170,000} + \text{Rs.30,000})}{\text{Rs.750,000}} = 26.7\%$$

Now, you can use ABC Clothing to determine the break-even operating expenses. If we know that the gross profit margin is 25 percent, the sales are Rs.750,000 and the current maturities of long-term debt are Rs.30,000, we can calculate the break-even operating expenses as follows:

$$\text{Break-even operating expenses} = (0.25 \times \text{Rs.7,50,000}) - \text{Rs.30,000} = \text{Rs.1,57,500}$$

Another question that arises in your mind is, how can you figure out what will be the breakeven point?

The answer for the same is breakeven point. Every company has one, though many are rather clueless as to what it is. For many, it's not even considered until that embarrassing meeting when the banker or investor gets that serious look in his or her eye and inquires about the company's breakeven point.

Strategies for Break-even Point

There are two key strategies to keep your breakeven point at a manageable level.

The first is to increase the company's overall gross margin. Simply put, increase your profit level on every sale. The most obvious way to accomplish this is to raise prices, but for some companies and products that is not an option. Instead they should look to either decrease variable costs or concentrate more heavily on the products with the highest gross margin.

The second way to manage the breakeven point is simply to cut overhead. Nobody likes to do it, but if the company fails, then

all holdings will be lost anyway. If managed early enough, a company can cut the fat without touching an ounce of the bone. Often there is plenty of unnecessary overhead to be cut without touching a single employee. How many dotcom jobs could have been saved, if companies had simply served generic coffee every morning instead of specially imported European blends? We all want our companies to be happy, comfortable places, but paying attention to our breakeven points can help alert us when it's time to give up some of our toys and which, if any, we can afford to keep.

Utility of the Break-even Analysis

- It serves as the most useful and important tool to study cost output-profit relationship at varying level of output.
- It is useful in reviewing pricing policies.
- It aids in planning capitalization of the enterprise.
- It provides the entrepreneurs decide whether to acquire or not assets involving additional fixed costs.

Shortcomings of the Break-even Analysis

1. The Break-Even Point (BEP) is based on some assumptions, such as sales price, costs, production, sales, etc. The technique will be only of financial value unless all these assumptions are well calculated. Besides, the technique is a preliminary and supplementary tool in the whole exercise of ratio analysis.
2. The technique is to provide cost-escalation as built-in safeguard against increase in prices.
3. The proper analysis of various costs into fixed costs and variable costs is very important. This is so because; some of the items will neither fall under fixed costs nor under variable costs. Hence, semi-variable costs may cast its effect on the BEP. BEP may not prove useful to rapidly growing enterprises and to enterprises that frequently change their product mix.
4. It has limited utility in case of multi products.
5. It does not due cognizance of factors like uncertainty and risk involved in estimates for costs, volume and profits.
6. It is not a patent toll for long Range Planning.

However, it is an important tool for the profitability analysis of the new projects.

Conclusion

The breakeven point may seem like Business 101, yet it remains an enigma to many companies and their management teams. Any company that ignores the breakeven point runs the risk of an early death and at the very least will encounter a lot of unnecessary headaches later on. If you've never computed your breakeven point, take a few minutes right now and perform some basic calculations for the sake of your investors, your employees, your customers and, most important, yourself. Doing a little easy math now, can prevent having to do a lot of hard math later.

If you sell a product:

1. Determine which items are your fixed costs. These cost the same regardless of your sales volume (rent, salaries, utilities, etc.).

2. Identify your variable costs. These are the ones incurred specifically to make and sell your product (materials, commissions, etc.).
3. Add items that are neither fixed nor variable (advertising, etc.).
4. Total these three categories.
5. Subtract your variable costs from your sales total to determine your contribution margin percentage. Your total fixed costs, divided by the contribution margin percentage, give you your sales break-even point, in dollars. To get your break-even point in products, divide the contribution margin by your total number of units produced. Then divide your total fixed costs by the contribution margin per unit.

If you sell a service:

1. Perform the first four steps.
2. Divide your total income by your total expenses. If your income exceeds your expenses, you are profitable.

Points to Ponder

- The break-even analysis.
- Importance of break-even point sales in amount and units of production.
- Utility of Break-even analysis.
- Development of strategy for balancing the break-even point.

Review Questions

1. Why we need to do break-even analysis?
2. Define Break-even Point?
3. How to calculate Break-even point of sales in units?
4. Explain what are questions arise during investment?
5. What the shortcomings of Break-even analysis?

Note

LESSON 19: PROJECT PROPOSALS

Learning Objectives

- To understand about project proposals.
- To know about various stages involved in project proposals.
- Role of Project Manager.

Introduction

You have learnt about the investment decisions, cash flow, various opportunities available and also the assessment when the projects will have the break-even. Further, how to have the track of situation about its development & growth by balancing with the break-even point and take necessary steps for further improvement in achieving the expectations.

In this lesson, you will know about various project proposals and how to assess the project as good or not and thereby take decisions regarding acceptability. The Project Proposal stage provides management with information for decision making by describing the project and identifying affected areas and personnel. The purpose of the Proposal Stage is to initiate a project and establish its priority relative to other projects.

Project Proposal Stage

This is an administrative step to provide management with needed information for decision-making. The purpose of the Project Proposal Stage is to initiate an information systems development project and establish its priority relative to other projects.

Process

The initiator of the project writes the Project Proposal, possibly with the assistance of other division and/or staff. The Project Proposal will include

- A general description of the project,
- Background information,
- Proposed technical approach
- An estimate of how long it will take to complete the Needs Assessment and Planning Stages.
- Identification of
 - Project manager,
 - Customer coordinator and
 - Resources who will be involved are to be made at this time in the
 - Needs Assessment Stage and
 - Planning Stage.

Role of Project Manager

1. Oversees the completion of the Project Proposal template.
2. When there is a cost that will be incurred by the customer to complete the Needs Assessment and Planning Stages, an attachment must be placed with the Project Proposal.

3. The attachment will provide an estimation of cost to the customer for each of the two stages.
4. He will then contact the client representative. The client representative will complete the Service Agreement (SA) for new development or amend the existing SA with a Service Level Estimate for the maintenance to an existing system.
5. He will request their supervisor to add the project (account & name) to the Time Accountability System.
6. He will instruct project members to use the Time Accountability System to record their time spent on tasks related to the project.
7. He will route the completed Project Proposal form to the managers listed in the Approval section.
8. He finally presents the document to the task force for review.
9. He will create an electronic project timeline that identifies the steps and tasks to produce the necessary deliverables and complete the Needs Assessment and Planning Stages.
10. He will ensure that each task must contain the estimated duration and assigned resources.
11. After creating the initial timeline, he will compare the time estimates to the time estimates from the project proposal and make any necessary adjustments.
12. Ensure the project team is completing all deliverables by the end of each stage.
13. Keep project schedule and cost estimates up-to-date.
14. Ensure regular updates to current/designated project reporting application and Time.
15. Meet regularly with project team members, including customer
16. Prepare monthly status reports for management
17. Manage project change

Role of Quality Assurance Analyst

- Receives the signed project proposal and
- Schedules a meeting with the task force.

All team members can update the timeline electronically, wherever such facilities are available. The electronic method allows the project manager to maintain the timeline automatically from the information supplied by the team members.

Tasks during Project Proposal Stage

The following steps are only indicative may not have to be completed in the same order.

1. Complete the Project Proposal Form.
2. Develop cost estimate for Needs Assessment and Planning Stages.
3. Obtain approval from customer management.

4. Obtain approval from individuals listed in the Approval section on the Project Proposal document.
5. Present the “signed” Project Proposal Form to the Task Force for review.
6. Send copy of approved/disapproved Project Proposal to all team members.
7. Complete the Service Agreement (SA).
8. Add the project to the Time Accountability System.
9. Develop a Project Schedule for Project Proposal, Needs Assessment and Planning stages.
10. Add project to current/designated project reporting application.

- Change management.
- Tasks during Project Proposal Stage.

Review Questions

1. Why we need to develop project proposal?
2. What are the roles of project manager during project proposal?
3. Discuss various steps involved in project proposal?
4. Explain change management is necessary for the project?

Appendix -1

Change Management

Agreements on modifications to the project between the customer and project staff should be documented. If changes exceed a specified time period or affect the project's scope, reevaluation of the project must be completed. If changes can be made in a fairly short time frame, a process must be in place to document the agreed upon modifications. At designated time periods the customer needs to be informed as to any scheduling adjustments, cost re-estimation, or changes to the project itself.

Ultimately, the project manager has to prepare final project proposal assessment sheet to have a better control as indicate in the format in the appendix -1.

Points to Ponder

- The project proposals and its steps.
- The role of project manager & quality assistant.

Final Project Proposal Assessment Sheet

Writer: _____

CRITERIA	YOUR POINTS	TOTAL POINTS
DESCRIPTION OF THE FINAL PROJECT - States how the Final Project meets the goals of the course. - Identifies which elements of technical communication (as defined by Markel in chapter 1) will be used and which will not. Provides suitable explanations. - Specifies what format the Final Project will take (brochure, manual, CD portfolio, schematics, web site, etc.). Explains why this is the best way to compose the Final Project.		2
PURPOSE OF THE FINAL PROJECT In one or two sentences, states clearly what aims or goals you envision for your Final Project.		1
AUDIENCE OF THE FINAL PROJECT In several concise sentences, identifies and describes a “real-world” audience for your Final Project.		1
FORMATTING, SYNTAX, GRAMMAR, PUNCTUATION, & SPELLING - Composed in a memo. - Uses appropriate headings. - Demonstrates effective sentence-level writing. - Contains few—if any—errors.		1
NOTE: A zero will be awarded for any assignment not complete, not on topic, or not handed in.		
TOTAL SCORE:		5

LESSON 20:

CAPITAL INVESTMENT DECISIONS

Learning Objectives

- To understand about Capital Budget.
- To know about capital investment decisions.
- Importance of capital investment decisions.
- Types of Capital Investment Decisions.
- Capital investment evaluation criteria.
- Classification of Capital Investment.

Introduction

In earlier units you have understood about the various aspects of investment decisions. Now, you are required to know about the long-term investments decisions. This is called as Capital investment decisions.

In this you will learn more about the capital investment decisions. The efficient allocation of capital is the most important finance function in the modern times. It involves decisions to commit the firm's funds to the long-term assets. Such decisions are of considerable importance to the firm since they tend to determine its value and size by influencing its growth, profitability and risk. In this chapter, we shall discuss the nature of investment decisions and the criteria for investment analysis. We shall also provide a comparison of the various investments criteria to indicate the most consistent criterion.

What are Capital Investment Decisions?

The investment decisions of a firm are generally known as the capital budgeting, or capital expenditure decisions. A capital budgeting decision may be defined as the firm's decision to invest its current funds most efficiently in the long-term assets in anticipation of an expected flow of benefits over a series of years. The long-term assets are those, which affect the firm's operations beyond the one-year period. The firm's investment decisions would generally include expansion, acquisition, modernization and replacement of the long-term assets. Sale of a division or business (divestment) is also analyzed as an investment decision. Activities such as change in the methods of sales distribution, or undertaking an advertisement campaign or a research and development programme have long term implications for the firm's expenditures and benefits, and therefore, they may also be evaluated as investment decisions. It is important to note that investment in the long-term assets invariably requires funds to be tied up in the current assets such as inventories and receivables. As such, investment in fixed and current assets is one single activity.

The following are the features of capital investment decisions:

1. The exchange of current funds for future benefits.
2. The funds are invested in long-term assets.
3. The future benefits will occur to the firm over a series of years

It is significant to emphasize that expenditures and benefits of an investment should be measured in cash. In the investment analysis, it is cash flow, which is important, not the accounting profit. It may also be pointed out that investment decisions affect the firm's value. The firm's value will increase if investments are profitable and add to the shareholders' wealth. Thus investments should be evaluated on the basis of a criterion, which is compatible with the objective of the shareholders' wealth maximization. An investment will add to the shareholders' wealth if it yields benefits in excess of the minimum benefits as per the opportunity cost of capital. In this chapter, we assume that the investment project's opportunity cost of capital is known. We also assume that the expenditures and benefits of the investment are known with certainty. Both these assumptions are relaxed in later chapters.

Why is Capital Investment Decisions Important?

Capital Investment decisions require special attention because of the following reasons:

- They have long-term implications for the firm, and can influence its risk complexion.
- They involve commitment of large amount of funds.
- They are irreversible decisions.
- They are among the most difficult decisions to make.

Growth

The effects of investment decisions extend into the future and have to be endured for a longer period than the consequences of the current operating expenditure. Firm's decision to invest in long-term assets has a decisive influence on the rate and direction of its growth. A wrong decision can prove disastrous for the continued survival of the firm; unwanted or unprofitable expansion of assets will result in heavy operating costs to the firm. On the other hand, inadequate investment in assets would make it difficult for the firm to compete successfully and maintain its market share.

Risk

Long-term commitment of funds may also change the risk complexion of the firm: If adoption of an investment increases average gain, but causes frequent fluctuations in its earnings, the firm will become more risky. Thus investment decisions shape the basic character of a firm.

Funding

Investment decisions generally involve large amount of funds, which make it imperative for the firm to plan its investment programmes very carefully and make advance arrangement for procuring finances internally or externally.

Irreversibility

Most investment decisions are irreversible. It is difficult to find a market of such capital items once they have been acquired. The firm will incur heavy losses if such assets are scrapped.

Complexity

Investment decisions are among the firm's most difficult decisions. They are an assessment of future events, which are difficult to predict. It is really a complex problem to correctly estimate future cash flows of an investment. The cash flow uncertainty is caused by economic, political, social and technological forces.

Types of Capital Investment Decisions

There are many ways to classify investments. Some classifications are as follows:

1. Expansion of 'existing business
2. Expansion of new business
3. Replacement and modernization.

Expansion and Diversification

A company adds capacity to its existing product lines to expand existing operations. For example, a fertilizer company may increase its plant size to manufacture more urea. Expansion of new business requires investment in new products and a new kind of production activity within the firm. If a package manufacturing company invests in new plant and machinery to produce ball bearings, which the firm has not manufactured before, this represents expansion of new business or diversification. Sometimes a company acquires existing firms to expand its business. In either case, the firm makes investment in the expectation of additional revenue. Investments in existing or new products may also be called as revenue-expansion investments.

Replacement and Modernization

The main objective of modernization and replacement is to improve operating efficiency and reduce costs. Cost savings will reflect in the increased profits, but the firm's revenue may remain unchanged. Assets become outdated and obsolete with technological changes. The firm must decide to replace those assets with new assets that operate more economically. If a cement company changes from semi-automatic drying equipment to fully automatic drying equipment, it is an example of modernization and replacement. Replacement decisions help to introduce more efficient and economical assets and therefore, are also called cost-reduction investments. However, replacement decisions, which involve substantial modernization and technological improvements, expand revenue as well as reduce costs.

Yet another useful way to classify Capital Investments is as follows:

1. Mutually exclusive investments
2. Independent investments
3. Contingent investments.

Mutually exclusive investments serve the same purpose and compete with each other

If one investment is undertaken, others will have to be excluded. A company may, for example, either use a more labour-intensive, semi-automatic machine, or employ a more capital-intensive, highly automatic machine for production. Choosing the semi-automatic machine precludes the acceptance of the highly automatic machine.

Independent investments serve different purposes and do not compete with each other

For example, a heavy engineering company may be considering expansion of its plant capacity to manufacture additional excavators and addition of new production facilities to manufacture a new product - light commercial vehicles. Depending on their profitability and availability of funds, the company can undertake both investments.

Contingent Investments are Dependent Projects

The choice of one investment necessitates that one or more other investments should also be undertaken. For example, if a company decides to build a factory in a remote, backward area, it may have to invest in houses, roads, hospitals, schools etc. for employees to attract the work force. Thus building of factory also requires investment in facilities for employees. The total expenditure will be treated as one single investment.

Capital Investment Evaluation Criteria

Three steps are involved in the evaluation of an investment:

1. Estimation of cash flows
2. Estimation of the required rate of return
3. Application of a decision rule for making the choice.

You have understood the first two steps while studying earlier units. Thus our discussion in this chapter is confined to the third step. Specifically, we focus on the merits and demerits of various decision rules.

The capital investment decision rules may be referred to as capital budgeting techniques, or investment criteria. A sound appraisal technique should be used to measure the economic worth of an investment project. The essential property of a sound technique is that it should maximize the shareholders wealth.

The following other characteristics should also be possessed by a sound investment evaluation criterion:

- It should consider all cash flows to determine the true profitability of the project.
- It should provide for an objective and unambiguous way of separating good projects from bad projects.
- It should help ranking of projects according to their true profitability.
- It should recognize the fact that bigger cash flows are preferable to smaller ones and early cash flows are preferable to later ones.
- It should help to choose among mutually exclusive projects that project which maximizes the shareholders' wealth.
- It should be a criterion, which is applicable to any conceivable investment project independent of others.

These conditions will be clarified as we discuss the features of various capital investment criteria in the following lessons.

A number of capital budgeting techniques are in use in practice. They may be grouped in the following two categories:

1. Discounted Cash Flow (DCF) Criteria

- Net Present Value (NPV)
- Internal Rate of Return (IRR)

- Profitability Index (PI)

2. Non Discounted Cash Flow Criteria

- Payback Period (PB)
- Accounting Rate of Return (ARR).

Points to Ponder

- Meaning of Capital Budget.
- Various capital investment decisions.
- DCF & Non-DCF techniques.
- Classification of Capital Investment.

Review Questions

1. How will you evaluate Capital investment?
2. What do you understand by Capital budget Technique?
3. Explain characteristics of sound capital investments criteria?
4. What is the basis for classifying Capital Investments?

Note

LESSON 21 : CAPITAL BUDGETING TECHNIQUES - NPV

Learning Objectives

- To understand about the Net Present Value.
- To know the importance of NPV.
- To know how to adopt NPV method in Cash Budget Technique.

Introduction

You have learnt about the capital investment decisions and different types in the earlier lesson. In this lesson, you will know the about capital budget and various techniques.

Capital

It is the fund required in any investment

Budgeting

The budget is derived from a French word “Bougette” representing a leather pouch into which funds are appropriated to meet the anticipated expenses. The process is called budgeting.

The institute of cost and works accountants (ICWA) has defined budget as “ a financial and / or quantitative statement prepared and approved prior to a definite period of time, of the policy to be pursued during that period for the purpose of attaining a given objective.”

In other words, it may be said that budget is a

- Statement of management policy for a finite period of time.
- Translates policy into actions to achieve the policy.
- Provides a standard of comparison with results actually achieved.
- It is not (purely) an accounting statement.

The budget is prepared on the basis of the forecast. Forecast is a mere estimate about what is likely to happen.

Budgeting is another concept, which you must know. It refers to the procedure followed to prepare the budget. Welch has said “Budgeting is the principal tool of planning and control offered to management by accounting functions.

It may be explained as

1. Process of planning, monitoring and control for costs, expenses and revenue.
2. Normally annual – broken down to weeks, months but sometimes longer.
3. Usually expressed financially.
4. Purpose – Increasing efficiency of money, labour, machinery etc.

You now recollect that there are number of capital budgeting techniques are in use in practice. They may be grouped in the following two categories:

1. Discounted Cash Flow (DCF) Criteria
 - Net Present Value (NPV)
 - Internal Rate of Return (IRR)

- Profitability Index (PI)

2. Non Discounted Cash Flow Criteria

- Payback Period (PB)
- Accounting Rate of Return (ARR).

The net present value criterion is the most valid technique of evaluating an investment project. It is generally consistent with the objective of maximizing the shareholders’ wealth.

Net present Value (NPV) Method

The net present value (NPV) method is the classic economic method of evaluating the investment proposals. It is one of the discounted cash flow (DCF) techniques explicitly recognizing the time value of money. It correctly postulates that cash flow arising at different time periods differs in value and are comparable only when their equivalents. Present values are found out.

The following steps are involved in the calculation of NPV:

1. Cash flows of the investment project should be forecasted based on realistic assumptions.
2. Appropriate discount rate should be identified to discount the forecasted cash flows.
3. The appropriate discount rate is the firm’s opportunity cost of capital, which is equal to the required rate of return expected by investors on investments of equivalent risk.
4. Present value of cash flows should be calculated using opportunity cost of capital as the discount rate.
5. Net present value should be found out by subtracting present value of cash outflows from present value of cash inflows. The project should be accepted if NPV is positive (i.e. NPV > 0).

You will understand better with an example.

Illustration

Assume that Project X costs Rs. 2,500/- now and is expected to generate year-end cash inflows as under: -

(Amt. In Rs.)

The opportunity cost of the capital may be assumed to be 10 per cent.

The net present value for Project X can be calculated by referring to present value table (Table C at the end of the book). The calculations are shown below:

$$NPV = \left[\frac{Rs\ 900}{(1 + .10)} + \frac{Rs\ 800}{(1 + .10)^2} + \frac{Rs\ 700}{(1 + .10)^3} + \frac{Rs\ 600}{(1 + .10)^4} + \frac{Rs\ 500}{(1 + .10)^5} \right] - Rs\ 2,500$$

NPV = (Rs 900 (PVF_{1,10}) + Rs 800 (PVF_{2,10}) + Rs 700 (PVF_{3,10}) + Rs 600 (PVF_{4,10}) + Rs 500 (PVF_{5,10})] - Rs 2,500

NPV = (Rs 900 X 0.909 + Rs 800 X 0.826 + Rs 700 X 0.751 + Rs 600 X 0.683 + Rs 500 X 0.620] - Rs 2,500

NPV = Rs 2,725 - Rs 2,500 = + Rs 225

Project X's present value of cash inflows (Rs 2,725) is greater than that of cash outflow (Rs 2,500). Thus, it generates a positive net present value (NPV = + Rs 225). Project X adds to the wealth of owners; therefore, it should be accepted.

The equation for the net present value can be written as follows:

$$\text{NPV} = \frac{C_1}{(1+k)} + \frac{C_2}{(1+k)^2} + \frac{C_3}{(1+k)^3} + \dots + \frac{C_n}{(1+k)^n} - C_0$$

$$\text{or NPV} = \sum_{t=1}^n \frac{C_t}{(1+k)^t} - C_0$$

Where, C_1, C_2 represents net cash inflows in year 1, 2 are the opportunity cost of capital. C_0 is the initial cost of the investment and n is the expected life of the investment. It should be noted that the cost of capital, k , is assumed to be known and is constant.

Why NPV is Important?

A question may be raised: why a financial manager should invest Rs 2,500 in Project X? Project X should be undertaken if it is best for the company's shareholders; they would like their shares to be as valuable as possible. Let us assume that the total market value of a hypothetical company is Rs 10,000. Which includes Rs 2,500 cash that can be invested in Project X. Thus the value of the company's other assets must be Rs 7,500. The company has to decide whether it should spend cash and accept Project X or to keep the cash and reject Project X. Clearly Project X is desirable since its PV (Rs 2,725) is greater than the Rs 2,500 cash. $\therefore$ Project X is accepted. The total market value of the firm will be: Rs 7,500 + PV of Project X = Rs 7,500 + Rs 2,725 = Rs 10,225; that is an increase by Rs 225. The company's total market value would remain only Rs 10,000 if Project X is rejected.

Why should the PV of Project X reflect in the company's market value? To answer this question, let us assume that a new company X with Project X as the only asset is formed. What is the value of the company? We know from our earlier discussion in Chapter 8 that the market value of a company's shares is equal to the present value of the expected dividends. Since Project X is the only asset of Company X, the expected dividends would be equal to the forecasted cash flows from Project X. Investors would discount the forecasted dividends at a rate of return expected on securities equivalent in risk to company X. The rate used by investors to discount dividends is exactly the rate which we should use to discount cash flows of Project X. The calculation of the PV of Project X is a replication of the process, which shareholders will be following in valuing the shares of company X. Once we find out the value of Project X, as a separate venture, we can add it to the value of other assets to find out the portfolio value.

The difficult part in the calculation of the PV of an investment project is the precise measurement of the discount rate. Funds

available with a company can either be invested in projects or given to shareholders. Shareholders can invest funds distributed to them in financial assets. Therefore, the discount rate is the opportunity cost of investing in projects rather than in capital markets. Obviously, the opportunity cost concept makes sense when financial assets are of equivalent risk as compared to the project

An alternate interpretation of the positive net present value of an investment is that it represents the maximum amount a firm would be ready to pay for purchasing the opportunity of making investment, or the amount at which the firm would be willing to sell the right to invest without being financially worse-off. The net present value (Rs 225) can also be interpreted to represent the amount the firm could raise at the required rate of return (10%), in addition to the initial cash outlay (Rs 2,500), to distribute immediately to its shareholders and by the end of the projects' life to have paid off all the capital raised and return on it. The point is illustrated by the calculations shown in Table below.

TABLE	INTERPRETATION OF NET PRESENT VALUE (Amt in Rs.)				
	Amount outstanding in the beginning	Return on outstanding amount at 10%	Total amount outstanding flows	Repayment from cash at the end	Balance Outstanding
1	2,725.00	272.50	2,997.50	900	2,097.50
2	2,097.50	209.75	2,307.25	800	1,507.25
3	1,507.25	150.73	1,657.98	700	957.98
4	957.98	95.80	1,053.78	600	453.78
5	453.78	45.38	499.16	500	(0.87)

Calculations in Table are based on the assumption that the firm chooses to receive the cash benefit resulting from the investment in the year it is made. Any pattern of cash receipts, such that, the net present value is equal to Rs 225, can be selected. Thus, if the firm raises Rs 2,500 (the initial outlay) instead of Rs 2,725 (initial outlay plus net present value) at 10 per cent rate of return, at the end of fifth year after having paid the principal sum together with interest, it would be left with Rs 363, whose present value at the beginning of the first year at 10 per cent discount rate is Rs 225. It should be noted that the gain to shareholders would be more if the rate of raising money is less than 10 per cent.

Acceptance Rule

It should be clear that the acceptance rule using the NPV method is to accept the investment project if its net present value is positive ($\text{NPV} > 0$) and to reject it if the net present value is negative ($\text{NPV} < 0$). The market value of the firm's share would increase if projects with positive net present values

- Accept $NPV > 0$
- Reject $NPV < 0$
- May accept $NPV = 0$

Evaluation of the NPV Method

1. It recognizes the time value of money—a rupee received today is worth more than a rupee received tomorrow.
2. It uses all cash flows occurring over the entire life of the project in calculating its worth. The NPV method relies on estimated cash flows and the discount rate rather than any arbitrary assumptions, or subjective considerations.
3. The discounting process facilitates measuring cash flows in terms of present values; that is, in terms of current rupee. Therefore, the NPV s of projects can be added. For example, $NPV(A + B) = NPV(A) + NPV(B)$. This is called the value-additively principle, and it implies that if we know the NPV s of separate projects, the value of the firm will increase by the sum of their NPV s. We can also say that if we know values of separate assets, the firm's value can simply be found by adding their values. The value-additively is an important property of an investment criterion because it means that each project can be evaluated, independent of others, on its own merit. We will see later on that the value-additively principle is not applicable in the alternative investment criteria.
4. The NPV method is always consistent with the objective of maximizing the shareholders' wealth. This is the greatest virtue of the method.

It should be noted that the ranking of investment projects as per the NPV rule is not, independent of the discount rates. Let us consider two projects A and B both costing Rs 50 each. Project A returns Rs 100 after one year and Rs 25 after two years. on the others hand, Project B returns Rs 30 after one year and Rs 100 after two years. At discount rates of 5 per cent and 10 per cent, the'NPV of projects and their ranking are as follows:

	NPVat 5%	Rank	NPVat 10%	Rank
Project A	67.92	II	61.57	I
Project B	69.27	I	59.91	II

Points to Ponder

- ## Review Questions

1. How do you say that NP

- Note

[illegible]

LESSON 22 : CAPITAL BUDGETING TECHNIQUES – IRR

Learning Objectives

- To understand about the IRR.
- To know the importance of IRR.
- To know how to adopt IRR method in Cash Budget Technique.

Introduction

In earlier lesson you have learnt about the one of the mainly used capital budgeting techniques i.e., NPV method. Another method, which is commonly used, is Internal Rate of Return method (IRR). Now, you will know more about the same.

Internal Rate of Return Method

The internal rate of return (IRR) method is another discounted cash flow technique, which takes account of the magnitude and timing of cash flows. Other terms used to describe the IRR method are yield of an investment, marginal efficiency of capital, rate of return over cost, time-adjusted rate of return and so on.

The concept of rate of return is quite simple to understand in the case of a one-period project. Assume that you deposit Rs 10,000 with a bank and would get back Rs 10,800 after one year. The true rate of return on your investment would be:

$$\begin{aligned} \text{Rate of return} &= \frac{10,800 - 10,000}{10,000} \\ &= \frac{10,800}{10,000} - 10,000 \\ &= 1.08 - 1 = 0.08 \text{ or } 8\% \end{aligned}$$

The amount which you would obtain in future (Rs 10,800) would consist of your investment (Rs 10,000) plus return on your investment (.08 x Rs 10,000):

$$\begin{aligned} 10,000 (1.08) &= 10,800 \\ 10,000 &= 10,800 (1.08) \end{aligned}$$

You may observe that i.e., rate of return of your investment (8 per cent) makes the discounted (present) value of your cash inflow (Rs 10,800) equal to your investment (Rs 10,000). We can now develop a formula for the rate of return (r) on an investment (Co) that generates a single cash flow after one period (C1) as follows:

$$\begin{aligned} r &= \frac{C_1 - C_0}{C_0} \\ &= \frac{C_1}{C_0} - 1 \end{aligned}$$

Equation (2) can be rewritten as follows:

$$\begin{aligned} \frac{C_1}{C_0} &= 1 + r \\ C_0 &= \frac{C_1}{(1 + r)} \quad \dots(3) \end{aligned}$$

From Equation (3) the rate of return can be defined as that rate which equates investment outlay with the present value of inflow received after one period. This also implies that the rate of return is the discount rate which makes NPV = 0. This rate may be referred to as the internal rate of return. There is no satisfactory way of defining true rate of return of a long-term asset. The internal rate of return (IRR) is the best available concept. We shall see that although it is a very frequently used concept in finance, yet at times it can be a misleading measure of investment worth.

The internal rate of return can be defined as that rate which equates the present value of cash inflows with the present value of cash outflows of an investment. In other words, it is the rate at which the net present value of the investment is zero. It is called internal rate because it depends solely on the outlay and proceeds associated with the investment and not on any rate determined outside the investment. It can be determined by solving the following equation for r :

$$\begin{aligned} C_0 &= \frac{C_1}{(1+r)} + \frac{C_2}{(1+r)^2} + \frac{C_3}{(1+r)^3} + \dots + \frac{C_n}{(1+r)^n} \\ C_0 &= \sum_{t=1}^n \frac{C_t}{(1+r)^t} \\ \sum_{t=1}^n \frac{C_t}{(1+r)^t} - C_0 &= 0 \quad \dots(4) \end{aligned}$$

It can be noticed that the IRR equation is the same as the one used for the NPV method with the difference that in the NPV method the required rate of return, k, is assumed to be known and the net present value is found, while in the IRR method the value of r has to be determined at which the net present value is zero.

Uneven Cash Flows: Calculating IRR by Trial and Error

The value of r in Equation (4) can be found out by trial and error. The approach is to select any discount rate to compute the present value of cash inflows. If the calculated present value of the expected cash inflow is lower than the present value of cash outflows a lower rate should be tried. On the other hand, a higher value should be tried if the present value of inflows is

higher than the present value of outflows. This process will be repeated unless the net present value becomes zero.

Illustration to explain the procedure to calculate the internal rate of return.

A project costs Rs.16,000/- and is expected to generate cash flows of Rs.8,000/- Rs.7,000/- and Rs.6000/- at the end of each year for next 3 years. You know that IRR is the rate at which project will have a zero NPV. As a first step, you try randomly a 20% discount rate. The project's NPV at 20 % is :

$$\begin{aligned} \text{NPV} &= - \text{Rs.16,000/-} + \text{Rs.8,000/-}(\text{PVF}_{1,0.20}) + \text{Rs.7,000/-}(\text{PVF}_{2,0.20}) + \text{Rs.6,000/-}(\text{PVF}_{3,0.20}) \\ &= - \text{Rs.16,000/-} + \text{Rs.8,000/-} \times 0.833 + \text{Rs.7,000/-} \times 0.694 + \text{Rs.6,000/-} \times 0.579 \\ &= - \text{Rs.16,000/-} + \text{Rs.14,996/-} = - \text{Rs. 1,004/-} \end{aligned}$$

A negative NPV of Rs.1,004/- at 20 % indicates that the project's true rate of return is lower than 20 per cent. Now let us try for 16 per cent as the discount rate. At 16 percent, the project's NPV is

$$\begin{aligned} \text{NPV} &= - \text{Rs.16,000/-} + \text{Rs.8,000/-}(\text{PVF}_{1,0.16}) + \text{Rs.7,000/-}(\text{PVF}_{2,0.16}) + \text{Rs.6,000/-}(\text{PVF}_{3,0.16}) \\ &= - \text{Rs.16,000/-} + \text{Rs.8,000/-} \times 0.862 + \text{Rs.7,000/-} \times 0.743 + \text{Rs.6,000/-} \times 0.641 \\ &= - \text{Rs.16,000/-} + \text{Rs.15,943/-} = - \text{Rs. 57/-} \end{aligned}$$

Since the project is still negative at 16 percent, a rate lower than 16 percent should be tried. When you select 15 percent as the trial rate, you will find that project's NPV is Rs.200/-

$$\begin{aligned} \text{NPV} &= - \text{Rs.16,000/-} + \text{Rs.8,000/-}(\text{PVF}_{1,0.15}) + \text{Rs.7,000/-}(\text{PVF}_{2,0.15}) + \text{Rs.6,000/-}(\text{PVF}_{3,0.15}) \\ &= - \text{Rs.16,000/-} + \text{Rs.8,000/-} \times 0.870 + \text{Rs.7,000/-} \times 0.756 + \text{Rs.6,000/-} \times 0.658 \\ &= - \text{Rs.16,000/-} + \text{Rs.16,200/-} = \text{Rs. 200/-} \end{aligned}$$

The true rate of return should lie between 15 – 16 percent. We can find out a close approximation of the rate of return by the method of linear interpolation as follows: -

		Difference
PV required	Rs.16,000	200
PV at lower rate, 15%	Rs.16,200	257
PV at higher rate, 16%	Rs.15,943	
r	= 15% + (16% - 15%)	200
		257
	= 15% + 0.8% =	15.8%

Level Cash Flows

An easy procedure can be followed to calculate the IRR for a project that produces level or equal cash flows each period.

Illustration

You assume that an investment would cost Rs. 20,000/-, which provides annual cash inflow of Rs 5,430/- for 6 years. If the opportunity cost of capital is 10 per cent, what is the investment's NPV?

The Rs 5,430 is an annuity for 6 years. The NPV can be found as follows:

$$\text{NPV} = - \text{Rs 20,000} + \text{Rs 5,430} (\text{PVAF}_{6,0.10})$$

$$= - \text{Rs 20,000} + \text{Rs 5,430} \times 4.355 = \text{Rs 3,648}$$

How much is the project's IRR?

The IRR of the investment can be found out as follows :

$$\text{NPV} = - \text{Rs 20,000} + \text{Rs 5,430} (\text{PVAF}_{6,r}) = 0$$

$$\text{Rs 20,000} = \text{Rs 5,430} (\text{PVAF}_{6,r})$$

$$\text{PVAF}_{6,r} = \frac{\text{Rs.20000}}{\text{Rs.5430}} = 3.683$$

The rate, which gives a PVAF of 3.683 for 6 years, is the project's internal rate of return.

Looking up PVAF in Table D (Table & formula book) across the 6 years row, you find it approximately under the 16 per cent column. Thus 16 per cent is the project's IRR, which equates the present value of the initial cash outlay (Rs 20,000) with the constant annual cash inflows (Rs 5,430 per year) for 6 years.

NPV Profile and IRR

The NPV of a project declines as the rate increases and for discount rates higher than the project's IRR, NPV will be negative. NPV profile of the project at various discount rates is shown as under: -

Project's NPV Profile

Discount rate (%)	NPV (Rs.)
0	12,580
5	7,561
10	3,648
16	0
20	- 942

Acceptance Rule

The acceptance or rejection rule, using the IRR method is to accept the project if its internal rate of return is higher than the opportunity cost of capital ($r > k$). k is also known as the required rate of return, or the cut-off or hurdle rate. The project shall be rejected if its IRR is lower than the opportunity cost of capital ($r < k$). The decision maker may remain indifferent if the internal rate of return is equal to the opportunity cost of capital. This is the IRR acceptance rule are:

- Accept $r > k$
- Reject $r < k$
- May accept $r = k$

Evaluation of IRR Method

Like NPV method, the IRR method recognizes the time value of money and it considers all cash flows occurring the entire life of the project to calculate its rate of return. It generally gives the same acceptance rule as the NPV method. It is consistent with the shareholders' wealth maximization objective. IRR rule can however, give misleading and inconsistent results when the NPV of a project does not decline with discount rates. It also fails to indicate a correct choice between mutually exclusive projects under certain situations. Unlike in case of NPV method, the value additivity principle does not hold when the IRR method is used – IRRs of project do not add. Thus, for the projects A and B, $\text{IRR (A)} + \text{IRR (B)}$ need not equal to IRR (A + B) . The

following example gives the clarity these characteristics of IRR method: -

Projects	C ₀	C ₁	NPV at 10%	IRR %
A	- 100	+ 120	+ 9.1	20.0
B	- 100	+ 168	+ 2.7	12.0
A + B	-250	+ 288	+ 11.8	15.2

You can see from the example that NPVs of projects add:

$$\text{NPV (A) + NPV(B) = NPV (A + B) = 9.1 + 2.7 = 11.8}$$

$$\text{IRR(A) + IRR(B) \neq IRR (A + B) = 20\% + 12\% \neq 15.2\%}$$

Points to Ponder

- Internal Rate of Return
- Calculation of IRR
- Differentiate IRR from NPV Method.

Review Questions

1. Define Internal Rate of Return?
2. How will you calculate IRR?
3. What are the shortcomings of IRR?
4. Differentiate IRR Method from NPV Method?

Note

LESSON 23 : CAPITAL BUDGETING TECHNIQUES – OTHER METHODS

Learning Objectives

- Profitability Index or Benefit-cost ratio.
- Payback Method
- Accounting rate of return (ARR) method.

Introduction

In earlier lesson you have learnt about the one of the mainly used capital budgeting techniques i.e., NPV method and Internal Rate of Return method (IRR). Now, you will know more about the same.

The other time adjusted method of evaluating the investment proposals are as follows: -

1. Profitability Index or Benefit-cost ratio
2. Payback Method
3. Accounting rate of Return method

Profitability Index or Benefit-cost Ratio

It is the ratio of the present value of cash inflows at the required rate of return, to the initial cash outflow of the investment. It may be gross or net, net being simply gross minus one. The formula to calculate benefit-cost ratio or profitability index is as under: -

$$\text{PI} = \frac{\text{PV of cash inflows}}{\text{Initial cost outlay}} = \frac{\text{PV}(C_t)}{C_0}$$

$$= \sum_{t=1}^n \frac{C_t}{(1+k)^t} + C_0 = 0$$

Acceptance Rule

The following are the Profitability Index (PI) rules:

- Accept $\text{PI} > 1$
- Reject $\text{PI} < 1$
- May accept $\text{PI} = 1$

When PI is greater than one, then the project will have positive NPV.

Evaluation of PI Method

Like NPV and IRR rules, PI is a conceptually sound method of appraising investment projects. It gives due consideration to the time value of money. It is a variation of the NPV method and requires the same computation as the NPV method. In PI method, since the present value of cash flows is divided by the initial cash outflow, it is a relative measure of a project's profitability.

Payback Method

The payback (PB) is one of most popular and widely recognized traditional methods of evaluating investment proposals. It is defined as the number of years required to recover the original cash outlay invested in the a project. If the project

generates constant annual cash inflows, the payback period can be computed by dividing the cash outlay by the annual cash inflow.

$$\text{Payback} = \frac{\text{Initial investment}}{\text{Annual cash inflow}} = \frac{C_0}{C}$$

Acceptance Rule

Many firms use the payback period as an accept or reject criterion as well as method of ranking projects. If the payback period calculated for a project is less than the maximum payback period set by management, it would be accepted; if not it would be rejected. As a ranking method, it gives highest ranking to the project, which has shortest payback period, and lowest ranking to the project with highest payback period. Thus if the firm has to choose between two mutually exclusive projects, project with shorter payback period will be selected.

Evaluation of Payback

The most significant merit of payback is that it is simple to understand and easy to calculate. The business executives consider the simplicity of method as a virtue, which is evident from their heavy reliance on it for appraising investment proposals in practice. Another merit is that it costs less than most of the sophisticated techniques, which require a lot of the analysis time and the use of computers.

Limitations

In spite of its simplicity, the payback may not be a desirable investment criterion since it suffers from a number of serious limitations:

1. It fails to take account of the cash inflows earned after the payback period. For example, consider the following projects X and Y:

	Cash Flows (Rs)					
Project	C_0	C_1	C_2	C_3	Payback	NPV @ 10% .
X	-4,000	0	4,000	2,000	2 yrs	+806
y	-4,000	2,000	2,000	0	2 yrs	-530

As per the payback rule, both the projects are equally desirable since both return the investment outlay in two years. If we assume an opportunity cost of 10 per cent, Project X yields a positive net present value of Rs 806 and Project Y yields a negative net present value of Rs 530. As per the NPV rule, Project X should be accepted and Project Y

rejected. Payback rule gave wrong results because it failed to consider Rs 2,000 cash flow in third year for Project X.

2. It is not an appropriate method of measuring the profitability of an investment project as it does not consider all cash inflows yielded by the project. Considering Project X again, payback rule did not take into account its entire series of cash flows.
3. It fails to consider the pattern of cash inflows, i.e., magnitude and timing of cash inflows. In other words, it gives equal weights to returns of equal amounts even though they occur in different time periods. For example, compare the following projects C and D where they involve equal cash outlay and yield equal total cash inflows over equal time periods:

	Cash Flows (Rs)					
Project	Co	C1	C2	C3	Payback	NPV @ 10%
C	- 5,000	3,000	2,000	2,000	2 yrs	+881
D	- 5,000	2,000	3,000	2,000	2 yrs	+798

Using payback period, both projects are equally desirable. But Project C should be preferable as larger cash inflows come earlier in its life. This is indicated by the NPV rule; project C has higher NPV (Rs 881) than Project D (Rs 798) at 10 per cent opportunity cost. It should be thus clear that payback is not a measure of profitability. As such, it is dangerous to use it as a decision criterion.

4. Administrative difficulties may be faced in determining the maximum acceptable payback period. There is no rational basis for setting a maximum payback period. It is generally a subjective decision.
5. The payback period method is not consistent with the objective of maximizing the market value of the firm's shares. Share values do not depend on payback periods of investment projects.¹

Despite its weakness, payback period method is popular in practice. Besides simplicity the causes for its popularity are:

- A company can have more favourable short-run effects on earnings per share by setting up a shorter standard payback period? It should, however, be remembered that this may not be a wise long-term policy as the company may have to sacrifice its future growth for current earnings.
- The riskiness of the project can be tackled by having a shorter standard payback period as it may ensure guarantee against loss. A company has to invest in many projects where the cash inflows and life expectancies are highly uncertain. Under such circumstances, payback may become important, not so

much as a measure of profitability but as a means of establishing an upper bound on the acceptable degree of risk.

- The emphasis in payback is on the early recovery of the investment. Thus, it gives an insight into the liquidity of the project. The funds so released can be put to other uses.

Let us re-emphasize that the payback is not a valid method for evaluating the acceptability of the investment projects. It can, however, be used along with the NPV rule as a first Step in roughly screening the projects. In practice, the use of DCF techniques has been increasing but payback continues to remain a popular and primary method of investment evaluation.

Discounted Payback Period

One of the serious objections to the payback method is that it does not discount the cash flows for calculating the payback period. Some people, therefore, discount cash flows and calculate the discounted payback period. The number of periods taken in recovering the investment outlay on the present value basis is called the discounted payback period. The discounted payback period still fails to consider the cash flows occurring after the payback period. Let us consider an example.

Projects P and Q involve the same outlay of Rs 4,000 each. The opportunity cost of capital may be assumed 10 per cent. The cash flows of the projects and their discounted payback periods are shown in the following table:

Table : Discounted Payback

Projects	Cash Flows (Rs.)					Simple payback period	Discounted payback period	NPV at 10%
	C ₀	C ₁	C ₂	C ₃	C ₄			
P	4000	3000	1000	1000	1000	2 years	--	--
PV of cash flows	4000	2727	826	451	683		2.6 years	+987
Q	4000	0	4000	1000	2000	2 years	--	--
PV of cash flows	4000	0	3304	751	1366		2.9 years	+1421

The projects are indicated of same desirability by the simple payback period. When cash flows are discounted to calculate the discounted payback period, Project P recovers the investment outlay faster, and therefore, would be preferred over Project Q, which takes more time to recover the investment outlay. Discounted payback rule is better as it does discount the cash flows until the outlay is recovered. But it does not help much. It does not take into consideration the entire series of cash flows. It can be seen in our example that if we use the NPV rule, Project Q is better.

Accounting Rate of Return Method

The accounting rate of return (ARR), also known as the return on investment (ROI), uses accounting information, as revealed by financial statements, to measure the profitability of an investment. The accounting rate of return is found out by dividing the average after-tax profit by the average investment. The average investment would be equal to half of the original investment if it is depreciated constantly. Alternatively, it can be found out dividing the total of the investment's book values after depreciation by the life of the project. The accounting rate of return, thus, is an average rate and can be determined by the following equation:

$$\text{ARR} = \frac{\text{Average income}}{\text{Average investment}}$$

In above equation average income should be defined in terms of earnings after taxes without an adjustment for interest viz. EBIT (I-T). Thus:

$$\text{ARR} = \frac{\left[\sum_{t=1}^n \text{EBIT}_t (1 - T) \right] / n}{(I_0 + I_n) / 2}$$

Where EBIT is earnings before interest and taxes, T tax rate, I_0 book value of investment in the beginning, I_n book value of investment at the end of n number of years.

Acceptance Rule

As an accept-or-reject criterion, this method will accept all those projects whose ARR is higher than the minimum rate established by management and reject those projects which have ARR less than the minimum rate. This method would rank a project as number one if it has highest ARR and lowest rank would be assigned to the project with lowest ARR.

Evaluation of ARR Method

The ARR method may claim some merits.

1. It is very simple to understand and use.
2. The ARR can be readily calculated from the accounting data; unlike the NPV and IRR rules, no adjustments are required to arrive at cash flows of the project,
3. The ARR rule incorporates the entire stream of income in calculating the projects profitability.

The ARR is a method commonly understood by accountants, and frequently used as a performance measure.

Shortcomings

As a decision criterion, however, it has serious shortcoming.

- It uses accounting profits, not cash flows, in appraising the projects. Accounting profits are based on arbitrary assumptions and choices and also include non-cash items. It is, therefore, inappropriate to rely on them for measuring the acceptability of the investment projects.
- The averaging of income ignores the time value of money. In fact, this procedure gives more weightage to the distant receipts.
- The firm employing the ARR rule uses an arbitrary cut-off yardstick. Generally, the yardstick is the firm's current return

on its assets (book-value). Because of this, the growth companies earning very high rates on their existing assets may reject profitable projects (i.e., with positive NPV s) and the less profitable companies may accept bad projects (i.e., with negative NPV s).

The ARR method continues to be used as a performance evaluation and control measure. But its use as an investment criterion is certainly undesirable. It may lead to unprofitable allocation of capital.

Points to Ponder

- Other capital budget evaluation methods other than NPV & IRR methods.
- Evaluate and take decision regarding the acceptance and rejection of any investment projects.
- Limitations of every capital budgeting method.

Review Questions

1. Name different capital budgeting techniques?
2. How you will accept or reject any project under Profitability Index, Payback and Accounting Rate of Return methods?
3. Differentiate between B-C Ratio method and Payback method of capital budget technique?

Note

LESSON 24 :

FINANCIAL STRUCTURE – CAPITAL

Learning Objectives

- To know about financial structure and its effect on financial decisions.
- To learn about the financial structure.
- To understand various sources of long-term sources of funds.
- To know about the various instruments through which, the company can raise funds.

Introduction

In earlier unit, you have learnt about the investment decision and capital budgeting. In this unit you will be more focusing on the financing decisions. This is because you as finance manager should be involved in all financial matters of the organization, since almost all the activities in the organization have financial implications.

Financing Decision

As a financial manager you have to plan for and mobilize the required funds from various sources when they are required and at an acceptable cost. This decision is called financing decision.

For this purpose, you have to keep liaison with the banks and financial institutions. You also deal with merchant banking agencies for procuring funds from the public through issue of shares, debentures and inviting public to subscribe to its fixed deposits. In deciding, how much to procure from various sources, you will weigh many considerations like cost of funds in the form of interest / dividend and cost of public issue in case of shares and debentures and the length of time for which funds will be available. Banks and other financial institutions, which give short-term and long-term loans generally, lay down their conditions. These conditions are aimed at ensuring the safety of the loans given by them and contain provisions restricting the freedom of the borrower to raise loans from other sources. Therefore, you as financial manager will try to balance the advantage of having funds available with the costs and loss of flexibility arising from the restrictive provisions of the loan contract.

Now, you will understand more about the financing decision in different steps. In the unit 5, you will know in detail about the short-term sources and its needed decision. In this unit, you will learn about the long-term sources and decision in respect of financial structure & the financial condition of the organization. You will also know about the cost of capital & its implications and effects of investments. Further, as finance manager, you will also know how to keep the stakeholder of the organization in tact, by means satisfying their expectations.

In this lessons, you will understand about the financial structure, which lay emphasis on product market, labour market, Stakeholders like shareholders and bondholders,

creditors, agencies involved during financial decision of any organization.

Financial Structure

Financial structure is the amount of debt versus equity funding you use in your business. Greater the proportion of borrowings, higher will be the return on equity. Increasing the level of borrowings has a multiplier effect on your returns. Financial structure comprises of short-term and long-term sources of funds for various activities of the business. The short-term sources are required for the day-to-day operation in the business. Whereas long-term sources are required for setting up of a firm, expansion, diversification, modernization and other similar capital expenditures.

The long-term sources are generally issue of securities, term loans, and internal accruals, supplier's credit scheme and equipment financing. In addition to these, you will also have the option of funding projects by way of deferred credit, unsecured loans and deposits and venture capital financing.

The important features of popular sources of long-term financing are as under: -

Types of Capital

Normally firms issue three types of capital, viz., equity, preference and debenture capital based on the risk, return and ownership pattern.

Equity Capital

Equity shareholders are the owners of the business. They enjoy the residual profits of the company after having paid the preference shareholders and other creditors of the company. Their liability is restricted to the amount of share capital they contributed to the company. Equity capital provides the firm the advantage of not having any fixed obligation for dividend payment but offers permanent capital with limited liability for repayment. However, the cost of equity capital is higher than other capital. Firstly, since the equity dividends are not tax-deductible expenses. Secondly, the high costs of issue. In addition to this since the equity shareholders enjoy voting rights, excess of equity capital in the firm's capital structure will lead to dilution of effective control.

Preference Capital

Preference shares have some attributes similar to equity shares and some to debentures. Like in case of equity shareholders, there is no obligatory payment to the preference shareholders and preference dividend is not tax deductible. However similar to debenture holders, wherein interest payment is earned a fixed rate of return for their dividend payments. In addition to this, the preference shareholders have preference over equity shareholders to the post-tax earnings in the form of dividends and assets in the event of liquidation.

Other features of the preference capital include the call feature, wherein the issuing company has the option to redeem the shares, (wholly or partly) prior to the maturity date and at a certain price. Prior to the companies Act 1956 companies could issue preference shares with voting rights. However, with the commencement of the companies Act 1956 the issue of preference shares with voting rights has been restricted to only in the following cases:

There are arrears in dividend for two or more years in case of cumulative preference shares.

Preference dividend is due for a period of two or more consecutive preceding year or

In the preceding six years including the immediately preceding financial year, if the company has not paid the preference dividend for a period of three or more years.

Types of preference capital:

It can be of two types in three categories.

1. Cumulative or Non-cumulative preference shares
2. Redeemable or perpetual preference shares.
3. Convertible or non-convertible preference shares.

For cumulative preference shares, the dividends will be paid on a cumulative basis, in case they remain unpaid in any financial year due to insufficient profits. The company will have pay up all the arrears of preference dividends before declaring any equity dividends. On the other hand, the non-cumulative shares don't enjoy such rights to dividend payments on cumulative basis.

Redeemable preference shares will be redeemed after a given maturity period while the perpetual preference share capital will remain with the company forever.

Debenture Capital

A debenture is a marketable legal contract whereby the company promises to pay its owner, a specified rate of interest for a defined period of time and to repay the principal at the specific date of maturity. Debentures are usually secured by a charge on the immovable properties of the company.

A trustee usually represents the interest of the debenture holders and this trustee (which is typically a bank or an insurance company or a firm of attorneys) is responsible for ensuring that the borrowing company fulfills the contractual obligations embodied on the contract. If the company issues debentures with a maturity period of more than 18 months, then it has to create a debenture redemption reserve (DRR), which should be at least half of the issue amount before the redemption commences. The company can also attach call and put options. With the call option the company can redeem the debentures at a certain price before the maturity date and similarly the put option allows the debenture holder to surrender the debentures at a certain price before the maturity period.

Types of Debentures

Debentures can be classified based on the conversion and security. A few types of debentures are as follows: -

Non-convertible Debentures (NCDs)

These debentures cannot be converted into equity shares and will be redeemed at the end of the maturity period.

Fully Convertible Debentures (FCDs)

These debentures will be converted into equity shares after a specified period of time at one stroke or in installments. These debentures may or may not carry interest till date of conversion. In the case of a fully established company with an established reputation and good, stable market price, FCDs are very attractive to the investors as their bonds are getting automatically converted into shares which may be at time of conversion be quoted much higher in the market compared to what the debenture holders paid at the time of FCD issue.

Partly Convertible Debentures (PCDs)

These are debentures, a portion of which will be converted into equity share capital after a specified period, whereas the NCD portion of the PCD will be redeemed as per the terms of issue after the maturity period. The non-convertible portion of the PCD will carry interest right up to redemption, whereas the interest on the convertible portion will be only up to the date of immediately preceding the date of conversion.

Secured Premium Notes (SPNs)

This is a kind of NCD with an attached warrant that has recently started appearing in the Indian market. This was first introduced by TISCO, which issued SPNs aggregating Rs.346.50 crore to existing shareholders on a right basis. Each SPN is of Rs.300 face value. No interest will accrue on the instrument during the first 3 years after allotment. Subsequently, the SPN will be repaid in 4 equal installments of Rs.75 each from the end of the fourth year together with an equal amount of Rs.75 with each installment. This additional Rs.75 can be considered either as interest (regular income) or premium on redemption (capital gain) based on the tax planning of the investor.

The warrant attached to the SPN gives the holder the right to apply for and get allotment of one equity share for Rs.100 per share through cash payment. This right has to be exercised between one and one-and-a-half year after allotment, by which time the SPN will be fully paid-up.

Apart from these sources by way of financial instruments, there are some new financial instruments, like

1. Non-voting Shares
2. Detachable Equity Warrants
3. Participating Debentures
4. Participating Preference Shares
5. Convertible debenture with options
6. Third Party Convertible Debenture
7. Mortgage-backed Securities
8. Convertible Debentures Redeemable at Premium
9. Debt-Equity Swaps
10. Zero-coupon Convertible Note.

Points to Ponder

- Mobilization of funds through the capital structure at the time of taking financial decision.
- Various types of financial instruments, while mobilizing the funds for various investment opportunities.

- The important aspects like product and labour markets effect the financial decisions in any investment opportunity along with effect of the creditors and other agencies.

Review Questions

1. Name different types capital through which funds are mobilized during financial decisions?
2. Define the following terms: -
 - Secured Premium Notes.
 - Cumulative Preference shares.
 - Perpetual preference shares.
3. Differentiate the following
 - Preferential share and Equity shares
 - Debenture bonds and equity shares
 - FCDs, NCDs and PCDs.

Note

LESSON 25 :

FACTORS EFFECTING FINANCIAL STRUCTURE

Learning Objectives

- To know the factors effecting the financial structure.
- To understand effects of product markets on debt.
- To learn about the creditors and agency costs.
- To know about the various agencies involved at different levels of investment process.

Introduction

In earlier lesson you have learnt about the how financial decisions will depend on the capital structure or vice versa during mobilization of funds for any investment. Development of financial structure also depends on the product and the labour market. In this lesson you know more about how the product and labour market will affect the financial structure during any financial decision for any investment. Further creditors and agency cost will also affect the financial structure of any corporate.

Financial Structure

Financial structure is the amount of debt versus equity funding you use in your business. Greater the proportion of borrowings, higher will be the return on equity. When we say debt in the financial structure will influx further funds for investments, which in turn brings-in enhanced turnover and thereby enhanced return.

In any manufacturing firm, the production manager controls a major part of the investment in the form of equipment, materials and men. He should so organize his department that the equipment under his control are used most productively, the inventory of work-in-process or finished goods and stores and spares is optimized and idle time and work stoppages are minimized. If the production manager can achieve this, he would be holding the cost of the output under control and thereby help in maximizing profits. He has to appreciate the fact that whereas the price at which the output can be sold is largely determined by factors external to the firm like competition, government regulations, etc. the cost of production is more amenable to his control. Similarly, he would have to make decisions regarding make or buy, buy or lease etc. for which he has to evaluate the financial implications before arriving at a decision. So, it becomes pertinent to the financial manager to look into the affairs of the product and labour market from the finance angle as it affects the cost of production as such the return on investment and to the investors.

Product Market & its Effect

The effect of product market on the debt will be as follows:

1. Basically, it is product in any investment matters. Nature of product and its market set the path for its demand for the product.

2. Based on the demand the corporate has to decide the level their product at a future date, this in fact gives anticipated market share of the corporate for that product.
3. Once the future market share for the product is decided, and then you can decide level of investment as a financial manager.
4. Level of investment will give the path to decide what infrastructure is requires by the corporate to meet this demand.
5. Further it also sets the path for deciding the holding level of raw material, semi-finished goods or finished goods.
6. This will decide what must be the cost required at the initial stage in terms for its machinery or holding level of inventory.
7. Accordingly, as a financial manager you have to have clarity in regard to product & its market.

Labour Market & its Impact

Another important aspect, which will effect the cost of production which in turn effect the profitability and return on investment is the labour. Broadly, labour requirement in any investment is classified into.

- Skilled labour
- Unskilled labour.

Factors affecting the labour market:

1. Place
2. Availability of skilled & unskilled labour
3. Cost of living of that particular place.
4. Other labour oriented activities in that place (determines the demand for labour which have bearing on the cost of labour.
5. Facilities like water, health services, education, etc., availability in that area (more so for the skilled labour)

Based on these various factors of the market, the cost of labour gets changed. This in turn affect the cost of production and thereby the profitability and return on investment. So as a finance manager you have look into various aspects during the financial decision. This in fact will affect the financial structure of the corporate.

Creditors

You will notice that during creating a financial structure for any financial decision on any investment creditors plays a vital role. Creditors may be of two types capital creditor and operational creditors. The creditor is defined as the entities, which provide funds, and helps in mobilization of funds for a fixed period of time at a fixed rate of return.

Capital Creditor

The entities which provide funds for the creation of capital which will be used as a long-term source, this is nothing but

debt capital like preference shareholders or bondholders. The provide funds with an anticipation of fixed dividend or interest. The details this debt capital and various types of the same is already known by you in earlier lesson.

Other creditors who provide funds and facilitate funds mobilization for any investment are fixed deposit holders and financial institutions. Normally, the funds mobilized thorough the fixed deposit will be utilized as a long-term source. There will also be the commitment for the corporate in terms of payment of interest on FDs regularly or at the end of maturity period as per the agreed rate for the period along with the principal amount for which the same has been kept. However, the corporate has to incur costs for raising such funds, due to various norms to be complied by the corporate as per SEBI guidelines.

Operational Creditors

Apart from the creditors, who provide funds for the long-term investments and fulfill the fund requirement of corporate, you also require funds for the operational aspects. These funds are made available either from supplier credits or the financial institutions.

You as financial manager may also look into the various aspects in raising funds from this route. They are as follows: -

- Rate of availability of funds.
- Quantum of availability.
- Time for which available.
- Terms & conditions of getting the funds.
- Time taken for getting the funds.
- Level of restrictions on freedom of the corporate.

The cost of getting these funds will affect the cost of production in turn the profitability, which ultimately decides about the return on investment.

Agency Costs

In any investment, you as financial manger have give equal importance to the agency costs. The agency cost is the cost, which will be incurred by the corporate starting from the initiation of investment decisions to the completion of the investment process. The various agencies involved at different levels can be classified as follows:

- a. Pre-Implementation
- b. Implementation
- c. Post-Implementation

Pre-Implementation

Pre-Implementation levels will involves all the agencies, which will facilitate for the investment process of the corporate. The agencies involved at this stage for which the corporate need to pay are

- Brokers
- Builders
- Supply agents, both machinery & raw materials
- Market survey agents
- Project developers
- Technical support agents.

Implementation

The agents who facilitate the corporate during the implementation are called implementation level agents. They include

- Supply agents.
- Distributing agents.
- Recovery agents.
- Settlement agents.

Post-Implementation

Agents who facilitate the corporate after the completion of investment process are called as post-implementation agents. They include

- Settlement agents.
- Brokers
- Facilitators.
- Arbitrators.
- Selling agents.

You have to note that some of the agents will operate at single or in all levels of implementation. However, the cost involved in the payment for these agents of the investment of the corporate will have its effect on the cost. So, care have to be taken by you to see that these costs are to be controlled, it may effect the return on invest which needs to be distributed to all stakeholder of the corporate.

Points to Ponder

- Product market
- Labour market
- Creditors
- Agencies

Review Questions

1. List various effects of product market on the financial decision?
2. How labour market will affect the return on investment?
3. List the major types of creditors and how they will effect the financial decision?
4. State the different level of agencies operating during investment process?
5. How the agencies at post-implementation agencies will effect the financial decision?

Note

LESSON 26 :

FINANCIAL DISTRESS – REMEDIAL OPTIONS

Learning Objectives

- To know about financial distress
- To understand various options available for the corporate during financial distress.
- To learn how to utilize various options like reorganization, restructuring, mergers, acquisitions, takeovers, etc.

Introduction

In earlier lessons, you have learnt about the building-up of financial structure based on the requirement of funds. The financial structure, which lay emphasis on product market, labour market, Stakeholders like shareholders and bondholders, creditors, agencies involved during financial decision of any corporate was also made known to you. A prudent financial decision is one, which takes care of any eventualities during the investment. This includes management of corporate during financial distress. So, in this lesson you will learn more about the financial distress and how as a financial manager you must be aware about various options available to the corporate during its financial distress and how to manage the same.

Corporate finance is about acquiring and investing funds to create value for shareholders. Sometimes value creation requires that a company change its product mix and the markets and industries in which it competes. On the financial balance sheet such a transformation is reflected in changes in the long-term assets of the company – in the mix of employees, factories, machines, products, brands, distribution channels and so on. Major changes within a corporation are often called restructurings. In this lesson you will have the description involving mergers, takeovers, divestitures, and going-private transactions, all of which involve change in who controls a set of assets. In takeover, a new management team acquires control of all of the target company's assets.

When the corporation sells or divests a subsidiary, it relinquishes control of those specific assets. Thus another way of thinking about corporate restructuring is as a market for the control of corporate assets. Financial economists refer to these restructuring transactions as the market for corporate control, mergers and acquisitions (M &A). All these activities takes place when there will be a financial distress.

Financial Distress

This is a condition which a corporate enters when all their expectation go hay-way due to various reasons. This effects all the stakeholders of the corporate like shareholders, bondholders, creditors, agents, employees and one and all. So, as a financial manager you have know about various aspects about the option available to the corporate during its financial distress. The various options available to corporate during their financial distress can be listed as follows:

1. Reorganization
2. Restructuring
3. Merger
4. Acquisition
5. Takeover

Reorganization

A process designed to revive a financially troubled or bankrupt firm. Reorganization involves the restatement of assets and liabilities, and communication with creditors in order to make arrangements for maintaining repayment.

Reorganization is an attempt to extend the life of a company facing bankruptcy through special arrangements and restructuring in order to minimize the possibility of past situations reoccurring.

Restructuring

A major business modification usually associated with personnel downsizing and asset revaluation. Companies experiencing a major restructuring are generally doing poorer than expected and wish to increase future earnings by writing down their assets.

Merger

Merger is a friendly combining of two companies. In most cases the boards of directors of the two companies discuss the mutual benefits of such a combination and then recommend the plan to shareholders for approval. Usually one company _exchanges its stock for the stock of the other, which is then retired. The relative market price of the two companies' stock determines the exchange rate of shares. Occasionally, cash or a combination of cash and stock may be exchanged. This decision is usually mutual between both firms.

Acquisition

Acquisitions involve the purchase of a company or business unit. Acquisitions can be friendly, as in a merger, or unfriendly, as in a hostile takeover. The vast majority of restructuring transactions are friendly purchases of a single business unit or subsidiary. Hostile takeovers of large companies by equally large bidders happen less frequently but get more attention.

The difference between the actual costs for acquiring a target firm versus the estimate made of its value before the acquisition. During a merger and acquisition, companies will first estimate the cost that they wish to pay for a target firm. As the entire M&A process takes many weeks, the price paid for the target firm may in fact be higher or lower than its market price at the time of completion because of economic fluctuations.

The Basics of Mergers and Acquisitions

M&A deals always catch headlines, but what do these deals really mean to investors? Learn what corporate restructuring is, why

companies do it, and why it sometimes doesn't work. Dawn raids, poison pills, and shark repellent: welcome to the dramatic world of mergers and acquisitions.

Takeover

A corporate action where an acquiring company makes a bid for an acquire. If the target company is publicly traded, the acquiring company will make an offer for the outstanding shares. A welcome takeover is usually referring to a favorable and friendly takeover. Friendly takeovers generally go smoothly because both companies consider it a positive situation. In contrast, an unwelcome or hostile takeover can get downright nasty!

Hostile Takeover

In a tender offer the bidding company makes its offer directly to the shareholders of the target firm. This action sometimes occurs if the target company's board has turned down the bidder's merger proposal. The bidder asks shareholders to tender, or sell, their shares at a stated price, usually 15% to 25% above the target firm's current share price. Shareholders then decide whether they prefer the bidder's cash offer or want to continue to own the shares of the company operated by the incumbent management team. If enough shareholders tender their shares, the bidder obtains voting control of the company, elects a new board, and replaces the managers. Hostile tender offers can become quite acrimonious, with both sides placing ads in the financial press explaining why their position is superior and why shareholders should or should not tender their shares.

Sometimes a takeover attempt that is strongly resisted by the target firm, then hostile takeover takes place. Hostile takeovers are usually bad news, as the employee moral of the target firm can quickly turn to animosity against the acquiring firm.

The Wacky World of M & A

Investing terms like "dawn raid," "poison pill," and "shark repellent" might be reminiscent of old James Bond movies, but there's nothing fictional about them. Owning stock in a company means you are part owner, and with mergers and acquisitions occurring every week, it is important to know what these terms mean for your holdings.

Mergers, acquisitions, and takeovers have been a part of the business world for centuries. In today's dynamic economic environment, companies often face decisions concerning these things. The job of management is to maximize shareholder value. In most cases, by using mergers and acquisitions, a company can develop a competitive advantage and ultimately increase shareholder value.

There are several ways that two or more companies can combine their efforts. They can partner on a project, mutually agree to join forces and merge, or one company can outright acquire another company. Here are some of the various intriguing ways that takeovers can take place:

Hostile Takeover - This is a takeover attempt that is strongly resisted by the target firm. These types of takeovers are usually bad news since the employee morale of the target firm can quickly turn to animosity against the acquiring firm. There is usually a reason why the acquirer had to resort to a hostile takeover rather than a friendly one.

Dawn Raid - Here a firm or investor buys, first thing in the morning when the stock markets open, a substantial amount of shares in a company. Usually a brokerage does the buying on behalf of the acquirer (the "predator") to avoid drawing attention to the buying. It builds up a substantial stake in its target (the "victim") at the current stock market price. Because this is done early in the morning, the target firm usually doesn't get informed about this until it is too late, when the acquirer has already scooped up controlling interest.

Saturday Night Special - This is a sudden attempt of one company to take over another by making a public tender offer. The name comes from the fact that this practice used to be done over the weekends.

Takeovers are announced practically everyday, but announcing them doesn't necessarily mean everything will go ahead as planned. In many cases the target company does not want to be taken over. What does this mean for investors? Everything! There are many strategies that management can use during M&A activity, and almost all of these strategies aim at affecting the value of stock in some way. Let's take a look at some more popular ways that companies can protect themselves from an unruly predator. These are all types of what is referred to as "shark repellent":

Golden Parachute

This measure discouraging an unwanted takeover offers lucrative benefits to top executives, who lose their job once their company is taken over by another firm. Benefits include items such as stock options, bonuses, severance pay, etc. Golden parachutes can be worth millions of dollars and can cost the firm a lot of money.

Greenmail

A spin-off of the term blackmail, greenmail occurs when an unfriendly company holds a large block of stock, who then forces the target company to repurchase the stock at a substantial premium to prevent the takeover.

Macaroni Defense

This is a defensive strategy by which the target company issues a large number of bonds that come with the guarantee that they will be redeemed at a high price if the company is taken over. Why is it called Macaroni Defense? Because if a company is in danger, the redemption price of the bonds expands, kind of like Macaroni in a pot! This is a highly useful tactic, but the target company must be careful it doesn't issue so much debt that it cannot make interest payments.

People Pill

Here, management threatens that in the event of a takeover, the entire management team will resign. This is especially useful if they are a good management team; losing them could seriously harm the company.

Poison Pill

With this strategy, the target company aims at making its own stock less attractive to the acquirer. There are two types of poison pills. The first is the "flip-in," which allows existing shareholders (except the acquirer) to buy more shares at a discount. The second is the "flip-over," which allows stock-

holders to buy the acquirer's shares at a discounted price after the merger. If investors fail to take part in the poison pill by purchasing stock at the discounted price, the outstanding shares will not be diluted enough to ward off a takeover.

Sandbag

Here, the target company stalls in the hope that another more favorable company will try to take them over. If management sandbags too long, they may not be spending much effort trying to run the company. Next time you read a news release that mentions your company is using a poison pill to defend it, you now know what it means. More importantly you'll also know that you have the opportunity to purchase more shares at a cheap price. Being a smart investor doesn't just mean knowing where and when to invest. It requires a full understanding of what it means to own a part of a company, and what decisions the company faces.

Busted Takeover

A Highly Leveraged Takeover

Consequently, some of the acquired company's assets must be sold off for the deal to go through. The proceeds from the sale of these assets are then used towards the purchase price of the acquisition.

Leveraged Buyout – LBO

A strategy involving the acquisition of another company using borrowed money (bonds or loans). The acquiring company uses its own assets as collateral for the loan in hopes that the future cash flows will cover the loan payments.

There is usually a ratio of 90% debt to 10% equity. Because of this high debt/equity ratio, the bonds are usually not investment grade and are referred to as junk bonds.

Debt/Equity Ratio

A measure of a company's financial leverage calculated by dividing long term debt by shareholders equity. It indicates what proportion of equity and debt the company is using to finance its assets.

Note: Sometimes investors only use interest bearing long-term debt instead of total liabilities.

$$= \frac{\text{Total Liabilities}}{\text{Shareholders Equity}}$$

A higher debt/equity ratio generally means that a company has been aggressive in financing its growth with debt. This can result in volatile earnings as a result of the additional interest expense.

Debt Reckoning

You learn about debt ratios and assessing a company's financial health., so, you could save a lot of money! In a post-Enron world, it's easy to forget that most of the companies that have been driven into bankruptcy are not reaping the consequences of accounting scandals, but of taking on too much debt.

For investors, there is no quick and easy way to get a handle on a company's debt situation. But as a starting point, debt ratios offer a valuable method for assessing a company's fundamental health. Looked at in context and over time, debt ratios can offer

valuable signals of deepening debt problems. Recognizing those situations can save investors a lot of money. In this article we'll examine one specific debt ratio in detail: the debt-to-equity ratio.

While debt ratios tell investors little about a company's growth prospects or earning performance, these ratios are vital tools for gauging balance sheet durability. If, for instance, a recession or downward cyclical phase is on the way, balance sheet strength becomes more important for investors. It can determine whether a company has a strong enough financial position to survive through a tough period. Financial markets tend to punish overextended firms at the start of a recession. The reason is simple:

Debt-leveraged companies have the hard task of paying their interest obligations out of a flat or declining level of income. Those firms that are unable to pay for their debt go bankrupt. Those that struggle to pay lose credit-worthiness and face further financing troubles down the road. Shrinking profits make debts that are otherwise manageable look horribly burdensome.

The debt-to-equity ratio offers one of the best pictures of a company's leverage. The formula is straightforward:

Total liabilities divided by total shareholders' equity

(shareholder equity = common stock plus firm profits or losses)

Quite simply, the higher the figure, the higher the leverage the company employs. Notice that this ratio uses all liabilities (short-term and long-term), and all owners' equity (both invested capital and retained earnings).

Here is one final warning to you : although the debt-to-equity ratio is a valuable number to examine, investors cannot focus only on it and without any other indicators. There are a number of frequently used debt ratios that show how much a company relies on debt financing. These include the debt ratio, the current ratio, interest coverage, etc. Together they vividly show how the amount of debt leverage can vary between healthy firms with low debt levels and plenty of cash to service it and troubled companies that are heavily leveraged and cash-poor.

Points to Ponder

- Reorganization
- Restructuring
- Merger
- Acquisition
- Takeover
- Debt Reckoning

Review Questions

1. What are various remedial options available for the corporate during financial distress?
2. Differentiate various types of takeovers?
3. Differentiate Hostile takeover and Busted takeover?
4. What do you understand by Saturday Night Special?
5. Define Debt Reckoning?
6. Brief explain the following

-
-
-
-
-

[illegible]

LESSON 27:

CORPORATE RESTRUCTURING OPTIONS

Learning Objectives

- To understand about the restructuring options
- To know about Divestitures.
- To learn from the past about the trends in restructuring of corporate.

Introduction

In earlier unit, you have learnt about the investment decision and capital budgeting. In this unit you will be more focusing on the financing decisions. As a step towards, this you need to know about the financial structure & the financial condition of the organization taking the decision. You also need to know about the cost of capital & its implications and effects of investments.

In earlier lessons of this chapter, you have understood about the financial structure, which lay emphasis on product market, labour market, Stakeholders like shareholders and bondholders, creditors, agencies involved during financial decision of any organization. A prudent financial decision is one, which takes care of any eventualities during the investment. This includes management of organization during financial distress. So, in this lesson you will learn more about the financial distress.

Divestitures

Mergers, acquisitions, and takeovers all entail an increase in assets. In the 1960s many corporations diversified into a range of unrelated markets. Today attitudes about diversification have changed. As companies narrow their focus, many are selling units or business segments that no longer fit into their strategic plan. With increased competition companies must pursue activities in which they have a comparative advantage. Divestitures help companies undo earlier diversification efforts, so they can focus on areas in which they can best compete.

There are several ways that a company can divest itself of a unit or subsidiary. It can announce that the unit is for sale, look for a buyer, or have an auction for the unit. A sale raises money for the parent company that can be used to support growth in its core business areas. Another method for selling a division of a company is an equity carve-out. In equity carve-out the parent company creates a new, publicly traded company out of one of its divisions and sells that stock to the public in an IPO (initial public offering). The parent may keep some portion of the stock. In some cases companies have carved out a division, kept 40% or 50% of the stock, and reabsorbed the division a few years later. Like a sale, an equity carve-out generates cash for the parent company.

If a company does not need cash, it can spin off a business unit to its share-holders. The division is turned into an independent company with publicly traded shares. Each shareholder receives shares in the new company proportionate to his or her ownership in the parent. If you owned 1 % of the parent company,

you would receive 1 % of the shares of the new company in a tax-free transaction.

Examples of spin-offs include AT&T spinning off Lucent Technologies, Whitman Corporation spinning off its Midas Muffler chain, Hilton spinning off its gam-bling and casino properties into Park Place Entertainment, and Campbell Soups spinning off Vlasic International, famous for making pickles.

IPO Roll-Ups

An IPQ roll-up is a transaction that simultaneously merges several private companies and creates a publicly traded holding company that sells its shares in an IPO. Roll-ups consolidate small businesses in fragmented industries, capturing economies of scale and giving them access to capital markets that small companies do not normally have. An interesting aspect of roll-ups is that the person or group that organizes or sponsors them does not need lots of capital to initiate the transaction. The proceeds from the IPO pay for the busi-nesses, organizing fees, and to finance growth. With good economic sense (the proper industry must be chosen) and lots of sweat equity, a person can head up a multimillion-dollar company with almost no personal financial investment. Roll-ups have become popular because a lot of small business owners are approaching retirement and the roll-up mechanism provides a good way to sell their business.

Going-Private Transactions

During the 1980s many dozens of publicly traded companies gave up their pub-lic status and returned to being privately held companies. These transactions are known as management buyouts (MBOs) or leveraged buyouts (LBOs). In buyouts, a small group of investors, usually includ-ing the company's managers, use a relatively small equity investment combined with enormous loans to buy all of a company's outstanding stock. Once the buy-out group owns the stock, they de-list the firm and make it a private, rather than a publicly traded, company, which is the origin of the term going-private trans-actions. The combination of high debt, with its threat of bankruptcy, and man-agerial stock ownership create powerful incentives for managers to improve the company's performance.

Alternatives to the Corporate Form

During the 1980s a few companies elected to change their organizational form from that of a corporation to some other form. Although such changes do not affect the control of assets, they do represent a significant change for the busi-ness, so they are usually included in discussions of restructurings. Tax savings stimulated some companies to create royalty trusts in the early 1980s. Later, a number of firms established master limited partnerships (MLPs). In royalty trusts and MLPs, pre-tax net operating profits go directly to shareholders (called unit holders in MLPs), where investors pay tax on the profits received. Managers operate a set of existing assets but rarely

retain funds to invest in growth. Often MLPs contain natural-resource assets such as a paper company's forestlands or an oil company's petroleum reserves. MLPs also hold specific assets or activities of some companies. For example, a number of research-and-development MLP exists. Companies want to separate the risk of their research efforts from the rest of the company and use the MLP for this separation. These transactions occur only occasionally, but they point out how corporate managers and their financial advisors constantly look for better ways to organize a company's activities.

Lessons from the History of the Market for Corporate Control

Restructurings often represent a major change in a company's direction and prospects, so they often arise from significant forces for change. Restructuring\ historical record shows that as market forces, investor concerns, laws, and tax codes change, the dominant type of restructuring transaction also changes. We cannot view restructuring as unique to a specific company, somehow separate from larger economic and technological trends. Rather, we should recognize re-structuring as a response to changing markets, the changing perceptions of investors, and changing opportunities. The recent history of corporate restructurings provides some insights into how corporations have responded to changes in market forces and may suggest how they will respond to future changes. Moreover, examining distinct periods in the evolution of the market for corporate control demonstrates some of the many reasons that companies have chosen to re-structure. As we progress through our historical overview, we will discuss the reasons underlying the type and magnitude of restructuring events in each era.

The 1960s: Synergistic Mergers

Restructurings have been occurring for hundreds of years. We will not go back that far to begin our overview of the market for corporate control. We begin in the 1960s, when most of the significant restructuring events were friendly mergers. Members of one company's board negotiated with the directors of the other company. When the two teams of directors reached a satisfactory agreement, they would recommend that their respective shareholders approve the transaction. In the 1960s shareholders were generally passive in corporate matters, so they only rarely challenged the recommendation of the board.

During the late 1960s and 1970s a wave of mergers and acquisitions occurred that created a number of diversified conglomerates. Investors recognized that firms participating in an array of markets or holding a diversified portfolio of businesses would have relatively stable earnings, because higher earnings in another offset poor earnings in one market. Moreover, economies of scale from centralizing routine organizational tasks, such as payroll, accounting, and cash management, generated savings that translated into profits. The search for synergistic mergers, combinations in which the value of the whole exceeded the value of the sum of the parts, became the restructuring rage? Business parodies joked that mergers would end only when a single, enormous corporation owned everything.

Synergistic gains motivated many transactions during the era of friendly mergers. Synergy implies that the combination of firms

itself creates value. Gains from synergy can arise from a variety of sources. One of the most commonly stated reasons for mergers and acquisitions in the 1960s was that synergistic gains arose from the risk reduction produced by diversification. Recall that as risk decreases, the discount rate investors apply to future cash flows also decreases. A lower discount rate implies a higher present value; all else equal, so diversifying mergers create gains through risk reduction. Many companies used this rationale to explain their acquisition activities. But investors quickly realized that by buying and selling stock, they could diversify their portfolios much more thoroughly and at lower cost than a company could via acquisitions. Moreover, owning shares in a diversified company forced investors to own the company's mix of industry exposure, which might not fit the risk preferences of all or even most investors. Gains from pure diversifying mergers were illusory at best.

If diversification does not explain mergers, why did so many smart executives spend time and money on these transactions? Very likely, the successful mergers and acquisitions of the 1960s relied on other sources of synergy to generate benefits. A few of the possible sources of gains include the following.

Increasing Market Power

Early mergers often saw companies in similar business lines join forces. Economists label such combinations horizontal mergers. Antitrust legislation prevents such combinations from resulting in monopolistic market power that creates the ability to raise prices indiscriminately. Nonetheless, larger firms may obtain some gains from their improved ability to negotiate with suppliers.

Growth for growth's sake is never a sufficient reason for a merger or acquisition. Nonetheless, there have almost certainly been acquisitions completed solely for this reason. Managers have incentives to run larger companies. CEO compensation plans often relate salary to the size or growth rate of the company. The pride and prestige of managing a larger company may also affect some acquisition decisions. Managers who increase firm size to realize personal gains almost always harm shareholders. When expansion increases shareholder wealth, those gains probably arise from economies of scope and scale, which we discuss next, rather than sheer size.

Economies of Scale

When companies merge, eliminating duplicated business functions can produce cost savings. If the total cost of performing a task increases more slowly than changes in volume, the production cost per unit falls. Economists call this condition decreasing average costs, or economies of scale. The costs of many business tasks, such as almost any computer-related task, behave this way.

Exploiting an Expertise (Economies of Scope)

Some mergers and acquisitions create value by exploiting economies of scope. Economies of scope arise when a company applies the same skill to a variety of products. For example, suppose that a company has a special skill in distributing items to drug stores. Currently the company manufactures and distributes only vitamins and diet supplements. The

company might consider adding to the lines of goods it distributes to further exploit its distribution expertise. Adding product lines has very little effect on the company's cost structure but increases its sales and profit potential. Thus, the firm might search for an acquisition candidate that would allow it to make fuller use of its distribution channels. Firms that can identify and exploit their special skills can earn profits that increase shareholders' wealth.

Vertical Integration

Vertically integrated companies own several links in the entire production and sales chain. An example of vertical integration is an oil refining company that invests in oil reserves (the input to production) and gas stations (the retail distribution channel). Vertical integration assures a supply of inputs or outlets and can help protect a company's proprietary information.

Surprisingly, little evidence exists supporting the value of vertical integration. Possibly the costs of vertical integration offset any gains from risk reduction. Integration requires companies to enter markets or industries in which managers may have little expertise. By integrating the production and sales activities, a company may miss important signals from suppliers or customers. For example, suppose an automobile manufacturer acquires a car-rental company to assure an outlet for a portion of its production.¹⁰ If consumers dislike a particular car model, sales figures may not show their displeasure because a certain level of sales are guaranteed. Rather than quickly responding to consumer preferences, the company may continue producing the car with its undesirable features because sales have not fallen sufficiently to alert them to the problem.

Vertically integrated companies may lose the benefits of economies of scale. If the division that supplies a company's inputs is not large enough to produce at the lowest cost volume, the cost of inputs produced internally may be higher than those purchased from optimally sized independent producers. The value of following a strategy of vertical integration depends on the frequency and cost of input shortages or the lack of outlets, and these benefits must be large enough to offset the various costs of integration.

During the 1960s and early 1970s many companies restructured to exploit economies of scale or scope. Such investments increased profits by decreasing costs or increasing revenues. Most restructurings were friendly, with corporate boards agreeing on the nature of the merger before announcing the proposal to shareholders. Although diversification was offered as the rationale for many combinations, in retrospect it probably contributed very little of benefit to these transactions.

The Late 1970s and 1980s: Competition, Hostile Takeovers, and Junk Bonds

Times and management theories change. In the late 1970s and 1980s several co-incidental events brought about a dramatic change in managers' and investors' attitudes about the organization of corporations and, thereby, the type of restructuring transactions that would increase shareholder wealth. Global competition increasingly affected U.S. corporations during this period. Increased competition requires companies to become more efficient. Some firms respond to this call for increased

efficiency by shedding subsidiaries and focusing on business areas in which they have substantial expertise. Others simply continue with business as usual.

During the 1980s, investors and the managers of competing companies quickly identified corporations that were not responding to this new level of competition. As the stock price of a noncompetitive company fell, it became an ideal target for acquisition. The incumbent managers of these unresponsive companies rarely agreed to a friendly merger, recognizing that there was little room for them in the successor firm. For a time the rules of civilized conduct applied, and spurned suitors left the targeted companies in peace. Eventually, however, the desire to acquire and reorganize inefficiently managed companies (and the potential profits from doing so successfully) overcame these conventional rules of business behavior. When the boards of the bidding and target companies could not negotiate a merger or when the target company's managers simply refused to consider any type of combination, bidders began to make offers directly to shareholders. Tender offers always include a premium over the target company's prefer stock price, and payment could be in cash, stock, or a combination of cash and securities. Tender offers give shareholders the opportunity to decide whether or not they want incumbent managers to continue running the company. Tender offers also show shareholders the value other, presumably highly informed, investors place on the company. These transactions differ from typically friendly mergers and are usually distinguished from mergers by denoting them as hostile takeovers, or hostile tender offers.

The emergence of hostile tender offers as a restructuring tool changed the U.S. business landscape. Suddenly inefficiently operated midsize and small corporations became the takeover targets of larger, acquisitive, bidders.

The 1980s were an extraordinary era in U.S. financial history. As the pace of restructuring quickened, multibillion-dollar takeovers regularly appeared as front-page news. Potential targets, with expensive legal help, designed takeover defenses, which, whether effective or not, brought a new vocabulary to corporate finance—poison pills, crown jewels, scorched earth, the Pac-Man defense, shark repellents, greenmail, golden parachutes, and white knights. In some cases the defensive maneuvers of potential targets required taking steps that increased corporate performance and, thereby, shareholder wealth; the threat of takeover was sufficient to cause managers to make decisions that increased shareholder wealth.

One of the most common strategies for improving performance was the sale of under performing divisions. A large manager-owner stockholding meant a bidder would have a very difficult (or expensive) time acquiring a controlling interest in the firm. As the 1980s progressed, changes in financial markets would solve the problem of finding sufficient funds to acquire very large companies. Besides the financial innovations developed in the 1980s, investment bankers also introduced new restructuring vehicles.

The most popular innovation was the

- Leveraged buyout (LBO) or
- Management buyout (MBO).

Buyouts are also known as going-private transactions; because once the transaction is completed the companies stock no longer trades publicly. In a buyout, a small investment group, which usually includes some of the company's managers, purchases all a company's outstanding stock using a small equity stake, supplemented by an enormous amount of debt. Debt-to-equity ratios of 10:1 or 15:1 were not uncommon in buy-outs. The company's assets secure much of the debt, which comes from banks as well as junk bonds, and some debt has provisions that allow the company to postpone interest and principal payments.

Many of the hostile takeovers and leveraged buyouts of the 1980s arose for one of the following two reasons.

1. Shifting Assets to More Highly Valued Uses

In many cases a bidder sees a way of making better use of a company's assets. By shifting assets to more highly valued uses, the bidder can pay a premium for the target company and also earn a profit.

2. Savings from Agency Cost Reduction

Replacing the CEO and other top managers is the key objective of some hostile takeovers. During the 1960s and 1970s many corporate boards simply rubber-stamped managements' recommendations without serious oversight. Without monitoring, some corporate managers became very complacent and others took on the trappings of royalty.

The Mid-1990s: The Era of Strategic Mergers and Acquisitions

The nature of the market for corporate control changed suddenly in 1990. The leveraged buyouts of the 1980s, however, showed investors the power of improved board oversight. Institutional investors, such as pension funds and insurance companies, began to demand that directors take a more active role in the governance of corporations. The large stockholdings of institutional investors meant that corporate boards had to listen to their concerns. Responding to these demands, boards became much more proactive. Directors of poorly performing corporations began to challenge the CEOs, asking about their plans for improving performance.

By the mid-1990s takeover activity was increasing, though the nature of the transactions differed significantly from those of the 1980s. Rather than raiders identifying and acquiring inefficiently operated companies, takeovers were motivated by strategic concerns. Companies tried to keep up with shifts in markets by identifying acquisition targets that would allow them to track those market changes. Companies can more quickly make these shifts by buying an existing company rather than by developing a subsidiary from scratch.

The prevalence of strategic restructurings in the 1990s reflects the need to maintain competitiveness as markets and technologies change. During the 1980s many inefficient companies were absorbed or dissolved. In the 1990s increased competition and stronger internal control systems reduced the need for disciplinary restructurings. The earlier restructuring transactions moved corporations to a reasonable level of efficiency; today's restructurings try to maintain that competitiveness and prepare companies to be competitive in the future.

The Twenty-First Century

Institutional Change, Consolidation, and a Global Quest for Shareholder Value. The importance of cross-border transactions certainly equals, and may exceed, that of domestic mergers and acquisitions.

- Institutional Changes throughout the globe.
- Creating Value for Shareholders.
- Time will tell whether these mega mergers make sense.
- All large control transactions share is the claim that the transaction will create value for shareholders.

Making Mergers Successful

Mergers and acquisitions are usually treated as financial decisions, but as we have stressed throughout the text, there are very few purely financial decisions. The vast majority of business decisions are best made using skills from several business disciplines. Mergers are no different. To increase the likelihood that a merger or acquisition will create value all of the various business areas must be involved. Here are just a few of the ways you can get benefit from the expertise of other business colleagues.

Identifying Acquisition Targets

This requires people from both marketing and strategic planning. The acquisition target must bring something valuable to the firm. It must move the firm along the desired strategic path.

As the Deal Progresses

Once a target is identified and an offer is made, the firm needs to determine which target employees are particularly important to retain. Investors are smart and will quickly see through claims about the value of the merged companies.

Upon Completion

Some involved in M & A area say that the 100 days following completion of an acquisition is a crucial period for integrating the new company.

Restructuring and actions that harm shareholders

Unjustified Growth

Overconfidence

Diversification and the Risk Aversion of Managers

Defensive Actions and Managerial Entrenchment

Points to Ponder

- Corporate restructuring options in any financial decision
- Modifying the financial structure as per the changes in the circumstance.
- Making successful mergers.

Review Questions

1. What is Divestitures?
2. What are the alternatives to corporate?
3. How to make a successful merger?
4. How restructuring will affect the shareholders?
5. Differentiate between Leveraged buyout (LBO) and Management buyout (MBO).

6. State the two reasons, why many of the hostile takeovers and leveraged buyouts of the 1980s arise?
7. What do you mean by Era of Strategic Mergers and Acquisitions?
8. What is the situation of the global corporate restructuring in the 21st century?
9. What do you mean by vertical integration?

Note

LESSON 28 : COST OF DEBT

Learning Objectives

- To understand the concept of cost of debt.
- To learn how to estimate the cost of debt.
- To know how to design the debt policy.
- To know the methods issuing debt instruments.
- To differentiate debt issued at par and with premium or discount.
- To know how to calculate the cost of term loan.

Introduction

Till now, you are familiar with the different sources of long-term sources. In any financial decision the cost of these long-term sources makes a lot of difference in terms of return on investment and the entire investment process itself.

Concept of the Cost Capital

The discount rate is the project's opportunity cost of capital (or simply the cost of capital) for discounting its cash flows. The project's cost of capital is the minimum acceptable rate of return on funds committed to the project. The minimum acceptable rate, or the required rate of return is a compensation for time and risk in the use of capital by the project. Since the investment projects may differ in risk, each one of them will have its own unique cost of capital.

The firm represents the aggregate of investment projects undertaken by it. Therefore, the firm's cost of capital will be the overall, or average, required rate of return on the aggregate of the investment projects. Thus the firm's cost of capital is of the same thing as the project's cost of capital. Can we use the firm's cost of capital for discounting the cash flows of the investment projects? The firm's cost of capital can be used for discounting the cash flows of those investment projects, which have risk equivalent to the average risk of the firm. As a first step, however, the firm's cost of capital can be used as a standard for establishing the required rates of return of the individual investment projects. The firm's cost of capital can be adjusted upward or downward to account for differential riskness of investment projects. That is, an investment project's required rate of return may be equal to the firm's cost of capital plus or minus a risk adjustment factor.

It should be recognized at the outset that the cost of capital is one of the most difficult and disputed topics in the finance theory. Financial experts express conflicting opinions as to the way in which the cost of capital can be measured. It should be noted that it is a concept of vital importance in the financial decision making. It is useful as a standard for:

- Evaluating investment decisions,
- Designing a firm's debt policy, and
- Appraising the financial performance of top management.

Investment Evaluation

The primary purpose of measuring the cost of capital is its use as a financial standard for evaluating the investment projects. In the NPV method, an investment project is accepted if it has a positive NPV. The project's NPV is calculated by discounting its cash flows by the cost of capital. In this sense, the cost of capital is the discount rate used for evaluating the desirability of the investment project. In the IRR method, the investment project is accepted if it has an internal rate of return greater than the cost of capital. In this context, the cost of capital is the minimum required rate of return on the investment project. It is also known as the cutoff, or the target, or the hurdle rate.

An investment project that provides a positive NPV when its cash flows are discounted by the cost of capital makes a net contribution to the wealth of shareholders. If the project has zero NPV, it means that its cash flows have yielded a return just equal to the cost of capital, and the acceptance or rejection of the project will not affect the wealth of shareholders. The cost of capital is the minimum required rate of return on the investment project that keeps the present wealth of shareholders unchanged. It may be thus noted that the cost of capital represents a financial standard for allocating the firm's funds, supplied by owners and creditors, to the various investment projects in the most efficient manner.

Designing Debt Policy

The debt policy of a firm is significantly influenced by the cost consideration. In designing the financing policy, that is, the proportion of debt and equity in the capital structure, the firm aims at minimizing the overall cost of capital. The cost of capital can also be useful in deciding about the methods of financing at a point of time. For example, cost may be compared in choosing between leasing and borrowing. Of course, equally important considerations are control and risk.

Performance Appraisal

Further, the cost of capital framework can be used to evaluate the financial performance of top management. Such an evaluation will involve a comparison of actual profitabilities of the investment projects undertaken by the firm with the projected overall cost of capital, and the appraisal of the actual costs incurred by management in raising the required funds.

Determining Component Cost of Capital

In the following pages you will know, the methods of computing the component costs of three major sources of capital: debt, preference shares and equity shares. Generally, the component cost of a specific source of capital is equal to the investors' required rate of return. But the investors' required rate of return should be adjusted for taxes in practice for calculating the cost of a specific source of capital to the firm. In the investment analysis, net cash flows are computed on after-

tax basis. Therefore, the component costs, used to determine the discount rate should also be expressed on an after-tax basis.

Cost of Debt

A company may raise debt in a variety of ways. It may borrow funds from financial institutions or public either in the form of public deposits or debentures (bonds) for a specified period of time at a certain rate of interest. A debenture or bond may be issued at par or at a discount or premium. The contractual or coupon rate of interest forms the basis for calculating the cost of any form of debt.

Debt Issued at Par

The before-tax cost of debt is the rate of return required by lenders. It is easy to compute before-tax cost of debt issued and to be redeemed at par; it is simply equal to the contractual or coupon rate of interest. For example, a company decides to sell a new issue of 7 - year 15 per cent bonds of Rs 100 each at par. If the company realizes the full face value of Rs 100 bond and will pay Rs 100 principal to bondholders at maturity, the before-tax cost of debt will simply be equal to the coupon rate of interest of 15 per cent.

Thus:

$$K_d = i = \frac{INT}{B_0} \quad (1)$$

Where K_d is the before-tax cost of debt, i is the coupon rate of interest, B_0 is the issue price of the debt and in Equation (1) it is assumed to be equal to the face value (F), and INT is the amount of interest. The amount of interest payable to the lender is always equal to:

Interest = Face value of debt x Coupon interest rate

$$INT = Fi$$

The before-tax cost of bond in the example is Rs15

$$K_d = \text{Rs-100} \times 0.15 \text{ or } 15\%$$

We could arrive at same results as above by using the formula for the internal rate of return: cash outflow are Rs 15 interest per year for 7 years and Rs 100 at the end of seventh year in exchange for Rs 100 now. Clearly, the before-tax cost of bond is the rate, which the investment should yield to meet the outflows to bondholders.

Debt Issued at Discount or Premium

You can write the formula for calculating the cost of debt as follows:

$$B_0 = \sum_{t=1}^n \frac{INT_t}{(1 + K_d)^t} + \frac{B_n}{(1 + K_d)^n}$$

Where B_0 is the repayment of debt on maturity and other variables as defined earlier Equation (2) can be used whether debt is issued at par or discount or premium, i.e., $B_0 = F$ or $B_0 > F$ or $B_0 < F$.

Cost of Term Loans

The cost of term loans will be simply equal to the interest rate multiplied by $(1 - \text{tax rate})$. The interest rate to be used here will be the interest rate applicable to the new term loan. The interest

is multiplied by $(1 - \text{Tax rate})$ as interest on term loans is also tax deductible.

$$K_t = I (1 - t) \quad \text{where, } I = \text{interest rate and } t = \text{tax rate}$$

Points to Ponder

- Cost of debt
- Terminology of debt instruments issued at par, at premium and at discount

Review Questions

1. How do you estimate the cost of debt?
2. Write the steps involved in issuing debt instruments at par, at premium or discount?
3. What is meant by cost of term loan and how the same is estimated?

Note

LESSON 29: COST OF CAPITAL

Learning Objectives

- To know how to estimate the cost of preference shares, capital, retained earnings.
- To understand various methods of estimating the cost of equity capital.

Introduction

In earlier unit, you have learnt about the investment decision and capital budgeting. In this unit you will be more focusing on the financing decisions. This is because you as finance a manger should be involved in all financial matters of the organization, since all most all the activities in the organization have financial implications.

Cost of Preference Capital

The measurement of the cost of preference capital poses some conceptual difficulty. In the case of debt, there is a binding legal obligation on the firm to pay interest, and the interest constitutes the basis to calculate the cost of debt. However, in the case of preference capital, payment of dividends is not legally binding on the firm and even if the dividends are paid, it is not a charge on earnings; rather it is a distribution or appropriation of earnings to preference shareholders. One may be, therefore, tempted to conclude that the dividends on preference capital do not constitute cost. This is not true.

The cost of preference capital is a function of the dividend expected by investors. Preference capital is never issued with an intention not to pay dividends. Although it is not legally binding upon the firm to pay dividends on preference capital, yet it is generally paid when the firm makes sufficient profits. The failure to pay dividends, although does not cause bankruptcy, yet it can be a serious matter from the common (ordinary) shareholders' point of view. The nonpayment of dividends on preference capital may result in voting rights and control to the preference shareholders. More than this, the firm's credit standing may be damaged. The accumulation of preference dividend arrears may adversely affect the prospects of ordinary shareholders for receiving any dividends, because dividends on preference capital represent a prior claim on profits. As a consequence, the firm may find difficulty in raising funds by issuing preference or equity shares. Also, the market value of the equity shares can be adversely affected if dividends are not paid to the preference shareholders and, therefore, to the equity shareholders. For these reasons, dividends on preference capital should be paid regularly except when the firm does not make profits, or it is in a very tight cash position.

Irredeemable Preference Share

The preference share may be treated as a perpetual security if it is irredeemable. Thus, its cost is given by the following equation (for a perpetuity):

$$k_p = \frac{DIV}{P_0}$$

where k_p is the cost of preference share, DIV is the expected preference dividend, and P_0 is the issue price of preference share.

Redeemable Preference Share

Redeemable preference shares (that is, preference shares with finite maturity) are also issued in practice. A formula similar to above Equation can be used to compute the cost of redeemable preference share:

$$P_0 = \sum_{t=1}^n \frac{INT_t}{(1 + K_d)^t} + \frac{B_n}{(1 + K_d)^n}$$

The cost of preference share is not adjusted for taxes because preference dividend is paid after the corporate taxes have been paid. Preference dividends do not save any taxes. Thus, the cost of preference share is automatically computed on after-tax basis. Since interest is tax deductible and preference dividend is not, the after-tax cost of preference is/ substantially higher than the after-tax cost of debt

Cost of Equity Capital

Firms may raise equity capital internally by retaining earnings. Alternatively, they could distribute the entire earnings to equity shareholders and raise equity capital externally by issuing new shares. In both cases, shareholders are providing funds to the firms to finance their capital expenditures. Therefore, the equity shareholders' required rate of return will be the same whether they supply funds by purchasing new shares or by foregoing dividends which could have been distributed to them. There is, however, a difference between retained earnings and issue of equity shares from the firm's point of view. The firm may have to issue new shares at a price lower than the current market price. Also, it may have to incur flotation costs. Thus, external equity will cost more to the firm than the internal equity.

Is Equity Capital Free of Cost?

It is sometime argued that the equity capital is free of cost. The reasons for such argument are that it is not legally binding for firms to pay dividends to ordinary shareholders. Further, unlike the interest rate or preference dividend rate, the equity dividend rate is not fixed. It is fallacious to assume equity capital to be free of cost. As we have discussed earlier, equity capital involves an opportunity cost; ordinary shareholders supply funds to the firm in the expectation of dividends (including capital gains) commensurate with their risk of investment. The market value of the shares determined by the demand and supply forces in a well functioning capital market reflects the return required by ordinary shareholders. Thus, the shareholders' required rate of return, which equates the present value of the expected dividends with the market value of the share, is the cost of

equity. The cost of external equity could, however, be different from the shareholders' required rate of return if the issue price is different from the market price of the share.

In practice, it is a formidable task to measure the cost of equity. The difficulty derives from two factors: First, it is very difficult to estimate the expected dividends. Second, the future earnings and dividends are, expected to grow over time. Growth in dividends should be estimated and incorporated in the computation of the cost of equity. The estimation of growth is not an easy task. Keeping these difficulties in mind, the methods of computing the cost of internal and external equity are discussed below.

Cost of Retained Earnings

The opportunity cost of the retained earnings (internal equity) is the rate of return on dividends foregone by equity shareholders. The shareholders generally expect dividend and capital gain from their investment. Thus, the shareholders' required rate of return or the cost of equity is the expected dividend yield and capital gains, Equation (7) can be used to calculate the cost of equity;

where k_e is the cost of equity, DIV expected dividend next year, P_0 share price today and g expected growth in dividends, The cost of equity is thus equal to the expected dividend yield (DIV / P_0) plus capital gain rate as reflected by expected growth in dividends (g). It may be noted that Equation (7) is based on the following assumptions:

- The market price of the ordinary share (P_0) is a function of expected dividends.
- The initial dividend DM is positive ($DIV_0 > 0$).
- The dividends grow at a constant rate g , and the growth rate (g) is equal to the return on equity (ROE) times the retention ratio (b): (that is $g = ROE \times b$).
- The dividend payout ratio i.e. $(1 - b)$ is constant.

The cost of retained earnings implies that if the firm would have distributed earnings to shareholders, they could have invested it in the shares of the firm or in the shares of other firms of similar risk at the market price (P_0) to earn a rate of return equal to k_e . Thus, the firm should earn a return on retained funds equal to k_e to ensure growth of dividends and share price. If a return less than k_e is earned on retained earnings, the market price of the firm's share will fall. It may be reemphasized that the cost of retained earnings will be equal to the shareholders' required rate of return.

Zero-growth

In addition to its use in constant and variable growth situations, the dividend valuation model can also be used to estimate the cost of equity of no-growth companies. The cost of equity of a share on which a constant amount of dividend is expected perpetually is given as follows:

$$k_e = \frac{DIV_1}{P_0}$$

The growth rate g will be zero if the firm does not retain any of its earnings; that is, the firm follows a policy of 100 per cent payout. Under such case, dividends will be equal to earnings, and therefore above equation can also be written as:

$$k_e = \frac{EPS_1}{P_0}$$

Which implies that in a no-growth situation, the expected earnings-price ratio may be used as the measure of the firm's cost of equity.

Cost of External Equity

The minimum rate of return, which the equity shareholders require, on funds supplied by them by purchasing new shares to prevent a decline in the existing market price of the equity share is the cost of external equity. The firm can induce the existing or potential shareholders to purchase new shares when it promises to earn a rate of return equal to:

$$k_e = \frac{DIV}{P_0} + g$$

Thus the shareholders' required rate of return from retained earnings and external equity is the same. The cost of external equity is, however, greater than the cost of internal equity for one reason. The selling price of the new shares may be less than the market price. In India the new issues of ordinary shares are generally sold at a price less than the market price prevailing at the time of the announcement of the share issue. Thus the formula for the cost of new issue of equity capital may be written as follows:

$$k_e = \frac{DIV}{P_0} + g$$

Where P_0 is the issue price of new equity. The cost of required earnings will be less than the cost of new issue of equity if $P_0 > I_0$.

Realized Yield Approach

According to this approach, the past returns on a security are taken as a proxy for the return required in the future by the investors. The assumptions behind this approach are that

1. The actual returns have been in line with the expected returns
2. The investors will continue to have the same expectations from the security.

As these assumptions generally do not hold well in real life, the results of this approach are normally taken as a starting point for the estimation of the required return.

The realized return over a n -year period is calculated as $(W_1 \times W_2 \times \dots \times W_n)^{1/n} - 1$

Where, W_t referred to as wealth ratio, is calculated as:

$$\frac{D_t + P_t}{P_{t-1}}$$

D_t = Dividend per share for year 't' payable at the end of year

P_t = Price per share at the end of year 't'

Capital Assets Pricing Model Approach

According to this approach, the cost of equity is reflected by the following equation:

$$K_i = R_f + b (R_m - R_f)$$

Where,

K_i	=	rate of return required on security i
R_f	=	risk-free rate of return
b	=	beta of security i
R_m	=	rate of return on market portfolio

Bond Yield plus Premium Approach

The logic behind this approach is that the return required by the investors is directly based on the risk profile of a company. This risk profile is adequately reflected in the return earned by the bondholders. Yet, since the risk borne by the equity investors is higher than that by the bondholders, the return earned by them should be higher. Hence this return is calculated as:

$$\text{Yield on the long-term bonds of the company} + \text{Risk premium.}$$

The risk premium is a very subjective figure, which is arrived at after considering the various operating and financial risks faced by the firm. Though these risks are already factored in the bond yield, since by nature equity investment is riskier than investments in bonds and is exposed to a higher degree of the firm's risks, they also have an impact on the risk premium.

Weighted Average Cost of Capital

Once the component costs have been calculated, they are multiplied by the weights of the various sources of capital to obtain a weighted average cost of capital (WACC). The composite or overall cost of capital is the weighted average of the costs of various sources of funds, weights being the proportion of each source of funds in the capital structure. It should be remembered that it is the weighted average concept, not the simple average. Which is relevant in calculating the overall cost of capital. The simple average cost of capital is not appropriate to use because firms hardly use various sources of funds equally in the capital structure.

The following steps are used to calculate the weighted average cost of capital:

- To calculate the cost of the specific sources of funds (i.e., cost of debt, cost of equity, cost of preference capital etc.)
- To multiply the cost of each source by its proportion in the capital structure.
- To add the weighted component costs to get the firm's weighted average cost of capital.

In financial decision-making, the cost of capital should be calculated on an after-tax basis. Therefore, the component costs to be used to measure the weighted cost of capital should be the after-tax costs. If we assume that a firm has only debt and equity in its capital structure, then its weighted average cost of capital (k_o) will be:

Where k_o is the weighted average cost of capital, $k_d (1 - T)$ and k_e are respectively the after-tax cost of debt and equity, D is the amount of debt and S is the amount of equity. In a general form, the formula for calculating WACC can be written as follows:

$$k_o = k_1 W_1 + k_2 W_2 + k_3 W_3 + \dots$$

Where $k_1, k_2, \dots$ are component cost and $W_1, W_2, \dots$ are weights of various types of capital employed by the company.

Weighted Marginal Cost of Capital

At the time of developing the concept of cost of capital, you have assumed that the risk profile and financing policy of the firm do not change. Now, the question that arises is if these assumptions hold, does the weighted average cost of capital remains unchanged irrespective of the magnitude of financing? Normally, it does not. Then the WACC increases with the level of financing required. The supplies of capital generally require a higher return as they supply more capital. A schedule showing the relationship between additional financing and the weighted average cost of capital is referred to as the weighted marginal cost of capital schedule.

The following steps are to be followed for determining the weighted marginal cost of capital schedule:

The cost of each individual source of finance for various levels of usage has to be estimated.

Given the ratio of different sources of finance in the new capital structure. Find out the levels of total new financing at which the cost of various sources would change. These levels called breaking points, can be found out as: =

Total new financing from that source at the breaking point

Proportion of that financing source in the capital structure

Calculate the weighted average cost of capital for various ranges of total financing between the breaking points.

List out the weighted average cost of capital for each level of total new financing. This is the weighted marginal cost of capital schedule.

Book-Value Vs. Market-Value Weights

The weighted cost of capital can be computed by using the book value or the market value weights. If there were a difference between book value and market value weights, the weighted average cost of capital would differ according to the weights used. The weighted average cost of capital calculated by using the book-value weights will be understated if the market value of the share is higher than the book value and vice-versa. Besides the simplicity of their use, the book-value weights are said to have the following advantages:

- Firms in practice set their capital structure targets in terms of book values.
- The book-value information can be easily derived from the published sources.

- Investors to evaluate the riskiness of the firms in practice analyze the book-value debt-equity ratios.

But the use of the book-value weights can be seriously questioned on theoretical grounds. The component costs are opportunity rates and are determined in the capital markets. The weights should also be market-determined. The book-value weights are based on arbitrary accounting policies followed to calculate retained earnings and value of assets. Thus they do not reflect economic values. It is very difficult to justify the use of the book-value weights in theory.

Market-value weights are theoretically superior to book-value weights. They presumably reflect economic values and are not influenced by accounting policies. They are also consistent with the market-determined component costs. The difficulty in using market-value weights is that the market prices of securities fluctuate widely and frequently.

A third alternative, which is justified below, is to use target weights in the calculation of the overall cost of capital for evaluating the investment projects.

Points to Ponder

- Cost of preference capital.
- Estimation of cost of equity capital.
- Methods adopted for cost of equity capital evaluation.
- Retained earnings and its calculations
- Understanding of concept of WACC and WMCC.

Review Questions

1. How will you estimate the cost of preference capital?
2. State the different methods used for evaluation of cost of equity capital?
3. What do you understand by Retained earnings and how it is calculated?
4. Differentiate between WACC and WMCC?
5. How Book value is different from Market-Value during estimation of cost of equity capital?
6. What do you understand by CAPM and how SML is drawn?

Note

LESSON 30 : FINANCIAL LEVERAGE

Learning Objectives

- To understand about Leverage & Financial Leverage
- To measure Financial Leverage and take appropriate financial decision.
- Effects on the return for the investors and other stakeholders
- EBIT-EPS analysis used in the financial leverage.

Introduction

In earlier unit, you have learnt about the investment decision and capital budgeting. In this unit you will be more focusing on the financing decisions. This is because you as finance a manger should be involved in all financial matters of the organization, since all most all the activities in the organization have financial implications.

Leverage

Leverage in the general sense means influence of power i.e., utilizing the existing resources to attain something else. Leverage in terms of financial analysis is the influence which independent financial variable has over the a dependent / related financial variable. When leverage is measured between two financial variables it explains how the dependent variable responds to a particular change in the independent variable. To explain further, let A be an independent financial variable and B its dependent variable, then the leverage which B has with A can be assessed by the percentage change in B to a percentage change in A.

$$LB / LA = \frac{\Delta B / B}{\Delta A / A}$$

where LB / LA = measure of the leverage which dependent B has with independent A

$$\begin{aligned}\Delta A &= \text{Change in A} \\ \Delta B &= \text{Change in B} \\ \Delta A / A &= \text{Percentage change in A} \\ \Delta B / B &= \text{Percentage change in B}\end{aligned}$$

Measure of Leverage

For your better understanding the importance of leverage in financial analysis, it is imperative to understand the three measures of leverage.

- Operating Leverage
- Financial Leverage
- Combined or Total Leverage.

In this lesson you will know much about the financial leverage. While operating leverage measures the change in the EBIT of a company to a particular change in output, the financial leverage measures the effect of the change in EBIT on the EPS of the company. Financial leverage also refers to the mix of debt and

equity in the capital structure of the company. The measure of financial leverage is the degree of financial leverage (DFL) and it can be calculated as under: -

$$DFL = (\text{Percentage change in EPS}) / (\text{Percentage change in EBIT})$$

$$DFL = (D \text{ EPS} / \text{EPS}) / (D \text{ EBIT} / \text{EBIT})$$

Substituting the equation with EPS, you will get,

$$DFL = \frac{\text{EBIT}}{\text{EBIT} - I - D_p} (1 - T)$$

Application and Utility of the Financial Leverage

Financial leverage when measured for various levels of EBIT will aid in understanding the behaviour of DFL and also explain its utility in financial decision-making. You will understand better with the following example: -

EBIT (Rs.)	DFL
50,000	- 0.40
1,00,000	- 1.33
1,75,000	∞
6,00,000	1.41
7,00,000	1.33
7,50,000	1.30

The DFL at EBIT level of Rs.1, 75,000 is undefined and this point is the financial break-even point. It can be defined as

$$\text{EBIT} = I + D_p / (1 - T)$$

You can also make following observation by studying the behaviour of DFL

Each level of EBIT has a distinct DFL

- DFL is undefined at the financial break-even point
- DFL will be negative when EBIT level goes below the financial break-even point.
- DFL will be positive for all values of EBIT that are above the financial break-even point. This will however start to decline as EBIT increases and will reach a limit of I.

By assessing the DFL, you can understand the impact of change in EBIT on the EPS of the company. In addition to this it also helps in assessing the financial risk of the firm.

Impact of Financial Leverage on Investor's Rate of Return

You can see with the help of a simple example, how financial leverage affects return on equity. A company needs a capital of Rs.10,000/- to operate. The shareholders of the company may bring in this money. Alternatively, a part of this money may also be brought in through debt financing. If the management raises Rs.10,000/- from shareholders, the company is not financially leverages and would have the following balance sheet.

Liabilities	Rs.	Assets	Rs.
Equity Capital	10,000	Cash	10,000

The company commences operations, which leads to the preparation of the following simplified version of its income statement.

Particulars	Rs.
Sales	10,000
Expenses	7,000
EBIT	3,000
Tax @ 50%	1,500
Net Profit	1,500

What is the return the company has earned on the owner's investment? You will see that the return on equity is 15%. The net profit of Rs.1,500/- may be paid fully or partly to the shareholders as dividends or may be retained to finance future activities of the company. Either way the return on equity is 15%.

What happens to the owner's rate of return if the management decides to finance a part of the required total investment of Rs.10,000/- through debt financing? The answer to this question depends upon

- The proportion of total investment which the management decides to finance through debt (Debt Equity Ratio the firm aspires to) and
- The interest rate on borrowed funds.

If the management has decided on a Debt Equity Ratio of 2 : 1, total borrowings will be amounting to Rs.10,000 X 2 = Rs.6,667/-. Assuming that the company is able to raise this amount at an interest of 15%, the company's balance sheet will appear as follows:

Liabilities	Rs.	Assets	Rs.
Equity capital	3,333	Cash	10,000
Debt Capital	6,667		
	10,000		10,000

The company now has an added financial burden of payment of interest on the amount it has borrowed. The income statement will now be as follows:

Particulars	Rs.
Sales	10,000
Expenses	7,000
EBIT	3,000
Interest charges	1,000
Profit before Tax (PBT)	2,000
Tax @ 50%	1,000
Net profit	1,000

The use of debt in the company's capital structure has caused the net profit to decline from Rs.1,500/- to Rs.1,000/-. But has the return on owner's capital decline? Return on equity now works out to 30%, as the owners have invested only Rs.3,333/- now which earned them Rs.1,000. What were the factors, which contributed to this additional return?

Though the company has to pay interest at 15% on the borrowed capital, the company's operations have been able to generate more than 15%, which is being transferred to the owners.

The reduction in PBT has brought about a reduction in the amount of tax paid, as interest is a tax deductible expense, to the extent of interest (1 – tax rate) i.e., Rs.500/-. The greater the tax rate, the more is the tax shield available to a company, which is financially leveraged.

As you have seen in the above example, a company may increase the return on equity by the use of debt i.e., the use of financial leverage. By increasing the proportion of debt in the pattern of financing, by increasing the debt –equity ratio, the company should be able to increase the return on equity.

Financial Leverage and Risk

If increased financial leverage leads to increased return on equity, why do companies not resort to ever increasing amounts of debt financing? Why do financial and other term lending institutions insist on norms for debt-equity ratio? The answer is that as the company becomes more financially leveraged, it becomes riskier, i.e., increased use of debt financing will lead to increased financial risk which leads to

- Increased fluctuations in the return on equity.
- Increase in the interest rate on debts.

Financial Leverage – Effect on the Shareholders' Return

The primary motive of a company in using financial leverage is to magnify the shareholders return under economic conditions. The role of financial leverage in magnifying the return of the shareholders is based on the assumptions that the fixed charges can be obtained at a cost lower than the firm's rate of return on net assets. Thus, when the difference between the earnings generated by assets financed by the fixed-charges funds and costs of these funds is distributed to the shareholders.. The EPS or ROE increases. However, EPS or ROE will fall if the company obtains the fixed-charges funds at a cost higher than the rate of return on the firm's assets. It should, therefore, be clear that EPS, ROE and ROI are the important figure for analyzing the impact of financial leverage.

EBIT-EPS Analysis

Normally, you assume EBIT as constant; however, EBIT for any firm is subjected to various influences. For example, because of the fluctuations in the economic conditions, sales of a firm change and as a result, EBIT also varies. In a given period, the actual EBIT of the firm may be more or less than the anticipated. It is therefore, useful to analyze the impact of the financial leverage on EPS (and (ROE) for possible fluctuations in EBIT.

Financial Plan I does not employ any leverages. As EBIT increases, EPS also increases. In fact, EPS increases with improved EBIT under all financial plans. What is important to note is that as the financial leverage is increased, EPS is further magnified.

Financial leverage works both ways. It accelerates EPS (and ROE) under favourable economic conditions, but depresses

EPS (and ROE) when the goings are not good for the firm. It can be seen that EPS or ROE is lower with debt in the capital structure in the poor years. With no leverage plan, if the firm's return on assets is positive, although low, the shareholders do obtain positive EPS or ROE.

But it becomes lower with more debt used, and even turns negative under very high leverage plan, such as Financial Plan IV. The unfavourable effect on EPS (and ROE) is more severe with more debt in the capital structure when EBIT (or r) is negative. The reasons again lie in the relationship between the return on assets and the cost of debt. If the cost of debt is more than the return on assets, EPS (or ROE) would depress with more leverage.

For any given level of variability in EBIT (or r) the increased financial leverage increases the degree of variability in EPS (or ROE). The indiscriminate use of financial leverage without taking into account the uncertainty surrounding EBIT (or r) can lead a firm into financial difficulties. More about the risk of the financial leverage

EBIT-EPS chart. One convenient and useful way of showing the relationship between EBIT and EPS for the alternative financial plans is to prepare till EBIT-EPS chart. The chart is easy to prepare since, for any given level of financial leverage, EPS is linearly related to EBIT. As noted earlier, the formula for calculating EPS is:

$$\begin{aligned} \text{EPS} &= \frac{(\text{EBIT} - \text{INT}) (1 - T)}{N} \\ &= \frac{(1 - T)}{N} [\text{EBIT} - \text{INT}] \end{aligned}$$

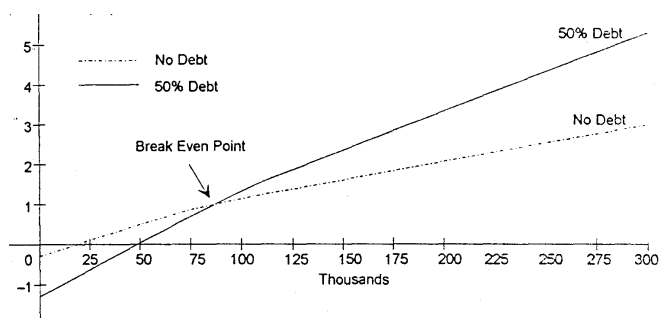


Fig. 1. EBIT-EPS Analysis

In figure 1, EBIT is shown on a horizontal line and EPS on a vertical line. If we identify any two points of EPS for two given levels of EBIT and join them in a straight line, we obtain EPS-line for a particular financial plan. The EPS-line can be similarly drawn for 100 per cent equity plan. We can see from Figure 1 that EPS-lines for all-equity and 50 per cent debt plans intersect at EBIT Rs 75,000. EPS under both plans are same at this level of EBIT. Figure 2, drawn similarly as Figure 1, shows four financial plans together.

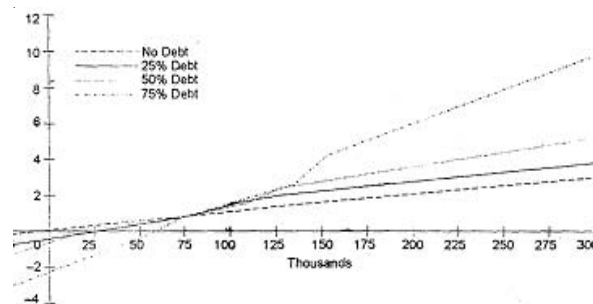


Fig. 2. EBIT-EPS Analysis

The steep solid lines in figure 2 show the relation between EBIT and EPS. The line relating EBIT and EPS becomes steeper with more debt in the capital structure, and steeper the line, the more the profit potential to the shareholders with increasing EBIT. If EBIT is declining, the loss to the shareholders will be magnified. The point of intersection of four EBIT-EPS lines in Figure 1 indicates the “indifference point” (also called the break-even point) at which EPS is same regardless of the level of the financial leverage. It can also be noticed from Figure 2 that below the “indifference point” the EBIT-EPS lines shift more towards the right when the level of financial leverage increases, indicating unfavourable effect because of a low rate of return on assets than the interest cost. The EBIT-EPS line shift more towards the left beyond the “indifference point” when the level of the, financial leverage increases.

This indicates favourable effect of leverage because the return on assets exceeds the interest rate. Thus you can conclude that the shareholders will benefit by the corporate financial leverage in terms of the increased EPS if return on assets is higher than the interest cost, and will have reduced EPS if return assets is lower than the interest cost; the shareholders' earnings will not be affected by the level of leverage if return on assets is just equal to the interest cost.

Points to Ponder

- Meaning of financial leverage.
- Calculation of Financial Leverage.
- Effects on the return for the investors and other stakeholders
- EBIT-EPS analysis used in the financial leverage.

Review Questions

1. What do you understand by the term Leverage?
2. What is Financial Leverage?
3. How financial leverage effects the return for the investors and stakeholders?
4. How do you measure the Financial Leverage?
5. Differentiate between Operating Leverage and Financial Leverage?
6. How Financial Leverage effects the financial decisions in any investment opportunity?
7. Explain EBIT-EPS analysis?

LESSON 31: DIVIDEND POLICY

Learning Objectives

- To know about why the corporate has to think of dividend policy.
- To understand various factors influencing dividend policy.
- To learn about the constraints of paying dividend.
- To know about the forms of stability of dividend.

Introduction

After understanding the various aspects of financial decision in respect of financial structure and cost of capital you have realized how that the shareholders who invest their funds as owners will always keep a watch on you, in fulfilling their aspirations of higher dividend and enhancement of share prices for the stock held by them. So automatically, you have certain questions raised in your mind as under: -

1. Do dividends affect the value of a share?
2. What does dividend decision involve?
3. Is It a Long-Term Financing Decision?
4. Is It a Wealth Maximization Decision?

Do Dividends Affect the Value of Share?

The theoretical views differ on this issue. On the one extreme, we have the view that dividends affect the value of the share. On the other extreme is the view that dividends are bad as they result into the payment of higher taxes (because of the difference in the ordinary income and capital gains tax rates), and thus, reduce the shareholders' wealth. We also have a moderate view, which asserts that because of the information and communication value of dividends, some dividends should be paid as it may have favourable effect on the value of the share.

How do companies set their dividend policies in practice? What factors do they consider in setting their policies? In this lesson you will exploring for getting answers to these questions.

What does Dividend Decision Involve?

As discussed in the previous chapter, a firm's dividend policy has the effect of dividing its net earnings into two parts: retained earnings and dividends. The retained earnings provide funds to finance the firm's long-term growth. It is one of the most significant sources of financing a firm's investments. Dividends are generally paid in cash. Thus, the distribution of earnings uses the available cash of the firm. A firm which intends to pay dividends and also needs funds to finance its investment opportunities will have to use external sources of financing, such as the issue of debt or new common shares. Dividend policy of the firm, thus, has its effect on both the long-term financing and the wealth of shareholders. As a result, the firm's decision to pay dividends may be shaped by the following two possible viewpoints.

Is It a Long-Term Financing Decision?

When dividend decision is treated as a financing decision, the

net earnings of the firm may be considered as a source of long-term funds. With this approach, dividends will be paid only when the firm does not have profitable investment opportunities. The firm grows at a faster rate when it accepts highly profitable investment projects. External equity could be raised to finance investments. But the retained earnings are preferable because, unlike external equity, they do not involve any floatation costs. The distribution of cash dividends causes a reduction in internal funds available to finance profitable investment opportunities and thus, either constrains growth or requires the firm to find other costly sources of financing. Thus, earnings may remain undistributed as part of a long-term financing decision. The dividends paid to shareholders represent a distribution of earnings that cannot be profitably reinvested by the firm. With this approach, dividend decision is viewed merely as a residual decision.

Is It a Wealth Maximization Decision?

One may argue that capital markets are not perfect; therefore, shareholders are not indifferent between dividends and retained earnings. Because of the market imperfections and uncertainty, shareholders may give a higher value to the near dividends than the future dividends and capital gains. Thus, the payment of dividends may significantly affect the market price of the share. Higher dividends increase the value of the shares and low dividends reduce the value. In order to maximize wealth under uncertainty, the firm must pay enough dividends to satisfy investors.

The management of a firm, while evolving a dividend policy, must strike a proper balance between the above-mentioned two approaches. When the firm increases the retained portion of the net earnings, shareholders' dividends decrease and consequently the market price may be adversely affected. But the use of retained earnings to finance profitable investments will increase the future earnings per share. On the other hand, when dividends are increased, though there may be a favourable reaction in the stock markets, but the firm may have to forego some investment opportunities for want of funds and consequently, the future earnings per share may decrease. Therefore, management should develop such a dividend policy, which divides the net earnings into dividends and retained earnings in an optimum way to achieve the objective of maximizing the wealth of shareholders. The development of such policy will be greatly influenced by investment opportunities available to the firm and the value of dividends as against capital gains to the shareholders. The other possible aspects of the dividend policy relate to the stability of dividends, the constraints on paying dividends and the forms of dividends.

Dividend Decision: Practical Considerations

The view that dividends are irrelevant is not correct, once we

modify the assumptions underlying this view to consider the realities of the world. In practice, every firm follows some kind of dividend policy. The typical dividend policy of most firms is to retain between one-third to half of the net earnings and distribute the remaining amount to shareholders. Companies generally specify dividends in terms of a dividend rate. They also tend to increase dividend rate if their profits increase substantially.

The questions, which relate to the dividend policy of a firm, are:

1. What are the preferences of shareholders? Do they want dividend income or capital gains?
2. What are the financial needs of the company?
3. How much should be paid out as dividends? What are the constraints on paying dividends?
4. Should the company follow a stable dividend policy?
5. What should be the form of dividends?

It is not easy to answer these questions. A number of factors will have to be evaluated to analyze each of these questions to evolve a long-term dividend policy for the firm. Broadly speaking, to develop a long-term dividend policy, the directors should aim at bringing a balance between the desires of shareholders and the needs of the company.

The factors, which generally influence the dividend policy of the firm, are as discussed below:

Shareholders' Expectations

Legally directors have considerable liberty to use their discretion to dispose of the earnings of the company. But shareholders are the legal owners of the company, and directors, appointed by them, are their representatives. Therefore, directors should give due importance to the expectations of shareholders in the matter of dividend decision. Shareholders' preference for dividends or capital gains depends on their economic status and the effect of tax differential on dividends and capital gains. A wealthy shareholder in a high income-tax bracket may be interested in capital gains as against current dividends. On the other hand, a retired person with small means, whose main source of income is dividend, would like to get regular dividend and may not be interested in capital gains.

Closely-held Companies

In case of a closely held company, management usually knows the expectations of shareholders. Therefore, they can easily adopt a dividend policy, which satisfies all shareholders. The body of the shareholders is a small, homogeneous group in a closely-held company. If most of the shareholders are in high tax brackets and have a preference for capital gains to current dividend incomes, the company can establish a dividend policy of paying less or no dividends and retaining the earnings within the company.

Widely-held Company

It is a formidable task to ascertain the preferences of shareholders in a wide-held company. The number of shareholders is very large and they may have diverse desire regarding dividends and capital gains. As a result, it is not possible in case of widely held company to follow a dividend policy, which equally satisfies all shareholders. The firm can follow a dividend policy, which

serves the purpose of the dominating group, but does not completely neglect the desires of others. Shareholders of a widely held company may be divided, for example, into four groups: small, retired, wealthy and institutional shareholders?

Small Shareholders

Small shareholders are not the frequent purchasers of the shares. They hold a small number of shares in a few companies. They generally purchase or sell shares on the advice of stockbrokers or friends. Their purpose sometimes is to receive dividend income, and sometimes, if possible, they may like to make capital gains. Thus, the small shareholders do not have a definite investment policy. They purchase shares only when their savings permit. This group rarely proves to be dominating in the body of shareholders in a company and it is not much concerned with the dividend policy of the company.

Retired and Old Persons

Retired and old persons generally invest in shares to get a regular income. They use their savings or pension funds to purchase shares. Therefore, these persons may select shares of the companies which have a history of paying regular and liberal dividends. However, a retired person who has some source of income and is in a high tax bracket may be interested in capital gains.

Wealthy Investors

Wealthy investors are very much concerned with the dividend policy followed by a company. They have a definite investment policy of increasing their wealth and minimizing taxes. These persons are in high tax brackets and the dividends received in cash by them would be taxed at a high rate. Therefore, they generally prefer a dividend policy of retaining earnings and distributing bonus shares. The wealthy shareholders' group is quite dominating in many companies as they hold relatively large blocks of shares and are able to influence the composition of the board of directors by their majority voting rights. On the dividend policies of these companies, this group will have a considerable influence.

Institutional Investors

Institutional investors purchase large blocks of shares to hold them for relatively long periods of time. Institutional investors, unlike wealthy shareholders, are not concerned with personal income taxes but with profitable investments. Most institutional investors avoid speculative issues, seek diversification in their investment portfolio and favour a policy of regular cash dividend payments.

It should be obvious from the above discussion that, in the case of a widely-held company, the interest of the various shareholders' groups are in conflict, it is not easy to reconcile these conflicting interests of the different types of shareholders. However, directors should consider following two points:

First, directors should adopt such a dividend policy, which gives some consideration to the interests of each of the groups comprising a substantial proportion of shareholders.

Second, the dividend policy, once established, should be continued as long as it does not interfere with the financing needs of the company.

A definite dividend policy, followed for a long period in the past, tends to create the clientele effect. That is, it attracts those investors who consider the dividend policy in accord with their investment requirements. If the company suddenly changes its dividend policy, it may work to the detriment of these shareholders, as they may have to switch to other companies to fulfill their investment needs. Thus, an established dividend policy should be changed only after having analyzed its probable effects on the existing share-holders. It should be changed slowly and not abruptly.

Financial Needs of the Company

The financial needs of the company may be in conflict with the desires of shareholders. Managerial prudence requires giving more weightage to the financial needs, of the company. However, retained earnings should be used as a source of financing only when the company has profitable investment opportunities. If shareholders themselves have better investment opportunities, the earnings should be distributed to them so that they may be able to maximize their wealth. Generally speaking, when the company has an internal rate of return greater than the return required by shareholders, it would be to the advantage of shareholders to reinvest earnings. When the company does not have highly profitable opportunities and earns a rate on investments, which is lower than the rate required by shareholders, it is not proper to retain earnings. It is sometimes argued that, even if the company has highly profitable investment opportunities, earnings should be distributed and funds should be raised externally to finance the investment. But the companies in practice prefer to retain earnings because issuing new share capital is inconvenient as well as involves flotation costs. If the company raises debt, the financial obligations and risk will increase. As a matter of fact, directors should neither follow a practice of paying 100 per cent dividends, nor a practice of retaining 100 per cent earnings. The company should have a minimum payout ratio. If the profitable investment opportunities do not exist, it can pay some "extra" dividend, but still retaining some earnings for the continued existence of the enterprise. Though shareholders are the owners of the company and directors should follow a policy desired by them, yet the directors cannot sacrifice the interests of other groups, such as debt-holders, employees, society and customers. Shareholders are the residual claimants to the earnings of the company. Directors must retain some earnings, whether or not profitable investment opportunities exist, to maintain the company as a sound and solvent enterprise. Only a sound company will be able to meet its debt commitments, give maximum facilities and good salaries to its employees, manufacture and supply better quality products for its customers and contribute towards the social overheads by way of taxes and donations.

Thus, depending on the needs to finance their investment opportunities, companies may follow different dividend policies. Mature companies that have few investment opportunities may generally have high payout ratios. Shareholders of such companies would be more interested in dividends, as they obtain return on their investments, than the company. The share prices of such companies are very sensitive

to dividend changes. The directors of these companies retain only a small portion of the earnings to meet emergent financial needs and to finance the occasional investment opportunities and distribute the rest. On the other hand, growth companies may have low payout ratios. They are continuously in need of funds to finance their fast growing fixed assets. The distribution of earnings will reduce the funds of the company. Therefore, sometimes the growth companies retain most of its earnings and declare bonus shares to satisfy the dividend requirements of shareholders. These companies would slowly increase the amount of dividends as the profitable investment opportunities start falling.

Constraints on Paying Dividends

Most companies recognize that the shareholders have a desire to receive dividends, although some of the shareholders are also interested in the capital gains. But the company's decision regarding the amount of earnings to be distributed as dividends depends on a number of factors. The factors, which restrict the firm's ability to declare and pay dividends, are below:

- Legal restrictions
- Liquidity.
- Financial condition.
- Access to the capital market.
- Restriction in loan agreements.
- Control.
- Investment opportunities.
- Inflation.

Legal Restrictions

The dividend policy of the firm has to be evolved within the legal framework and restrictions. The directors are not legally compelled to declare dividends. For example, the Companies Act provides that dividend shall be declared or paid only out of the current profits or past profits after providing for depreciation. However, the Central Government is empowered to allow any company to pay dividend for any financial year out of the profits of the company without providing for depreciation. The Central Government shall give such relief only when it is in the public interest. The dividend should be paid in cash, but a company is not prohibited to capitalize profits or reserves (retained earnings) for the purpose of issuing fully paid bonus shares (stock dividend). It has been held in some legal cases (for example, *Lubbock v. The British Bank of South America*) that capital profits should not be distributed as dividends unless the distribution is permitted by the company's Articles of Association and the profits have been actually realized.

The legal rules act as boundaries within which a company can operate in terms of paying dividends. Acting within these boundaries a company will have to consider many financial variables and constraints in deciding the amount of earnings to be distributed as dividends.

Liquidity

The payment of dividends means cash outflow. Although a firm may have adequate earnings to declare dividend, it may not have sufficient cash to pay the dividends. Thus, the cash position of the firm is an important consideration in paying

dividends; the greater the cash position and overall liquidity of a company, the greater its ability to pay dividends. A mature company is generally liquid and is able to pay large amount of dividends. Such a company does not have much investment opportunities, nor are the funds tied up in permanent working capital and, therefore, has a sound cash position. On the other hand, a growing firm faces the problem of liquidity. Even though it makes good profits it needs funds for its expanding activities and permanent working capital. Because of the insufficient cash or pressures on liquidity in case of a growth firm, management may not be able to declare dividends.

Financial Condition

The financial condition or capability of a firm depends on its use of borrowings and interest charges payable. A high degree of financial leverage makes a company quite vulnerable to changes in earnings and also, it becomes quite difficult to raise funds externally for financing its growth. A highly levered firm is, therefore, expected to retain more to strengthen its equity base. However, a company with steadily growing earnings and cash flows may follow a high dividend payment policy in spite of high amount of debt in its capital structure.

Access to the Capital Market

A company, if not sufficiently liquid, can pay dividends if it is able to raise debt or equity in the capital markets. If it is well established and has a record of profitability it will not find much difficulty in raising funds in the capital markets. Easy accessibility to the capital markets provides flexibility to the management in paying dividends as well as in meeting the corporate obligations. Thus, at last growing firm, which has a tight liquidity position, will not face any difficulty in paying dividends if it has access to the capital markets. A company which does not have sound cash position and is also unable to raise funds, will not be able to pay dividends. Thus, greater the ability of the firm to raise funds in the capital markets, the greater will be its ability to pay dividends even if it is not liquid.

Restriction in Loan Agreements

Lenders may generally put restrictions on dividend payments to protect their interests when the firm is experiencing liquidity or profitability difficulties. As such the firm agrees as part of a contract with a lender to restrict dividend payments. As an example, a loan agreement may prohibit payment of dividends as long as the firm's debt-equity ratio is in excess of 1.5 : 1 or when the liquidity ratio is less than 2 : 1 or may require the firm to pay dividends only when some amount of current earnings has been transferred to a sinking fund established to retire debt. These are some of the examples of the restrictions put by lenders on the payment of dividends. When these restrictions are put, the company is forced to retain earnings and have a low payout.

Control

The objective of maintaining control over the company by the existing management group or the body of shareholders can be an important variable in influencing the company's dividend policy. When a company pays large dividends, its cash position is affected. As a result, the company will have to issue new shares to raise funds to finance its investment programmes.

The control of the existing shareholders will be diluted if they do not want or cannot buy additional shares. Under this circumstance, the payment of dividends may be withheld and earnings may be retained to finance the firm's investment opportunities.

Investment Opportunities

We have discussed earlier that a dividend policy is greatly influenced by the financial needs of the company. A growing firm gives precedence to the retention of earnings over the payment of dividends in order to finance its expanding activities. When investment opportunities occur infrequently, the company may not be justified in retaining the earnings at least during those periods when such opportunities do not exist. If the company retains earnings during such periods, the retained funds would either be reinvested in short-term securities yielding nominal returns or remain idle. This will have an impact of reducing the wealth of shareholders. The better course, when the investment opportunities arise infrequently, is to follow policy of paying dividends and raise external funds when the investment opportunities occur.

Inflation

In an indirect way inflation can act as a constraint on paying dividends. Our accounting system is based on historical costs. Depreciation is charged on the basis of original costs at which assets were acquired. As a result, when prices rise, funds saved on account of depreciation would not be adequate to replace assets or to maintain the capital intact. Consequently, to maintain the capital intact and preserve the earnings power of the firm earnings would be retained.

Stability of Dividends

Stability or regularity of dividends is considered as, a desirable policy by the management of most companies. Shareholders, also generally favour this policy and value stable dividends higher than the fluctuating ones. All other things being the same, stable dividend may have a positive impact on the market price of the share.

Stability of dividends sometimes means regularity in paying some dividend annually, even though the amount of dividend may fluctuate from year and may not be related with earnings. There are a number of companies which have records of paying dividend for a long unbroken period. More precisely, stability of dividends refers to the amounts paid out regularly.

Three distinct forms of such stability may be distinguished:

- Constant dividend per share or dividend rate
- Constant payout
- Constant dividend per share plus extra dividend

Constant Dividend Per Share or Dividend Rate

A number of companies follow the policy of paying a fixed amount per share or fixed rate on paid-up capital as dividend every year. Irrespective of the fluctuations in the earnings, this policy does not imply that the dividend per share or dividend rate will never be increased. When the company reaches new levels of earnings and expects to maintain it, the annual dividend per share may be increased.

It is easy to follow this policy when earnings are stable. If the earnings pattern of a company shows wide fluctuations, it is difficult to maintain such a policy. With earnings fluctuating from year to year, it is essential for a company, which wants to follow this policy to build up surpluses in years of higher than average earnings to maintain dividends in years of below average earnings. In practice, when a company retains earnings in good years for this purpose, it earmarks this surplus as reserve for dividend equalization. These funds are invested in current assets like marketable securities, so that they may easily be converted into cash at the time of paying dividends in bad years.

The dividend policy of paying a constant amount of dividend per year treats ordinary shareholders somewhat like preference shareholders without taking into account the firm's or shareholders' investment opportunities. Those investors who have dividends as the only source of their income prefer the constant dividend policy. They are hardly concerned about the changes in share prices. In the long run, such behavior helps to stabilize the market price of the share.

Constant Payout

The ratio of dividend to earnings is known as payout ratio. Some companies may follow a policy of constant payout ratio, i.e., paying a fixed percentage of net earnings every year. With this policy the amount of dividend will fluctuate in direct proportion to earnings. If a company adopts a 40 per cent payout ratio, then 40 per cent of every rupee of net earnings will be paid out. For example, if the company earns Rs. 2 per share, the dividend per share will be Re 0.80 and if it earns Rs. 1.50 per share the dividend per share will be Re 0.60.

This policy is related to a company's ability to pay dividends. If the company incurs losses no dividends shall be paid regardless of the desires of shareholders. Internal financing with retained earnings is automatic when this policy is followed. At any given payout ratio, the amount of dividends and the additions to retained earnings increase with increasing earnings and decrease with decreasing earnings. This policy simplifies the dividend decision, and has the advantage of protecting a company against over or under payment of dividend. It ensures that dividends are paid when profits are earned, and avoided when it incurs losses.

Small Constant Dividend Per Share Plus Extra Dividend

Under the constant dividend per share policy; the amount of dividend is set at a high level, and the companies with stable earnings usually adopt this policy. For companies with fluctuating earnings, the policy to pay a minimum dividend per share with a step-up feature is desirable. The small amount of dividend is fixed to reduce the possibility of ever missing a dividend payment. By paying extra dividend (a number of companies pay an interim dividend followed by a regular final dividend) in periods of prosperity, an attempt is made to prevent investors from expecting that the dividend represents an increase in the established dividend amount. This type of a policy enables a company to pay constant amount of dividend regularly without a default and allows a great deal of flexibility for supplementing the income of shareholders only when the company's earnings are higher than the usual, without committing

itself to make larger payments as a part of the future fixed dividend. Certain shareholders like this policy because of the certain cash flow in the form of regular dividend and the option of earning extra dividend occasionally.

We have discussed three forms of stability of dividends. Generally when we refer to a stable dividend policy, we refer to the first form of paying constant dividend per share. A firm pursuing a policy of stable dividend may command a higher price for its shares than a firm, which varies dividend amount with cyclical fluctuations in the earnings.

Significance of Stability of Dividends

The stability of dividends has various advantages from the point of view of shareholders as well as company. They are:

- Resolution of investors' uncertainty.
- Investors' desire for current income.
- Institutional investors' requirements.
- Raising additional finances.

Resolution of Investors' Uncertainty

We have argued in the previous chapter that dividend has informational value and resolves uncertainty in the minds of investors. When a company follows a policy of stable dividends, it will not change the amount of dividend if there are temporary changes in its earnings. Thus, when the earnings of a company fall and it continues to pay same amount of dividend as in the past, it conveys to investors that the future of the company is brighter than suggested by the drop in earnings. Similarly, the amount of dividend is increased with increased earnings level only when it is possible to maintain it in future. On the other hand, if a company follows a policy of changing dividend with cyclical changes in the earnings, shareholders would not be certain about the amount of dividend. This may lead them to require a higher discount factor to be applied to the company than if a stable dividend policy were followed.

Investors' Desire for Current Income

There are many investors, such as old and retired persons, women, children etc., who desire to receive regular periodic income. They invest their savings in the shares with a view to use dividends as a source of income to meet their living expenses. These investors, who desire to receive a regular dividend income, will prefer a company with stable dividends to one with fluctuating dividends.

Institutional Investors' Requirements

Shares of the companies are not only purchased by individuals but also by financial, educational and social institutions and unit trusts. In India financial institutions, such as IFCI, IDBI, LIC, and UTI are some of the largest investors in corporate securities. Every company is interested to have these financial institutions in the list of their investors. These institutions generally invest in the shares of those companies, which have a record of paying regular dividends. These institutional investors may not prefer a company, which has a history of adopting an erratic dividend policy. Thus, to cater to the requirement of institutional investors, a company prefers to follow a stable dividend policy.

Raising Additional Finances

A stable dividend policy is also advantageous to the company in its efforts to raise external finances. Stable and regular dividend policy tends to make the share of a company as quality investment rather than a speculation. Investors purchasing these shares intend to hold them for long periods of time. The loyalty and goodwill of shareholders towards a company increases with stable dividend policy. They would be more receptive to an offer by the company for further issues of shares. A history of stable dividends serves to spread ownership of outstanding shares more widely among small investors, and thereby reduces the chance of loss of control. The persons with small means, in the hope of supplementing their income, usually, purchase shares of the companies with a history of paying regular dividends. A stable dividend policy also helps the sale of debentures and preference shares. The fact that the company has been paying dividend regularly in the past is sufficient assurance to the purchasers of these securities that no default will be made by the company in paying their interest or preference dividend and returning the principal sum. The financial institutions are the largest purchasers of these securities. They purchase debentures and preference shares of those companies, which have a history of paying stable dividends.

Danger of Stability of Dividends

The greatest danger in adopting a stable dividend policy is that once it is established, it cannot be changed without seriously affecting investors' attitude and the financial standing of the company. If a company, with a pattern of stable dividends, misses dividend payment in a year, this break will have an effect on investors more severe than the failure to pay dividend by a company with unstable dividend policy. The companies with stable dividend policy create a 'clientele', which depends on dividend income to meet their living and operating expenses. A cut in dividend is considered as a cut in 'salary.' Because of the serious depressing effect on investors due to a dividend cut, directors have to maintain stability of dividends during lean years even though financial prudence would indicate elimination of dividends or a cut in it. Consequently, to be on the safe side, the dividend rate should be fixed at a conservative figure so that it may be possible to maintain it even in lean periods of several years. To give the benefit of the company's prosperity, extra or interim dividend, can be declared. When a company fails to pay extra dividend, it does not have as depressing an effect on investors as the failure to pay a regular dividend.

Points to Ponder

- Develop a dividend policy,
- Forms of stability of dividend policy
- Danger of stable dividend policy.
- Constraints on Paying Dividends
- The factors restricting the firm's ability to declare and pay dividends.

Review Questions

1. How do dividends affect the value of a share?
2. State the various advantages of stability of dividends?
3. Why stable dividend policy is dangerous?

4. List the various constraints on Paying Dividends?
5. What are the preferences of shareholders? Do they want dividend income or capital gains?
6. What are the financial needs of the company?
7. How much should be paid out as dividends? What are the constraints on paying dividends?
8. Should the company follow a stable dividend policy?
9. What should be the form of dividends?

Note

[illegible]

LESSON 32 :

FORMS OF DIVIDEND

Learning Objectives

- To know various forms of dividend.
- To understand the advantages of bonus shares to the investor and company.
- To know, when share split and reverse split is effected.

Introduction

In the earlier lessons you have learnt about the various financial decisions to be made by you as a financial manager in respect of financial structure and the cost of capital. In this lesson you will understand that how dividend decisions will effect the financial decision. Various forms of dividends are necessarily known by you to take proper dividend decision which will effect your financial decision at large.

Forms of Dividends

The usual practice is to pay dividends in cash. Other option is payment of the bonus shares or stock dividend. In this section we shall also discuss the stock split. The stock split is not a form of dividend; but its effects are similar to the effects of the bonus shares.

Cash Dividend

Most companies pay dividends in cash. Sometimes cash dividend may be supplemented by a bonus issue (stock dividend). A company should have enough cash in its bank account when cash dividends are declared. If the company does not have enough bank balance at the time of paying cash dividend, arrangement should be made to borrow funds. When the company follows a stable dividend policy, it should prepare a cash budget for the coming period to indicate the necessary funds, which would be needed to meet the regular dividend payments of the company. It is relatively difficult to make cash planning in anticipation of dividend needs when an unstable policy is followed.

The cash account and the reserves account of a company will be reduced when the cash dividend is paid. Thus, both the total assets and the net worth of the company are reduced when the cash dividend is distributed. The market price of the share drops in most cases by the amount of the cash dividend distributed.

Bonus Shares

An issue of bonus share represents a distribution of shares in addition to the cash dividend (known as stock dividend in the U.S.A.) to the existing shareholders. This has the effect of increasing the number of outstanding shares of the company. The shares are distributed proportionately. Thus, a shareholder retains his proportionate ownership of the company. For example, if a shareholder owns 100 shares at the time when a 10 per cent (i.e., 1:10) bonus issue is made, he will receive 10 additional shares. The declaration of the bonus shares will increase the paid-up share capital and reduce the reserves and

surplus of the company. The total net worth is not affected by the bonus issue. In fact, a bonus issue represents a recapitalization of the owners' equity portion, i.e., the reserves and surplus. It is merely an accounting transfer from reserves and surplus to paid-up capital.

Advantages of Bonus Shares

Prima facie the bonus shares do not affect the wealth of the shareholders. In practice, however, it carries certain advantages both to shareholders and the company.

Shareholders

The following are advantages of the bonus shares to Shareholders:

- Tax benefit.
- Indication of higher future profits.
- Future dividend may increase.
- Psychological value.

Tax Benefit

One of the advantages to shareholders in the receipt of bonus shares is the beneficial treatment of such dividends with regard to income taxes. When a shareholder receives cash dividend from company, this is included in his ordinary income and taxed at ordinary income tax rate. But the receipt of bonus shares by the shareholder is not taxable as income. Further, the shareholder can sell the new shares received by way of the bonus issue to satisfy his desire for income and pay capital gain taxes, which are usually less than the income taxes on the cash dividends. The shareholder could sell a few shares of his original holding to derive capital gains. But selling the original shares are considered as a sale of principal by some shareholders. They do not mind selling the shares received by way of the bonus shares as they consider it a windfall gain and not a part of the principal.

Indication of Higher Future Profits

Shareholders normally interpret the issue of bonus shares as an indication of higher profitability. When the profits of a company do not rise, and it declares a bonus issue, the company will experience a dilution of earnings as a result of the additional shares outstanding. Since a dilution of earnings is not desirable, directors usually declare bonus shares only when they expect rise in earnings to offset the additional outstanding shares. The bonus shares, thus, may convey some information, which may have a favourable impact on value of the shares. But it should be noticed that the impact on value is that of the growth expectation and not the bonus shares, which simply conveys the information.

Future Dividend may Increase

If a company has been following a policy of paying a fixed amount of dividend per share and continues it after the

declaration of the bonus issue, the total cash dividends of the shareholders will increase in future. For example, a company may be paying a Re 1 per share dividend and pays 1 : 10 bonus shares with the announcement that the cash dividend per share will remain unchanged. If a shareholder originally held 100 shares, he will receive additional 10 shares. His total cash dividend in future will be Rs 110 (Rs 1 x 110) instead of Rs 100 (Re 1 x 100) received in the past. The increase in the shareholders' cash dividend may have a favourable effect on the value of the share. It should be, however, realized that the bonus issue per se has no effect on the value of the share.

Psychological Value

The declaration of the bonus issue may have a favourable psychological effect on shareholders. The receipt of bonus shares gives them a chance to sell the shares to make capital gains without impairing their principal investment. They also associate it with the prosperity of the company. Because of these positive aspects of the bonus issue, it is usually received positively by the market. The sale of the shares, received by way of the bonus shares, by some shareholders widens the distribution of the company's shares. This tends to increase the market interest in the company's shares; thus supporting or raising its market price.

Company

The bonus share is also advantageous to the company. The advantages are:

- Conservation of cash.
- Only means to pay dividend under financial difficulty and contractual restrictions.
- More attractive share price.

Conservation of Cash

The declaration of a bonus issue allows the company to declare a dividend without using up cash that may be needed to finance the profitable investment opportunities within the company. The company is, thus, able to retain earnings and at the same time satisfy the desires of shareholders to receive dividend. We have stated earlier that directors of a company must consider the financial needs of the company and the desires of shareholders while making the dividend decision. These two objectives are often in conflict. The use of bonus issue represents a compromise, which enables directors to achieve both these objectives of a dividend policy. The company could retain earnings without declaring bonus shares issue. But the receipt of bonus shares satisfies shareholders psychologically. Also their total cash dividend can increase in future, when cash dividend per share remains the same.

Only means to pay dividend under financial difficulty and contractual restrictions

In some situations, even if the company's intention is not to retain earnings, the bonus issue is the only means to pay dividends and satisfy the desires of shareholders. When a company is facing a stringent cash situation, the only way to replace the cash dividend is the issue of bonus shares. The declaration of the bonus issue under such a situation should not convey a message of the company's profitability, but financial difficulty. The declaration of the bonus issue is also

necessitated when the restrictions to pay the cash dividend are put under loan agreements. Thus, under the situations of financial stringency or contractual constraining the bonus issue is meant to maintain the confidence of shareholders in the company.

More Attractive Share Price

Sometimes the intention of a company in issuing bonus shares is to reduce the market price of the share and make it more attractive to investors. If the market price of a company's share is very high, it may not appeal to small investors. If the price could be brought down to a desired range, the trading activity would increase. Therefore, the bonus issue is used as a means to keep the market price of the share within a desired trading range.

Limitations of Bonus Shares

Bonus shares are considered valuable by most shareholders. But they fail to realize that the bonus shares do not affect their wealth and therefore, in itself it has no value for them. The declaration of bonus shares is a method of capitalizing the past earnings of the shareholders. Thus, it is a formal way of recognizing something (earnings), which the shareholders already own. It merely divides the ownership of the company into a large number of share certificates. Bonus shares represent simply a division of corporate pie into a large number of pieces. In fact, the bonus issue does not give any extra or special benefit to a shareholder. His proportionate ownership in the company does not change. As discussed in the preceding section, the chief advantage of the bonus share issue is that it has a favourable psychological impact on shareholders. The issue of bonus shares gives an indication of the company's growth to shareholders. Therefore, they welcome the distribution of bonus shares.

The disadvantage of bonus issues from the company's point of view is that they are more costly to administer than cash dividend. The bonus issue can be disadvantageous if the company declares periodic small bonus shares. The investment analysts do not adjust the earnings per share for small issues of bonus shares. Only the significant issues of bonus shares are adjusted by them. When the earnings per share are not adjusted, the measured growth in the earnings per share will be less than the true growth based on the adjusted earnings per share. As a result, the price-earnings ratio would be distorted downwards.

Share Splits

A share split is a method to increase the number of outstanding shares through a proportional reduction in the par value of the share. A share split affects only the par value and the number of outstanding shares; the shareholders' total fund remains unaltered.

Bonus Share Vs. Share Split

As with the bonus share the total net worth does not change and the number of outstanding shares increases with the share split. However, with the share split, the number of outstanding shares increases substantially. The bonus issue and the share split are similar except for the difference in their accounting treatment. In case of the bonus shares, the balance of the

LESSON 33 : DIVIDEND MODELS & ITS DECISIONS

Learning Objectives

- To understand how to take dividend decisions.
- To know various dividend models giving different approaches to the dividend policy.

Introduction

Till now, you are familiar various dividend policy decisions as well as the various forms in which you can give the dividend and satisfy the shareholders as a financial manager. In this lesson you will understand how the dividend decisions were taken earlier and is there any models, which can be followed by you during the course of your tenure financial decision maker?

Dividend Decision

There are basically two options, which a firm has while utilizing its profits after tax. Firms can either plough back the earnings by retaining them or distribute the same to their shareholders. The first option suits those firms, which need funds to finance their long-term projects. However, such projects should have enough growth potential and sufficient profitability. On the other hand, the second option of declaring cash dividends from the profits after tax will lead to maximization of the shareholders wealth.

The returns to the shareholders either by way of the dividend receipts or capital gains are affected by the dividend policies of the firm. This is mainly due to the fact that the dividend policy decides the retention ratio and pay-out ratio (dividend as a percent of profits). Furthermore, the dividend policy of the firm gains importance especially due to unambiguous relationship that exists between the dividend policy and the equity returns. Thus, a firm's decision should meet the investors' expectations.

A few models, which studied this relationship and the dividend policies of firms, are given below:

- Traditional Position
- Walter Model
- Gordon Model
- Miller & Modigliani Position
- Rational Expectations Model

Traditional Position

The traditional approach to the dividend policy, which was given by B.Graham and D.L.Dodd lays a clear emphasis on the relationship between the dividends and the stock market. According to this approach, the stock value responds positively to higher dividends and negatively when there are low dividends.

The following expression, given by traditional approach, establishes the relationship between market price and dividends using a multiplier: -

$$P = m (D + E / 3) \dots(1)$$

Where,

P	=	Market price
m	=	Multiplier
D	=	Dividend per share
E	=	Earning per share

Limitations

The traditional approach, further states that the P / E ratios are directly related to the dividend payout ratio i.e., a high dividend payout ratio will increase the P / E ratio and vice-versa. However, this may not be true in all situations. A firm's share price may rise even in case of a low payout ratio, if the earnings are increasing. Here the capital gains for the investor will be higher than the cash dividends. Similarly for a firm having a high dividend payout ratio with a slow growth rate there will be a negative impact on the market price (because of lower earnings). In addition to this there may be a few investors of the company who would prefer the dividends to the uncertain capital gains and few who would prefer taxed capital gains. These conflicting factors that have not been properly explained from the major shortcomings of the dividend policy given by the traditional approach.

Walter Model

Similar to the traditional approach, the dividend policy given by James E Walter also considers that dividends are relevant and they do affect the share price. In this model he studied the relationship between the internal rate of return (r) and the cost of capital of the firm (k_e), to give a dividend policy that maximizes the shareholders' wealth.

The model studies the relevance of the dividend policy in three situations:

$$\begin{aligned} r &> k_e \\ r &< k_e \\ r &= k_e \end{aligned}$$

According to the Walter model, when the return on investment is more than the cost of equity capital, the firm can retain the earnings, since it has better and more profitable investment opportunities than the investors. It implies that the returns the investor gets when the company re-invests the earnings will be greater than what they earn by investing the dividend income. Firms which have their $r > k_e$ are the growth firms and the dividend policy that suits is the one which has a zero pay-out ratio. This policy will enhance the value of the firm.

In the second case, the return on investment is less than the cost of equity capital and in such situation the investor will have a better investment opportunity than the firm. This suggests a dividend policy of 100% payout. This policy of a full pay-out ratio will maximize the value of the firm.

Finally, when the firm has a rate of return that is equal to the cost of equity capital, the firms' dividend policy will not affect

the value of the firm. The optimum dividend policy for such firm will range between zero to a 100% pay-out ratio, since the value of the firm will remain constant in all the cases.

Assumptions

The relevance of the dividend policy as explained by the Walter's Model is based on a few assumptions, which are as follows: -

Retained earnings are the only source of finance available to the firm, with no outside debt or additional equity used.

r and k_e are assumed to be constant and thus additional investments made by the firm will not change its risk and return profiles,

Firm has an infinite life,

For a given value of the firm, the dividend per share and the earnings per share remain constant.

According to Walter, the market price of the share is taken as the sum of the present value of the future cash dividends and capital gains. His formula is based on the share valuation model and is arrived at in the following manner: -

Step 1: Market per share price of the firm is given as

$$P = D / (k_e - g)$$

Thus we have

$$k_e = D / P + g$$

$$\text{Since, } g = D P / P$$

$$\text{We have, } k_e = D / P + D P / P$$

$$\text{But since } D P = \frac{r (E - D)}{k_e}$$

(Since retained earnings is the only sources of finance), substituting the same,

We have

$$P = \frac{D + r (E - D) / K_e}{k_e - k_e}$$

Where,

P	=	market price per share
D	=	Dividend per share
E	=	Earning per share
r	=	Internal rate of return
k_e	=	Cost of equity capital
$D P$	=	Change in the price
g	=	Growth rate of earnings

Limitations

Most of the limitations for this model arise due to the assumptions made. They are: -

- The first assumption of exclusive financing by retained earnings make the model suitable only for all equity firms.
- Secondly, Walter assumes the return on investments to be constant. This again will not be true for firms making high investments
- Finally, Walter model on dividend policy ignores the business risk of the firm, which has a direct impact on the value of the firm. Thus, k cannot be assumed to be constant.

Gordon's Dividend Capitalization Model

Yet another model that has given importance to the dividend policy of the firm is the Gordon Model. Myron Gordon used the dividend capitalization approach to study the effect of the firms' dividend policy on the stock price.

Assumptions

The following are the assumptions based on which Gordon based the dividend policy model for firms: -

- The firm will be an all-equity firm with the new investment proposals being financed solely by the retained earnings.
- Return on Investment (r) and the cost of equity capital (k_e) remain constant
- Firm has an infinite life.
- The retention ratio remains constant and hence the growth rate also is constant ($g = br$)
- $k > br$ i.e., cost of equity capital is greater than the growth rate.

Gordon model assumes that the investors are rational and risk averse. They prefer certain returns to uncertain returns and thus put a premium to the certain returns and discount the uncertain returns. Thus, investors would prefer current dividends and avoid risk. Retained earnings involve risk and so the investor discounts the future dividends. This risk will also affect the stock value of the firm.

Gordon explains this preference for current income by the bird-in-hand argument. Since, a bird-in-hand is worth two in bush, the investors would prefer the income that they earn currently to that income in future which may be or may not be available. Thus, investors would prefer to pay a higher price for the stocks, which earn them current dividend income and would discount those stocks, which either postpones / reduce the current income. The discounting will differ depending on the retention rate (percentage of retained earnings) and the time.

Gordon's dividend capitalization model gave value of the stock as

$$P = \frac{E (1 - b)}{k_e - br}$$

Where,

P	=	share price
E	=	Earning per share
b	=	Retention ratio
$(1 - b)$	=	Dividend pay-out ratio
k_e	=	Cost of equity capital (cost of capital of the firm)
br	=	Growth rate (g) in the rate of return on investment
g	=	Growth rate of earnings

So, Gordon correlated that the firm's share value is positive with the payout ratio when $r_e > k_e$ and decreases with an increase in the payout ratio when $r > k_e$. Thus, firms with rate of return greater than the cost of capital should have a higher retention ratio and those firms, which have a rate of return less than the cost of capital, should have a lower retention ratio. The

dividend policy of firms which have a rate of return equal to the cost of capital will however not have any impact on its share value.

Miller & Modigliani Position

Miller and Modigliani have propounded the MM hypothesis to explain the irrelevance of a firm's dividend policy. This model, which was based on a few assumptions, sidelined the importance of the dividend policy and its effect thereof on the share price of the firm. According to the model, it is only the firm's investment policy that will have an impact on the share value of the firm and hence should be given more importance.

Critical Assumptions

It is pertinent to look into the assumptions on which this model is based, before going into the details of the model.

- The essence of a perfect market is that all investors are rational. In perfect market condition there is easy access to information and the flotation and the transaction costs do not exist. The securities are infinitely divisible and hence no single investor is large enough to influence the share value.
- It is assumed that there are no taxes, implying that there are no differential tax rates for the dividend income and the capital gains.
- There is a constant dividend policy of firm, which will not change the risk complexion nor the rate of return even in cases where the investments are funded by the retained earnings.
- It was also assumed that the investors are able to forecast the future earnings, the dividends and the share value of the firm with certainty. This assumption was however, dropped out of the model.

Based on these assumptions and using the process of arbitrage Miller and Modigliani have explained the irrelevance of the dividend policy. The process of arbitrage balances or completely offsets two transactions, which are entered into simultaneously. Arbitrage can be applied to the investments function of the firm. As mentioned earlier, firms have two options for utilizing its after tax profits (i) to retain the earnings and plough back for investment purposes (ii) distribute the earnings as cash dividends. If the firm selects the second option and declares dividend then it will have to raise capital for financing its investment decisions by selling new shares. Here, the arbitrage process will neutralize the increase in the share value due to the cash dividends by the issue of additional shares. This makes the investor indifferent to the dividend earnings and the capital gains since the share value of the firm depends more on the future earnings of the firm, than on its dividend policy. Thus, if there are two firms having similar risk and return profiles the market value of their shares will be similar in spite of different payout ratios.

Symbolically the Model is given as

In the first step the market price of the shares is equal to the sum of the present values of the dividend paid and the market price at the end of the period.

$$P_0 = \frac{1}{(1 + k_e)} (D_1 + P_1)$$

Where,

P_0	=	Current market price of the share
P_1	=	Market price of the share at the end of the period. (t = 1)
D_1	=	Dividends to be paid at the end of the period (t = 1)
k_e	=	Cost of equity capital.

With no external financing the total value of the firm will be as follows:

$${}_nP_0 = \frac{1}{(1 + k_e)} ({}_nD_1 + {}nP_1)$$

Where,

n	=	No. of shares outstanding
-----	---	---------------------------

Now, if the firm finances its investment decisions by rising additional capitals issuing n_1 new shares at the end of period (t = 1), then the capitalized value of the firm will be the sum of the dividends received at the end of the period and the value of the total outstanding shares at the end of the period less the value of the new shares, remains as it is. Thus we have,

$${}_nP_0 = \frac{1}{(1 + k_e)} ({}_nD_1 + (n + n_1) P_1 - n_1 P_1)$$

Firms will have to raise additional capital to fund their investment requirements, if its investment requirements is more than its retained earnings, additional equity capital ($n_1 P_1$) after utilizing its retained earnings is as follows:

$$n_1 P_1 = I - (E - {}_nD_1)$$

Where,

I	=	Total Investments required
${}_nD_1$	=	Total dividend paid
E	=	Earnings during the period

$$(E - {}_nD_1)$$

Simplifying the above equation we get,

$$n_1 P_1 = I - E + {}_nD_1$$

Substituting this value of the new shares in the equation we get,

$${}_nP_0 = \frac{1}{(1 + k_e)} ({}_nD_1 + (n + n_1) P_1 - I + E - {}_nD_1)$$

$${}_nP_0 = \frac{(n + n_1) P_1 - I + E}{(1 + k_e)}$$

Thus, according to the MM model, the market value of the share is not affected by the dividend policy and this is explicitly shown above.

Critical Analysis of the Assumptions

The MM approach to the irrelevance of dividend has been based on a few assumptions, which need to be evaluated critically especially since a perfect market and absence of flotation costs and transaction costs are situation, which do not happen in reality. Few assumptions have been critically viewed by as under: -

This assumption cannot be true, since in the real world the tax rate for the dividend income is higher than the tax rate applicable to the capital gains

The proceeds, which the firm gets from the issue of securities, will be net off the issue expenses. The total issue expenses, which include the underwriting expenses, brokerages, other marketing costs will be around 10 – 15 % of the total issue in India. With the costs of mobilizing capital from the primary market being high, these costs cannot be ignored.

This is an unrealistic assumption, since investors do have to incur certain transaction cost like the brokerage expenses while they dispose off their shares. Thus, if the investors are to equate the capital gains to the dividend income, they should sell off the shares at a higher price. In addition to this, the inconvenience and the uncertainty involved in the share price movements make the investors prefer current income by way of dividends to plough back their profits.

Sometimes the market conditions do effect the investment decisions of the firm. For instances, though a firm has profitable investment opportunities, the bad market condition may not allow it to mobilize the funds. IN such cases, the firms will have to depend on the retained earnings and have a low dividend pay-out ratio. IN still other cases, there may be certain sub-standard investment opportunities in which the firm will invest just because there is an easy access to funds from the market.

If the company has to raise funds from the market, it will have to sell the shares to the new shareholders at a price that is less than the prevailing market price. Thus, with the shares being under priced the firm will have to sell more shares to replace the dividend amount.

Rational Expectations Model

According to the rational expectation model, there would be no impact of the dividend declaration on the market price of the share as long as it is at the expected rate. However, it may show some adjustments in case the dividends declared are higher or lower than the expected level. For instance, when a firm declares dividends higher than what was expected, it would result in an upwards movement of the share price as there would be expectations of higher earnings and similarly expectations model suggests that alterations in the market price will not be necessary where the dividend meet the expectations and only in case of unexpected dividend will there be a change in the market price as stated above.

- Models in Dividend decisions
- Assumptions and Limitations in various models.

1. What are the options available for Dividend decision?
2. List the various models available for taking dividend decisions?
3. What do you mean by traditional approach to the dividend policy and what are its limitations?
4. How Walter Model is different from Gordon's Dividend Capitalization Model?
5. Explain in detail about Gordon's Dividend Capitalization Model?
6. Analyze critically of the Assumptions of MM model in dividend decision?
7. What are the critical assumptions of Miller & Modigliani Model?
8. What is the need for Rational Expectations Mo

[illegible]

LESSON 34

FINANCIAL RATIOS

Learning Objectives

- To understand what ratio means and the significance of ratios to different categories of people
- To learn various classifications of ratios and computation of the same.

Introduction

Financial Analysis can be undertaken by the management or owners of the firm, or by external people viz., creditors, investors and others. The nature of analysis and ratios used depend upon the perspective / need of the person computing the ratios.

Sl. No.	Analysed by	Area of Interest	Area of Finance
1	Trade Creditors	Firm's ability to meet short term claims	Liquidity Position
2	Lenders of the Long term debt	-Firm's ability to pay interest and repay principal amount -Relationship between various sources of funds -Future Solvency and Profitability	-Long-term Solvency -Consistency in Profitability -Capital Structure Relationships -Projected Financial Statements
3	Investors	Steady growth in earnings	-Present & future profitability -Influence of financial structure on earning ability and risk
4	Management	Most effective & efficient use of the resources and sound financial condition	Every aspect of financial analysis

Meaning of Ratio

The term Ratio refers to the numerical or quantitative relationship between two related items / variables / accounting figures.

Ratios help to make qualitative judgement. Ratios may be express as

- Proportion of numbers e.g. 2:1; or
- A 'Rate' or 'Time' e.g. 2 times; or
- Percentages e.g. 200%; or
- Fraction or Quotient / decimal e.g. $\frac{2}{3}$ or 0.667

Ratio analysis makes related information comparable. Ratios are helpful in ascertaining the financial condition of the firm, strengths and weaknesses of the firm.

A single ratio like an absolute figure is not of much use. Ratios need to be compared –

- Over a period of time i.e. trend or time series analysis
- With competitors, industry standards, selected firms of the industry (Cross-sectional analysis)
- Rule of thumb

Following are the steps involved in Ratio Analysis:

- Determination of the objective of the analysis
- Selection of relevant data from the financial statements
- Comparison of ratios computed for different points of time or comparison with firms of the same industry (inter-firm comparison) etc.
- Interpretation of the ratios

Types / Classification of Ratios

Traditionally ratios are classified based on statement/s used

- Balance Sheet Ratios
- Profit & Loss a/c Ratios
- Composite or Mixed or Inter-statement ratios

However, Functional Classification leads to a more meaningful discussion, understanding and analysis

- Liquidity Ratios
- Activity or Efficiency or Performance or Turnover Ratios
- Profitability Ratios
- Leverage or Solvency / Capital Structure Ratios

i Liquidity Ratios

Liquidity Ratios measure the ability of the firm to meet current / short-term obligations. They establish a relationship between cash and other current assets to current obligations. A firm should strike a proper balance between high liquidity with more cash balance (which means less profitability) and low liquidity (with problems of failure to meet obligations etc.)

Liquidity Ratios include (i) Current Ratio, (ii) Liquid or Acid Test Ratio or Quick Ratio, (iii) Cash Ratio or Absolute Liquid Ratio, (iv) Defensive-interval Ratio

- a. Current Ratio: Current Ratio establishes relationship between the current assets and current liabilities and measures the ability of the firm to meet current liabilities.

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

This ratio indicates the rupees of current assets available for each rupee of current liability / obligation. The higher the ratio, larger is the ability to meet current obligations and greater is the safety of funds of short-term creditors. Depending upon the industry the ratio may vary between 1.5 to 3.5 though the rule of thumb is 2.

- b. Liquid Ratio or Acid Test Ratio or Quick Ratio: Liquid Ratio measures the ability to meet the current liabilities from the current assets which are readily or quickly convertible into cash (Current Assets less Inventory & Pre-paid expenses). Inventory is not readily convertible into cash and hence to be excluded.

$$\text{Liquid Ratio} = \frac{\text{Liquid or Quick Assets}}{\text{Current Liabilities}}$$

It is called Acid Test Ratio since it is more severe and stringent test. Rule of thumb is 1.

- c. Cash Ratio / Absolute Liquid Ratio: Absolute Liquid Assets are considered here. Receivables have doubts about their realisability in time and hence they are excluded here.
Cash & Bank Balances + Marketable Securities

$$\text{Cash Ratio} = \frac{\text{Cash \& Bank Balances + Marketable Securities}}{\text{Current Liabilities}}$$

- d. Defensive-internal Ratio

This ratio measures the ability to meet projected daily operating expenditure

$$\text{Defensive-internal Ratio} = \frac{\text{Liquid Assets}}{\text{Projected daily cash requirements}}$$

Where,

Projected Cash Operating Expenditure may be ascertained by adding Cost of goods sold + Selling & administrative and other cash expenses less depreciation & other non-cash expenditure. The resultant figure will be in number of days.

i Leverage / Capital Structure Ratios

The process of magnifying the shareholders' return through the employment of debt is called "Financial Leverage" or "Trading on Equity". These ratios are called as financial ratios also. Leverage Ratios which are used to ascertain long-term solvency of a firm have two aspects:

- A. Debt Repaying capacity measured through Structural Ratios
- B. Interest paying capacity measured through Coverage Ratios

A. Structural Ratios

These ratios examine the soundness of the capital structure

- a. Debt-Equity Ratio

This ratio indicates the relative proportions of debt and equity in financing & claims against the assets of the firm.

$$\text{D/E Ratio} = \frac{\text{Debt}}{\text{Equity}} \text{ or } \frac{\text{Total Debt or Outsiders Funds or External Equities}}{\text{Shareholders' Equity or Net Worth or Internal Equities}}$$

The Debt Equity Ratio may relate only Long-term debt, in which case the formula is as below:

$$\text{D/E Ratio} = \frac{\text{Long-term Debt}}{\text{Shareholders' Equity}}$$

- b. Debt to Total Assets

The Proportion of the assets that are financed with debt is indicated by this ratio

$$\frac{\text{Debt}}{\text{Total Assets}}$$

- c. Debt to Capitalization Ratio

This is a link between the outsiders long-term debt and long-term funds in the firm.

$$\frac{\text{Debt}}{\text{Capital Employed (Total Debt + Net Worth)}} \text{ or } \frac{\text{Total Debt}}{\text{Net Assets (Total Assets - Current Liabilities)}}$$

- d. Proprietary Ratio

The Proportion of Total Assets financed by owners is indicated by this ratio.

$$\frac{\text{Net Worth}}{\text{Total Assets}}$$

- e. Capital Gearing Ratio

$$\frac{\text{Net Worth}}{\text{Fixed income bearing funds (debentures, preference capital, loans)}}$$

B. Coverage Ratios

These Ratios measure the ability of the firm to cover or meet the obligations of paying interest on its debt. They reflect the ability of the firm to service the claims of long-term creditors.

- a. Interest Coverage Ratio / Debt Service Ratio

It measures the debt servicing capacity of the firm.

$$\text{Interest Coverage} = \frac{\text{Earnings Before Interest \& Taxes (EBIT)}}{\text{Interest}}$$

Since taxes are calculated after interest, the earnings before taxes is taken. Since the resultant figure gives the number of times interest covered by the EBIT, it is also known as "Times interest earned ratio".

From the point of view of creditors, the larger the ratio, more assured is the payment of interest. For the firm a too high ratio means that the firm is too conservative in using

debt and not using credit to the best advantage of shareholders. On the contrary, low ratio is a dangerous indication of excessive debt and not able to assure payment of interest to creditors.

b. Total Cash Flow Coverage

Total Cash Flow Coverage Ratio is used for relating cash Resources to the fixed financial obligations.

$$\frac{\text{EBIT} + \text{Lease Payments} + \text{Depreciation} + \text{Non Cash Expenses}}{\text{Interest} + \text{Lease Payments} + \frac{\text{Preference Dividend} + \text{Loan Repayment}}{1 - \text{Tax Rate}}}$$

Since only the after-tax earnings is available to repay principal, the principal repayment is converted to a before-tax basis by dividing it by $1 - \text{Tax Rate}$.

iii Activity or Efficiency or Turnover Ratios

The efficiency with which the assets are utilized or converted into sales, is evaluated with these ratios. The speed at which assets are converted into sales, is reflected in these ratios.

a. Inventory or Stock Turnover Ratio

Inventory Turnover Ratio measures the velocity of the movement of goods. It measures the relationship between the Cost of Goods Sold and the Inventory level.

$$\text{Inventory Turnover} = \frac{\text{Cost of Goods Sold}}{\text{Average Inventory}}$$

Cost of goods sold = Opening stock + Purchases and Manufacturing Cost - Closing Stock

$$\text{Average Inventory} = \frac{\text{Opening Stock} + \text{Closing Stock}}{2}$$

The average time taken for clearing the stocks is calculated as follows:

$$\text{Inventory conversion period} = \frac{\text{Number of days in a year}}{\text{Inventory Turnover Ratio}}$$

Generally, high inventory ratio implies good inventory management. But it may indicate under-investment which means loss of business opportunities. Similarly, a low inventory turnover may be indicative of over-investment which is still worse evil. Hence what is required is the optimum inventory level.

The Inventory Turnover Ratio can be extended to the components of inventory: (i) Raw materials and (ii) Work-in-progress. This may examine the efficiency with which the raw materials are converted into work-in-progress and work-in-progress to finished goods. Raw Materials inventory is related to Materials Consumed and work-in-progress to the Cost of Goods Manufactured.

b. Debtors or (Accounts) Receivables Turnover Ratio and Average Collection Period

Debtors Turnover Ratio measures the efficiency at which the sales are converted into cash

$$\text{Debtors Turnover} = \frac{\text{Credit Sales}}{\text{Average Trade Debtors}} \quad \text{or} \quad \frac{\text{Total Sales}}{\text{Debtors}}$$

Average Collection Period:

Average Collection Period represents the average number of days for which a firm has to wait for converting a receivable into cash.

$$\text{Average Collection Period} = \frac{\text{Trade Debtors}}{\text{Average Sales per day}}$$

$$\text{Average Sales per day} = \frac{\text{Sales}}{\text{No. of days in a year}}$$

Hence,

$$\frac{\text{Trade Debtors}}{\text{Sales}} \times \text{No. of days in a year} \quad \text{OR}$$

$$\text{Average Collection Period} = \frac{\text{No. of days in a year}}{\text{Debtors Turnover}} \quad \text{or} \quad \frac{\text{No. of days in a year}}{\frac{\text{Sales}}{\text{Trade Debtors}}}$$

To supplement the information, Ageing Schedule may be used to ascertain the quality of the trade debtors to identify the areas of problems in debtors.

c. Assets Turnover Ratios

Assets Turnover Ratios measure the efficiency of the assets in generating sales

i. Total Assets Turnover

To ascertain the ability to generate sales from all financial resources

$$\frac{\text{Cost of Goods Sold or Sales}}{\text{Total Assets}}$$

ii. Fixed and Current Assets Turnover Ratio

To ascertain the adequacy of investment in fixed and current assets

$$\text{Fixed Assets Turnover Ratio} = \frac{\text{Cost of Goods Sold or Sales}}{\text{Fixed Assets}}$$

$$\text{Current Assets Turnover Ratio} = \frac{\text{Cost of Goods Sold or Sales}}{\text{Current Assets}}$$

iii. Working Capital Turnover Ratio

To ascertain the efficiency with which working capital is utilized in the business

$$\text{Working Capital Turnover Ratio} = \frac{\text{Cost of Goods Sold or Sales}}{\text{Net Working Capital}}$$

iv Profitability Ratios

All the stakeholders are interested in assessing the profitability of a firm, more interested are the owners and

the management. Profit is the final measure of performance of a firm. Profit is the basic thing for survival and growth of a business.

Profitability Ratios can be related to sales as well as investments.

A. Profitability in relation to Sales

Profitability ratios can be related to Sales in respect of Gross Profit Margin, Net Profit Margin and Expenses.

a. Gross Profit Margin to Sales ratio

$$\text{Gross Profit Margin} = \frac{\text{Gross Profit}}{\text{Sales}} \times 100 \quad \text{or} \quad \frac{\text{Sales} - \text{Cost of Goods Sold}}{\text{Sales}} \times 100$$

High G/P ratio is a sign of good management and low G/P ratio is a cause of worry to unless there is improvement in managing.

b. Net Profit Margin to Sales ratio

Net Profit Margin ratio is calculated as below:

$$\text{Net Profit Ratio} = \frac{\text{Net Profit after tax}}{\text{Sales}} \times 100$$

c. Expenses Ratios

These ratios establish relationship between various expenses and sales:

i. Cost of Goods Sold Ratio

$$\frac{\text{Cost of Goods Sold}}{\text{Net Sales}} \times 100$$

ii. Administrative Expenses Ratio

$$\frac{\text{Administrative Expenses}}{\text{Net Sales}} \times 100$$

iii. Selling & Distribution Expenses Ratio

$$\frac{\text{Selling \& Distribution Expenses}}{\text{Net Sales}} \times 100$$

iv. Operating Ratio

$$\frac{\text{Operating Cost}}{\text{Net Sales}} \times 100 \quad \text{or} \quad \frac{\text{Cost of Goods Sold} + \text{Operating Expenses}}{\text{Net Sales}} \times 100$$

B. Profitability in relation to Investments

These ratios relate Profit to Investments. Return On Investments (ROI) is related to three categories of Assets, Shareholders' Equity / Funds and Capital Employed:

a. Return on Assets

$$\text{ROA} = \frac{\text{Net Profit After Taxes} + \text{Interest}}{\text{Total Assets}} \quad \text{or} \quad \frac{\text{EBIT} (1 - T)}{\text{Total Assets}}$$

b. Return on Capital Employed (ROCE)

$$\text{ROCE} = \frac{\text{Net Profit after taxes}}{\text{Capital Employed}} \\ \text{(Non Current Liabilities} + \text{Owners' Equity)}$$

c. Return to Shareholders

i. Return on Total Shareholders' Equity

$$\frac{\text{Net Profit after Tax}}{\text{Shareholders' Funds}}$$

Shareholders' Funds = Equity Share Capital + Preference Share Capital + Reserves & Surplus – Accumulated losses, if any)

ii. Return on Equity Capital (ROE)

$$\text{ROE} = \frac{\text{Profit after tax} - \text{Preference Dividend}}{\text{Shareholders' Equity or Net Worth}}$$

iii. Earnings Per Share (EPS)

EPS is the amount equity holders can get on every share held

$$\text{EPS} = \frac{\text{Net Profit available to equity holders} \\ \text{(Net Profit after tax} - \text{Preference Dividend)}}{\text{Number of Equity Shares}}$$

iv. Dividend Per Share (DPS)

The earnings distributed to the shareholders as cash dividends

$$\text{DPS} = \frac{\text{Earnings paid to shareholders}}{\text{Number of Equity Shares}}$$

v. Dividend Payout Ratio

$$\text{Payout Ratio} = \frac{\text{Dividend Per Share}}{\text{Earnings Per Share}} \quad \text{or} \quad \frac{\text{DPS}}{\text{EPS}}$$

Or

$$\frac{\text{Total Dividend to Equity holders} \\ \text{(Cash Dividend)}}{\text{Total Net Profit belonging to Equity holders}}$$

vi. Earnings Yield and Dividend Yield

The yield is expressed in terms of the market value per share

$$\text{Earnings Yield} = \frac{\text{EPS}}{\text{Market Value Per Share}} \\ \text{(Earnings-Price Ratio)}$$

$$\text{Dividend Yield} = \frac{\text{DPS}}{\text{Market Value Per Share}}$$

vii. Price-Earnings (P/E) Ratio

P/E Ratio assesses a firm/s performance as expected by the investors. It is the reciprocal of Earnings Yield or Earnings Price Ratio.

$$\text{P/E Ratio} = \frac{\text{Market Price Per Share}}{\text{Earnings Per Share}}$$

viii. Market to Book Value Ratio

$$\frac{\text{Market Price Per Share}}{\text{Book Value Per Share}}$$

Points to Remember

- Du Pont analysis helps in understanding how the return on total assets is influenced by the net profit margin and the total assets turnover ratio.
- Several ratios and indicators have been used in some kind of a multivariate model in order to assess corporate excellence.

Review questions

1. Discuss briefly how ratios help different people in making financial decisions.
2. What are the major classification of ratios? How these ratios help in improving the financial performance of the organization?
3. Determine sales of a firm with the following data:
 Current ratio = 1.8
 Acid-test ratio = 1.5
 Current liabilities = 7,50,000
 Inventory turnover ratio = 6 times
4. A firm's current assets and current liabilities are 1,800 and 1,250 respectively. How much can it borrow on a short term basis without reducing the current ratio below 1.30.

Note

LESSON 35 : WORKING CAPITAL MANAGEMENT

Learning Objectives

- To understand about various concepts of working capital and its components, its significance, cash conversion cycle
- To learn about working capital financing strategies and integrated working capital policy
- To learn about major sources of working capital

Introduction

Capital required for a business can be classified under two main heads:

- i. Fixed Capital
- ii. Working Capital

Fixed capital / Long term funds is required to meet long term obligations namely purchase of fixed assets such as plant & machinery, land, building, furniture etc. Any business requires funds to meet short-term purposes such as purchase of raw materials, payment of wages and other day-to-day expenses. These funds are called Working capital. In short, Working Capital is the funds required to meet day-to-day operations of a business firm. And hence study of Working capital is considered to be very significant. An inefficient management of working capital leads to not only loss of profits but also to the closure of the business firm.

There are two concepts of Working capital namely,

1. Gross Working Capital (GWC)
2. Net Working Capital

Generally working capital refers to the gross working capital and represents funds invested in total current assets of the firm.

$$GWC = CA$$

where CA = Current Assets

Net Working Capital is often referred to as circulating capital and represents the excess of current assets over current liabilities. Current liabilities are short-term obligations which are to be paid in the ordinary course of the business within a short period of one accounting year.

$$NWC = CA - CL$$

where Current Assets - Current Liabilities

Net working capital is positive when current assets exceed current liabilities. It is negative when current liabilities exceed current assets.

Components of Working Capital

Constituents of Current Assets in the order of decreasing liquidity

1. Cash and bank balance
2. Investments (marketable securities)
3. Government and other Trustee securities (other than for long term purpose e.g. Sinking Fund, Gratuity Fund etc.)

4. Fixed deposits with banks (maturing within one year)
5. Receivables arising out of sales other than deferred receivables (including bills purchased and discounted by bankers)
6. Inventories:
 - a. Raw materials
 - b. Work-in-process
 - c. Stores and spares
 - d. Finished goods
7. Advance payment for tax
8. Pre-paid expenses
9. Advances for purchase of raw materials, components and consumable stores
10. Deposits kept with public bodies for normal business operation maturing within the normal operating cycle
11. Money receivable from contracted sale of fixed asset during the next 12 months

Constituents of Current Liabilities

1. Short term borrowings (including bills purchased and discounted from banks and others)
2. Unsecured loans maturing within one year
3. Public deposits maturing within one year
4. Sundry creditors (trade) for raw materials and consumable stores and spares
5. Interest and other charges due for payment
6. Advance/Progress payments from customers
7. Deposits from dealers, selling agents etc.
8. Instalments on term loans, deferred payments credits, debentures and long term deposits payable within one year
9. Statutory liabilities:
 - a. Provident fund dues
 - b. Provision for taxation
 - c. Sales tax, excise
 - d. Statutory obligations towards workers
10. Miscellaneous current liabilities:
 - a. Dividends
 - b. Liabilities for expenses
 - c. Gratuity payable within one year
 - d. Any other payment due within 12 months

Significance of Working Capital

Gradual increase in profit results in capital growth of the firm. Increasing sales volume results in increasing profits. Sales is done on either cash or credit basis. Sale of goods will not be converted into cash immediately when sales is done on credit

basis. Therefore additional capital is required to have uninterrupted business operations as the sales amount gets locked up in current assets like accounts receivable, inventory etc. The firm needs extra funds for carrying out regular operation on day-to-day basis till accounts receivables are converted into cash and hence the need for working capital.

The Cash Conversion Cycle

The most liquid asset is cash in hand and cash at bank. The time required to complete the following cycle of events in case of a manufacturing firm is called the cash conversion cycle or the operating cycle.

1. Conversion of cash into raw materials
2. Conversion of raw materials into work in process
3. Conversion of work in process into finished goods
4. Conversion of finished goods into debtors and bills receivables through sales
5. Conversion of debtors and bills receivables into cash

Operating cycle in case of a trading firm consists of following events

1. Cash into inventories
2. Inventories into accounts receivable
3. Accounts receivable into cash

Service and financial firms has the operating cycle consisting of

1. Conversion of cash into debtors
2. Conversion of debtors into cash

Level of Current Assets

The level of investment in current assets determines the working capital policy. A business firm can adapt any of the following working capital policies:

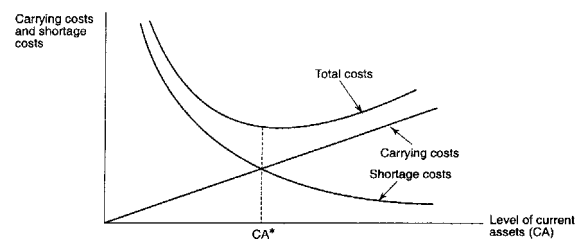
1. Conservative working capital policy
2. Aggressive working capital policy
3. Moderate working capital policy

Under Conservative approach, the firm carries high investment in current assets such as cash, marketable securities and carries large amount of inventories and grants generous terms of credit to customers resulting in a high level of debtors. The consequences of conservative working capital policy are quick deliveries to customers and more sales due to generous credit terms.

Under Aggressive working capital policy, investment in current assets is very low. The firm keeps less amount of cash and marketable securities, manages with less inventories and tight credit terms resulting in low level of debtors. The consequences of aggressive working capital policy are frequent production stoppages, delayed deliveries to customers and loss of sales.

A trade off between two costs namely carrying cost and shortage cost determines the optimal level of current assets. Costs that rise with current assets i.e. that cost of financing a higher level of current assets form carrying costs. Shortage costs are in the form of disruption in production schedule, loss of sales and loss of goodwill.

The optimum level of current assets is denoted by the total costs (= carrying costs + shortage costs) minimized at that level.



Working capital financing strategies

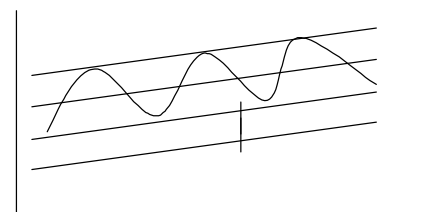
After determining the level of current assets, the firm must determine how these should be financed.

Investment in current assets can be broken into two parts

1. Permanent current assets
2. Temporary current assets

A firm requires a certain amount of current assets to meet even the minimum level of sales where as temporary current assets reflects a variable component that moves in line with seasonal fluctuations.

Several strategies are available for financing capital requirements.



Capital requirements and their financing

The fixed proportion of working capital should be generally financed from the fixed capital sources while the temporary or variable working capital requirements of a firm may be met from the short term sources of capital. Based on this idea, we have 3 strategies possible.

Strategy A

Long term financing is used to meet fixed asset requirement as well as peak working capital requirement. When the working capital requirement is less than its peak level, the surplus is invested in liquid assets(cash & marketable securities)

Strategy B

Long term financing is used to meet fixed asset requirements, permanent working capital requirement, and a portion of fluctuating working capital requirement.

During seasonal upswings, short term financing is used. During seasonal downswings, surplus is invested in liquid assets.

Strategy C

Long term financing is used to meet fixed asset requirement and permanent working capital requirement. Short term financing is used to meet fluctuating working capital requirement.

Integrated Working Capital Policy

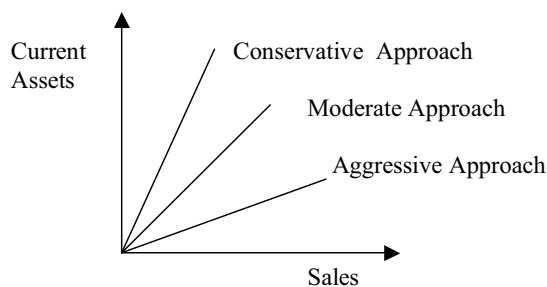
Working capital requirement of a firm is determined by considering two questions in mind.

1. What should be the level of current assets in relation to sales?
2. What should be the ratio of long term and short term financing?

Current Asset Policy

Question 1 tell us what should be the level of current assets to be maintained by the firm and hence it gives rise to three types of current asset policy namely

- a. Conservative approach : is carrying more amount of current assets in relation to sales which results in more carrying costs, relaxed credit terms and period and hence less turnover.
- b. Moderate approach : is always maintaining required amount of current assets depending upon sales.
- c. Aggressive Approach :is managing with less current assets in relation to sales and hence may result in more turnover, stringent credit terms and may lead to loss of customer goodwill and low liquidity.



Current Assets Policy graph

Current financing policy

Question 2 is about how the currents should be financed, either long term or short term financing. Current assets being financed using long term funds namely Equity shares, Debentures results in more costs of capital and relatively less profits whereas short term financing namely short term loans from banks and financial institutions, overdrafts, trade credit, commercial paper etc. results in more profits and relatively less costs

A judicious mix of Current Asset policy and Current Asset Financing Policy gives rise to an integrate Working Capital policy.

Current Asset Financing policy	Conservative	Moderate Working Capital Policy	Conservative Working Capital Policy
	Aggressive	Aggressive Working Capital Policy	Moderate Working Capital Policy
		Aggressive	Conservative
		Current Asset Policy	

Thus we arrive at the following :

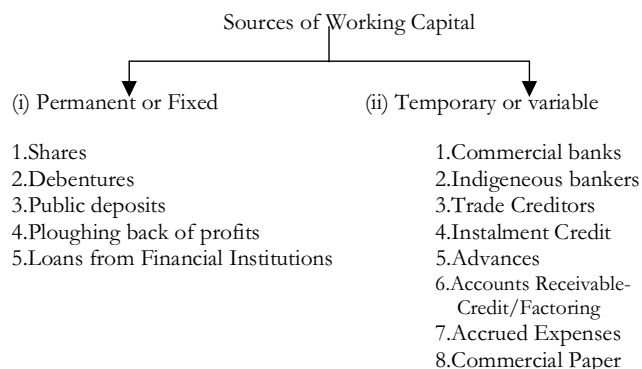
Aggressive Working capital is a mix of Aggressive Current Asset policy and Aggressive Current Asset Financing Policy. Firms with short operating cycle can adopt this policy generally.

Conservative Working capital is a mix of Conservative Current Asset policy and Conservative Current Asset Financing Policy. Firms having long operating cycle namely manufacturing companies can adopt this policy to decouple production and distribution.

Moderate Working capital is a mix of Aggressive Current Asset policy and Conservative

Current Asset Financing Policy or a mix of Conservative Current Asset policy and Aggressive Current Asset Financing Policy.

Sources for Financing Working Capital



Sources of Permanent or Fixed or Long Term Working Capital

1. Shares and Debentures

A firm can issue various types of shares such as equity shares, preference shares and deferred shares. According to Companies Act 1956, a public company cannot issue deferred shares. Equity shares do not have any fixed commitment charge. Dividend to equity shareholders is paid after fixed rate of interest to debentures holders and dividend at fixed rate to preference shareholders are met. Repayment of capital at the time of winding up of the company is done in the same priority and hence equity shareholders are eligible to residual income.

2. Debentures

A debenture is an instrument issued by the company acknowledging its debt to its holder. It is a long term borrowing and the debenture holders are the creditors of the company. Interest on debentures is a charge against profit and loss account. Debentures may be of different kinds such as secured, unsecured, redeemable, irredeemable, convertible and non convertible. Interest on debentures is a tax deductible expense.

3. Public Deposits

Public deposits are the fixed deposits accepted by a business enterprise directly from the public. Non-banking concerns cannot borrow by way of public deposits more than 25% of its paid up capital and free reserves.

4. Ploughing Back of Profits

It refers to the reinvestments by concern of its surplus earnings in its business. And is an internal source of finance. It is the cost free source of finance, gains confidence of the public and there is no dilution of control.

But excessive resort to ploughing back of profits may lead to monopolies, misuse of funds, over capitalization and speculation etc.

5. Loans from Financial Institutions

Financial institutions such as Commercial banks, Life Insurance Corporation, Industrial Finance Corporation of India, State Financial Corporation (LIC), State Industrial Development Corporation (SIDC), Industrial Development Bank of India (IDBI) etc. provide short term medium term and long term loans. This source of finance is more suitable to meet the medium term demands of working capital. Interest is charged on such loans at a fixed rate and the amount of the loan is to be repaid by way of instalments in a number of years.

Sources of Short term Working Capital (Temporary or Variable)

1. Indigenous Bankers

Private money lenders were the major source of finance prior to the establishment of commercial banks. They used to charge very high rates of interest. Even today some business houses depend upon indigenous bankers for their working capital requirement.

2. Trade Credit

Trade credit refers to the credit extended by the suppliers of goods in the normal course of business. The credit worthiness of a firm and the confidence of its suppliers are the main basis of securing trade credit.

The main advantages of trade credit as a source of short term finance are

- i. It is an easy and convenient method of finance
- ii. It is flexible as the credit increases with the growth of the firm
- iii. It is informal and spontaneous source of finance

Major disadvantage of this method is charging of higher prices by the suppliers and loss of cash discount.

3. Instalment Credit

In this method, the assets are purchased and the possession of goods is taken immediately but the payment is made in instalments over a pre determined period of time. Normally, interest is charged on the unpaid price or it may be adjusted in the price.

4. Advances

Generally firms manufacturing industrial products having long production cycle prefer to take advances from their customers against their orders. It is the cheapest short term source of finance, thus enables firms to minimize their investment in working capital.

5. Factoring or Accounts Receivable Credit

A commercial bank may provide finance by discounting the bills or invoices of its customers. Thus a firm gets immediate payment for sales made on credit.

A factor is a financial institution which offers services relating to management and financing of debts arising out of credit sales.

Factor render various services including maintenance of sales ledger, collection of accounts receivables, credit control and protection from bad debts, provision of finance and rendering of advisory services to their clients. Factoring may be on a recourse basis, where the risk of bad debts is borne by the client or on a non-recourse basis, where the risk of credit is borne by the factor.

Major disadvantages of factoring are:

- i. The high cost of factoring as compared to other sources of short term finance
- ii. The perception of financial weakness about the firm availing factoring services
- iii. Adverse impact of tough stance taken by factor, against a defaulting buyer, upon the borrower resulting into reduces future sales

6. Accrued Expenses

Accrued expenses are the expenses which have been incurred but not yet due and hence not yet paid also. It represents a liability that a firm has to pay for the services already received by it. It includes wages and salaries, interest and taxes. It is a cost free source of financing.

Wages and salaries are usually paid on monthly, fortnightly or weekly basis for the services already rendered by employees.

Income tax is paid periodically much after the profits have been earned. Interest is paid periodically while the funds are used continuously by the firm.

The amount of accruals varies with the change in the level of activity of a firm. When the activity level expands, accruals also increase and hence they provide a spontaneous source of financing.

7. Deferred Incomes

Deferred incomes are incomes received in advance before supplying goods or services. These funds increase the liquidity of a firm. Firms having good demand for its products and services can demand deferred incomes.

8. Commercial Paper

Commercial Paper represents unsecured promissory notes issued by firms to raise short term funds. Only large companies enjoying high credit rating and sound financial health can issue commercial paper to raise short term funds.

RBI has laid a number of conditions to determine eligibility of a company for the issue of a commercial paper. Only a company which is listed on the stock exchange has a net worth of at least Rs.10 crores and a maximum permissible bank finance of Rs.25 crores can issue commercial paper not exceeding 30 percent of its working capital limit.

It is sold at a discount from its face value and redeemed at face value on its maturity. Hence the cost of raising funds in this way, is a function of the amount of discount and the period of maturity and no interest rate is provided by the RBI for this purpose.

Commercial papers are normally bought by investors including banks, insurance companies, unit trusts and firms to invest surplus funds for a short period.

A credit rating agency called CRISIL has been set up in India by ICICI and UTI to rate commercial papers.

Commercial paper is a cheaper source of raising short term finance as compared to bank credit. However, it can be used as a source of finance only by large companies enjoying high credit rating and sound financial health.

Main disadvantage is that it cannot be redeemed before the maturity date even if the issuing firm has surplus funds to pay back.

Points to Remember

- A deposit made by one company with another, normally for a period upto six months, is referred to as an inter- corporate deposit. Three types of deposits are call deposits, 3-months and 6-months deposits.
- The cost of trade credit depends on the terms of credit offered by the supplier.

Review Questions

1. Define the concept of working capital. What are the components of working capital?
2. What is the significance of working capital. Explain the cash conversion cycle.
3. Explain working capital financing strategies.
4. What are the sources of financing working capital? Explain.

Note

LESSON 36 :

BANK CREDIT

Learning Objectives

- To study major forms of working capital finance provided by the banks
- To note down basic principles of bank financing –3Cs
- To know the recommendations made by Tandon and Chore committees
- To learn about credit authorization scheme

Bank Credit Working Capital Finance by Commercial Banks)

Bank credit is the primary institutional source of working capital finance in India.

The different forms in which the bank normally provide loans and advances are as follows:

- a. Loans
- b. Cash credits
- c. Overdrafts
- d. Purchasing and discounting of bills

a. Loans

When a bank makes an advance in lump-sum against some security it is called a loan. Commercial banks generally provide short term loans up to one year for meeting working capital requirements. The term loans may be either medium-term or long term loans. In case of a loan, a specified amount is sanctioned by the bank to the customer.

The entire loan amount is paid to the borrower either in cash or credit to his account. The borrower is required to pay interest on the entire amount of the loan from the date of the sanction. A loan may be repayable in lump sum or instalments. Interest on loans is calculated at quarterly rests and where repayments are stipulate in instalments, the interest is calculated at quarterly rests on the reduced balances.

b. Cash Credits

A cash credit is an arrangement by which a bank allows his customer to borrow money up to a certain limit against some tangible securities or guarantees. The interest is charged on the daily balance and not on the entire amount of the account. Hence it is the most favourite mode of borrowing by industrial and commercial concerns.

c. Overdrafts

Overdraft means an agreement with a bank by which a current account holder is allowed to withdraw more than the balance to his credit up to a certain limit. The interest is charged on daily overdrawn balances.

The main difference between cash credit and overdraft is that overdraft is allowed for a short period and is a temporary accommodation whereas the cash credit is allowed for a longer

period. Overdraft either can be clean overdraft, partly secured or fully secured.

d. Purchasing and Discounting of bills:

In this case, a bank lends without any collateral security. The seller draws a bill of exchange on the buyer of goods on credit. Such a bill may be either a clean bill or a documentary bill which is accompanied by documents of title to goods such as railway receipts. The bank purchases the bills payable on demand and credits the customer's account with the amount of bill less discount. At the maturity of the bills, bank presents the bill to its acceptor for payment. In case the bill discounted is dishonoured by non-payment, the bank recovers the full amount of the bill from the customer along with expenses in that connection.

Basic Principles of Bank Financing -3Cs

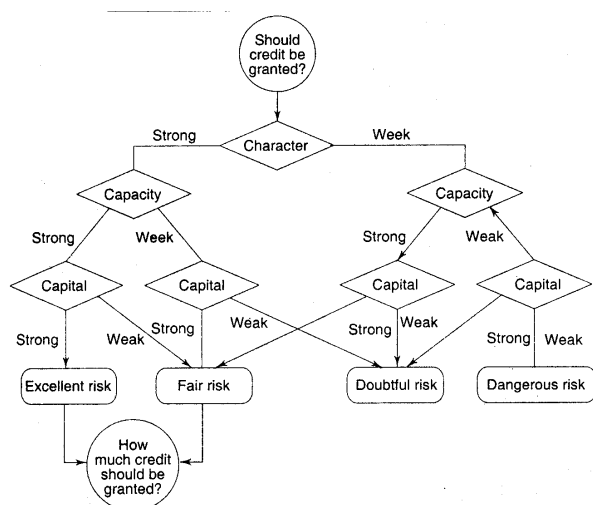
The traditional approach to credit analysis calls for assessing a prospective customer in terms of the '5Cs of credit'

1. Character: The willingness of the customer to honour his obligations. It reflects integrity, a moral attribute that is considered very important by credit managers.
2. Capacity: The ability of the customer to meet credit obligations from the operating cash flows
3. Capital: The financial reserves of the customer. If the customer has difficulty in meeting his credit obligations from its operating cash flows, the focus shifts to its capital.
4. Collateral: The security offered by the customer in the form of pledged assets
5. Conditions: The general economic conditions that affect the customer

A firm may rely on the following sources to obtain information on the 5Cs.

1. Financial Statements: provide useful insights into the creditworthiness of the customer. Ratios such as current ratio, acid-test ratio, debt-equity ratio, EBIT to total assets ratio and return on equity
2. Bank References: To ensure higher degree of candour, the customer's banker may be approached indirectly through the bank of the firm granting credit.
3. Experience of the firm: If the firm had previous dealings with the customer, then it is worth: How prompt has the customer been in making payments? How will has the customer honoured his word in the past? Where the customer is being approached for the first time, the impression of the company's sales personnel is useful.
4. Prices and Yield on securities: Higher the price earnings multiple and lower the yield on bonds, other things being equal, lower will be the credit risk.

For the sake of simplicity, only 3Cs namely Character, Capacity and capital are considered. For judging a customer on these dimensions, the credit analyst may use quantitative measures (like financial ratios) and qualitative assessments (like 'trustworthiness')



Tandon Committee Report

Reserve Bank of India setup a committee under the chairmanship of Shri P.L. Tandon in July 1974.

The terms of reference of the Committee were:

1. To suggest guidelines for commercial banks to follow up and supervise credit from the point of view of ensuring proper end use of funds and keeping a watch on the safety of advances;
2. To suggest the type of operational data and other information that may be obtained by banks periodically from the borrowers and by the Reserve Bank of India from the leading banks;
3. To make suggestions for prescribing inventory norms for the different industries, both in the private and public sectors and indicate the broad criteria for deviating from these norms ;
4. To make recommendations regarding resources for financing the minimum working capital requirements ;
5. To suggest criteria regarding satisfactory' capital structure and sound financial basis in relation to borrowings ;
6. To make recommendations as to whether the existing pattern of financing working capital requirements by cash credit/overdraft system etc., requires to be modified, if so, to suggest suitable modifications.

The committee was of the opinion that:

- i. Bank credit is extended on the amount of security available and not according to the level of operations of the customer
- ii. Bank credit instead of being taken as a supplementary to other sources of finance is treated as the first source of finance. Although the Committee recommended the

continuation of the existing cash credit system, it suggested certain modifications so as to control the bank finance.

The banks should get the information regarding the operational plans of customer in advance so as to carry a realistic appraisal of such plans and the banks should also know the end use of bank credit so that the finances are used only for purposes for which they are lent.

The recommendations of the committee regarding lending norms have been suggested under alternatives. According to the first method, the borrower will have to contribute a minimum of 25% the working capital gap from long-term funds, i.e., owned funds and term borrowing; this will give minimum current ratio of 1.17 : 1

Under the second method of the borrower will have to provide a minimum. of 25% of the total current assets from long-term funds; this will give a minimum current ratio of 1.33:1. In the third method, the borrower's contribution from long-term funds will be to the extent of the entire core current assets and a minimum of 25% of the balance current assets, thus strengthening the current further.

Illustration:

	Rs.
Total current assets required	40,000
Current liabilities other than bank borrowings	10,000
Core current assets	5,000
1st Method	
Total current assets required	40,000
Less current liabilities	10,000
Working Capital Gap	30,000
Less 25% from Long term sources	7,500
Maximum permissible bank borrowings	22,500
2nd Method	
Current Assets required	40,000
Less 25% to be provided from long term funds	10,000
Less Current Liabilities	30,000
Maximum permissible bank borrowings	20,000
3rd Method	
Current assets	40,000
Less Core Current assets	5,000
	35,000
Less 25% to be provided from long term funds	8,750
	26,250
Less current liabilities	10,000
Maximum permissible bank borrowings	16,250

Computation of Maximum Permissible Bank Finance (MPBF) :

The Tandon Committee had suggested three methods for determining the maximum permissible bank finance(MPBF). They are

Method 1: $MPBF = 0.75(CA - CL)$

Method 2: $MPBF = 0.75(CA) - CL$

Method 3: $MPBF = 0.75(CA - CCA) - CL$

Where CCA=Core Current Assets- this represents the permanent component of working capital

Illustration:

The following are the data of Mercury Company Limited

Current Assets	Rs.(in millions)
Raw material	18
Work in process	5
Finished goods	10
Receivables(including bills Discounted)	15
Other current assets	2
	—
	50
	—
Current Liabilities	
Trade Creditors	12
Other current liabilities	3
Bank borrowings(including Bills discounted)	25
	—
	40
	—

MPBF for Mercury Company Limited as per above methods are:

Method 1: $0.75(\text{CA}-\text{CL}) = 0.75(50-15) = \text{Rs.}26.25$ million

Method 2: $0.75(\text{CA})-\text{CL} = 0.75(50)-15 = \text{Rs.}22.5$ million

Method 3: $0.75(\text{CA}-\text{CCA})-\text{CL} = 0.75(50-20)-15 = \text{Rs.}7.5$ million

Method 2 is adopted and the minimum current ratio works out to be 1.33.

Under special consideration, within the MPBF so arrived at in terms of the present guidelines, banks/consortia/syndicates are required to restrict sanction of cash credit limits to borrowers up to a certain portion (25% of the MPBF). Where borrowers desire to avail of bank credit for the balance portion (75% of the MPBF), or any part thereof, this will be considered on merit by banks/consortia/syndicates in the form of a short term loan repayable on demand for working capital purpose for a stipulated period.

Chore Committee Report

The Reserve Bank of India in March, 1979 appointed another committee under the chairmanship of Shri K.B. Chore to review the working of cash credit system in recent years with particular reference to the gap between sanctioned limits and the extent of their utilisation and also to suggest alternative type of credit facilities which should ensure greater credit discipline.

The important recommendations of the Committee are as follows:

1. The banks should obtain quarterly statements in the prescribed format from all borrowers having working capital credit limits of Rs. 50 lacs and above.
2. The banks should undertake a periodical review of limits of Rs. 10 lacs and above.
3. The banks should not bifurcate cash credit accounts into demand loan and cash credit components.
4. If a borrower does not submit the quarterly returns in time the banks may charge penal interest of one per cent on the total amount outstanding for the period of default.

5. Banks should discourage sanction of temporary limits by charging additional one per cent interest over the normal rate on these limits.
6. The banks should fix separate credit limits for peak level and non-peak level, wherever possible.
7. Banks should take steps to convert cash credit limits into bill limits for financing sales.

Credit Authorization Scheme

Under the scheme, all selected commercial banks were advised to obtain the RBI's prior authorization before sanctioning any credit limit of Rs.1 crore or more to an single party or any limit that would take the total limits enjoyed by such party from the entire banking system (including co-operating sector) as a whole to Rs.1 crore or more. The appraisal on the part of the banks related to need of funds, security for the advance and suitability of terms and conditions is taken for granted. Normally authorization is given within 3 days of the receipt of application. Rejections are few and mainly related to the regular activity of the borrower, intermediate trading activities and double financing.

Since 1965, the Credit Authorisation Scheme (CAS) has been playing a significant role as an effective instrument of credit regulation.

Rationalisation of the Scheme

RBI advised commercial banks in June 1970 to collect and examine data relating to

- a. Utilisation of existing credit limits by borrowers
- b. Total working capital requirements and bank finance permissible together with the borrower's ability to meet the gap between the two.
- c. Comparative financial position for the last 3 years
- d. Cash flow
- e. In respect of term loans, project cost and sources of financing the project.

CAS Form of Application

The forms prescribed under the Credit Authorization Scheme, particularly relating to the assessment of working capital requirements, considers the following important aspects of lending.

- a. Examination of its operation cycle and total working capital requirement is necessary, as finance is required by a company for its total current operations and not merely inventory or receivables.
- b. Assessment of the company's peak requirement of current assets as balance sheet changes indicate only the movement in funds between 2 years.
- c. Adjustments for other available sources of finance such as sundry creditors, advances received etc. so as to arrive at
 - i. networking capital requirements of the borrowing company
 - ii. the peak level of current assets to maintain a given level of production

- d. Security and drawing power(after providing usual margins) to arrive at permissible bank finance
- e. Margin requirement i.e the gap between the total working capital requirement and credit limits available (including those proposed as per the application submitted to the RBI) to be met out of current working capital
- f. margin requirement and credit limits available (including as per the application submitted to RBI) to meet the current working capital surplus and/or other long term sources of finance.

The objective of the Credit Authorization Scheme is in the nature of credit planning and control than credit appraisal.

Points to Remember

- While banks can devise their own information and reporting system they largely follow the system recommended by the Chore committee.
- The lack of legal hassles and bureaucratic red tape makes an intercorporate deposit transaction very convenient.

Review Questions

1. How does banks help financing working capital requirements of an organization?
2. Explain basic principles of bank financing.
3. What are the recommendations made by Tandon Committee. Explain with illustration.
4. Explain Chore committee recommendations.
5. What is Credit Auhorisation Scheme? What are the important aspects of lending need to be considered

Note

“The lesson content has been compiled from various sources in public domain including but not limited to the internet for the convenience of the users. The university has no proprietary right on the same.”



Rai Technology University

ENGINEERING MINDS

Rai Technology University Campus

Dhodballapur Nelmangala Road, SH -74, Off Highway 207, Dhodballapur Taluk, Bangalore - 561204

E-mail: info@raitechuniversity.in | Web: www.raitechuniversity.in